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Reunião de Pesquisadores em Doença
de Alzheimer e Desordens Relacionadas

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BIOMARKERS

TECHNICAL AREA: BIOMARKERS

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A highly sensitive Alzheimer's disease screening method through an AI-assisted blood biomarkers panel

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Background: Alzheimer's Disease (AD) is the leading cause of dementia, affecting an estimated 55 million people globally. Despite its high incidence, up to 75% of dementia cases remain undiagnosed, particularly in low- and middle-income countries. Current diagnostic methods are invasive, costly, and remain largely inaccessible. **Objective:** Our project aimed to develop an accurate, accessible blood biomarker panel for AD diagnosis using cutting-edge Machine Learning (ML). **Methods:** We analyzed a comprehensive blood biomarker data from 800 individuals participating in the Alzheimer's Disease Neuroimaging Initiative (ADNI), divided in three groups—Cognitively Normal, Mild Cognitive Impairment (MCI), and AD. To identify optimal diagnostic signatures, 187 potential biomarkers (proteomic, routine labs, epidemiological) were analyzed via various ML techniques, identifying 20 promising markers. These were then tested in a 30-subject validation cohort in Belo Horizonte, Brazil: 15 diagnosed within the AD disease spectrum with confirmed CSF pathology, and 15 healthy controls. **Results:** Initial ADNI models achieved an accuracy of 91%, with 99% sensitivity and 97% specificity. However, validation in the 30-patient cohort showed lower performance, likely due to sample size. Subsequent model refinement to this sample size led to an optimal 5-feature panel demonstrating an accuracy of 83%, sensitivity of 93%, specificity of 78%, and an F1-score of 85%. **Discussion:** Our non-invasive and scalable AI-powered blood panel could significantly improve AD diagnosis and patient management, especially in primary care. **Conclusion:** Our work aims to improve AD diagnosis with a highly sensitive, accessible, affordable blood test, enhancing diagnostic rates and global patient outcomes.

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Plasma Aβ42/pTau217 and pTau217 accurately predict CSF AD biomarker profiles in a Brazilian SCD and MCI cohort

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Introduction: Early detection of Alzheimer's disease (AD) is essential, yet validation of blood-based biomarkers in genetically diverse populations remains limited. This study assessed the utility of plasma biomarkers in predicting CSF-defined AD pathology in a Brazilian cohort with Subjective Cognitive Decline (SCD) and Mild Cognitive Impairment (MCI). **Methods:** Seventy-eight participants underwent neuropsychological testing, MRI, and lumbar puncture. CSF Aβ42, p-Tau181, and t-Tau were measured using Roche's Elecsys immunoassay. Plasma Aβ42, Aβ40, p-Tau181, p-Tau217, and t-Tau were quantified via SIMOA HD-X. Logistic regressions evaluated plasma biomarkers as predictors of CSF p-Tau181/Aβ42 and t-Tau/Aβ42 ratios, adjusting for age, sex, and APOE ε4 status. ROC curves assessed classification performance. **Results:** All models using CSF p-Tau181/Aβ42 and t-Tau/Aβ42 as outcomes were statistically significant when plasma Aβ42/pTau217 or pTau217 was used independently as a predictor. APOE ε4 was also a significant covariate in models including pTau217. For altered CSF t-Tau/Aβ42, AUCs were 0.89 for both plasma Aβ42/pTau217 (cutoff=43.42; sensitivity=80.0%; specificity=95.3%) and pTau217 (cutoff=0.152; sensitivity=80.0%; specificity=88.4%). For altered CSF p-Tau181/Aβ42, AUCs were 0.88 (Aβ42/pTau217) and 0.83 (pTau217), with no significant difference (DeLong test, p=0.91). **Discussion and Conclusion:** Plasma Aβ42/pTau217 and pTau217 are highly accurate and statistically equivalent biomarkers of CSF-defined AD pathology. These findings support their clinical utility for early detection and monitoring of Alzheimer's disease in genetically diverse populations.

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Plasma p-Tau217 accurately identifies amyloid burden in a Brazilian cohort with low education

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Blood-based biomarkers (BBMs) have emerged as promising tools to enhance Alzheimer's disease (AD) diagnosis. Despite two-thirds of dementia cases occurring in the Global South, research on BBMs has predominantly focused on populations from the Global North. This geographical disparity hinders our understanding of BBM performance in diverse populations. To address this, we evaluated the diagnostic properties of AD BBMs in a real-world memory clinic from Brazil, one of the largest countries in the Global South. We measured blood and cerebrospinal fluid (CSF) biomarkers—amyloid-β (Aβ)40, Aβ42, phosphorylated tau (p-tau) 217, neurofilament light (NfL) chain, and GFAP—in 59 individuals. Sample comprised 20 cognitively unimpaired (CU) individuals, 22 with AD dementia, and 17 with vascular dementia (VaD). We compared BBM levels across diagnostic groups and assessed their discriminative ability

for AD. Notably, individuals with VaD and AD had lower educational levels (6.8 ± 3.0) compared to CU individuals (17.3 ± 6.9). Among the BBMs tested, plasma p-tau217 demonstrated the best performance, exhibiting high accuracy in differentiating CU from AD (AUC 0.96) and A β pathology (AUC 0.98). However, the ability of AD BBMs to distinguish between AD and VaD was lower than expected (AUC from 0.52 to 0.79), particularly when compared to studies from the Global North. Our findings highlight the potential utility of BBMs for AD diagnosis in real-world settings within Brazil. However, they also underscore the need for proper implementation and validation of these biomarkers within these populations to ensure accurate and reliable results.

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Logopenic variant primary progressive aphasia with suspected non-Alzheimer pathology

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Case Presentation: An intellectually active, highly educated octogenarian developed progressive language difficulties, including speech pauses and impaired sentence repetition. Comprehension of isolated words, as well as reading and writing abilities, remained preserved—consistent with the clinical profile of the logopenic variant of primary progressive aphasia (lvPPA). Notably, the initial symptom was selective auditory verbal agnosia for Portuguese, his second language, while comprehension in English, his native language, remained intact. Structural MRI and FDG-PET imaging revealed no evidence of temporoparietal atrophy or cortical hypometabolism. Plasma biomarkers associated with Alzheimer's disease (A β 42/40 ratio and phosphorylated tau-217) were within normal limits.

Discussion: LvPPA is commonly associated with Alzheimer's pathology. However, this case exhibits an atypical presentation marked by bilingual asymmetry, negative Alzheimer's biomarkers, and preserved neuroimaging. Such findings raise the possibility of alternative underlying pathologies. Primary age-related tauopathy and argyrophilic grain disease are considered potential etiologies, both of which can mimic lvPPA. **Final**

Considerations: This case illustrates the importance of a multimodal diagnostic approach, combining clinical, imaging, and biomarker data. The presence of bilingual asymmetry highlights the need to account for language dominance and acquisition in neurodegenerative syndromes. Furthermore, in the era of disease-modifying therapies targeting Alzheimer's pathology, the accurate differentiation between Alzheimer's and non-Alzheimer's PPA variants becomes crucial, emphasizing the growing need for validated biomarkers for non-Alzheimer's neurodegenerative conditions.

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Cognitive stimulation combined with education for active aging of healthy older adults using the moove sensory digital platform

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Introduction: Education for aging, combined with technologies and cognitive games, promotes well-being, social engagement and cognitive stimulation in older adults. **Objective:** To promote access to cognitive stimulation games, combined with education on aging, through educational workshops for older adults. **Methods:** A workshop comprising 10 weekly 90-min sessions was run. Data collection was conducted online and involved application of a sociodemographics questionnaire. The protocol included the Cognitive Function Instrument (CFI), Geriatric Anxiety Inventory (GAI-SF), Geriatric Depression Scale (GDS-15), Satisfaction with Life Scale (SL), Social Support Scale (MOS-SSS), and Lubben Social Network Scale (LSNS). Sessions were run by two monitors who held graduate degrees in Gerontology. A total of 31 older adults took part (mean age=67; SD=6.40 years). Most participants were women (96.77%) and had completed higher education (77.42%). **Results:** On the post-test, participants had a mean score for cognitive performance of 416.43 points (scale 0-1000), most notably for impulsivity control (55.93 points), followed by reaction (43.93) and decision (41.07). There was a reduction in anxious and depressive symptoms, and improved self-perceived performance in cognitive skills. **Conclusion:** The digital cognitive games platform proved feasible and well accepted, showing potential for expanding self-awareness about aging and promoting cognitive and psychosocial health in older adults.

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Alzheimer's disease and type 2 diabetes mellitus: overlapping of miRNAs, genes and biological pathways through systematic reviews and machine learning

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Introduction: Alzheimer's disease (AD) and type 2 diabetes mellitus (T2DM) are common conditions, with higher predisposition for AD in diabetic patients. MicroRNAs (miRNAs) are non-coding RNA, which can regulate gene expression. In this study, we aimed to identify circulating miRNAs commonly expressed in AD and T2DM, as well as their target genes and associated biological pathways, using systematic review (SRs) and

machine learning (ML). **Methods:** Two independent SRs were conducted to identify differentially expressed miRNAs in AD and T2DM. Searches covered major databases, following Cochrane Handbook and PRISMA guidelines. Data was analyzed to identify commonly regulated miRNAs in both diseases, with target genes extracted from miRTarBase. Pathway enrichment analysis was performed using EnrichR, and relevant pathways were ranked using ML approaches. **Results:** From the SRs (AD: 49 studies; T2DM: 104 studies), 21 miRNAs were identified as commonly expressed in both diseases. Gene extraction revealed 337 target genes for downregulated miRNAs and 233 for upregulated miRNAs, as well as 265 and 243 pathways associated to these genes, respectively. Using ML, the most important pathways were the T cell receptor Ras-extracellular signal (TCR-RAS), and the extracellular matrix (ECM) pathway. **Conclusions:** TCR-RAS signaling cascade is related to T cell homeostasis and self-tolerance. ECM pathway, a mixture of collagens, fibronectins, laminins, and proteoglycans, provides structure to cells and tissues, cells communication, proliferation, adhesion, differentiation, and apoptosis. These findings suggest that differentially expressed microRNA, and their shared signaling pathways, may offer insights into their pathophysiology and suggest potential therapeutic targets for T2DM and AD.

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New biomarkers for Alzheimer's disease: a cohort of patients with memory complaints

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Background: This work presents the BioAz cohort, a longitudinal, observational study that has been conducted since July 2024 at UNIFESP. The cohort, composed of over 40 researchers, has been recruiting individuals over the age of 55 with memory complaints. BioAz was designed to follow 500 individuals aged 55 and older over a 5-year period, conducting baseline clinical assessments and complementary exams, with annual follow-ups throughout the 60 months of monitoring. **Objectives:** The main goals of the cohort are to identify new biomarkers for Alzheimer's disease (AD), validate existing biomarkers, and develop predictive risk models through machine learning and artificial intelligence. **Methods:** BioAz is divided into three phases. Phase 1, also called the clinical phase, involves baseline neurological, neuropsychological, speech-language, and nutritional assessments. Phase 2, referred to as the complementary phase, includes additional tests such as plasma and CSF biomarkers, metabolomics, genomics, OCT, audiometry, and brain MRI. Phase 3, known as the follow-up phase, consists of annual follow-ups repeating part of the baseline evaluations. **Results:** The cohort has already recruited over 400 participants and enrolled more than 100, with results from Phase 1 and some preliminary data from Phase 2 to be presented at this conference. **Discussion and Conclusion:** The methods and preliminary results of this project aim to be presented at this conference in order to share current progress and encourage similar studies that could expand the Brazilian sample.

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Identification of glial-related CSF biomarkers for Alzheimer's disease and Parkinson's disease

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Glial cells play important roles in the pathophysiology of Alzheimer's disease (AD) and Parkinson's disease (PD). Although glial dysfunction is thought to contribute to disease progression, few cerebrospinal fluid (CSF) biomarkers capture these changes. Identifying novel glial-derived CSF biomarkers may improve our understanding of AD and PD pathology and support disease monitoring. Thus, we selected astrocyte-, oligodendrocyte-, oligodendrocyte precursor cell (OPC)-, and microglia-enriched and enhanced genes from single-cell/nucleus transcriptomic datasets. These gene lists were cross-referenced with CSF proteomic data (SomaLogic) from two AD cohorts (cohorts 1 and 2) and one PD cohort (cohort 3). Linear regression models assessed differential protein expression in CSF between cognitively unimpaired (CU) and cognitively impaired (CI) individuals, as well as between amyloid-beta (Aβ)-positive and alpha-synuclein-positive (α-syn+) individuals, within each glial cell type. In cohorts 1 and 3, differentially abundant proteins were clustered based on their pseudo-progression across CU and CI individuals, stratified by Aβ (pTau181/Aβ42 cut-off=0.028) or α-syn status. We identified 47 astrocyte-, 94 oligodendrocyte/OPC-, and 94 microglia-related proteins altered in CI individuals compared to CU. In Aβ+ individuals, 52 astrocyte-, 99 oligodendrocyte/OPC-, and 117 microglia-related proteins were differentially expressed. Similarly, in α-syn+ individuals, 143 glial proteins were altered across astrocytes, oligodendrocytes/OPCs, and microglia. Clustering analysis categorized proteins into four distinct groups per cell type and cohort. In total, we identified 485 glial-related CSF proteins altered in AD or PD. These results revealed distinct glial signatures across disease stages, supporting their potential as biomarkers of neurodegenerative progression and glial heterogeneity in AD and PD.

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How age-related noradrenergic dysregulation impacts pupillary response in anti-saccade and prosaccade tasks

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Aging is associated with multifaceted neurobiological changes that impact cognitive and oculomotor functions, including dysregulation of the locus coeruleus noradrenergic (LC-NA) system. This study investigated the effects

of aging on the LC-NA system using task-evoked pupil dilation as a physiological marker of cognitive demand during antisaccade and prosaccade tasks. The sample included 93 women and 43 men, who underwent cognitive assessments and performed the eye-tracking tasks. We observed a significant negative association between prosaccade task-evoked pupil dilation and age ($R=-0.30$, $p=0.002$). In the antisaccade task-evoked pupil dilation, the association with age remained after controlling for anticholinergic burden ($R=-0.02$, $p=0.020$). Also, we observed an age-related increase in errors in antisaccade ($R=0.37$, $p<0.000$) and prosaccade ($R=0.48$, $p<0.000$) tasks. Furthermore, prosaccade task-evoked pupil dilation was positively associated with visuospatial abilities ($R=0.211$, $p=0.033$), with better performers exhibiting greater pupillary dilation. Finally, increased number of errors in the prosaccade and antisaccade tasks were associated with worse general cognition, executive function, visuospatial abilities, and memory. These findings suggest that dysregulation of the LC-NA system is associated with age-related cognitive decline and this association can be partially explained by increased anticholinergic burden. The study highlights task-evoked pupil dilation as a tool for monitoring age-related to age-related cognitive decline.

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Complete blood count and systemic inflammation indexes in individuals with Alzheimer's disease: a case-control study

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Background: Alzheimer's disease is a neurodegenerative condition, characterized by an subclinical inflammatory response that leads to systemic neutrophil hyperactivation and increased migration of lymphocytes. The complete blood count (CBC) indexes are non-expensive and accessible tools that can be used to measure the inflammatory state of the individual. Therefore, we aim to compare the new blood count indexes in individuals with AD and cognitively healthy individuals. **Methods:** Sixty-one individuals with AD and 59 controls were evaluated. The diagnosis of dementia due to AD was based on NIA-AA with biomarkers of the cerebrospinal fluid compatible with the diagnosis. Ratios and inflammation indexes were calculated as follows: $NLR=Neutrophils/Lymphocytes$; $PLR=Platelets/Lymphocytes$; $LMR=Lymphocytes/Monocytes$; $dNLR=Neutrophils/Global\ Leukocytes - Neutrophils$; $SIRI=Neutrophils \times Monocytes/Lymphocytes$; $AISI=Neutrophils \times Monocytes \times Platelets/Lymphocytes$; and $SII=Platelets \times Neutrophils/Lymphocytes$. Statistical analyses were performed using SPSS V.21.0 software. Significant value of $p<0.05$. **Results:** There was no significant difference in age between the AD and control groups ($p=0.068$). Neutrophil count ($p=0.010$), neutrophil/lymphocyte ratio ($p=0.020$) and SIRI ($p=0.040$) were significantly higher, while lymphocyte count ($p=0.018$) and platelet count ($p=0.038$) were significantly lower in the AD compared to the control groups. Monocyte count, platelet/lymphocyte ratio, AISI and SII were not different between the groups. **Conclusion:** The data suggest that the CBC indices may reflect the subclinical inflammatory state of AD and may be potential and accessible markers for the disease.

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Analysis of biomarkers in the CSF of six patients with cerebral amyloid angiopathy in a neurology reference hospital

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Background: Many studies have demonstrated the pattern of biomarkers ($A\beta 42$, $A\beta 40$, t -tau and p -tau) in the cerebrospinal fluid (CSF) of patients with Cerebral Amyloid Angiopathy (CAA). Therefore, we need to validate CSF biomarkers to improve the diagnosis and earlier recognition of CAA. **Objective:** Analyze the CSF biomarkers of six patients with a previous diagnosis of CAA to confirm whether they have the pattern of alteration described in the literature. **Methods:** The liquor of six patients diagnosed with CAA probable or possible (based on the Boston 2.0 criteria), including one of them with a compatible brain biopsy, was analyzed for the biomarkers mentioned above. **Results:** The results obtained demonstrated that two of the six patients had CSF compatible with the typical CAA pattern described in the literature, showing low $A\beta 40$ and $A\beta 42$ values with a normal $A\beta 42/A\beta 40$ ratio. In addition, the values of tau protein are high, mainly total tau (t -tau). **Discussion:** CAA is characterized by the progressive deposition of $A\beta$ (amyloid- β) in the small cortical and leptomeningeal vessel walls of the brain. The diagnosis is made using the Boston 2.0 criteria which include clinical and MRI findings. However, this criteria only detects late pathological consequences of CAA. **Conclusion:** CSF biomarkers are an important tool to enable earlier diagnosis of CAA.

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Salivary biomarkers for Alzheimer's disease: a systematic review and meta-analysis of diagnostic performance

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Introduction: Fluid biomarkers have transformed the diagnosis of Alzheimer's disease (AD). While blood-based markers demonstrate high diagnostic accuracy, saliva has emerged as a promising—yet controversial—alternative biological matrix. This meta-analysis evaluates the diagnostic performance of salivary biomarkers in AD. **Methods:** We systematically searched four databases (Embase, PubMed, Scopus, and Web of Science) from inception to 2025. Eligible studies were assessed for demographic, pre-analytical, and analytical characteristics, including sensitivity, specificity, and area under the curve (AUC). Meta-analysis was performed using a random-effects model in accordance with PRISMA guidelines. **Results:** Among 3.118 screened studies, 18 met the inclusion

criteria. A total of 46 biomarkers were investigated, with amyloid-beta ($A\beta$) being the most frequently studied ($n=5$), followed by lactoferrin ($n=4$) and phosphorylated tau (p-tau) species ($n=4$). The pooled AUC for distinguishing AD from cognitively unimpaired (CU) individuals was 0.70 (95%CI 0.50–0.91) for $A\beta$ and 0.88 (95%CI 0.71–1) for lactoferrin. However, heterogeneity was high for both biomarkers ($I^2=92\%$ and 89.9% , respectively). Pre-analytical procedures (e.g., centrifugation, storage, sample handling) varied widely across studies, and ELISA was the predominant analytical method (64.7%). **Discussion:** The diagnostic performance of salivary $A\beta$ and lactoferrin exhibited substantial heterogeneity, likely due to inconsistent pre-analytical methodologies. Standardization of sample collection, processing, and analysis is critical for advancing salivary biomarkers in AD diagnosis.

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Increased serum urea levels related to cognitive performance in patients with Alzheimer's disease: preliminary findings

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Abstract: Alzheimer's disease (AD) is a neurodegenerative disorder characterized by memory loss and cognitive decline. Urea is a chemical compound resulting from protein metabolism, and its cycle plays important role in several diseases. **Objective:** To investigate associations between peripheral urea levels and cognitive function in patients with AD. **Methods:** This cross-sectional study included 14 patients with mild cognitive impairment or mild/moderate dementia due to AD, aged over 60 years (71.4% women; mean age 76.7 years). Serum urea levels were obtained from medical records (in samples collected up to 15 days after the cognitive assessment), and global cognitive status was measured using the Mini-Mental State Examination (MMSE). Descriptive statistics and correlation tests were performed with a significance level of $p<0.05$. **Results:** The mean MMSE score was 19.8 ± 5.7 points. The mean urea levels were 44.6 ± 17 mg/dL, which showed a negative and significant correlation with MMSE scores ($r=-0.597$; $p=0.024$). **Discussion:** Recent studies suggest that there is altered urea cycle enzyme expression and altered levels of urea cycle metabolites in the AD brain. Our preliminary results corroborate these studies, showing that serum urea levels are inversely correlated to cognitive performance in AD patients. **Conclusion:** Higher serum urea levels are correlated to worst global cognitive performance in this AD sample. Compromised urea cycle could be involved in pathophysiology of AD.

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Plasma TGF- β 1 are inversely correlated with P-Tau levels in cerebrospinal fluid in Alzheimer's disease

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Background: Alzheimer's disease (AD) is characterized by the deposition of $A\beta$ plaques and neurofibrillary tangles (NFT), constituted by hyper-phosphorylated protein tau (P-Tau). The transforming growth factor β (TGF- β) canonical Smad3 signaling is reduced in aged individuals, compromising the protective activation and facilitating cytotoxic reactivation of microglia, potentiating microglia-mediated neurodegeneration. **Objective:** To investigate the relationship between plasma TGF- β 1 levels and cerebrospinal fluid (CSF) markers in patients with AD. **Methods:** This study included 32 patients with AD, recruited at specialized outpatient clinics of a public university hospital. AD diagnosis was based on the NIA-AA criteria. The TGF- β 1 levels were obtained in citrate platelet-poor plasma using the Human TGF- β 1 kit (R&D Systems). $A\beta$, T-Tau and P-Tau levels were measured by ELISA test (Fujirebio®). **Results:** The group presented a mean age of 70.9 ± 8.7 years. The TGF- β levels were 120.8 ± 63.1 pg/mL. The mean values of the CSF markers were: $A\beta$ - 549.7 ± 141.4 pg/mL, T-Tau - 717.9 ± 303.7 pg/mL, and P-Tau - 101.2 ± 41.2 pg/mL. Plasma TGF- β levels were significantly and inversely correlated with P-tau levels in CSF ($r=-0.440$, $p=0.019$), but not with T-tau or $A\beta$ levels. **Discussion:** The results suggest that the neuroprotective TGF- β 1 levels are reduced in patients who present hyperphosphorylation of Tau, indicating a close relationship between impaired anti-inflammatory action and NFT formation. **Conclusion:** Plasma TGF- β 1 could be an accessible marker to appraise the formation of P-tau in CSF and its decrease may be related to the pathophysiology of AD.

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MiRNA-30c-5p in behavioral variant frontotemporal dementia: a new disease biomarker?

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Background: Frontotemporal dementia (FTD) is a neurodegenerative disease that accounts for 10% of dementia cases. MicroRNAs (miRNAs) are small non-coding RNAs that regulate gene expression and have been widely investigated as potential biomarkers for several diseases. The overexpression of miRNA-30c-5p is associated with the regulation of inflammation, apoptosis, and microglial survival, but it has not yet been investigated in FTD. **Objective:** To investigate the expression of miRNA-30c-5p in patients with behavioral variant FTD (bvFTD) compared to cognitively healthy controls. **Methods:** This study included nine patients with bvFTD and 10 cognitively healthy individuals matched by age. miRNA extraction was per-

formed from serum samples using the NucleoSpin® miRNA Plasma kit, and cDNA synthesis was conducted with the TaqMan® Advanced miRNA cDNA Synthesis Kit (Thermo Scientific). miRNA-30c-5p expression analysis was performed using the comparative 2- $\Delta\Delta$ CT method, considering a fold change greater than 2 as significant, normalized by the endogenous control miRNA-484. **Results:** The mean age was 63.6 $\pm$ 8.7 years for the bvFTD group and 58.2 $\pm$ 11.7 years for controls. miRNA-30c-5p expression was increased by 2.29-fold in bvFTD patients compared to the control group. **Discussion:** This difference supports the involvement of inflammation and microglial apoptosis in bvFTD pathophysiology, in addition to the potential of miRNA-30c-5p as a biomarker of the disorder. **Conclusion:** The results indicate that miRNA-30c-5p is upregulated in patients with bvFTD and could be useful to investigate inflammatory and apoptotic processes in this disease.

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Relationship between $\alpha\beta$, T-Tau and P-tau in cerebrospinal fluid with the score in the Mini Mental State Examination in Alzheimer's disease

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Background: Alzheimer's disease (AD) is the most frequent cause of dementia. AD neuropathology is characterized by accumulation of amyloid- β (A β) plaques and neurofibrillary tangles composed by hyperphosphorylated tau (P-Tau) protein, leading to decrease levels of A β and increased Tau total (T-Tau) and P-tau in cerebrospinal fluid (CSF). In neuropathological studies, tau accumulation is considered the best marker of cognitive impairment. **Objective:** To investigate correlations between A β , T-Tau and P-tau CSF levels with Mini-Mental State Examination (MMSE) scores in AD patients. **Methods:** Thirty-two AD patients were recruited at outpatient clinics of a public university. Diagnosis of AD was based on NIA-AA criteria. MMSE was used to evaluate global cognitive status. A β , T-Tau and P-Tau levels were measured by ELISA test (Fujirebio®) according to the manufacturer recommendations. **Results:** AD group presented a mean age ($\pm$ SD) of 70.9 $\pm$ 8.7 years, 53% were women. Mean MMSE score was 19.2 $\pm$ 6.3. Mean levels of the CSF markers were: A β - 549.7 $\pm$ 141.4 pg/mL, T-Tau - 717.9 $\pm$ 303.7 pg/mL, and P-Tau - 101.2 $\pm$ 41.2 pg/mL. A β levels were positively correlated with MMSE scores ($r=0.398$, $p=0.029$). T-tau and P-tau did not display significant correlations ($p=0.783$ and 0.773 , respectively). **Discussion:** Lower A β levels in CSF were related to worst global cognitive performance in this group of patients. **Conclusion:** A β level was associated with cognitive impairment in this AD patients' sample, in contrast to tau pathology, suggesting that CSF A β may be a good marker of global cognitive performance in this group.

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Combined use of imaging and plasma tau biomarkers for predicting clinical progression in Alzheimer's disease

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Imaging and fluid biomarkers are important tools for Alzheimer's disease (AD) diagnosis and staging. Tau positron emission tomography (PET) imaging reliably predicts the risk of clinical progression, while plasma phosphorylated tau at threonine 217 (p-tau217) has been proposed as an alternative. We aimed to test the utility of combining these biomarkers for predicting the risk of clinical progression in cognitively unimpaired (CU) individuals. We assessed 211 CU participants from the A4 Study placebo group with PET for amyloid- β (A β) plaques ([18F]Florbetapir) and tau tangles ([18F]Flortaucipir), plasma p-tau217, and neuropsychological testing. A β -positive (A+) individuals were classified as p-tau217-positive (A+Tp-tau217+), tau PET-positive in the medial temporal lobe (A+TMTL+), or the neocortex (A+TNEO+). Cox proportional hazard ratio (HR) models showed a gradual increase in the risk of clinical progression in the A+TMTL+ (HR=3.80, $p<0.001$) and A+TNEO+ (HR=5.39, $p<0.001$) groups compared to the A-T- (reference) group. We found that the A+Tp-tau217+ group showed a similar risk of clinical progression compared to A+TMTL+ individuals (HR=3.40, $p<0.01$ vs. HR=3.85, $p<0.001$), but a lower risk compared to the A+TNEO+ group (HR=2.68, $p<0.05$ vs. HR=5.45, $p<0.001$). Model fit indexes corroborated that adding p-tau217 did not significantly improve the predictive performance of the models, as demonstrated by negligible differences in Harrell's C-index and Akaike Information Criterion. Our results suggest that while p-tau217 captures the risk of clinical progression, it does not seem to significantly improve the prognostic value of tau PET. These findings further support imaging-based biomarkers—particularly tau PET—as the most reliable prognostic tools for early detection of AD.

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Neck circumference as a parameter risk of sarcopenia in patients with Alzheimer's disease: preliminary findings

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Background: Alzheimer's disease (AD) is a neurodegenerative condition characterized by cognitive impairment. Sarcopenia is a condition that leads to loss of muscle mass and function. AD and sarcopenia are age-related conditions

most commonly observed in older adults, and their association is related to a worst prognosis. Neck circumference is an easily measured anthropometric related to cardiovascular diseases. **Objective:** To investigate the use of neck circumference as an anthropometric parameter to identify sarcopenia in patients with AD. **Methods:** Fourteen patients with mild cognitive impairment or mild/moderate dementia due to AD were included in the study (mean age 76.3 ± 9.5 years). Cognitive performance was screened using the ACE-R test. Sarcopenia was evaluated using the SARC-F method, which includes muscle strength, difficulty walking, climbing steps and rising from a chair, and the frequency of falls, with each parameter rated from 0 (no difficulty) to 2 (great difficulty or disability). $P < 0.05$ were considered significant. **Results:** Analysis of the SARC-F score indicated that 14.2% of patients presented sarcopenia (score ≥ 4). The neck circumference mean was 35.4 ± 3.8 cm. A significant and positive correlation was observed between the SARC-F score and neck circumference ($r = 0.536$; $p = 0.048$). **Discussion:** Neck circumference is a simple, non-invasive measure that can be used to assess cardiovascular risk. In addition, our results suggest that higher neck circumference are also correlated to sarcopenia in AD patients. **Conclusion:** Neck circumference can be a useful tool for identifying AD individuals at higher risk for sarcopenia and cardiovascular complications.

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The role of miRNA-125a-5p as a biomarker in patients with frontotemporal dementia: a case-control study

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Background: Frontotemporal dementia (FTD) is a neurodegenerative disorder related to anterior frontal or temporal lobes, leading to progressive deficits in behavior and cognitive domains. MicroRNAs (miRNAs) are small non-coding RNAs involved in gene regulation. MiRNAs can be detected in peripheral blood, offering a less invasive and more accessible alternative to analysis of cerebrospinal fluid (CSF). **Objective:** To investigate the expression of miRNA-125a-5p in patients with behavioral variant FTD (bvFTD) and in cognitively healthy controls. **Methods:** The study evaluated 16 patients with bvFTD and 20 controls matched by age and sex. The miRNA was extracted from serum samples using the kit NucleoSpin® miRNA Plasma. The cDNA was obtained using the kit TaqMan® Advanced miRNA cDNA Synthesis and qPCR amplification. The expression analysis was performed using the comparative $2^{-\Delta\Delta CT}$ method, normalized by the endogenous control miRNA-484. We considered a fold change greater than two as significant. **Results:** The mean age was 65.2 ± 8.2 years for the bvFTD group and 63.1 ± 10.8 years for controls. Expression analysis revealed that the miRNA-125a-5p levels were 2.3-fold higher in the bvFTD group compared to the controls. **Discussion:** miRNA-125 is known to regulate synaptic plasticity and its overexpression is associated with memory deficits and tau hyperphosphorylation. The findings suggest that the differential expression of miRNA-125a-5p is related to the bvFTD pathophysiology. **Conclusion:** The upregulation of miRNA-125a-5p in bvFTD patients supports its involvement in disease-related molecular mechanisms and reinforces its potential as a minimally invasive biomarker for this disorder.

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APOE rs769449 polymorphism associates with increased tau pathology in Alzheimer's disease

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Single-nucleotide polymorphisms (SNPs) in the apolipoprotein E (APOE) gene are well-established genetic risk factors for Alzheimer's disease (AD). However, the role of the APOE rs769449 in modulating AD risk remains unknown. We investigated the influence of rs769449 on amyloid- β ($A\beta$)-driven tau pathology and its interplay with APOE $\epsilon 4$ carriership. We included 422 individuals from the ADNI cohort with available APOE genotyping, $A\beta$ and tau PET, and clinical status. The interaction effect between rs769449 and $A\beta$ on tau pathology was assessed using a linear regression model adjusted for age, sex, APOE $\epsilon 4$, and clinical diagnosis. Participants were further categorized by rs769449 allele dosage, clinical impairment, and APOE $\epsilon 4$ carriership. We found that the tau PET signal was significantly elevated in $A\beta$ -positive rs769449 carriers ($p = 1.645 \times 10^{-2}$). The interaction between rs769449 and $A\beta$ on tau pathology was markedly higher in carriers ($\beta = 0.22$, $p = 1.195 \times 10^{-2}$), with homozygous individuals showing notably increased tau burden ($\beta = 1.25$, $p = 1.68 \times 10^{-5}$). This effect was particularly pronounced among cognitively impaired individuals ($\beta = 1.17$, $p = 6.49 \times 10^{-3}$), but not observed in the unimpaired group ($\beta = 0.34$, $p = 0.44$). Moreover, rs769449 homozygotes and APOE $\epsilon 4$ carriers showed markedly greater $A\beta$ -driven tau pathology ($\beta = 1.27$; $p = 1.39 \times 10^{-5}$). Our results suggest that rs769449 carriership is associated with increased vulnerability to $A\beta$ -driven tau pathology, particularly in cognitively impaired individuals. In conclusion, these findings suggest a potential intragenic SNP interaction within APOE locus, with rs769449 emerging as a promising biomarker for AD.

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Cerebrospinal fluid biomarkers in patients with initial movement disorder: clinical profile and diagnostic implications

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Background: Cerebrospinal fluid (CSF) biomarkers are increasingly used to support the diagnosis of neurodegenerative diseases, particularly Alzheimer's disease (AD). In atypical cases presenting initially with movement disorders, CSF analysis may provide diagnostic insights. **Objective:** To

analyze the clinical characteristics and CSF biomarker profiles, according to the 2024 Alzheimer's Association (AA) criteria, in patients presenting initially with movement disorders. **Methods:** This was a retrospective study with prospective biomarker data collection from patients evaluated at a specialized neurology clinic. Included patients presented with cognitive impairment associated with movement disorders and underwent CSF biomarker testing due to diagnostic uncertainty, following the 2024 guidelines of the Brazilian Academy of Neurology (ABN). Patients were classified as AD (Group A1) or non-AD (Group A2) based on CSF profile using AA-2024 criteria. Data were analyzed using the t-test and Mann-Whitney test (SPSS v26). **Results:** Seventeen patients were included: 9 in Group A1 and 8 in Group A2. Mean age was significantly higher in Group A1 (62 years; range 45–79) compared to Group A2 (50 years; range 42–65), with $p=0.037$. Mini-Mental State Examination (MMSE) scores were similar: 19.2 (A1) vs. 19.6 (A2). Sex distribution was predominantly female in both groups. No other significant differences were found. **Conclusion:** Except for age at symptom onset, no significant clinical differences were observed between AD and non-AD biomarker profiles. These findings suggest CSF biomarker analysis may aid in clarifying diagnoses in patients with atypical presentations such as early movement disorders.

TECHNICAL AREA: BIOMARKERS
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Biomarker derived from the cortex fractal geometry helps revealing early structural deviations in Alzheimer's disease

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The primate cerebral cortex follows a universal fractal morphology with dimension $d=2.5$, as predicted by a biomechanical model proposed by Mota and Herculano-Houzel. Building on this framework, we introduce a novel structural biomarker that detects deviations from normative trajectories in cortical folding. We analyzed 287 T1-weighted MRI scans from a Brazilian aging cohort (40 Alzheimer's Disease [AD], 69 Mild Cognitive Impairment [MCI], and 178 cognitively normal [NC] individuals; all with ≥ 8 years of education). Cortical surfaces were reconstructed using FreeSurfer v6, and fractal properties were assessed via a coarse-graining method (5 minutes/subject runtime). Group comparisons using ANOVA revealed significant deviations in fractal structure in both AD and MCI relative to NC. Furthermore, three in five individuals who progressed from MCI to AD showed early detectable changes in fractal geometry. These findings highlight the potential of the biomarker for early, non-invasive detection of neurodegeneration in Alzheimer's disease.

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Differentially expressed proteins in blood plasma of individuals with Alzheimer's disease

through hyphenated mass spectrometry: a case-control study

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Background: Proteomic analysis is promising in identifying potential biomarkers with the potential to differentiate individuals with and without amyloid pathology. **Objective:** To discover differentially expressed proteins (DEPs) in the blood plasma of individuals with Alzheimer's Disease (AD) through mass spectrometry. **Methods:** 37 participants were included, 17 with probable AD according to clinical criteria and 20 controls. 5 ml of peripheral blood was collected from the participants. Proteomic analyses were performed using hyphenated mass spectrometry. The PROGENESIS QI software was used for data analysis and statistical analysis was performed using the OmicScope package in Python. **Results:** The sample consisted of 55.3% women, with a mean age of 75.1 years in the AD group, and 69.6 years in the controls. 47% of the AD group was in the advanced dementia stage. Between mild AD and controls, 33 DEPs were identified (adjusted $p \leq 0.05$), of which 21 were upregulated and 12 were downregulated. Among moderate AD and controls, 13 DEPs were identified (adjusted $p \leq 0.05$), and all were upregulated. Between advanced AD and controls, 6 DEPs were found (adjusted $p \leq 0.05$), of which 4 were upregulated and 2 were downregulated. Most of the detected DEPs are related to inflammatory and immunological activity, as well as lipid metabolism. **Discussion:** DEPs involved in inflammatory processes were identified in all AD subgroups, from mild to severe, in agreement with data in the literature. **Conclusion:** Our results corroborate the implications of deregulated inflammatory pathophysiology of AD.

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CD163 correlates with core Alzheimer's biomarkers in dementia

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Neuroinflammation, marked by glial activation and inflammatory mediator release, plays a key role in Alzheimer's disease (AD) pathophysiology. CD163, a transmembrane receptor expressed in immunosuppressive cells, regulates immune responses and is elevated in AD, yet its association with core AD biomarkers remains unclear. This study investigates the relationship between CD163 protein levels and astrogliosis, $A\beta$, and tau biomarkers across the AD continuum. We included 170 cognitively unimpaired (CU), 411 mild cognitively impaired (MCI), and 138 individuals with dementia (Dementia) from ADNI with available fluid biomarkers, blood transcriptomics, and neuroimaging data at baseline. Generalized linear mixed models were performed in R to assess the association of CD163 protein levels with AD biomarkers, accounting for age and sex. Akaike Information Criterion (AIC) comparison was used to evaluate the model used to analy-

ze CD163 as a function of biomarkers for each group. No significant differences in CD163 protein levels or gene expression were observed between diagnostic groups. In CU individuals, CSF CD163 was positively associated with total-tau, A β 42, and pTau181. In MCI, CD163 correlated positively with A β -PET, pTau181, total-tau, and negatively with MMSE scores. In dementia, CD163 levels were again positively associated with total-tau, A β 42, and pTau181. No associations were found between CD163 and hippocampal volume. AIC analyses indicated more consistent biomarker relationships in the dementia group. The strong positive correlations between CD163 and core AD biomarkers in dementia suggest that CD163 may serve as a marker of advanced neuroinflammation and neurodegeneration, highlighting its potential relevance in AD progression.

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Ageing and neurodegeneration: a study of plasma biomarkers and neuroimage in a Brazilian cohort

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Background: Plasma neurofilament light chain (NfL) is a valuable tool for diagnosing and monitoring neurodegenerative diseases. Although reference values of this biomarker are established in Europe and the US, Brazil and Latin America lack local standards and clinical validation. **Objective:** This study aimed to establish local reference values for plasma NfL and other biomarkers of neurodegeneration in healthy older Brazilians. **Methods:** This study included 77 cognitively healthy older adults from a longitudinal dementia study at IDOR, with MRI scans at three time points analyzed using FreeSurfer v7.2.0. Plasma biomarkers were measured with the SIMOA HD-X platform. **Results:** We determined normal values and the annual rate of increase for plasma NfL in the cohort (Pearson $r = 0.4993$, $p < 0.001$). Correlations with Grey matter volume (Pearson $r = -0.29$, $p < 0.0001$), White matter volume (Pearson $r = -0.21$, $p = 0.0022$), Mean Cortical Thickness (Pearson $r = -0.22$, $p < 0.001$) and hippocampus volume (Pearson $r = -0.39$, $p < 0.0001$) were also measured. Analyses of clinical and sociodemographic data showed that NfL negatively correlates with education levels and eGFR. **Conclusions:** Plasma NfL is a promising biomarker for neurodegeneration, showing strong associations with aging and reduced brain volume, including gray and white matter, hippocampus, and cortical thickness. Establishing reference values for the Brazilian population is essential for clinical application, supporting its use in diagnosing and monitoring neurodegenerative diseases.

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Astrocyte reactivity: a key mediator of Alzheimer's disease risk following critical illness

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Survivors of a critical illness are at high risk of developing neurodegeneration and long-term cognitive impairment. Although common, this condition is poorly understood. Astrocyte reactivity, a heterogeneous response of challenged astrocytes, is a common characteristic of Alzheimer's disease. Here, we aimed to evaluate the impact of critical diseases on astrocyte reactivity, neurodegeneration, and cognition. Individuals from an intensive care unit ($n=53$) and healthy controls ($n=44$) aged 40 years or older were recruited. The Simplified Acute Physiology Score III (SAPS III) was used to predict mortality in ICU patients. GFAP and NfL levels in plasma were measured with SIMOA. Mini-Mental State Examination (MMSE) and the Montreal Cognitive Assessment (MOCA) were used for cognitive screening. Statistical analyses were conducted using generalized linear models, which accounted for age and sex. The critical disease group presented higher GFAP ($F=15.094$, $p < 0.001$) and NfL levels ($F=12.32$, $p < 0.001$) compared to controls. GFAP levels were positively associated with SAPS III ($\beta=0.0351$, 95%CI 0.011, 0.0588, $p=0.004$), while NfL levels were not ($\beta=0.00347$, 95%CI -0.027 – 0.0339 , $p=0.824$). MMSE ($F=5.371$, $p=0.029$) and MOCA ($F=11.75$, $p=0.002$) showed a significant difference between groups; however, neither test was associated with biomarker levels. Critical diseases influence astrocyte reactivity and neurodegeneration processes, suggesting that they may be associated with neuropathological processes. Astrocyte reactivity biomarkers were associated with mortality prediction scores, highlighting the crucial role astrocytes have in mediating brain-periphery communication.

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Automated brain volumetry and visual classification of hippocampal atrophy in magnetic resonance imaging

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Introduction: Visual assessment of medial temporal atrophy (MTA) using a five-point scale is a quick and low-cost method to evaluate hippocampal degeneration in Alzheimer's disease. Automated volumetry offers objective and continuous measurements. Understanding how these approaches align can improve diagnostic accuracy. **Objective:** To compare visual MTA scores with automated hippocampal volumetry and analyze their correlation. **Method:** This retrospective study analyzed 677 MRI scans from the ADNI database (CN=366, MCI=233, AD=78). T1-weighted images were processed using FreeSurfer to obtain normalized hippocampal volumes. Two neuroradiologists independently rated MTA scores for both hemispheres. Pearson

correlation and linear regression models adjusted for age, sex, intracranial volume assessed associations. Inter-rater agreement was evaluated using weighted Kappa. Outliers were defined as individuals with hippocampal volume higher than two standard deviations from their mean MTA score. **Results:** Mean age was 71.1±7.4 years; 55% were female. Hippocampal volume declined with increasing cognitive impairment and age. Each point increase in MTA score corresponded to ~250–293 mm³ volume reduction. Adjusted R² ranged from 0.57 to 0.60. Inter-rater agreement was substantial (Cohen's Kappa ≈0.61–0.65; weighted Kappa ≈0.83–0.85). Spearman's correlation between raters exceeded 0.85. A total of 62 individuals (9.2%) were outliers, showing discrepancies between MTA scores and volumetry. **Conclusion:** Visual and volumetric assessments are generally aligned but not interchangeable. Volumetry did not enhance visual scoring overall but helped identify outliers. Future work may establish volumetric thresholds to support clinical interpretation.

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Diagnostic performance of capillary p-Tau217 in Alzheimer's disease: results from the DROP-AD project

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Background: Phosphorylated tau at Thr 217 (pTau217) is the most accurate blood biomarker for detecting Alzheimer's disease (AD) pathology. The DROP-AD project evaluates biomarker quantification in capillary blood as an alternative to overcome logistical challenges and broader biomarker access. **Objective:** To evaluate the diagnostic performance of capillary pTau217. **Methods:** In this study, 203 participants (mean age 71.8 years; 59.2% female) from five European centers provided paired venous plasma and dried plasma from finger-prick collection. A subset self-collected a second sample without supervision. After processing of capillary cards, samples were analyzed using the ALZPath assay (Simoa platform). CSF biomarkers were available in 151 individuals. Amyloid positivity (Aβ+) was defined using, exploratorily, plasma pTau217 (≥0.42 pg/mL) or, as gold standard, CSF Aβ42/pTau181 (<11.94). Cognitive assessments (MMSE/CDRglobal) were available for all participants. **Results:** Capillary and venous samples showed strong correlation (n=203, rs=0.779, P<0.001). Self-collected and study personnel-collected samples yielded equivalent results (n=45, P=0.97). Capillary pTau217 detected abnormal venous pTau217 with an AUC of 0.921. A capillary cutoff of 0.017 pg/mL (90% specificity) resulted in a PPV of 0.915 and an NPV of 0.714. For CSF Aβ42/pTau181, plasma and capillary pTau217 had AUCs of 0.982 and 0.884, respectively. Finger-prick pTau217 reflected cognitive performance (MMSE r=-0.37, p<0.001) similarly to plasma (MMSE r=-0.41, p<0.001). **Conclusion:** Finger-prick collection reliably measures pTau217, showing promising performance for detecting AD pathological changes. This simple, self-executable, temperature-independent method is promising for use in areas with limited blood-processing infrastructure, and may open new windows for Alzheimer's biomarker phenotyping in the Global South.

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Neurofilament light chain in frontotemporal dementia

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Background: Frontotemporal dementia (FTD) is a complex neurodegenerative disorder with early onset, often leading to diagnostic challenges due to symptom overlap with psychiatric and other neurological conditions. Neurofilament light chain (NfL), a marker of axonal damage, has emerged as a promising non-invasive biomarker for FTD diagnosis and monitoring. However, its accuracy, prognostic value clinical utility require further investigation. **Objective:** To evaluate the diagnostic accuracy and prognostic value of plasma NfL in FTD, to explore its association with disease severity, progression, clinical applicability. **Methods:** An integrative review was conducted, searching PubMed for original studies published between 2020–2025, in English or Portuguese. Studies evaluating plasma or serum NfL in FTD for diagnostic or prognostic purposes were included. Exclusions comprised reviews, meta-analyses, protocols, irrelevant articles. The selection process was managed using Rayyan[®]. **Results:** Plasma NfL levels were consistently higher in FTD patients (41–86 pg/mL) compared to controls (<20 pg/mL) across 14 studies. NfL demonstrated high diagnostic accuracy (AUCs 0.85–0.93) in differentiating FTD from controls, Alzheimer's disease, psychiatric disorders. Higher NfL levels correlated with faster clinical decline, increased brain atrophy, shorter survival. Variability was influenced by age, genotype, and disease stage, indicating a need for standardized cutoffs. **Discussion:** Plasma NfL is a valuable biomarker reflecting neuroaxonal damage in FTD, aiding in diagnosis and monitoring. Its ability to differentiate FTD from other conditions and predict disease progression highlights its clinical utility. While promising, further research is needed to establish standardized cutoffs and fully understand the impact of biological variability on its application. **Conclusion:** Plasma NfL is a promising, minimally invasive biomarker with strong potential to improve FTD diagnosis, monitoring, and personalized management. Its consistent elevation in FTD and correlation with disease progression support its role as a reliable indicator of neurodegeneration. Continued research will refine its clinical application.

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Correlation between blood-based biomarkers and clinical staging of Alzheimer's dementia: focus on neurodegeneration and astrocytic activation

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Background: Blood-based biomarkers (BBM) have emerged as promising tools for the pathological diagnosis of Alzheimer's disease (AD). **Objective:** Correlate BBM (NFL, GFAP) with clinical staging in patients without cognitive decline, with mild cognitive impairment (MCI), and with AD. **Methods:** Cross-sectional study including patients aged ≥ 55 years, classified as cognitively normal, MCI, or clinical diagnosis of mild to moderate AD (CDR 1 and 2). Patients with advanced dementia (CDR 3), stroke, or active psychiatric disorders were excluded. All participants underwent clinical evaluation, neuropsychological testing, and plasma biomarker collection, using Single Molecule Array (SIMOA). **Results:** 102 patients were evaluated: 33% without cognitive decline, 23.5% with MCI, 31.4% with mild dementia, and 11.8% with moderate dementia. The mean age was 76.7 ± 7.2 years. Significant elevation in NFL was observed in the moderate dementia group ($p < 0.05$; $\epsilon^2 = 0.28$; 95%CI 0.24–1.00). There was a positive correlation between GFAP and CDR sum of boxes ($p < 0.05$; $\rho = 0.40$; 95%CI 0.12–0.63). **Discussion:** Elevation of NFL supports the presence of neurodegeneration in more advanced stages of dementia. The significant association and positive correlation between GFAP levels and the CDR sum of boxes suggest astrocytic activation in the progression of dementia. **Conclusion:** The findings suggest the potential use of plasma NFL and GFAP in the clinical staging of AD.

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Correlations between plasma irisin and dementia biomarkers in a Brazilian cohort

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Background: Physical exercise (PE) is regarded as a potential non-pharmacological preventative and interventional strategy to slow down the decline of cognition, both in clinically healthy individuals and patients with Alzheimer's disease (AD). Irisin, produced by skeletal muscle during PE has shown to play a role in the nervous system, and it is speculated that it may have a neuroprotective role. **Objective:** We aim to evaluate through linear regression models which variables have an impact on plasma irisin levels in a dementia cohort. **Methods:** Using a well-characterized dementia cohort established at the D'Or Institute, we gathered biomarker and demographic data from a group of 30 cognitively unimpaired and cognitively impaired older adults (aged 55+) diagnosed with amnesic mild cognitive impairment (aMCI) or Alzheimer's disease. Through Hierarchical Multiple Linear Regression with irisin as the dependent variable, we combined biomarker data with demographic variables such as age, physical activity, sex, education, BMI and diagnosis, in search of the best-performing model. Biomarker data was acquired using the Single Molecule Array (SIMOA) platform, and all measurements were made in plasma. Statistical analysis was performed using Rstudio. **Results:** Models that included either the ptau181/ab42 or ptau217/ab42 ratios as well as physical exercise as predictor variables reached significance with the dependent variable. **Conclusion:** Results indicate that both pTau181/ab42 and pTau217/ab42 ratios influence the positive association between plasma irisin and physical activity. In contrast, demographic and clinical variables such as age, sex, BMI, education and diagnosis did not show statistically significant associations.

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Plasma exosomal miRNAs as good biomarkers candidates for canine cognitive dysfunction

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Alzheimer's Disease (AD) and Canine Cognitive Dysfunction Syndrome (CCDS) are progressive neurodegenerative disorders characterized by the accumulation of amyloid- β and phosphorylated tau. These aggregates cause similar brain damage in both diseases, making the dog a good model for AD studies. We isolated miRNAs from plasmatic exosomes of dogs suspected with CCDS, based on clinical signs and molecular findings, to better understand CCDS pathophysiology. Dogs aged from 8 to 14 years old were selected based on anamnesis, neurological exams, and a high score in tutor-completed questionnaires. Detailed neurological and deep pain evaluations were performed in both experimental and control groups. Blood was collected to isolate plasma. Exosomes were obtained by ultracentrifugation at 1000.000 g for 1.5 hours, repeated twice. miRNAs were extracted, converted to cDNA, and quantified by RT-PCR. Common signs in the experimental group included those summarized by the acronym DISHA: Disorientation, altered social Interaction, disrupted Sleep-cycle, Housesoiling and Anxiety. No differences were observed in total miRNA levels between CCDS and healthy dog plasma. However, miRNA29 and miRNA21 showed a significantly higher expression in exosomes from CCDS dog's plasma compared to healthy dogs, similar to results previously shown in patients with neurological disorders. These results underscore the importance of studying the role of exosomes and their cargo in neurodegenerative disorders.

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Use of the centiloid scale in PET imaging of amyloid with [18F] Flortbetaben

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Alzheimer's disease (AD) is the leading neurodegenerative disorder and the main cause of dementia. Its neuropathological hallmarks include β -amyloid (A β) plaque deposition and intraneuronal tau accumulation, detectable via cerebrospinal fluid biomarkers or PET imaging using specific radiotracers. Amyloid PET quantification can be interpreted through SUVR (cut-off: 1.42) or the Centiloid Scale (CL), which ranges from 0 to 100 and classifies A β burden as low (<20 CL), intermediate-high (20CL–50 CL), or high (>50 CL). The aim is to analyze the distribution of CL values in [18F]Flortbetaben (FBB) PET images in individuals of cognitive decline. A retrospective analysis was performed on 266 anonymized PET-FBB scans acquired in list mode on a Discovery D600 PET/CT, allowing static and dynamic reconstructions with appropriate corrections. T1-weighted MRI scans from a 3T GE Signa HDxT scanner were used for co-registration and normalization to the Centiloid atlas. Image processing was done

in PMOD using the PNEURO tool. The CL implementation followed the GAAIN image processing pipeline. A total of 266 participants, 89 had <20 CL, 42 had between 20 and 50 CL and 135 had more than 50 CL. SUVR values were also analyzed. This distribution between aging stages allowed for the evaluation of how different A β cut-off points can affect the diagnosis. This analysis of CL distributions among individuals at different stages of cognitive decline highlights how classification thresholds may influence AD diagnosis. The CL offers a standardized and reliable method for quantifying amyloid deposition, supporting its use in monitoring disease progression and treatment response.

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Diagnostic accuracy of plasma p-tau217 and A β 42/40 ratio across the Alzheimer's disease continuum

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Introduction: Alzheimer's disease and Mild Cognitive Impairment are characterized by cognitive decline. Early diagnosis is crucial, and blood-based biomarkers offer a promising alternative. **Methods:** We searched MEDLINE (via PubMed), Embase, Web of Science, and Cochrane Library up to March 2025, following Preferred Reporting Items for Systematic Reviews and Meta-Analysis for diagnostic test accuracy (PRISMA-DTA) 2018 protocol. We included studies evaluating the diagnostic accuracy of plasma p-tau217 and A β 42/40 for Alzheimer's disease, using CSF or amyloid PET as comparison. Risk of bias was assessed using QUADAS-2. Statistical analysis was performed in STATA 14.2 using metandi and midas. **Results:** The diagnostic odds ratio was 52, and the area under the summary receiver operating characteristic (ROC) curve was 0.94, suggesting strong overall diagnostic performance. Considerable heterogeneity was observed, with an inconsistency index (I²) of 98 percent. Sensitivity and specificity estimates varied between studies, and a strong threshold effect was identified, indicating that differences in diagnostic thresholds likely contributed to the observed variation. **Conclusion:** Plasma p-tau217 and A β 42/40 ratio demonstrate strong potential for improving early diagnosis. These biomarkers could significantly impact timely interventions and help slow disease progression.

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Towards an early diagnosis platform of Alzheimer' disease using Raman spectroscopic ophthalmoscopy

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Background: Raman Spectroscopy (RS), which enables molecular identification through its vibrational modes, is a robust and sensitive methodology in the biochemical characterization of neural tissue. The retina, as a densely synaptic neural tissue accessible by optical techniques, has emerged as a promising site for noninvasive assessment of Alzheimer's Disease (AD). In addition, experimental models have shown that retinal amyloid plaques precede those in the cortex and hippocampus. Based on these premises, our group has developed a Raman-based ophthalmoscope for early diagnosis of AD. **Objective:** To identify RS fingerprints of amyloid plaques in cortical and hippocampal tissues of bitransgenic mice, as a basis for a diagnostic platform designed to detect both amyloid aggregation and neurodegeneration in the retina prior to clinical onset of AD, with potential application in clinical trials. **Methods:** An experimental model using APPswe/PS1 Δ E9 bitransgenic mice was developed. Plaques from 6- and 12-month-old mice were analyzed using RS for topographic and molecular characterization. Vibrational signatures related to amyloid aggregation and associated neurodegeneration were obtained. **Results:** RS identified multiple signatures in the core (amide I, III, B, phenylalanine) and halo (C-H₂ and C-H₃), enabling detailed plaque characterization. Amide B emerged as a rapid and promising marker of amyloid aggregation, while lipid-associated C-H signals defined halo expansion over time without further amyloid aggregation. Soluble beta-amyloid protofilaments were observed around the core and beneath lipid clusters. **Conclusion:** RS reveals specific molecular events underlying AD and offers a novel, noninvasive, and promising approach for early diagnosis.

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Biomarkers and neuroimaging in Lewy body dementia: an integrative review of diagnostic advances

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Lewy Body Dementia (LBD) is the second most common cause of neurodegenerative dementia, clinically characterized by visual hallucinations, cognitive fluctuations, REM sleep behavior disorder, and parkinsonism. The underlying pathology involves α -synuclein accumulation in the central nervous system, impairing neuronal function. Although LBD accounts for 10–15% of dementia cases, diagnosis remains challenging due to overlapping symptoms with other disorders, such as Alzheimer's disease. Advances in biomarkers and neuroimaging hold promise for earlier and more accurate diagnosis. This study aimed to evaluate the role of biomarkers and neuroimaging techniques in the diagnosis and clinical monitoring of LBD. An integrative review was conducted using the PubMed database. The keywords "Lewy Body Dementia," "Neuroimaging," "Biomarkers," "Diagnosis," and "Screening" were combined using the Boolean operator "AND." Articles published between 2019 and 2024 in Portuguese, English, and Spanish were considered. A total of 25 articles were selected from an initial pool of

13,123. Alpha-synuclein was the most relevant biomarker, associated with disease progression, despite detection challenges. Neuroimaging modalities such as PET, SPECT, and fMRI proved crucial in differential diagnosis. PET distinguished LBD from Alzheimer's, while SPECT detected early dopaminergic deficits. Combining biomarkers with neuroimaging enhanced diagnostic accuracy, particularly in early disease stages. Early diagnosis of LBD is essential to improve quality of life and treatment outcomes. Biomarkers in body fluids, skin biopsies, and nasal swabs show diagnostic potential. Functional neuroimaging aids in disease monitoring and differentiation. Larger multicenter studies are needed to validate these approaches.

TECHNICAL AREA: BIOMARKERS
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Impact of GLP-1 agonists on the progression of Alzheimer's disease: a systematic review

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Introduction: Alzheimer's disease (AD) is the most common form of dementia and a major public health concern. Its rising incidence has increased interest in therapies targeting pathophysiological mechanisms. GLP-1 receptor agonists (GLP-1 RAs), initially developed for type 2 diabetes (T2DM), have demonstrated potential neuroprotective effects. **Objective:** This systematic review evaluated the association between GLP-1 RAs and the incidence, progression, or cognitive symptoms of AD. **Methods:** A comprehensive search was conducted in PubMed, Scopus, Embase, and Web of Science until February 2024. Eligible studies assessed the impact of GLP-1 RAs on cognition, dementia risk, or AD-related biomarkers. Two reviewers independently selected studies and extracted data. The protocol was registered in PROSPERO (CRD42025105947). **Results:** A total of five studies met inclusion criteria: two randomized controlled trials, two systematic reviews with meta-analyses, and one additional systematic review. Most studies reported a reduction in dementia risk associated with GLP-1 RA use, with odds ratios ranging from 0.54 to 0.72. Cognitive improvements were observed with liraglutide and exenatide in individuals with mild cognitive impairment or AD. Proposed mechanisms include enhanced insulin sensitivity, anti-inflammatory properties, and decreased levels of neurodegenerative biomarkers. **Conclusion:** These findings suggest that GLP-1 RAs may reduce AD risk and improve cognitive outcomes, particularly in patients with T2DM. Their mechanisms indicate a possible disease-modifying role. However, further well-designed controlled studies are needed to establish the clinical efficacy and safety of GLP-1 RAs for neuroprotection in AD.

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Lifetime cumulative estrogen exposure influences Tau deposition in women

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Evidence suggests that sex influences tau deposition in the human brain and affects its relationship with Alzheimer's Disease (AD) outcomes. This study evaluated whether lifetime estrogen exposure influences tau deposition in women. 456 participants from the HEAD study (mean age: 66 years) were analyzed using plasma Tau-PET (Flortaucipir, MK6240), Aβ-PET, and p-Tau 217 (n = 352). Women were classified according to years of estrogen exposure (menarche to menopause or current age, including hormone therapy use, if applicable): low (<40 years), regular (40–45), and high (>45). Regression models adjusted for age, clinical classification, and amyloid load assessed sex differences and the influence of estrogen exposure on tau deposition. Women with regular exposure showed greater tau uptake in Braak III-IV and V-VI regions compared to men, a difference not observed without stratification. The interactions between Aβ-PET and sex were stronger in these regions, especially in women with regular exposure. The interaction between p-Tau 217 and sex was significant in Braak I-II regions only in women with exposure <35 or >40 years, but not in those with regular exposure. We conclude that cumulative estrogen exposure influences tau deposition in women in a non-linear fashion. These findings highlight the role of hormones in the pathophysiology of Alzheimer's disease and the need for further studies on the hormonal impact throughout life.

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Performance of plasma biomarkers for diagnosis and prediction of dementia in a Brazilian cohort

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Background: Dementia is a growing concern throughout the developing world and is severely underdiagnosed among the Brazilian population. Despite remarkable progress in the biomarker field in recent years, local testing and validation of plasma biomarkers of AD and dementia is still lacking in Brazil and Latin America. **Methods:** In this longitudinal cohort study of 145 participants, the diagnostic performance of plasma biomarkers was assessed based on clinical diagnosis and CSF biomarker positivity. Follow-up data of up to 4.7 years were used to determine biomarker performance in predicting diagnostic conversions. The study was conducted at the Memory Clinic at the D'Or Institute for Research and Education (IDOR) in Rio de Janeiro. Participants were native Brazilians of at least 60 years of age. They were categorized as cognitively unimpaired (n=49) or cognitively impaired (CI; n=96), with CI participants further classified as aMCI, Alzheimer's disease (AD), Lewy body dementia, or vascular dementia. Plasma Tau, Aβ40, Aβ42, NFL, GFAP, pTau231, pTau181 and pTau217 were measured using a SIMOA HD-X instrument. CSF biomarker data was available for 36% of the sample. **Results:** The pTau217/Aβ42 ratio was the top-performing biomarker for discriminating participants based on CSF-biomarker positivity (ROC AUC=0.98), followed by pTau217 alone (ROC AUC=0.94). **Conclusions:**

In this first biomarker profiling of a Brazilian dementia cohort, plasma pTau217 confirmed its potential as a clinically useful diagnostic tool. This study comprises an initial step towards local validation and adoption of dementia biomarkers in Brazil and Latin America.

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Plasma biomarkers are related to white matter integrity in the Alzheimer's disease continuum.

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Background: Plasma biomarkers have been highlighted as an innovative tool for the early diagnosis of Alzheimer's disease (AD), offering a less invasive and more economical alternative to cerebrospinal fluid biomarkers. It is necessary to understand the relationship between blood biomarkers and other markers related to AD, such as neuroimaging alterations that reflect neurodegeneration. **Objective:** This study aimed to investigate if plasma biomarkers are correlated with white matter integrity. **Methods:** 63 older adults (42 with mild cognitive impairment, 8 with subjective cognitive decline, and 13 controls) underwent to magnetic resonance imaging exam and blood sample collection. Plasma biomarkers (Aβ40, Aβ42, tTau and pTau181) concentrations were analyzed in SIMOA. White matter integrity was analyzed using diffusion tensor imaging in tracts associated with AD. Higher fractional anisotropy (FA) values indicate greater neuronal integrity, whereas higher axial diffusivity (AxD) and radial diffusivity (RD) values are indicative of increased axonal degeneration. Spearman's correlation was used. **Results:** higher concentrations of pTau-181, Aβ42, Aβ40, pTau-181/Aβ42 and Aβ40/pTau181 were correlated with increased AxD. Additionally, lower Aβ42/pTau181 ratios were linked to increased AxD. Higher tTau, Aβ42, tTau/Aβ40 and Aβ40/pTau181 were associated with increased RD. Conversely, lower Aβ40/tTau ratios correlated with higher RD. Regarding FA, we found a positive correlation with Aβ40/tTau and a negative correlation with tTau, pTau181/Aβ40, and tTau/Aβ40. **Discussion:** Our results suggest that higher plasma concentrations of AD biomarkers are associated with increased axonal degeneration. **Conclusion:** plasma biomarkers reflect changes in white matter structure, supporting their potential role as indicators of neurodegeneration in AD.

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Accuracy of the figure memory test of brief cognitive screening battery to diagnose Alzheimer's disease in older adults with mild cognitive impairment

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Background: The Brief Cognitive Screening Battery (BCSB) has recently been proposed for use in primary care, demonstrating good accuracy for dementia diagnosis, however, its accuracy for the etiological diagnosis of Alzheimer's disease (AD) is yet to be established. **Objective:** To assess the accuracy of the Delayed Recall of Figure Memory Test from the BCSB (DR-FMT) for the biological diagnosis of Alzheimer's disease in older adults with mild cognitive impairment. **Methods:** This retrospective cross-sectional study included 96 older adults with amyloid PET, comprising 31 with mild cognitive impairment (MCI) and 65 with cognitively unimpaired (CU) status. The mean age was 72.5 and 73.7, respectively. They were evaluated using the BCSB and Mini-Mental State Examination (MMSE), Rey Auditory Verbal Learning Test (RAVLT), Logical Memory (LM), and Rey Complex Figure. The biological diagnosis of AD was determined from amyloid PET (core biomarker 1 according to Alzheimer's Association criteria). **Results:** Positive amyloid PET was found in eight participants with MCI (25.8%), and twelve CU controls (18.5%). In the MCI group, DR-FMT significantly predicted the biological diagnosis of AD (AUC=0.893, 95%CI 0.717–1.000), with a cutoff of ≤7.0 yielding 75% sensitivity, 78.0% specificity, and 77.4% accuracy. The negative predictive value was 90%. **Conclusion:** The DR-FMT demonstrates good accuracy for the diagnosis of Alzheimer's disease in older adults with MCI, outperforming other cognitive measures such as RAVLT delayed recall.

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Cerebrospinal fluid biomarkers: a diagnostic comparison between the 2018 and 2024 criteria

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Background and Objectives: Cerebrospinal fluid (CSF) biomarkers have become tools for the diagnosis of Alzheimer's disease (AD). Two main diagnostic frameworks—the ATN system proposed by the NIA-AA in 2018 and the updated version released by the Alzheimer's Association (AA) in 2024—offer distinct classification criteria. This study aimed to compare diagnostic outcomes and assess potential clinical differences between the analyzed groups. **Methods:** We performed a retrospective analysis of prospectively collected data from 97 patients treated at a specialized neurology outpatient clinic, according to the 2024 Brazilian Academy of Neurology (ABN) guidelines. Patients were classified based on CSF biomarker profiles into two groups, according to the 2018 and 2024 criteria. Variables assessed included sex, clinical

presentation, age at symptom onset, and Mini-Mental State Examination (MMSE) scores. Statistical analysis was performed using the t-test, Fisher's exact test, and Mann-Whitney test (SPSS v26, Windows). **Results:** Among the 97 patients, 33 (34%) met the 2018 criteria and 59 (60%) met the 2024 criteria for AD diagnosis ($p=0.0003$). Patients classified only under the 2024 criteria showed similar demographic and cognitive profiles when compared to the 2018 group. Neuropsychiatric symptoms were present in 6 patients (23%) in the 2024 group, while 7 patients (21%) in the 2018 group presented amnesic syndrome. **Conclusions:** The 2024 criteria identified a higher number of patients as having AD based on CSF biomarkers, although no relevant clinical differences were observed between the groups.

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Clinical profile of patients presenting with rapidly progressive dementia and correlation with cerebrospinal fluid biomarkers for Alzheimer's disease

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Background: Cerebrospinal fluid (CSF) biomarkers are valuable in diagnosing Alzheimer's disease (AD) and may assist in the differential diagnosis of rapidly progressive dementia (RPD). **Objective:** To evaluate the clinical profile of patients with RPD who underwent CSF biomarker testing for AD in a specialized neurology clinic. **Methods:** All patients underwent lumbar puncture for CSF biomarker analysis according to the 2024 Alzheimer's Association (AA) criteria. Sixteen patients were included and classified into two groups: AD (G1) and non-AD (G2) based on CSF biomarker profiles. Statistical analysis was performed using the t-test with Welch's correction (SPSS, version 26). **Results:** The mean age at symptom onset was 63 years (range 35–81), with 56% female patients. Mini-Mental State Examination (MMSE) scores ranged from 0 to 14 (mean 8.4). Eleven patients (69%) were classified as G1 and five (31%) as G2. No statistically significant differences were observed between groups regarding age at onset, sex, or MMSE scores. Additionally, no significant differences were found when comparing the classification results of the 2024 AA criteria with the 2018 NIA-AA criteria. **Conclusion:** No significant differences in clinical profiles were identified between AD and non-AD groups in patients presenting with RPD.

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Association between peripheral chemokine levels and cognitive deficits in mild acute ischemic stroke: a cross-sectional study

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Background: Ischemic stroke, a leading cause of death and disability worldwide, frequently results in cognitive decline. The inflammatory response to cerebral ischemia involves chemokine release and increased blood-brain barrier permeability, exacerbating brain injury and neuroinflammation. However, evidence regarding pathophysiological mechanisms and markers that can predict cognitive symptoms following mild to moderate stroke remains scarce. **Objective:** We aimed to investigate plasma levels of chemokines and their relationship with cognitive abilities in post-stroke patients compared to healthy controls. **Methods:** Forty-three participants were divided into post-stroke ($n=22$) and control ($n=21$) groups. Stroke severity was assessed using the NIH Stroke Scale (NIHSS) within seven days. Cognitive function was evaluated using the Mini-Mental State Examination (MMSE) and Frontal Assessment Battery (FAB). Plasma levels of CCL2, CCL3, CCL4, CCL5, CCL11, CXCL8, and macrophage migration inhibitory factor (MIF) were measured via ELISA. Linear regression explored associations between chemokine levels and clinical variables. **Results:** Groups were age-, sex-, and education-matched. Post-stroke participants scored lower on FAB subtests and total FAB score. Decreased plasma levels of CCL3, CCL4, and CXCL8 were found in the post-stroke group compared to controls. CCL2, CCL11, and MIF were similar between groups. Higher levels of CCL3 and CCL4 were associated with reduced NIHSS scores. CCL4 also correlated with total FAB score, while CXCL8 correlated with immediate memory. **Conclusion:** Decreased plasma levels of chemokines were associated with mild to moderate stroke and may be useful to predict cognitive decline within a therapeutic window for preventing long-term cognitive sequelae.

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Analysis of CSF biomarkers and clinical profile of patients with suspected normal pressure hydrocephalus followed in a neurology service

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Introduction: Normal pressure hydrocephalus (NPH) represents a diagnostic challenge. It is characterized by a classic triad of symptoms, including gait disturbances, urinary incontinence, and dementia. Patients may not present with classic clinical and radiological findings. Cerebrospinal fluid (CSF) has been important in the diagnosis of NPH, aiding in the differentiation of secondary hydrocephalus, as well as in the evaluation of the presence of other associated neurodegenerative diseases. **Objective:** To analyze the results of CSF biomarkers and the clinical characteristics of patients with suspected NPH. **Methods:** This is a retrospective study with prospective analysis of biomarker data collected from patients treated at a specialized neurology outpatient clinic in cases of diagnostic uncertainty, following the recommendations of the 2024 Brazilian Academy of Neurology (ABN) guidelines. Patients with suspected NPH who underwent tap test and CSF biomarker analysis were selected. **Results:** Six patients were selected; 33.3% met the criteria for Alzheimer's disease (AD) according to the 2024 AA guidelines and 50% of these had a positive response to the tap test. Among those with a negative test for AD,

75% responded to the tap test. All selected patients had a negative T profile. None of the patients met the criteria for AD according to the 2018 ATN. The patients evaluated had symptom onset in the post-senile period, initially presenting amnesic symptoms. **Conclusions:** Biomarkers can be an important tool in the differential diagnosis of patients with suspected PHN, especially when there is overlapping pathologies.

BASIC SCIENCE

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Association of cognitive decline and peripheral inflammatory markers in Long COVID

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Background: COVID-19 has been extensively associated with persistent cognitive and immunological changes, even months after acute infection. However, the association between these parameters in long COVID (LC) remains poorly understood. **Objective:** This study aimed to investigate associations between cognitive performance and peripheral inflammatory cytokine levels in individuals with LC. **Methods:** 24 subjects with LC-related cognitive impairment (6–12 months after SARS-CoV-2 infection) and 25 healthy (naïve) individuals matched by sex, age, and education were included in this study. Cognitive performance of LC individuals was assessed using Trail Making Test (TMT), Hopkins Verbal Learning Test (HVL), Stroop, Digit Span, verbal fluency (animals), Symbol Digit, and Boston Naming Test. Plasma concentrations of the interleukins (IL) IL-4, IL-6, IL-10, IL-18, and Tumor Necrosis Factor (TNF) were measured using enzyme-linked immunosorbent assay (ELISA). **Results:** LC individuals showed cognitive deficits in both attention and memory domains. Specifically, 71% had selective attention and inhibitory control impairment in Stroop (56% severe), 67% sustained and alternating attention deficits in TMT (56% severe), and 63% episodic memory impairment in HVL (40% moderate). Plasma levels of IL-6, IL-10, IL-18, and TNF were decreased in the LC group compared to controls. In addition, higher levels of IL-4, IL-18, and TNF correlated with better executive function performance. **Conclusion:** The results indicate cognitive deficits in LC, primarily affecting attention and memory domains, accompanied by reduced peripheral cytokine levels. Our findings also suggest that plasma levels of IL-4, IL-18, and TNF may be useful to predict long-term cognitive performance in individuals with LC.

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Early disruptions in the reelin pathway dynamics in a rat model of autosomal dominant Alzheimer's disease

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A rare RELN variant has been associated with cognitive resilience in PSEN-1-mutated autosomal dominant Alzheimer's disease (ADAD). Interestingly, children carrying PSEN1 mutations exhibit brain hyperconnectivity and structural abnormalities, suggesting developmental disruptions. However, whether the reelin signaling is altered during neurodevelopment in ADAD remains unclear. Thus, we investigated the expression of Reln, Mapt, Psd95, and Cldn5, genes associated with the reelin pathway, in rat embryos harboring ADAD mutations. For this, brain tissue was collected at gestational day 13.5 from wild-type (WT) and TgF344-AD (TG) embryos (e) of WT and TG progenitors (p). Gene expression was quantified using RT-qPCR, and gene interactions were assessed through Pearson correlation and linear regression analyses ($p < 0.05$). Overall, no differences in gene expression were observed. However, correlation analysis revealed positive associations within the reelin pathway in groups with at least one transgenic trait. In WTpTG, Reln correlated with Mapt ($\beta = 0.37$, $r^2 = 0.85$), while Mapt correlated with Psd95 in TGp-WT ($\beta = 0.85$, $r^2 = 0.89$). In TGpTG, Reln correlated with both Mapt ($\beta = 0.99$, $r^2 = 0.88$) and Psd95 ($\beta = 0.79$, $r^2 = 0.94$). Additionally, Mapt and Psd95 were significantly correlated ($\beta = 0.75$, $r^2 = 0.95$). Cldn5 showed no associations. Although expression levels remained unaffected, the altered correlation patterns suggest early modulation of the reelin pathway by ADAD mutations. The absence of Cldn5 associations supports pathway specificity. In conclusion, these findings of gene interaction dynamics suggest that reelin signaling may be disrupted from early embryonic stages in ADAD.

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Irisin attenuates astrocyte activation and rescues memory in Alzheimer model mice

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Alzheimer's disease (AD) is neurodegenerative disease characterized by memory failure, and severe cognitive impairment. Physical exercise stimulates neuroprotective pathways and has been reported to alleviate memory impairment in AD. Irisin, an exercise-induced hormone, is secreted following proteolytic cleavage of fibronectin type-III-domain-containing. Irisin protects synapses and rescue memory in mouse models of AD. The aim of the current study was to evaluate the possible role of irisin in astrocytes within the framework of AD pathophysiology. We exposed primary cortical astrocyte cultures to irisin (25 nM), and the expression of neurotrophic factors, $\alpha 5\beta 5$ integrin receptor, and LDH5 α were determined by RT-PCR. MAPK activity was determined by Western blot. Cognitive impairment was investigated in C57BL/6 mice that received an intracerebroventricular infusion of A β oligomers (A β Os, 10 pmol) in the presence or absence of irisin (30 pmol). The expression of neurotrophic factors was determined in the hippocampus and frontal cortex of the same animals 7 days after the A β Os or irisin infusion by RT-PCR. We demonstrate that irisin promotes the expression of brain-derived neurotrophic factor (BDNF), nerve growth factor (NGF), glial cell line-derived neurotrophic factor (GDNF), $\alpha 5$ subunits of integrin receptors and LDH5 α , and stimulates transient activation of ERK 1/2 in astrocytes. We further show that irisin attenuates memory impairments in an AD mouse model, and increases hippocampal expression of BDNF, GDNF and NGF in vivo. Our findings support the idea that physiological protection by irisin against synaptotoxic A β Os can be mediated by modulation of astrocyte-derived factors.

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NLRP3 inflammasome in Alzheimer's disease: strengthening the link between neuroinflammation and hippocampal selective vulnerability

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Background: Neuroinflammation is recognized as a key contributor to the early stages of neurodegenerative diseases. Experimental models have demonstrated the deleterious role of inflammasomes in Alzheimer's disease (AD). However, neuropathological studies evaluating NLRP1 and NLRP3 pathways in human AD brains are limited, and their association with cognitive decline remains unclear. **Methods:** We analyzed hippocampal postmortem samples from individuals with pure AD neuropathological change (ADNC) and cognitively healthy controls from an admixed Latin American cohort. Immunohistochemistry and morphometric analyses quantified the expression of NLRP1, NLRP3, caspase-1, ASC, gasdermin D, IL-1 β , IL-18, amyloid- β , and hyperphosphorylated tau (p-tau) in the cornu ammonis (CA), dentate gyrus (DG), and subiculum (SUB). We performed 3D reconstructions to visualize the spatial distribution of NLRP3/ASC and AD pathology. Ordinal logistic regression assessed associations between the inflammatory proteins and pathological burden, while linear regression evaluated correlations with cognitive performance (CDR-SB). **Results:** NLRP3, ASC, caspase-1, IL-1 β , and IL-18 were significantly overexpressed in the CA and SUB of ADNC cases compared to controls. The NLRP3 pathway was associated with amyloid and tau pathology and higher CDR-SB scores, particularly in CA and SUB regions. No significant association was observed for NLRP1. **Discussion and Conclusion:** Our findings suggest that NLRP3 inflammasome activation contributes to hippocampal vulnerability, pathological progression, and cognitive decline in AD. These results highlight NLRP3 as a potential target for therapeutic strategies for modulating neuroinflammation in AD. Alzheimer's disease, neuroinflammation, inflammasome, neuropathology

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Plasma levels of CX3CL1, NSE, and S100B reflect cognitive impairments and stroke severity in acute ischemic stroke

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Introduction: Ischemic stroke (IS) leads to abrupt cerebral hypoperfusion, triggering neuroinflammation and neuronal injury. Post-stroke cognitive impairment and dementia significantly contribute to global morbidity and mortality, and early therapeutic intervention is essential to limit long-term deficits. Identifying reliable plasma biomarkers may help stratify IS severity and predict cognitive outcomes. **Objectives:** This study aimed to investigate the association between markers of neurochemical and neuronal injury with acute IS and cognitive dysfunction. **Methods:** We assessed 21 patients with mild to moderate IS and 22 matched healthy controls. Within seven days of stroke onset, participants underwent clinical evaluation (NIHSS) and cognitive testing (MMSE, FAB, and VMT). Plasma levels of S100B, CX3CL1, BDNF, NSE, APP, and NFL were quantified. **Results:** Compared to controls, IS patients had impairments of frontal functions (FAB, $p=0.002$), immediate memory ($p=0.001$), learning ($p=0.045$), recall ($p=0.044$), and recognition ($p=0.037$). Increased CX3CL1 ($p=0.049$) and NSE ($p=0.002$), and decreased S100B ($p=0.026$) plasma levels were found in IS patients. Higher CX3CL1 and NSE levels correlated with worse FAB scores, NSE was also associated with deficits in immediate memory and learning. Higher S100B levels correlated with stroke severity. **Conclusions:** CX3CL1, NSE, and S100B emerge as promising plasma biomarkers for assessing acute IS severity and predicting early impairment in executive functions. These findings highlight the potential utility of neurochemical and neuronal damage markers in guiding clinical decision-making during the acute phase of ischemic stroke. Stroke; neurochemical markers; neuronal injury; cognition

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Correlation between anthropometric parameters and cognitive function in patients with Alzheimer's disease: preliminary findings

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Background: Alzheimer's disease (AD) is an age-related condition marked by memory loss and cognitive decline. Sarcopenia, also age-related, involves the progressive loss of muscle mass and function. Both conditions negatively impact health and quality of life. Studying their association is a key to improving patient care and outcomes. **Objective:** To investigate anthropometric parameters and their correlation with cognitive function

in patients diagnosed with AD. **Methods:** This cross-sectional study included patients aged 60 or older with mild cognitive impairment or mild/moderate dementia due to AD treated at a specialized outpatient clinic. Anthropometric measurements - neck, arm, waist, and calf circumferences, and weight - followed WHO guidelines, and cognitive function was assessed using the ACE-R test and Mini-Mental State Examination (MMSE) score. Statistical analysis included descriptive measures and correlation tests with significance at $p < 0.05$. **Results:** Fourteen patients (71.4% women; mean age 76.7 years) were evaluated. Mean MMSE score was 19.8 ± 5.7 . These anthropometric variables were analyzed for correlation with MMSE and ACE-R tests. Only calf circumference showed a significant positive correlation with ACE-R score ($r = 0.618$; $p = 0.019$). **Discussion:** Considering calf circumference as an indicator of muscle mass loss in sarcopenia, its correlation with cognitive function highlights the importance of detecting muscle impairment to help predict cognitive decline in patients with AD. **Conclusion:** The findings suggest a positive correlation between calf circumference and ACE-R score. Exploring the link between cognitive function and sarcopenia in AD patients may enhance care and quality of life by guiding monitoring and treatment.

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Determining the levels of brain opioid receptors in Alzheimer's disease

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Background: While opioid neuromodulation has been mostly investigated in the context of pain and addiction, mounting results sustain their importance for memory and cognition. The opioid receptor family comprises three major receptors: δ -opioid (DOR), κ -opioid (KOR), and μ -opioid (MOR). MOR is a G protein-coupled receptor that both blocks cAMP through Gi, and activates the MAPK and β -arrestin pathways. MOR is activated by endogenous opioid peptides (endorphins and enkephalins) but also by potent synthetic (and addictive) compounds used to treat severe pain, including morphine, oxycodone, and fentanyl. However, despite its importance for cognitive processes, whether components of opioid signaling are disrupted in Alzheimer's disease (AD) is unknown. **Objective:** Evaluate the levels of opioid receptors and their ligands in AD and control brains, and in the APP/PS1 mouse model of AD. **Methods:** qRT-PCR and ELISA were used to determine the levels of opioid receptors and ligands. **Results:** We determined that MOR mRNA levels, but not KOR or DOR, are reduced in APP/PS1 mice, compared to their wild type littermates. We also found that MOR mRNA levels are reduced in the AD hippocampus, but not prefrontal cortex. Using ELISA, we found a differential regulation of the endogenous ligands in AD. We observed a decrease of β -endorphin levels in the AD Prefrontal Cortex and an increase of Leu-enkephalin in the AD Hippocampus. **Conclusions:** Our results suggest potential deregulations of brain μ -opioid signaling components in AD, further analyses are necessary to characterize the activity, localization and role of the opioid system in AD.

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Investigating the role of exercise linked-irisin and extracellular vesicles in Alzheimer's disease

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Exercise is known to prevent and improve Alzheimer's disease (AD), and upregulates irisin, a myokine derived from FNDC5, previously shown to mediate the beneficial actions of exercise on memory in AD mouse models. Exercise causes an overall increase in plasma extracellular vesicles (EVs), which are membranous particles that circulate and carry various biomolecules, including myokines. Interestingly, EVs modulate some of the physiological roles of FNDC5/irisin in peripheral tissues. However, the involvement of EV-associated FNDC5/irisin in the brain has not been investigated. Herein, we hypothesize that exercise stimulates free and EV-associated irisin levels and that EVs transporting irisin reach the brain, resulting in beneficial effects in cognition. To test this, we investigated whether EVs from mice carry FNDC5/irisin and whether intravenous injection of these EVs exerts neuroprotective effects. Our results show that FNDC5/irisin increases in plasma EVs of exercised mice or mice that overexpress FNDC5. Furthermore, we report that the intravenous treatment with EVs obtained from exercised wild-type mice, but not from exercised FNDC5 KO mice, elevates the BDNF cortical expression and rescues memory in an AD mouse model. Similarly, EVs obtained from mice overexpressing FNDC5/irisin also rescued memory in a transgenic AD mouse model. These findings suggest that EVs play a critical role in mediating FNDC5/irisin communication between the periphery and the brain, providing insight into the role of EVs as carriers of FNDC5/irisin during exercise and suggesting a therapeutic potential of EVs carrying irisin to restore cognition in AD.

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Impact of early auditory deprivation on the hippocampus of swiss mice

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Background: In rodents, low-threshold hearing begins on the 14th post-natal day; however, the maturation of auditory pathways continues thereafter, with the refinement of neuronal properties resulting from auditory stimulation. Evidence suggests that early acoustic deprivation impairs both auditory and cognitive processing. In humans, auditory maturation occurs from the 38th embryonic week until 12 months of age, and auditory experience during this period is crucial for the development of auditory and processing pathways. Evidence shows that hearing loss is associated with dementia, such as Alzheimer's disease, but the mechanisms underlying cognitive deficits remain unclear. **Objective:** To investigate the effects of early acoustic deprivation on hippocampal cognitive and motor functions. **Me-**

thods: Swiss mice were divided into two groups: the Plug group (bilateral auditory canal occlusion on the 14th postnatal day with silicone elastomer for 30 days) and the Sham group (control). The efficacy of occlusion was verified by Auditory Brain Response testing. Morphological analysis of the hippocampus was performed using the Golgi-Cox impregnation method and evaluation of dendritic spine densities. Sensorimotor pathways were analyzed using the prepulse inhibition (PPI) of the acoustic startle reflex. Hippocampal function was assessed by the Object Recognition Test. **Results:** The Plug group exhibited a significantly higher auditory threshold, increased startle reflex latency, reduced PPI-mediated reflex inhibition, and deficits in the Object Recognition Test, suggesting auditory and hippocampal impairment. **Conclusion:** Early auditory deprivation has deleterious effects on the development of auditory pathways and cognitive functions.

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Alzheimer's disease brain- derived extracellular vesicles induce cognitive impairment in mice

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Alzheimer's disease (AD) is a neurodegenerative disorder, being the main form of dementia in the elderly. The deposition of β -amyloid peptide (A β) and hyperphosphorylation of tau are the major histopathological hallmarks in the brain of AD patients. Extracellular vesicles (EVs) are lipid nanoparticles secreted by many cell types. EVs carry in their lumen several biomolecules from their cell of origin and can play an important role in intercellular communication. Previous work has shown that EVs can play an important role in the spread of neurodegenerative diseases. Brain-derived EVs from AD patients contain increased levels of neurotoxic A β species. In this work, we investigate whether EVs derived from post mortem brain tissue from individuals diagnosed with AD and from APP/PS1 transgenic mice, respectively, induce changes in the behavior and memory of C57BL/6 mice, when injected via intrahippocampal route. We detect the presence of APP in AD brain-derived EVs, as well as in microvesicles isolated from the brain of APP/PS1 mice. Furthermore, our results suggest that C57BL/6 animals injected with AD brain-derived EVs were able to induce a cognitive impairment after 14 and 21 days of injection. Together, our data indicate the ability of EVs to carry pathological molecules associated with AD, as well as to induce memory impairment that recapitulate features of AD. This evidence makes EVs an important study target for AD, in addition to pointing them out as a potential target for new therapeutic approaches in neurodegenerative diseases in the future.

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Early excitatory/inhibitory imbalance in embryos of an autosomal dominant Alzheimer's disease rat model

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Impairments in the excitatory/inhibitory (E/I) balance have been observed in Alzheimer's disease (AD) individuals. Intriguingly, autosomal dominant AD (ADAD) mutation carriers present brain hyperconnectivity during childhood, suggesting an early E/I imbalance in AD. However, it remains unclear at what stage of development this E/I dysfunction emerges. We aimed to investigate the expression of genes involved in E/I balance during neurodevelopment in an ADAD rat model. We collected brain tissue from wild-type (WT) and TgF344-AD (TG) embryos (e) at gestational day 13.5. We quantified brain gene expression of the solute carrier family 1 member 3 (Slc1a3; excitatory cells), parvalbumin (Pvalb) and somatostatin (Sst)—markers of GABAergic inhibitory interneurons, glial fibrillary acidic protein (Gfap; astrocytes), and integrin subunit alpha-M (Itgam; microglia) using RT-qPCR (2- $\Delta\Delta$ Ct method). Statistical comparisons between WTe and TGe were conducted with Student's t-test. Pearson's correlation examined gene-gene associations. No significant differences were observed in the expression of Slc1a3, Pvalb, Sst, Gfap, and Itgam between WTe and TGe (p-values ranging from 0.12 to 0.97). In the WTe group, significant correlations were demonstrated between Pvalb and Itgam, Slc1a3 and Gfap, and Sst and Slc1a3 gene expression ($r=-0.67$, -0.77 and 0.73 ; $r^2=0.45$, 0.60 and 0.53 , respectively $p<0.05$), while no significant correlation was observed for TGe. Our results suggest that there is an early E/I imbalance in ADAD embryos. Further analyses should be conducted at advanced gestational days to better understand the effect of ADAD mutations on the E/I balance during neurodevelopment.

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Association of the FNDC5 rs1746661 single nucleotide polymorphism with reduced brain glucose metabolism in elderly humans

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Fibronectin type III domain—containing protein 5 (FNDC5) and its derived hormone, irisin, have been associated with metabolic control in humans, with described FNDC5 single nucleotide polymorphisms being linked to obesity and metabolic syndrome. Decreased brain FNDC5/irisin has been reported in subjects with dementia due to Alzheimer's disease. Since impaired brain glucose metabolism develops in ageing and is prominent in Alzheimer's disease, here, we examined associations of a single nucleotide polymorphism in the FNDC5 gene (rs1746661) with brain glucose metabolism and amyloid- β deposition in a cohort of 240 cognitively unimpaired and 485 cognitively impaired elderly individuals from the Alzheimer's Disease Neuroimaging Initiative. In cognitively unimpaired elderly individuals harbouring the FNDC5 rs1746661(T) allele, we observed a regional reduction in low glucose metabo-

lism in memory-linked brain regions and increased brain amyloid- β PET load. No differences in cognition or levels of cerebrospinal fluid amyloid- β 42, phosphorylated tau and total tau were observed between FND5 rs1746661(T) allele carriers and non-carriers. Our results indicate that a genetic variant of FND5 is associated with low brain glucose metabolism in elderly individuals and suggest that FND5 may participate in the regulation of brain metabolism in brain regions vulnerable to Alzheimer's disease pathophysiology. Understanding the associations between genetic variants in metabolism-linked genes and metabolic brain signatures may contribute to elucidating genetic modulators of brain metabolism in humans.

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Low dose ketamine modulates microglial reactivity in an Alzheimer's disease mouse model

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Alzheimer's disease (AD) is a neurodegenerative disease that affects memory and cognition, causing dementia. Its pathology is characterized by accumulation of tau aggregates, β -amyloid peptides, in insoluble plaques and soluble oligomers, and neuroinflammation. Microglia are central nervous system immune cells that respond to potentially harmful stimuli by releasing cytokines and increasing their phagocytic capacity. In AD, chronic activation of microglial response leads to tissue damage and persistent neuroinflammation. Ketamine is an NMDA receptor antagonist, with antidepressant properties in low doses. Among its mechanisms, ketamine has been proposed to exert anti-inflammatory activity. Here, we investigated if microglial functions are modulated by low dose ketamine in APP/PS1 mice. Transgenic APP/PS1 mice chronically treated with low dose ketamine displayed an increased number of microglia around amyloid plaques in the hippocampus. Morphological analysis by the Sholl method revealed that ketamine induces morphological changes in APP/PS1 and WT microglia, but in opposite directions. Ketamine treatment was associated with a decrease in the average length of microglial processes in APP/PS1, but with an increase in WT mice. Gene expression analyses by qPCR suggest that ketamine induces expression of interferon gamma (IFNG), previously associated with the microglial response in AD, in WT animals, but not in APP/PS1. Our results indicate that microglial function is modulated by low dose ketamine by mechanisms that may interact with AD pathology. However, further experiments are required to confirm these findings and identify precise molecular mechanisms.

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Linking cognitive deficits to neuroinflammation in long COVID: a translational approach

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Background: Long COVID-19 is marked by persistent symptoms following acute SARS-CoV-2 infection. Emerging evidence links COVID-19 to neurodegeneration, suggesting a role in dementia development and progression. **Objectives:** We aimed to evaluate cognitive performance and plasma levels of glial and neurochemical markers in individuals with long COVID (6–12 months post-infection) compared to age- and sex-matched healthy controls (<65 years; n=25/group). We also sought to investigate these markers in the brains of C57Bl/6 mice (n=6/group) inoculated via intranasal route with the betacoronavirus MHV-A59. **Methods:** Plasma levels of GFAP, TREM2, CX3CL1, BDNF, GDNF, and NGF were measured by ELISA. Cognitive function was assessed using a comprehensive neuropsychological battery: digit span, verbal fluency, Trail Making Test (TMT), Stroop, Hopkins Verbal Learning Test (HVLT), Boston Naming, and symbol-digit tests. The prefrontal cortex and hippocampus of mice were collected at 5, 30, and 60 days post-infection (dpi) for marker analysis. **Results:** Each group included 23 women (92%). Median age was 39 (controls) and 35 (long COVID). Long COVID participants had more severe impairments in TMT, HVLT, and Stroop, along with decreased plasma levels of TREM2, CX3CL1, GDNF, and NGF. Cognitive deficits correlated with GFAP, TREM2, GDNF, and NGF levels. Mice exhibited similar neurochemical changes in the prefrontal cortex, but not in the hippocampus. **Conclusion:** Plasma levels of glial markers and neurotrophic factors may be useful in predicting post-COVID impairment of executive functions. The MHV-A59 model may provide a unique platform for investigating neuropsychiatric long-term sequelae unleashed by long COVID.

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The eukaryotic elongation factor 2 kinase participates in microglial activation and amyloid clearance in Alzheimer's disease

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Increasing evidence suggests important roles for soluble A β oligomers (A β Os) and, more recently, tau oligomers (TauOs) in the pathogenesis of Alzheimer's disease (AD). Although several cellular events underlying AD pathophysiology are well established, little is known about the mechanisms that lead to the loss of microglial homeostasis, which re-

presents a key factor in disease progression. We and others have previously demonstrated that global protein synthesis is attenuated, and phosphorylation of both eukaryotic initiation factor 2a (eIF2a) and eukaryotic elongation factor 2 (eEF2) are increased in AD patients and mice. However, the role of these translational pathways in AD microglia remains elusive. In the present work, we analyzed the regulation of microglial protein synthesis in AD humans and mouse models. Spatial transcriptomic data indicate that human microglia have increased expression of genes related to ribosome metabolism in symptomatic AD patients. Among several translational factors, we found that eEF2 phosphorylation (p-eEF2) is particularly increased in the cortical microglia of 5xFAD mice and primary microglia cultures exposed to either A β O or TauOs, via distinct mechanisms. Moreover, the pharmacological inhibitor of the eEF2 kinase, A4849-54, prevented, at least in part, A β O- and TauO-induced cell morphology changes and NF κ B nuclear translocation in vitro. Lastly, chronic intraperitoneal treatment with A4849-54 reduced microglial reactivity and amyloid plaque area in 5xFAD female mice. Taken together, these findings point to an important role of microglial p-eEF2 in the regulation of the immune response in AD.

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Impact of deuterated polyunsaturated fatty acids in neuroinflammation-associated cognitive dysfunction

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Neuroinflammation, oxidative stress and lipid peroxidation comprise key events in neurodegenerative diseases and are associated with neuronal damage, synaptic loss and cognitive decline. Recent studies have demonstrated that deuterated polyunsaturated fatty acids (D-PUFA) are able to favor resolvin pathways (e.g. lipoxins) in detriment of the prostaglandin pathway, and therefore promote anti-inflammatory effects. Our group has shown that lipoxin A4 protects against cognitive decline in animal models and is reduced in the liquor of patients with dementia. Here we investigate whether a diet enriched with D-PUFAs is neuroprotective against neuroinflammation induced by LPS and in transgenic animal model of Alzheimer's disease. Three-month-old C57BL/6 male and female mice were fed with D-PUFA enriched diet or control diet (H-PUFA) for two months. Then, mice received intraperitoneal injections of LPS or vehicle. Behavior was tested by novel object recognition (NOR), displaced object recognition (DOR) and open field tasks. Five-month-old 5xFAD animals were fed with D-PUFA diet for 2 months and behavior was tested by NOR, DOR and open field. Our results demonstrated that, whereas animals fed with control diet that received LPS presented deficits in NOR and DOR test, animals fed with D-PUFA diet that received LPS did not present memory impairment. 5xFAD fed with D-PUFA diet were protected from memory impairment in contrast with animals fed with control diet that presented deficit. Brain tissues and plasma were collected for biochemistry analysis. Our findings suggest that D-PUFA diet might be an interesting nutritional approach to mitigate neuroinflammation and cognitive decline.

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Epigenetic upregulation of FNDC5/irisin via CRISPR-dCas9 as a therapeutic approach for Alzheimer's disease

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Alzheimer's disease (AD) is a progressive neurodegenerative disorder affecting over 50 million people globally, with no effective treatment. It is well established that exercise improves brain health and protects against AD, partly by increasing levels of muscle-derived molecules such as FNDC5/irisin in mouse models and humans. During exercise, the cleavage of the membrane protein FNDC5 releases circulating irisin, which can cross the blood-brain barrier. Some studies have also shown that FNDC5 is expressed in the brain. Given this, we hypothesize that epigenetic reprogramming using CRISPR-dCas9 to upregulate FNDC5/irisin could mimic exercise benefits, and represent an alternative therapy for AD. Here, we evaluated the ability of CRISPR-dCas9-VPR to upregulate FNDC5 expression in human iPSC-derived neural progenitor cells (NPCs). We tested single-guide RNAs (sgRNAs) targeting the regulatory region of FNDC5. sgRNAs and mRNAs were transfected with cationic lipids. FNDC5 mRNA expression was measured by TaqMan RT-qPCR, and irisin levels were assessed by ELISA. The specificity of the top 3 leading sgRNAs were assessed using RNA sequencing. The results showed that the sgRNA1 is the most effective for upregulating FNDC5 mRNA up to 380X-fold in NPCs. RNA sequencing showed high specificity with minimal putative off-target effects for the leading sgRNA1. Finally, irisin quantification from the culture supernatants confirmed the increase in protein levels. These results support CRISPR-dCas9-mediated FNDC5/irisin upregulation in vitro, and if we observe similar effects in vivo, it might represent a strategy to mimic exercise-induced neuroprotection and potentially ameliorate AD progression and cognitive decline.

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The role of glial cells in neuroinflammation in the context of Alzheimer's disease

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Background: Alzheimer's disease (AD) is the most prevalent form of dementia worldwide. A key feature of its pathogenesis is the accumulation of β -amyloid oligomers (A β O), which are neurotoxic agents associated with synaptic loss, mitochondrial dysfunction, and impaired proteostasis. Neuroinflammation also plays a crucial role in AD progression, particularly through the activation of microglia and astrocytes into pro-inflammatory states. This study investigates the involvement of these glial cells in AD pathophysiology. **Method:** Morphological and functional changes in glial cells were analyzed using both in vitro and in vivo experimental AD models. First, we examined the activation of primary microglia and astrocytes in vitro after A β O exposure. Second, we assessed in vivo glial activation following intracerebroventricular (icv) injection of A β O in mice. To evaluate inflammation and glial responses, we used RT-PCR and immunocytochemistry to measure cytokine expression

and glial markers. **Results:** Preliminary data indicate increased expression of activation markers such as GFAP, GLAST, IBA1, and F4/80, along with complement proteins like C1q, seven days after A β O infusion. Immunocytochemistry revealed enhanced nuclear translocation of NF- κ B in microglia exposed to A β Os, suggesting activation of inflammatory pathways. **Conclusion:** These findings support the involvement of microglia and astrocytes in AD, showing persistent glial activation in vitro and in vivo. Chronic glial activation may lead to sustained release of cytokines, complement components, and chemokines, contributing to A β accumulation and further neuronal damage, ultimately exacerbating AD pathology.

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Effects of melatonin, Vitamin D, and their combinations in a murine model of sleep deprivation and β A1-42 peptide-induced dementia

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Alzheimer's disease is the most common cause of dementia, and sleep disturbances may precede its clinical diagnosis by years, although the mechanisms involved are not yet fully understood. This study aimed to evaluate the effects of combining memantine (Mem) and donepezil (Done) with melatonin (Mela) or vitamin D (VitD) in a murine model of paradoxical sleep deprivation (PSD) associated with β A1-42-induced dementia. Male and female mice (3 months old) were subjected to PSD for 21 days (18h/day). On day 10, β A1-42 or ACSF was administered intracerebroventricularly. From day 11, animals received by gavage the following combinations: Done+Mem, Done+Mem+Mela, Done+Mem+VitD, or water. Behavioral tests (Y-maze and radial maze) were conducted from days 21 to 27. On day 27, animals were euthanized, and the frontal cortex and hippocampus were collected for cytokine analysis. The PSD+A β 1-42 model caused short- and long-term learning deficits in both sexes. The Done+Mem+Mela combination improved long-term memory in females, while Done+Mem+VitD reversed both short- and long-term memory deficits in both sexes. The model also induced neuroinflammation, with increased TNF- α (female hippocampus), decreased IL-4 (male hippocampus), and reduced IL-10 (female frontal cortex and hippocampus). Done+Mem+Mela reduced neuroinflammation by decreasing TNF- α and IL-1 β (female hippocampus), increasing IL-4 (male frontal cortex and female hippocampus), and IL-10 (male hippocampus). Done+Mem+VitD also reduced neuroinflammation by decreasing TNF- α (hippocampus of both sexes), increasing IL-4 (male hippocampus), and IL-10 (male frontal cortex and female hippocampus). In conclusion, Done+Mem+VitD was the most effective combination in reversing memory impairments and reducing neuroinflammation.

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β -amyloid oligomers stimulate proteasome activity in cultured astrocytes

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The ubiquitin-proteasome system (UPS) is crucial for maintaining cellular homeostasis by regulating protein degradation. Dysregulation of the UPS has been implicated in various diseases, including Alzheimer's disease (AD). Despite the growing interest in studying glia in the context of neurodegenerative diseases, little is known on the function of the UPS in glial cells in AD models. This study aims to evaluate whether and how β -amyloid oligomers (A β Os) modulate the activity of the UPS and its main components in cultured astrocytes. A β O-exposed astrocyte cultures exhibited increased chymotrypsin-like and caspase-like activities. No significant differences were found in levels of the α 1 subunit of the 20S proteasome (20SC2 protein) comparing vehicle- or A β O-exposed cells. Because the catalytic activity of the proteasome can be independent of ubiquitin, the amount of ubiquitinated proteins in astrocyte culture homogenates was quantified by western blotting. No differences were observed between cultures exposed or not to A β Os, indicating that proteolytic function occurs independently of ubiquitin. Immunolabeling for proteasomal protein 20SC2 showed no difference in mean fluorescence intensity values for 20S C2, supporting the notion that the increase in activity is not due to an increase in proteasome levels. These results suggest that A β Os enhance proteasome activity, but not total proteasome levels, in primary cultured astrocytes. Furthermore, activity appears to be independent of ubiquitin, suggesting alternative functional pathways related to the regulation of proteolysis.

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Evaluation of the peptide neurocidentalis in a mouse model of Alzheimer's disease

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β -amyloid (β A) aggregation is a neuropathological hallmark of Alzheimer's disease and a key target to develop new drugs. The wasp venom bioinspired peptide Octovespin previously demonstrated cognitive benefits in mice after intracerebroventricular administration and showed anti-aggregative activity against β A-42 in vitro. However, clinical use requires penetration of the blood-brain barrier and good absorption. So, this study evaluates the neuroprotective effects of Neurocidentalis, a synthetic peptide derived from octovespin, in a β A-42 induced mouse model of Alzheimer's disease by intranasal administration. Male swiss mice underwent neurosurgery for injection of three μ L of β A-42 or PBS into the dentate gyrus. Treatment began one-hour post-surgery and continued for 30 days. The experimental groups were SHAM, β A, memantine (20 mg/kg) and the peptide in three doses (0.2 μ g/2 μ L, 2 μ g/2 μ L and 20 μ g/2 μ L) and they underwent Open Field Test (OFT), Novel Object Recognition test (NORT) and Morris Water Maze (MWM) in the final week of treatment to evaluate their cognition. Memantine was the only group to show locomotor abnormalities in the OFT, probably due to drug toxicity. NORT results showed that peptide-treated groups had no difference from either SHAM or the β A group, which showed negative discrimination. MWM showed that β A animals failed to learn the platform location, whereas the SHAM group reduced latency over time. However, the peptide-treated groups didn't demonstrate cognitive benefits, possibly due to the peptide's poor solubilization requiring dilution in 20% DMSO, which may have caused toxicity. That way, improving the peptide solubilization may enhance its cognitive benefits.

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Amyloid plaque topography obtained with Raman spectroscopy of bitransgenic mice

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Background: Although beta-amyloid aggregation is considered a primary event in AD, neurodegeneration is a sustained, multifactorial process whose relationship with amyloid aggregation is not yet fully understood. Raman Spectroscopy (RS), which enables molecular identification through its vibrational modes, is a robust methodology in the biochemical characterization of neural tissue. **Objective:** To characterize the vibrational topography of amyloid plaques based on main fingerprints of amyloid aggregation and neurodegeneration, as identified by RS in the core and halo regions, respectively. **Methods:** Plaques from cortex and hippocampus of 6- and 12-month-old APPswePS1ΔE9 bitransgenic mice were analyzed with RS. A detailed topographical characterization of the core and halo was obtained based on the spatial distribution of major vibrational fingerprints. **Results:** Two core subregions were identified: a nuclear core marked by phenylalanine and a superficial core marked by amide III. The halo was characterized by C-H, with progressively decreasing intensity with increasing distance from the core. Surrounding the core, a zone of low density of beta amyloid signals was obtained. Lipid halo presented as lipid discontinuous patches separated from the core. **Discussion:** Vibrational findings suggest a densely packed amyloid aggregation in nuclear core, a potentially soluble, less compact aggregation in superficial core, and a dynamic equilibrium between core surface and amyloid oligomers in the surrounding tissue. **Conclusion:** Topographic analysis indicates that, at least beyond a certain stage, plaque expansion is not primarily driven by amyloid aggregation in the core, but rather reflects a continuous and self-sustained neurodegenerative process.

EPIDEMIOLOGY

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Comorbidities and dementia: a clinical profile from a Brazilian public health setting

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Background: Hypertension, hearing loss, and low education are key risk factors for dementia, especially in low-income populations. Protective factors include job stability, higher income in old age, and ongoing intellectual activity. Stroke, psychiatric disorders and alcoholism are also associated with an increased risk. These risks are compounded by limited access to healthcare and education. **Objective:** To assess the prevalence and association of comorbidities with dementia severity. **Methods:** A cross-sectional study was conducted with 62 patients from a cognitive neurology center in Brazil. Data were collected from medical records from 2023. Variables

included age, sex, comorbidities, education, dementia level (CDR), symptom duration, syndromic and etiologic diagnosis. Statistical analysis included Shapiro-Wilk, One-Way ANOVA with Bonferroni post-test, and chi-square test via Monte Carlo simulation, using Stata® 14.0. Significance level was $p < 0.05$. Ethics approval was obtained. **Results:** Among the patients, 37.7% had one comorbidity and 32.8% had two. The most frequent conditions were hypertension (61.3%), diabetes (27.4%), and depression (19.4%). **Discussion:** The study highlights a high prevalence of modifiable comorbidities among individuals with dementia. Additionally, hypothyroidism was also noted as a relevant comorbidity among patients with dementia in our sample. However, the overall evidence regarding the association between thyroid dysfunction and dementia remains inconclusive in recent large-scale analyses. **Conclusion:** Managing modifiable risk factors is crucial, especially in low-income, low-education populations, to help reduce dementia risk and improve outcomes

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Dementia and healthcare services use among older Brazilians: the Brazilian longitudinal study of aging (ELSI-Brazil study)

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Objectives: To describe and compare healthcare service use between individuals with and without dementia, adjusting for sociodemographic and clinical characteristics. **Methodology:** A sample of ELSI-Brasil, comprising 5,249 participants aged 60 and over. Health services use were compared between people living with dementia (PLWD) ($n=362$) and without dementia ($n=4,007$). Logistic regressions models were used to estimate these association (odds ratios fitted with 95% (Cis)). Statistically significant results were test for interaction with urban ($n=4,197$) and rural zone ($n=789$). Results PLWD were older, with less education and, less likely to have a partner, however, across whole use of healthcare service. In fully adjusted models, dementia was strongly associated with any home care visits (OR=2.58; 95%CI 1.74–3.84), number of home care visits (OR=0.40; 95%CI 0.11–0.69), hospitalization in past 12 months (OR=1.97; 95%CI 1.38–2.80), number of hospitalizations (OR=0.13; 95%CI 0.03–0.23), and length of hospital stay (OR=0.8; 95%CI 0.18–1.42). Test of interaction showed that the association of dementia with number of home care visits (coef 0.82, 95%CI 0.60–1.04) and length of hospital stay (coef 0.82, 95%CI 0.59–1.04) was stronger in rural area compared to urban. **Discussion and Conclusion:** PLWD exhibit higher healthcare service utilization. Association of dementia with number of home care visits and increased length of hospital stay was stronger among those living in rural compared to urban areas suggesting that increased service demand reflects both the clinical progression of dementia and structural barriers to outpatient care.

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Dimorphic changes in mortality among antipsychotic users with Alzheimer's disease: a real-world data

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Introduction: Antipsychotic (AP) use in Alzheimer's disease (AD) patients is linked to increased mortality. However, potential sex-based differences in outcomes remain insufficiently explored. This study evaluated mortality risk and survival differences between male and female AD patients using APs. **Methods:** We performed a retrospective cohort analysis using TriNetX, a global health research network with real-world data. AD patients prescribed APs were stratified by sex. Mortality risk, odds ratios (OR), risk ratios (RR), and hazard ratios (HR) were calculated. Kaplan-Meier survival analyses and log-rank tests assessed survival probability over time. **Results:** The female AP user cohort (n=112,566) exhibited lower mortality risk (25.2%) than male AP users (n=74,251; 27.6%). The risk difference was -2.4% (95%CI -2.8% to -2.0%, p<0.001), with an RR of 0.913 and OR of 0.883. Median survival was longer in females (3,018 days) versus males (2,372 days), with survival probability at the study's end of 19.1% vs. 26.3%. HR was 0.823 (95%CI 0.809–0.838, p<0.001). In patients with extended follow-up, mortality remained significantly lower in females (17.2%) versus males (27.2%), with a risk difference of -10% and HR of 0.544 (95%CI 0.533–0.554, p<0.001). Median survival reached 6,770 days in females versus 3,383 in males. **Conclusion:** Female AD patients using antipsychotics consistently demonstrated lower mortality and longer survival compared to males. These findings highlight the importance of incorporating sex as a biological variable in treatment decisions and risk assessments related to AP use in AD.

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Factors associated with the burden of caregivers of people with dementia in Brazil: a cross-sectional study

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Background: Family caregivers of people with dementia (PwD) often face high psychological demands. Identifying factors associated with caregiver burden may inform targeted support strategies. **Objective:** To investigate sociodemographic and clinical factors associated with caregiver burden among PwD and their caregivers in Brazil. **Methods:** This cross-sectional study involved domiciliary interviews with 140 dyads of PwD and their caregivers across 17 municipalities in Brazil's five geographic regions. Associations with Zarit Burden Interview scores were examined using linear regression. **Results:** Caregivers were predominantly women (86.4%), with a mean age of 57.8 years (SD=12.6) and a mean caregiving duration of 1.7 years (SD=0.76). The most common education level was 9–11 years of schooling (28.6%). Care recipients were mostly female (69.3%), with a mean age of 81.3 years (SD=7.9).

Greater caregiver burden was significantly associated with more severe neuropsychiatric symptoms ($\beta=0.205$; $p=0.016$) and with advanced dementia (CDR=3 vs. 1: $\beta=9.666$; $p=0.009$). Compared to partners, sons and daughters showed a nonsignificant trend toward lower burden ($\beta=-5.046$; $p=0.330$), while siblings and other relatives reported significantly lower burden ($\beta=-13.373$; $p=0.021$). **Discussion:** Caregiver burden appears to be more strongly associated with the clinical presentation of dementia and the emotional closeness of the caregiving relationship than with individual caregiver characteristics. **Conclusion:** Interventions that address neuropsychiatric symptoms and offer targeted support to caregiving partners may help mitigate burden.

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Main occupation complexity and later-life cognitive function among older adults in Brazil

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Background: Occupation complexity during adulthood may contribute to cognitive reserve in later life. Research on occupational complexity and cognitive function has focused on high-income countries, where there is a large proportion of individuals with complex occupations.

Objective: Investigate the association between main occupation complexity and cognitive function in the context of low- and middle-income countries where there is greater variance in occupation complexity and where there is a higher proportion of lower-complexity occupations.

Methods: The sample included adults 50 years or older from the 2018 wave of the Brazilian Longitudinal Study of Aging (n=4,314). Participants were classified according to occupational complexity as unskilled (agricultural workers, artisans), semiskilled (customer service workers, trades workers), and skilled (engineers, managers, teachers). We used linear regression models to estimate the association between occupational complexity and cognitive performance adjusted for sociodemographic variables, chronic diseases, and health behaviors. **Results:** Participants had a mean age of 62.7±8.8 years, 48.8% were women, 49.6% had unskilled occupations. Participants with skilled occupations performed better on memory tests than those with unskilled occupations ($\beta=0.19$, 95% confidence interval [CI]=0.02–0.36, $p=0.023$). **Discussion:** Higher occupational complexity may contribute to later-life cognitive reserve in the memory domain. However, there was no significant association between the degree of occupational complexity and the other domains of cognition. **Conclusion:** The results of this study suggest that increasing opportunities for higher-complexity occupations in Brazil may benefit efforts to prevent cognitive impairment in this region.

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Motoric cognitive risk syndrome: prevalence and associated sociodemographic and clinical factors in primary care settings in Brazil

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Background: Motoric Cognitive Risk Syndrome (MCR) is a pre-dementia syndrome characterized by subjective memory complaints and slow gait speed. The syndrome is associated with increased risks of hospitalization and mortality. Studies on MCR remain scarce in low—and middle-income countries, particularly in Latin America. **Objective:** The aim of the present study was to investigate the prevalence of MCR in a community-sample in Brazil and its associations with sociodemographic and clinical factors. **Methods:** Cross-sectional study conducted in four public health/primary care settings in Belo Horizonte, Brazil. Sample size calculation indicated that 310 subjects aged 60+ years should be included. Sociodemographic, clinical, motor (Unified Parkinson's Disease Rating Scale – UPDRS – subsection III) and cognitive (Addenbrooke's Cognitive Examination-Revised - ACE-R) features were assessed. **Results:** The final sample consisted of 355 individuals, with mean age of 71.4 years, 69.6% women. Prevalence of MCR was 11.3%. In multivariate logistic regression analysis, history of falls in the past year (OR=14.4; 95%CI 5.5–37.4), polypharmacy (OR=4.3; 95%CI 1.8–10.3), and higher scores in UPDRS-III (OR=1.1; 95%CI 1.03–1.17) remained significantly associated with MCR. **Discussion:** The present study found a prevalence of 11.3% of MCR among older adults seen at primary care level in Belo Horizonte, Brazil. MCR was associated with polypharmacy, falls and parkinsonian signs. **Conclusion:** These findings are consistent with previous studies and may provide further insights into this preclinical condition, which is easily identifiable at low-cost at primary care/public health settings.

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Multimorbidity and cognitive trajectories in resource-limited settings: real-world evidence from a Brazilian public memory clinic

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Background: Dementia co-occurs with comorbidities that accelerate cognitive decline. Low- and middle-income countries (LMICs) face higher dementia burden due to limited resources, prevalent comorbidities, and population aging. We aimed to identify clinical comorbidities associated with dementia and assess how comorbidity clusters influence cognitive decline. **Methods:** We analyzed 500 adults evaluated over 10 years at a tertiary memory clinic in Brazil's public health system. Participants were classified as subjective cognitive decline (SCD), mild cognitive impairment (MCI),

or dementia. One-way ANOVA and Fisher's exact test compared baseline characteristics. Unpenalized and ridge logistic regressions tested associations between ten binary comorbidities and dementia, adjusting for age and sex. Cluster analysis informed linear mixed-effects models of Mini-Mental State Examination (MMSE) decline. **Results:** Dementia patients were older (74±10.3 years) and less educated (3.1±3.7 years). Hypertension and low education affected over 65 % of participants. Stroke (OR 3.53; 95%CI 1.65–7.57) and alcohol use (OR 6.56; 95%CI 1.69–25.43) predicted dementia diagnosis; stroke remained significant after excluding vascular dementia. Ridge regression confirmed these associations and identified dyslipidemia (OR 1.13 [1.11–1.14]), neurosensory losses (OR 1.34 [1.30–1.38]), and low education (OR 1.32 [1.28–1.36]) as additional predictors. Four clusters emerged: cerebrovascular, cardiometabolic, neurosensory, and substance abuse. A higher cerebrovascular burden predicted a 2.5-fold faster MMSE decline. (–0.43 points/month; p<0.01); neurosensory and substance abuse clusters showed similar trends. **Conclusion:** In this LMIC sample, multimorbidity—especially cerebrovascular, neurosensory, and substance abuse burdens—was linked to dementia risk and accelerated cognitive decline. Low education increased cognitive vulnerability.

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Profiling underdiagnosed dementia in older Brazilian population: Insights from the ELSI-Brasil study

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Introduction: Dementia underdiagnosis is a public health challenge worldwide. Herein, we aimed to estimate the prevalence of underdiagnosed dementia and identify associated sociodemographic factors. **Methods:** Data from the ELSI-Brasil Study (Wave 1, 2015–2016) were used. Individuals were classified with dementia based on a global cognitive composite score (including orientation, fluency, and memory recall scores) and functional impairment, defined as dependency in at least one of four instrumental daily living activities. Individuals were classified into underdiagnosed dementia when previous diagnosis was unreported. Prevalence of underdiagnosed accounted for complex sampling design and sociodemographic factors were evaluated. **Results:** The overall prevalence of dementia was 6.8%; of underdiagnosed dementia, 4.16% (95%CI 3.04–5.29). The Northeast region had the highest rate (8.41%), while the Midwest had the lowest (1.77%). Black participants had the highest prevalence (7.81%) and White, the lowest (2.48%). Prevalence was 11.06% in those with the least education (<2 years) and 6.73% in the lowest income quartile (=7 years) and income (>R\$5,515/month) groups, respectively. Prevalence of underdiagnosed dementia between people with dementia was 77.3%. **Conclusion:** Underdiagnosed dementia is notably prevalent among older adults in Brazil, with disparities across sociodemographic factors, especially education and income. These findings underscore the need for public health initiatives to improve early detection and reduce diagnostic inequalities across the Brazilian population.

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Public knowledge on dementia and its risk factors: a cross-sectional study in a Brazilian urban setting

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Background: The growing prevalence of dementia, especially Alzheimer's disease, is often masked by its misinterpretation as normal aging. This contributes to underdiagnosis and delayed care, particularly regarding modifiable risk factors. **Objective:** The main objective is to assess public knowledge about dementia and its associated risk factors in a Brazilian urban population. **Methods:** An online survey was administered during a public health outreach event in a Brazilian urban center. The questionnaire included sociodemographic questions and five objective items about dementia and its risk factors. **Results:** Ninety-one adults aged 18–78 participated. Older adults were more likely to believe that forgetfulness is a normal part of aging ($p=0.037$) and to conflate dementia with Alzheimer's disease ($p=0.003$). Depression and physical inactivity were the most recognized risk factors ($\geq 82\%$ among those ≥ 65). Social isolation was cited by 88% of older adults and 71% of youth. **Discussion:** Recognition of chronic conditions (e.g., diabetes, hypertension, obesity) was higher among individuals with greater income. Young participants linked excessive screen time to cognitive decline (28.5%). Brain injury from illicit drug use was widely acknowledged across age groups ($\geq 60\%$). **Conclusion:** Findings highlight widespread misconceptions among older adults and an emerging awareness in youth. Socioeconomic status influenced access to information and health perceptions. Educational interventions tailored to age and income levels are critical for improving public understanding, fostering early diagnosis, and reducing the societal impact of dementia.

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All-cause mortality and dementia among older adults: the ELSI-Brasil study

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Background: There were 57 million dementia cases worldwide in 2019, with 61% in low- and middle-income countries (LMICs). Although dementia increases mortality risk in older adults, there is a lack of community-based data from LMICs. **Objective:** To estimate mortality rates and identify risk factors that contribute to mortality in older adults with dementia in Brazil. **Methods:** We used data on 5,249 individuals aged 60 years and older from the Brazilian Longitudinal Study of Aging. Baseline data was collected in 2015–2016, and vital status was ascertained

between 2015 and 2021. Dementia status was defined using an algorithm or by self-reported diagnosis of Alzheimer's Disease. Mortality rates were estimated and multivariate cox regression analysis was used to assess hazard ratios (HR) for mortality. **Results:** Participants with dementia ($n=392$) were more likely to die during follow-up than people without dementia after adjusting for sex and age ($HR=2.65$, 95%CI 2.08–3.39). In the fully adjusted models, men, those with more years of education and worse functionality had a higher mortality risk. Those who had a partner and better cognition had a lower risk. **Discussion:** Better cognitive abilities, having a partner and functional independence decreased mortality risk even with advancing age. A higher mortality for those with more years of education may be due to late diagnosis as a result of better cognitive reserve. **Conclusion:** Understanding factors associated with mortality among people with dementia locally is essential to enable strategies integrating medical, social, and policy-driven solutions to improve longevity.

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A cross-city comparison of dementia underreporting: Recife vs. Amsterdam, 2022

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Background: Dementia is a neurocognitive syndrome characterized by a progressive deterioration of higher cortical functions, including memory, spatial and temporal orientation, language, numerical reasoning, and judgment. These deficits compromise personal autonomy and self-sufficiency, profoundly impairing an individual's functional capacity and social integration. **Objective:** To compare the prevalence of dementia between Recife (Brazil) and Amsterdam (Netherlands) and to analyze the impact of health policies and registry systems on the accuracy of dementia case reporting in both cities. **Methods:** Data were obtained from the IBGE, DATASUS, Centraal Bureau voor de Statistiek (CBS), and the Dementia Care and Support Registry (DZO) for the year 2022, using descriptors such as "Dementia," "Epidemiology," "Recife," and "Amsterdam." **Results:** In 2022, Recife had a population of 1,488,920 inhabitants, while Amsterdam had 918,117. Among these, reported cases of dementia totaled 62 in Recife and 8,829 in Amsterdam. By age group: <65–79 years: 51 cases in Recife vs. 4,046 in Amsterdam; ≥ 80 years: 11 cases in Recife vs. 4,783 in Amsterdam. The prevalence of dementia was nearly 143 times higher in Amsterdam compared to Recife. **Conclusion:** The underreporting of dementia in Recife is due to deficiencies in notification systems and record-keeping, reflecting broader systemic challenges in Brazil. In contrast, significant investments by the Dutch Ministry of Health have enhanced the accuracy of dementia case documentation, enabling a better understanding of the condition and more targeted therapies. The Netherlands serves as a global model for dementia management, underscoring the need for Brazil to improve its reporting and registry processes.

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Racial and sex differences in the association between anticholinergic burden and cognitive decline: longitudinal findings from the ELSA-Brazil study

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The impact of anticholinergic burden on cognition is a debated topic, with evidence linking it to cognitive decline in previous studies. However, few studies examine how these associations may differ by race and gender, particularly in racially diverse countries like Brazil. The Longitudinal Study of Adult Health (ELSA-Brazil), with its large and diverse cohort, offers a unique opportunity to investigate these differences. This study examined racial and sex differences in the association between ACB and cognitive decline over eight years. ELSA-Brazil is a multicenter, prospective cohort study of civil servants aged 35–74 from six Brazilian cities. Participants with missing data on medication use, cognitive tests, or covariates were excluded. ACB was measured at baseline using the ACB scale. Cognitive performance was assessed across three waves (2008–2010, 2012–2014, and 2017–2019) using tests of memory, verbal fluency, and executive function. Linear mixed-effects models were used to examine associations between ACB and cognitive decline. Of 15,105 participants, 13,065 were included in the analysis. The average age was 51.7 years; 55% were women and 53% identified as white. Higher ACB was significantly associated with declines in global cognition, memory, and executive function among white participants. In men, ACB was linked to declines in all cognitive domains. No significant associations were found for Black/brown participants or women. Higher ACB was differently linked to cognitive decline by race and sex. These associations were more pronounced in white and male participants, highlighting the importance of considering racial and sex differences in the cognitive effects of anticholinergic medications.

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Association between risk factor profiles for dementia and cognitive performance: a cross-national analysis

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Background: Currently, around 50 million people live with dementia, projected to rise to 130 million by 2050. It is estimated that 40% of cases could be prevented or delayed by managing risk factors related to cardiovascular health, lifestyle, and psychosocial conditions. **Methods:** This study analyzed data from the Brazilian Longitudinal Study of Aging (ELSI, n=4,990) and the Mexican Health and Aging Study (MHAS, n=8,618), focusing on adults aged 50 and older. Participants were categorized into four dementia risk groups: low risk, cardiovascular risk, psychosocial risk, and lifestyle risk, using latent class analysis. The association between these groups and cognitive performance was assessed through linear regression, adjusted for sociodemographic

factors. **Results:** In Brazil, the psychosocial risk group showed poorer global cognition ($\beta=-0.534$, $p<0.001$) compared to the low-risk group, particularly in memory and verbal fluency. In Mexico, the psychosocial risk group also exhibited lower scores in global cognition ($\beta=-0.185$, $p<0.001$) and in memory and verbal fluency. **Conclusion:** Different risk profiles are associated with cognitive performance in Brazil and Mexico, with psychosocial factors having the most significant negative impact on cognition. These results highlight the importance of tailored preventive strategies.

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Sociodemographic profile of hospitalization for Alzheimer's disease in the regions of Brazil between 2015 and 2024

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Background: Alzheimer's disease (AD) is the most common form of dementia worldwide, affecting approximately 1 in 9 individuals over the age of 65. As the risk increases with age, the number of people living with AD is expected to rise significantly in the coming years. **Objective:** To analyze the sociodemographic profile of hospitalizations due to AD in Brazil. **Methods:** A cross-sectional observational study using data from the SIH/SUS system. **Results:** From 2015 to 2024, a total of 21,479 hospitalizations due to AD were recorded in Brazil. The numbers remained stable over time, with a decline during the COVID-19 pandemic and a peak in 2023 (2,629 cases). The Southeast region accounted for 12,672 cases, followed by the South (4,162), Northeast (3,055), Central-West (1,080), and North (510). Most hospitalizations involved white individuals (11,237; 52.3%), followed by brown (4,951; 23.0%) and Black individuals (1,455; 6.8%). Other categories had lower frequencies, while 3,568 cases (16.6%) had no reported race/ethnicity. **Discussion:** This pattern is linked to late diagnosis and lack of community support reflecting aging populations in regions with fragile healthcare infrastructure. White patients predominated, and the high missing data reveal racial disparities in access and reporting. **Conclusion:** The findings highlight significant regional and racial inequalities, with a concentration of cases in the Southeast and underrepresentation of marginalized groups. There is an urgent need to strengthen preventive and community-based care strategies, promoting equity and dignity in aging.

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Accuracy of the revised Addenbrooke Cognitive Examination and Mini-Mental State Examination in a quilombola community with low education attainment

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Background: Illiteracy may overestimate screening test interpretation for older adults with suspected cognitive decline. In rural areas of Brazil, the illiteracy rate and the lack of valid cognitive instruments may postpone the diagnosis of cognitive disorders. **Objective:** To determine the diagnostic accuracy of the Addenbrooke's Cognitive Examination (ACE-R) and Mini-Mental State Exam (MMSE) as well as the prevalence of mild cognitive impairment (MCI) for older adults with low education in a quilombola community placed in Northeast Brazil, compared to cognitively healthy. **Methods:** 204 participants were collected from a sociodemographic and clinical evaluation and were subsequently subjected to the ACE-R and MMSE. Among them, 25 subjects had MCI, and 179 were cognitively healthy. **Results:** The prevalence of MCI was 12.3%; optimal accuracy yielded highest values for the comparison between controls and MCI for both ACE-R—Area Under the Curve (AUC)=0.96—and MMSE (AUC=0.96). **Conclusions:** Our findings suggest cutoff scores for interpreting ACE-R in a non-dementia sample with low educational attainment, considering factors related to the ecological context, particularly living in hinterland areas. Such results can help improve diagnostic accuracy for diagnosing MCI and healthy controls and ease early therapeutic interventions.

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Discrepancy between Delphi consensus estimates of dementia prevalence and the dispensation of Alzheimer's medications in Brazil

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The global acceleration of population aging, particularly in developing countries, has increased the prevalence of age-related conditions, such as dementia, posing significant challenges to public health systems. Epidemiological research is essential to quantify this burden and to inform effective care and prevention strategies. To assess the discrepancy between dementia prevalence estimates from the Delphi Consensus and the dispensing of Alzheimer's medications in Brazil and its regions. This cross-sectional analysis used data from the 2023 Delphi Consensus, in which 15 specialists from Brazil's five macro-regions estimated dementia prevalence. These estimates were compared with the distribution of cholinesterase inhibitors recorded in the Neurological Disease Observatory of the Brazilian Academy of Neurology. Only 4.2% of individuals with dementia and 7.0% of those with Alzheimer's disease are estimated to be receiving cholinesterase inhibitors in Brazil. The Southeast (8.6% for Alzheimer's; 5.2% for dementia) and South (6.8%; 4.1%) regions showed the highest treatment coverage, while the Northeast (2.1%; 1.2%), Center-West (1.8%; 1.1%), and North (0.5%; 0.3%) regions exhibited the lowest, reflecting substantial regional disparities. The North and Center-West regions, which show the lowest treatment rates, also have the fewest public pharmacies (33 and 6, respectively), large geographic areas, lower Human Development Index, lower education levels, and physician shortages. These factors likely contribute to delayed diagnosis

and limited access to treatment. Despite the availability of Alzheimer's medications in Brazil's public healthcare system, there is a substantial gap between estimated dementia prevalence and actual treatment rates, with marked regional disparities largely influenced by socioeconomic and structural inequalities.

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Consultation psychiatry in general hospital during COVID-19 pandemic: sociodemographic and clinical pattern of older inpatients evaluated

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Background: Mental health of population worldwide was seriously affected during COVID 19 pandemic, including older adults. **Objective:** This study aimed to investigate the sociodemographic and clinical pattern of older inpatients evaluated in Consultation Psychiatry of general hospital during COVID-19 pandemic. **Methods:** A cross-sectional study was conducted using records of older inpatients evaluated in Consultation Psychiatry, of a general university hospital, between March 2020 and December 2023. ICD criteria were used to define the psychiatric diagnoses. Results. Of the 1267 patients referred to Consultation Psychiatry, 177 (13.9%) were older people (≥60 years old), from which 162 records were evaluated. The mean age was 68.4 (SD=7.1; 60–93), 60.5% were female. Between the referrals, 44.2% were from internal medicine ward, 26% were from general emergency unit and 22.9% were from surgery ward. The principal psychiatric diagnoses were: depression (principally depressive episode) (26%), adjustment and anxiety disorders (17.5%), bipolar disorder (10.7%), delirium (9%), alcohol use disorders (7.3%) and dementia (5.1%). Men were proportionally more diagnosed with alcohol use disorders (18.7 vs. 1%), delirium (10.9 vs. 7.1%) and depression (32.8 vs. 21.4%); women with adjustment and anxiety disorders (21.9 vs. 12.5%), bipolar disorder (15.3 vs. 4.6%) and dementia (6.1 vs. 3.1%). **Conclusions:** The proportion of older patients referred to Consultation Psychiatry was much lower than of younger patients and more patients with neuropsychiatric symptoms related to dementia were expected to be diagnosed between older patients. Was there influence of pandemic on findings?

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Alzheimer's disease in Brazil: an ecological study of the temporal trend of hospitalization rates from 2014 to 2024

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Background: Alzheimer's Disease (AD) is a neurodegenerative condition affecting elderly people and burdening Brazil's Unified Health System. A

previous study reported 59,573 hospitalizations for AD in Brazil from 2013 to 2023. Despite its relevance, recent studies on the trends of these hospitalizations are scarce. This research contributes to reflecting on the burden of AD and how socioeconomic, demographic, and structural factors influence hospitalization rates. **Objective:** To analyze the temporal trend of hospitalization rates for Alzheimer's disease in Brazil from 2014 to 2024. **Methods:** Ecological time series study using secondary data from DATASUS. Hospitalizations for AD across Brazilian macro-regions were analyzed. Rates per 100,000 inhabitants were calculated in Microsoft Excel®, and simple linear regression was performed using the Statistics Kingdom platform, adopting a 5% significance level ($p \leq 0.05$). **Results:** Nationally, no statistically significant increase was observed ($p = 0.136$). However, significant upward trends were found in the Northeast ($p = 0.0003$), South ($p = 0.003$), North ($p = 0.022$), and Central-West ($p = 0.028$). A non-significant trend was observed in the Southeast. The national average rate was 0.75 per 100,000, with the highest rates in the South (1.25). **Discussion:** The increase in AD hospitalizations reflects complex epidemiological, sociodemographic, and structural factors. The North, with no prior significant trend (2010–2019), recently showed growth likely linked to the COVID-19 pandemic's impact on chronic conditions, along with new policies expanding diagnosis and care. **Conclusion:** Most regions showed significant growth in hospitalizations for AD. Further studies are needed to better understand the associated factors.

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Age and dementia severity: a cross-sectional analysis from a Brazilian cognitive neurology center

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Background: Dementia prevalence in Brazil varies across studies. The EL-SI-Brasil study found a prevalence of 5.8% in individuals aged 60 and older, rising with age—from 3.2% (60–64 years) to 42.8% (90+ years). **Objective:** To describe the age profile of dementia patients and associations with severity. **Methods:** A cross-sectional study was conducted with 62 patients from a cognitive neurology center. Data were collected from medical records from 2023, including variables such as age, sex, comorbidities, education, dementia level (CDR), symptom duration, and diagnosis. Statistical analysis was performed using One-Way ANOVA, Bonferroni post-test, and chi-square tests ($p < 0.05$). **Results:** The mean age was 68.66 years (± 10.04), with an even sex distribution. The most common age group was 61–70 years (37.1%). Dementia severity increased with age: questionable dementia was more common in younger patients (44–60 years: 41.18%), mild in those aged 61–70 (50%), moderate in 71–80 (36.84%), and severe in those over 81 (33.33%; $p = 0.001$). **Discussion:** Age is a key risk factor for both the development and progression of dementia. Younger patients showed milder impairment, while older individuals had more advanced decline. These findings align with global data: although age-standardized prevalence remains stable, the total number of cases is expected to rise sharply due to population aging. This highlights age as a major driver of the global dementia burden. **Conclusion:** Older age is closely linked to increased dementia risk and severity. Early detection and targeted care are key to lessening its impact.

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Sociodemographic and gender-based analysis of dementia patients at Santa Casa de Belo Horizonte

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Background: Dementia prevalence increases with age and is influenced by factors such as education and comorbidities. **Objective:** To describe the sociodemographic characteristics, risk factors, and etiological distribution of dementia syndromes. **Methods:** A cross-sectional study was conducted with 62 patients from a cognitive neurology center. Data were collected from medical records from 2023, including variables such as age, sex, comorbidities, education, dementia level (CDR), symptom duration, and diagnosis. Statistical analysis was performed using One-Way ANOVA, Bonferroni post-test, and chi-square tests ($p < 0.05$). **Results:** The sample included 31 women and 31 men. The mean age was 68.66 years (± 10.04), ranging from 44 to 93 years. The most prevalent age group was 61–70 years (37.10%), followed by 71–80 years (27.42%). Most patients (70.97%) had only 1 to 4 years of schooling. Regarding symptom duration, 96.77% reported onset more than one year before the first specialist consultation. **Discussion:** While more women presented with moderate dementia and more men with questionable impairment, these differences were not statistically significant. However, evidence from The Lancet Public Health suggests female sex is an independent risk factor, even after adjusting for education, possibly reflecting biological and social mechanisms. **Conclusion:** Although no sex-based differences were observed in this cohort, broader evidence indicates increased vulnerability among women. Early detection and equitable access to care are essential, particularly in low-education populations, to reduce the burden of dementia.

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Diagnostic delay in patients with neurocognitive syndromes assessed at a referral outpatient center

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Background: The syndromic and etiological diagnosis of neurocognitive syndromes depends on the family-patient binomial and on health services. In this context, delayed diagnosis might result in overload of the health system and impact on quality of life. **Objective:** To investigate the time between onset of cognitive symptoms and the first medical consultation. **Methods:** This is a descriptive cross-sectional study conducted at a referral outpatient center. Sixty-two patients were included, presenting: dementia, epileptic, parkinsonian, pyramidal and cerebellar syndromes, non-dementia cognitive impairment, mild cognitive impairment and mild behavioral impairment. Data were collected from medical records and presented as absolute frequency and percentage. Significance was assessed by the chi-square test via Monte Carlo

simulation. **Results:** 60 patients (96.77%) reported onset of cognitive symptoms more than 1 year before the first consultation. When comparing the time from symptom onset to first consultation (if ≤ 1 year or >1 year) and the clinical dementia rating (CDR), there was no significant difference between the groups. **Discussion:** Almost all patients had an interval between symptom onset and first consultation of more than one year. The factors that can influence this delay are diverse, including variables in social, medical and institutional spheres. Furthermore, this time was not associated with the level of severity of the cognitive syndrome, assessed with CDR. **Conclusion:** Delayed diagnosis of cognitive syndromes is common. The dissemination of cognitive screening methods, including in primary care, may be interesting for early detection and referral to specialized services.

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Determinants of Alzheimer's disease dementia diagnosis: findings from the ELSI-Brasil study

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Background: In Brazil, the number of people with dementia (PWD) is projected to triple by 2050. However, diagnostic rates remain low, limiting access to care and support. **Objective:** To identify sociodemographic and clinical factors associated with the likelihood of receiving a diagnosis of Alzheimer's disease dementia (ADD) in older adults. **Methods:** We analyzed data from 5,249 participants aged ≥ 60 years in the ELSI-Brasil cohort, a nationally representative sample. Dementia was defined using an algorithm that integrated cognitive and functional impairment with informant reports. Logistic regression models assessed associations between a self-reported diagnosis of ADD and predictors including age, sex, marital status, education, ethnicity, urbanicity, comorbidities, healthcare use, and performance on memory and verbal fluency tests. **Results:** Among individuals identified as having dementia, older age (OR 1.08; 95%CI 1.02–1.15) and higher memory test scores (OR 3.83; 95%CI 2.03–7.23) were associated with increased likelihood of diagnosis. No significant associations were observed for the other variables. **Discussion:** The association with older age may reflect both increased prevalence and delayed detection. The unexpected link between better memory performance and diagnosis may be influenced by recall bias or limitations of the diagnostic algorithm. **Conclusion:** In this nationally representative sample, ADD diagnosis was more likely in older adults and those with higher memory scores. These findings highlight the need for enhanced screening strategies and earlier diagnostic efforts in Brazil's aging population.

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Early diagnosis and risk factor management in mild cognitive impairment: impact on cognitive outcomes in Brazilian memory clinic

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Background: Mild Cognitive Impairment (MCI) is a clinical condition that may represent an intermediate stage between normal aging and dementia. Emerging evidence suggests that early identification and proactive management can stabilize or improve cognition, particularly in specialized care settings. **Objective:** To evaluate cognitive trajectories in patients with MCI compared to those with Alzheimer's disease (AD) or mixed dementia, and to assess the impact of early diagnosis and risk factor control on cognitive outcomes. **Methods:** This retrospective study analyzed 50 patients followed in a tertiary memory clinic in São Paulo, Brazil. Participants were classified as MCI (n=20), AD (n=20), or mixed dementia (n=10). Cognitive performance was assessed with the Mini-Mental State Examination (MMSE) at baseline and after an average follow-up of 17 months. Statistical analyses included linear regression and the Wilcoxon rank-sum test. **Results:** MCI patients demonstrated a mean improvement of +2 MMSE points, while patients with dementia exhibited significant cognitive decline ($p < 0.001$). Regression analysis identified MCI diagnosis as an independent predictor of cognitive improvement ($\beta = +2.01$; $p < 0.001$). Depressive symptoms and education level were not significant predictors. The model explained 42% of the variance in cognitive change. **Discussion:** Early MCI diagnosis, coupled with structured follow-up and targeted risk factor management, was associated with meaningful cognitive improvement. This underscores the potential of specialized interventions to modify disease trajectories. **Conclusion:** Implementing early detection and intervention strategies in memory clinics can effectively promote cognitive stability or improvement, potentially delaying progression to dementia.

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Impact of social determinants of health in cognitive impairment: an ELSI-Brasil study

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Background: Social Determinants of Health (SDH) are increasingly recognized as significant risk factors for dementia. However, evidence in Low and Middle-Income Countries is limited. **Objective:** Evaluate the association between SDH and cognitive performance among adults and older adults in Brazil. **Methods:** ELSI-Brasil Wave 2 (2019–2021) was evaluated. SDH included economic stability, education, healthcare access, neighbourhood safety, and social community. A global cognitive composite score was calculated. Individuals with cognitive scores below -1 SD from the mean were classified as cognitively impaired (CI). We evaluated the association between SDH and cognitive scores using a logistic regression model to predict CI ($R^2 = 0.28$). **Results:** A total of 9,069 individuals were included (mean age 63.5 ± 10.1 , 58.6% women). All 5 SDH were more prevalent among CI individuals compared to cognitively unimpaired ($p < 0.001$), especially regarding economic stability (45, 10%), educational access (25.3, 78%), healthcare access (85.5, 74.5%), neighbourhood safety (75.9, 35.3%), and social community context (12.5, 11.3%). SDH were associated with cognitive scores in the regression model ($p < 0.001$); and education was the strongest predictor ($\beta -0.55$, $p < 0.001$). **Discussion:** Five SDHs were evaluated as being as-

sociated with CI in Brazil, with low education being the most important predictor. These findings highlight SDH as an urgent public health target for aging populations. **Conclusion:** Economic stability, educational access, healthcare access, neighbourhood safety, and social community context were predictors of CI. Further analyses should assess each SDH in other Latin American populations.

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The importance of assessing dementia risk factors in Brazil: insights from a memory clinic cohort and comparison with international data

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Background: Modifiable risk factors are major contributors to dementia. Although their relevance is well established in international guidelines, data on their distribution and impact in Brazilian populations remain scarce. **Objective:** To assess the prevalence of major modifiable risk factors in a Brazilian memory clinic cohort and explore their associations with different dementia subtypes, providing local evidence to inform prevention strategies. **Methods:** We retrospectively analyzed data from 118 patients evaluated at a cognitive disorders clinic in Brazil. The following risk factors were assessed: hypertension, diabetes mellitus, dyslipidemia, alcohol use, smoking, hypothyroidism, mood disorder, hearing impairment, and visual impairment. Associations with dementia subtypes were evaluated using multinomial logistic regression. **Results:** Prevalence of risk factors was as follows: hypertension 62%, mood disorder 56%, dyslipidemia 38%, smoking 41%, diabetes mellitus 38%, visual impairment 32%, hearing impairment 19%, hypothyroidism 12%, and alcohol use 15%. While the overall presence of risk factors did not significantly distinguish between patients with confirmed dementia diagnoses and those under investigation, regression analyses identified specific patterns of association across dementia subtypes. **Discussion:** Our findings reveal a substantial burden of modifiable risk factors among Brazilian patients with cognitive impairment, consistent with international data. However, local patterns of association highlight the importance of context-specific prevention strategies. **Conclusion:** Systematic risk factor assessment should be integrated into dementia care pathways in Brazil. Larger national studies are needed to refine understanding of these relationships and inform public health policies.

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Does education predict cognition and function in Brazil? Insights from a heterogeneous Brazilian memory clinic cohort

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Brazil is a very diverse country, with average low levels of education. Educational achievement is known to influence cognitive performance and may modulate functional decline in neurodegenerative diseases. Thus, we aimed to investigate the associations between years of schooling, cognitive performance and functional status in patients from a Brazilian memory clinic. We conducted a cross-sectional analysis of 80 patients evaluated at a specialized outpatient memory clinic. Cognitive function was assessed using the Mini-Mental State Examination (MMSE) and functional status was measured with the Katz Index of Independence in Activities of Daily Living. Educational attainment was recorded in years of formal schooling. Pearson's correlation coefficients were calculated to assess the relationships between education, MMSE and Katz scores. The mean educational level in our sample was 6.93 years. In our sample, with a mean age of 72 years, and 44 (55%) female, the correlations between years of education and MMSE scores ($r=0.14$, $p=0.26$), between education and Katz scores ($r=-0.13$, $p=0.30$), and between MMSE and Katz scores ($r=0.18$, $p=0.14$) were all weak and not statistically significant. These findings suggest that in this heterogeneous clinical population, functional and cognitive outcomes are influenced by multiple factors beyond educational attainment. Our results indicate that educational attainment alone may not serve as a strong predictor of cognitive performance or functional status. These findings underscore the need for multidimensional and individualized approaches in the assessment and management of cognitive impairment.

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Race and ethnicity in phase 3 Alzheimer's disease clinical trials: a call for diversity and inclusion

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Introduction: Alzheimer's disease (AD) affects over 57 million people globally, disproportionately impacting women and minority populations, yet racial and ethnic minorities remain underrepresented in Phase 3 clinical trials. **Methods:** We conducted a comprehensive search of PubMed, Web of Science, Cochrane Library, and ClinicalTrials.gov to identify Phase 3 AD treatment trials reporting race and ethnicity data. **Results:** From 49 trials analyzed (1998–2024), we found that 69% reported race/ethnicity, with reporting rates improving from 43% in the 2000s to 73% in the 2010s and 2020s. **Discussion:** Minority populations remained severely underrepresented across trials, with little inclusion of data from Latin America or Africa. Biological differences, including variation in APOE ϵ 4 risk profiles and biomarker expression among racial groups, reaffirm the necessity of diverse representation to ensure equitable and effective the-

rapeutics. However, limitations such as inconsistent demographic reporting, regional bias toward North American and European populations, and methodological variability constrain generalizability. **Conclusion:** Our findings highlight critical gaps in trial diversity and emphasize the urgent need for regulatory reforms, targeted recruitment strategies, and the development of biomarkers validated across populations. Ensuring inclusive participation is essential for advancing equitable Alzheimer's care and accurately capturing the disease's heterogeneity across diverse racial and ethnic groups.

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Dementia mortality patterns in young elderly in Brazil between 2000 and 2023: an ecological time series study

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Background: Dementia syndromes are on the rise worldwide, especially in underdeveloped countries. **Objective:** To describe the patterns of mortality due to dementia in elderly individuals aged 60 to 69 in Brazil over the last 23 years. **Methods:** Ecological study conducted by the Mortality Information System of the Department of Information and Informatics of the Unified Health System. Deaths due to Alzheimer's Disease (AD), Vascular Dementia (VD) and Unspecified Dementia (UNES) between 2000-2023 were analyzed. Sex, race, marital status, education, region and state of death were assessed. **Results:** 15,943 deaths, 83.3% AD, 14% DNE, and 2.7% DV. The highest rates of mortality were recorded in 2023 (8.06%), 2022 (7.33%), and 2021 (6.63%), with an average growth of 9% per year. Males led the number of deaths, with 7,994 cases (50.13%). Whites accounted for 67.34% of deaths, followed by browns with 22.61%. Approximately 21.53% of the sample had between 1 and 3 years of schooling, and 48.45% were married. The region with the highest mortality rate was the Southeast (54.70%), with emphasis on the states of São Paulo (28.68%), Minas Gerais (12.50%), and Rio de Janeiro (11.28%). **Discussion:** Studies on the prevalence of dementia in young elderly people, although scarce in the literature, are essential for the creation of public policies to support this population. **Conclusion:** Deaths were concentrated in men, married individuals, whites and residents of the Southeast region of Brazil, with emphasis on AD.

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Knowledge about dementia and Alzheimer's disease in the Brazilian and Latin American population (PLATCODEM)

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Background: The general population's lack of knowledge about dementia and Alzheimer's disease (AD) is concerning; 80% of lay people believe dementia is a normal part of aging. Due to the scarcity of studies on this subject in Latin America (LA), this research is relevant. **Objectives:** To analyze the knowledge about dementia and AD in a Brazilian and Latin American sample. **Methods:** Cross-sectional observational study using a "Google Form" questionnaire, via electronic link, with Portuguese and Spanish similar versions. The questionnaire comprised 36 statements based on the Dementia Knowledge Assessment Scale (DKAS), the Dementia Knowledge Assessment Tool version 2 (DKAT2), and modifiable risk factors for dementia. **Results:** A total of 1,867 responses (1,146 Brazil; 721 LA). In Brazil, 901 (78.6%) were female, 1,008 (87.95%) had more than 12 years of education, and 729 (63.6%) were in intellectual occupations. In LA, 563 (78.08%) females, 390 (54%) >12 years of education, 675 (93.61%) in intellectual occupations. The mean questionnaire score was 21.78 points in Brazil and 21.05 in LA. Regarding the 21 questions on modifiable dementia risk factors, the mean score was 13.46 in Brazil and 12.61 in LA. **Discussion:** There is a predominance of females, high education levels, and intellectual professions. The knowledge level is moderate and consistent with the literature. Both groups exhibited similar mean scores on the overall questionnaire and on modifiable risk factors for dementia. **Conclusion:** The study demonstrates that population knowledge about dementia and AD is insufficient and highlights the need for broad educational interventions.

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Impact of pharmacological treatment on hospitalization risk and survival in frontotemporal dementia: a TriNetX-based study

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We conducted a retrospective study using TriNetX, an international database with real-world clinical information. We included 141 individuals diagnosed with probable frontotemporal dementia (FTD) within the last five years, from 39 healthcare centers, excluding cases with other neurodegenerative diseases. All underwent neuroimaging examinations. We used logistic regression models and survival analysis to compare clinical outcomes between the groups. Of the participants, 88 (62%) did not receive pharmacological treatment after diagnosis or used medications outside those analyzed. Benzodiazepines were the most prescribed class and

the most common first-line treatment, with 26 patients on monotherapy, followed by atypical antipsychotics (7) and Alzheimer's medications (6). Lorazepam was the most prescribed drug. After matching for age, sex, and comorbidities, the models indicated a higher risk of hospitalization, emergency room visit, or need for intensive care in individuals exposed to inadequate treatments (RR 1.7; 95% CI: 1.1–2.6; $p=0.0123$). Survival analysis revealed a significant difference between the groups: patients not exposed to inadequate medications had a median survival time two years longer than those exposed ($\chi^2 = 7.9$; $p = 0.0049$). Conclusion: Benzodiazepines were the most common pharmacological intervention in FTD, but their use, as well as that of typical antipsychotics and Alzheimer's drugs, was associated with worse outcomes. More evidence-based prescribing strategies may reduce these risks.

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Characterization of older adults with subjective cognitive decline and mild cognitive impairment: data from Brazilian Aging and Memory Study cohort

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Introduction: Cognitive disorders are increasing with population aging. Identifying factors associated with age-related cognitive decline is essential for developing effective future treatments. **Objective:** To describe the demographic, clinical, and cognitive characteristics of the Brazilian Aging and Memory Study (BRAMS) cohort. **Methods:** A retrospective and cross-sectional study, analyzing baseline data from the BRAMS cohort. Demographic, medical, anthropometric data, cognitive assessments, cognitive complaint scales (Cognitive Function Instrument – CFI), and anxiety and depression symptom scales (Hospital Anxiety and Depression Scale – HADS) were used. Participants with major untreated psychiatric disorders were excluded. **Results:** The cohort comprises 203 participants, divided into three groups: cognitively unimpaired (CU) ($n=23$), subjective cognitive decline (SCD) ($n=64$), and mild cognitive impairment (MCI) ($n=116$). SCD participants were older on average (76.4 ± 7.8) compared to those with MCI (73.0 ± 6.8). MCI group had lower schooling (11.1 ± 5.0) compared to the CU (14.0 ± 5.0) and SCD (13.5 ± 4.9) groups. There was no sex difference, and White participants were the majority in all groups (over 70%). Among cardiovascular risk factors, only dyslipidemia frequency differed significantly (73% in SCD, 47.8% in CU, 50.9% in MCI). The SCD and MCI groups had the highest scores on the CFI (2.48 ± 2.28 in both groups). The MCI group scored highest on the depression subscale of the HADS (3.96 ± 3.3). The CFI showed a positive correlation with the HADS-D ($r = 0.322$; $p=0.004$), but with no cognitive evaluation. **Conclusion:** The MCI group was associated with risk factors linked to cognitive decline (lower education, dyslipidemia, and depressive symptoms).

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Alzheimer's disease and inequality: profile of hospital costs, mortality, and length of stay across different educational levels

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Background: Socioeconomic factors such as educational level and income significantly influence hospital outcomes in Alzheimer's disease (AD). Understanding the impact of social inequality and education is essential to guide equitable care strategies. **Objective:** To analyze hospital costs, mortality, and length of stay related to AD across different educational levels, highlighting social disparities in hospital outcomes in Brazil. **Methods:** This ecological, cross-sectional epidemiological study used secondary data on AD hospitalizations from DATASUS/TABNET (2014–2024), stratified by education, sex, race/color, and age group. **Results:** From 2014 to 2024, 3,075 AD deaths were registered in Brazil: 1,997 (64.9%) women and 1,078 (35.1%) men. The Southeast accounted for 1,921 deaths (62.5%), followed by South (635; 20.6%), Northeast (301; 9.8%), Midwest (165; 5.4%), and North (53; 1.7%). Standardized mortality rates were highest in the Southeast for men (27.14/100,000) and women (26.71/100,000), lowest among women in the North (10.95/100,000). Blacks (30.45/100,000) and browns (27.77/100,000) had highest rates; Indigenous lowest (7.14/100,000). **Discussion:** Lower education associates with higher mortality and hospitalization due to reduced cognitive reserve and delayed diagnosis. Social inequalities, including income, race/color, and limited specialized care, worsen outcomes. **Conclusion:** Alzheimer's disease reflects structural inequalities in Brazil's health system. Lower education is associated with higher mortality and longer hospital stays, likely due to delayed diagnosis and limited access to care. Underdiagnosis is suggested in less developed regions. These findings highlight the urgent need for equitable policies and expanded specialized care access.

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Prevalence and correlates of cognitive decline and dementia in older adults in primary care

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Background: Population aging has led to a significant increase in the prevalence of dementia, impacting public health, especially in primary care. In

Brazil, data on the prevalence of dementia in primary care settings remain scarce, making it challenging to formulate appropriate policies. **Objective:** To estimate the prevalence of dementia among older adults treated in primary care units in São José dos Campos and to identify associated factors. **Methods:** An observational, cross-sectional study was conducted with data from 1,739 older adults (age ≥ 60) treated in 14 primary health units. Information was collected through structured interviews, including sociodemographic and behavioral variables, clinical characteristics, depressive symptoms (PHQ-2), and cognitive screening using the 10-CS. Descriptive analyses and logistic regression were performed. **Results:** Of 1,732 participants with complete data, 598 (34.5%) had cognitive decline (10-CS ≤ 7) and 448 (25.9%) had dementia (10-CS ≤ 5). Dementia prevalence was 18.9, 32.7, and 55% among those aged 60–69, 70–79, and ≥ 80 , respectively. Dementia was associated with lower education, higher age, and the use of sleep aids. No associations were found with sex, marital status, alcohol use, smoking, income, comorbidities, or depression. **Discussion:** These findings reveal a high prevalence of dementia in primary care and highlight the impact of aging and low education. Results must be interpreted with caution due to limitations related to the specificity of the short screening test. **Conclusion:** Findings support the need for public policies that optimize screening, prevention, and multidisciplinary care for older adults within primary care settings.

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Impact of education on mortality from Alzheimer's disease in the last 20 years in Brazil: a time series study

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Introduction: The accumulation of cognitive reserve, which is directly proportional to the individual's education level, protects neural tissue from pathological processes, such as dementia. **Objective:** To investigate the influence of educational level on Alzheimer's disease (AD) mortality in Brazil over the past 20 years. **Method:** Epidemiological study of deaths from AD (2013–2023), based on the Mortality Information System of the Department of Information Technology of the Unified Health System. **Results:** There were 315,684 deaths from AD, with progressive and greater growth in 2023 ($n=29,937$; 9.48%). The majority occurred in individuals ≥ 75 years of age ($n=279,743$; 88.6%), especially with 1–3 years of schooling ($n=73,744$; 26.4%). This level of schooling was the most frequent in the period ($n=82,356$; 26.1%), predominating among women (65.7%), whites (74%) and widows (55.2%). In 2023, deaths at 4–7 years of age exceeded those at 1–3 years of age (+5.4%). São Paulo (29%), Minas Gerais (12%) and Rio de Janeiro (10%) stood out, with a higher concentration in the Southeast (53.9%). In the Northeast, deaths without education exceeded those with 1–3 years of education by 20.3% **Discussion:** The high mortality among elderly individuals with low education, especially women, whites and widowers in the Southeast, suggests an impact of education on cognitive health. **Conclusion:** Low educational attainment was associated with higher mortality from Alzheimer's disease, suggesting that lower education levels may reduce cognitive reserve and increase vulnerability among older adults.

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Epidemiological profile of deaths from Parkinson's disease in the last decade among young elderly people in Brazil: an ecological study

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Introduction: Parkinson's disease (PD) is a neurodegenerative condition characterized by tremors, stiffness and bradykinesia, with the most affected population composed of the elderly. **Objective:** To describe the mortality profile of Parkinson's disease in young elderly individuals in Brazil over the last decade. **Method:** Cross-sectional study using data on deaths from PD extracted from SIM/DATASUS (2013–2023). Data are stratified by gender, age, education and ethnicity. **Results:** There were 18,429 deaths from PD between 2003–2023 in young elderly (60–79 years), with progressive and greater growth in 2023 ($n=2,318$; 12.5%). The majority occurred between 75–79 years (46.9%), especially with 1–3 years of schooling (23%), predominating among men (62.5%), whites (68.9%) and married (50.2%). Between the years 2020–2023, deaths with 4–7 years of schooling exceeded those with 1–3 years (+15.9%). The states of São Paulo (22.8%), Minas Gerais (11.5%) and Rio Grande do Sul (10%), with the highest concentration in the Southeast (46.7%) stood out. In the Northeast, deaths without schooling exceeded those of 1–3 years by 19.6%. **Discussion:** There was a progressive increase in deaths from PD in young elderly people, with a predominance of men, white, married and with low education. The higher occurrence in the 75–79 age group reinforces the progression of the disease with age, confirming previous studies. **Conclusion:** The findings highlight the need for public policies aimed at early diagnosis and equitable care of people with PD.

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Dementia: the diagnostic gap between the onset of symptoms and medical care - analysis and implications

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Background: Early diagnosis of neurological diseases is crucial. However, delays in seeking medical attention are common due to the insidious onset of symptoms, which negatively impacts quality of life. **Objective:** To evaluate the time between symptom onset and the first medical consultation, correlating it with age of onset (pre-senile vs. senile) and characteristics of initial symptoms. **Methodology:** Fifty-one medical records were analyzed,

excluding those with incomplete data. Age of onset and time to seek care (months) were calculated. Patients were categorized as having pre-senile (<65 years) or senile onset (≥65 years). Descriptive statistics and comparative analyses (Mann-Whitney U, Kruskal-Wallis) were applied. **Results:** Median time to seek care was 16.49 months (range: 0.0–296.76). The pre-senile group (n=24) had a median delay of 19.45 months; the senile group (n=27), 16.49 months. Amnesic symptoms were most prevalent (78.43%). Visuospatial and praxis impairments (median: 44.66 months) and executive dysfunction (median: 35.66 months) were associated with longer delays. In contrast, behavioral changes (median: 14.93 months) and neuropsychiatric symptoms (e.g., aggressiveness, median: 16.84 months) were associated with shorter delays. **Discussion:** There is substantial variability in time to seek care. Although age of onset shows no marked difference in delay, more disruptive symptoms—particularly behavioral—prompt earlier medical attention than subtle cognitive impairments. **Conclusion:** Delays in seeking care are frequent. Strategies to raise awareness of less overt symptoms and to educate professionals are essential to promote early diagnosis and improve management.

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Initial cognitive profile of patients in a tertiary cognitive neurology outpatient clinic in Fortaleza

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Background: Characterizing the initial cognitive profile in tertiary neurology clinics is key to guiding diagnosis and care. The Mini-Mental State Examination (MMSE), when analyzed by domains, offers insight into specific patterns of impairment. combined with sociodemographic and clinical data, it allows better understanding of patient presentations. **Objective:** To describe the initial cognitive profile of patients seen in a tertiary cognitive neurology clinic, focusing on MMSE domains and performance on a Brief Cognitive Battery (BCB), in relation to age, education, and diagnosis. **Methods:** Retrospective descriptive study of 60 patients seen between July 2024 and June 2025. Variables included age, education, clinical diagnosis, MMSE subscores, and BCB performance. Descriptive statistics and group comparisons were conducted. **Results:** Mean age was 72.4 years, mostly with low to intermediate education. MMSE total score averaged 14.6 (SD=6.8), with deficits most pronounced in recall (0.7/3), attention/calculation (1.6/5), and orientation. In the BCB, naming averaged 8.08, immediate memory 5.04, delayed recall 4.65, and recognition 8.25. Higher education correlated with better performance. **Discussion:** Most patients presented with marked cognitive impairment at first evaluation. The degree and distribution of deficits, especially in memory and attention, suggest delayed access to adequate care and limited cognitive reserve, often associated with low education. These findings underscore the importance of early detection and access to adequate care. **Conclusion:** The pattern of impairment reflects late-stage presentation, emphasizing the need to strengthen cognitive screening and referral strategies in earlier stages of care.

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Regional disparities in adverse effects of anti-amyloid antibodies: a multinational study on ARIA in Alzheimer's disease patients around the world

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Background: Anti-amyloid monoclonal antibodies such as aducanumab, lecanemab, and donanemab represent new treatment options for Alzheimer's disease (AD). However, real-world data on their safety, particularly concerning amyloid-related imaging abnormalities (ARIA) remain limited outside clinical trials. **Objective:** To assess ARIA-related outcomes and patient characteristics among AD patients treated with anti-amyloid therapies across the United States (US), Latin America (LATAM), and Asia-Pacific (APAC) regions. **Methods:** We conducted a retrospective, multi-regional study using TriNetX, a global federated health network. Patients diagnosed with AD (ICD-10 G30) and treated with aducanumab, lecanemab, or donanemab were compared with matched untreated controls using 1:1 propensity score matching. Primary outcomes included ARIA-H (intracerebral hemorrhage), ARIA-E (cerebral edema), encephalopathy, seizures, and delirium. **Results:** Clinical characteristics differed markedly by region. APAC patients were oldest at diagnosis, while US patients had the highest burden of comorbidities such as hypertension (≈49%), diabetes (≈20%), and cerebrovascular disease. LATAM showed the lowest comorbidity prevalence. Racial distribution reflected regional demographics, with APAC almost entirely Asian, the US predominantly White, and LATAM mostly non-White. ARIA events were most frequently reported in the US, likely due to superior access to MRI and standardized imaging protocols. In the US cohort, anti-amyloid therapy was associated with increased ARIA-H (5.6 vs. 1.5%) and ARIA-E (4.2 vs. 0.9%), but lower rates of encephalopathy, seizures, and delirium compared to controls. **Conclusion:** Anti-amyloid therapies increase ARIA risk in real-world settings. Global disparities in comorbidity burden and ARIA detection highlight the urgent need for standardized monitoring and infrastructure across regions.

SPEECH THERAPY

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Language impairment in sporadic and familial (type 8) amyotrophic lateral sclerosis: a longitudinal study

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Background: Patients with sporadic amyotrophic lateral sclerosis (sALS) and with familial type 8 ALS (ALS8), linked to p.P56S VAPB mutation, have language impairment with reduced performance in naming, syntac-

tic comprehension, grammatical expression, and orthographic processing. Objective: We aimed to investigate whether language decline has prognostic value in patients with ALS. Methods: First, we performed a cross-sectional evaluation. We included two groups of participants: patients with sALS (n=17); patients with ALS8 (n=22). The groups were matched for age, sex, and education level. All participants underwent a comprehensive language battery, including the Boston Diagnostic Aphasia Examination, the reduced Token Test, letter fluencies (FAS), categorical fluency (animals), word definition from the Cambridge Semantic Memory Research Battery, and a narrative discourse analysis. We also applied the Addenbrooke's Cognitive Exam - Revised Version (ACE-R), the Hospital Anxiety and Depression Scale (HADS), and the ALS Functional Rating Scale-Revised (ALSFRS-R). The patients were reassessed 10–65 months after the baseline visit with ALSFRS-R, to follow up the progression rate of the disease. Results: We observed that the letter fluency index was able to predict functional declining in the motor domain in patients with sALS ($R^2=0.573$, $p=0.002$). The use of phonological cues (Boston Naming Test) in patients with ALS8 was able to predict global functional declining in this group ($R^2=0.600$, $p<0.001$). Conclusion: Our data suggest that language decline may have prognostic value in cases of sALS and ALS8.

TECHNICAL AREA: SPEECH THERAPY

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Lexical-semantic performance and neuroimaging in primary progressive aphasia

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Background: deterioration in lexical-semantic performance can manifest in PPA. Few studies have demonstrated this performance in different subtypes of APP and related them to their neuroimaging data. Objective: To analyze lexical-semantic performance in patients with different subtypes of Primary Progressive Aphasia and its correlation with neuroimaging data. Methods: This is an observational, descriptive, cross-sectional study. To date, 31 patients (aged 52–77, education 4–21 years) have been assessed using word definition and verbal fluency tests, followed by subsequent analysis of the correlation of these preliminary results with demographic and neuroimaging data. The subjects were divided into 3 groups, according to the variant presented: agrammatic/non-fluent (n=3), semantic (n=18) and mixed (n=10). Results: Lexical-semantic performance showed no differences between the patient groups for semantic verbal fluency ($F=0.568$, $p=0.573$), FAS ($F=0.829$, $p=0.448$) and definition ($F=0.411$, $p=0.668$), nor in relation to the affected hemisphere for semantic fluency ($F=2.087$, $p=0.144$), FAS ($F=0.137$, $p=0.873$) and definition ($F=0.669$, $p=0.522$). Discussion: The results presented show that there was no significant difference in lexical-semantic performance in the population studied, both for comparisons and analysis of the tests related to the affected hemisphere, considering neuroimaging data. Conclusion: There were no significant correlations between the variables analyzed in the sample studied. Further studies with a larger sample size are needed to obtain more robust and insightful results on the implications of the condition.

TECHNICAL AREA: SPEECH THERAPY

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Speech-language intervention in a case of posterior cortical atrophy

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Case Presentation: Female patient, 58 years old, with 24 years of education, who began presenting the first linguistic-cognitive symptoms in 2019 while finishing her doctoral thesis, at which time the family noticed difficulty in organizing her studies and signs of sadness. The medical diagnosis of Posterior Cortical Atrophy (PCA) was made in 2022. In the first speech-language assessment, linguistic deficits were observed, such as difficulty in constructing grammatically correct sentences, along with limitations in initiating and maintaining a conversation; lexical-semantic deficits that affected the patient's communication, including frequent pauses, anomias, paraphasias, and paraphrases. Oral comprehension was mildly impaired. Attention, memory, and visuospatial difficulties were also observed. The aim of this study was to evaluate the effect of a speech-language intervention program in a patient with PCA. Discussion: The speech-language intervention took place weekly for one year and was designed based on a combination of linguistic-cognitive strategies reported in previous studies. For the pre-, during, and post-intervention evaluations, the following protocols were applied: the Addenbrooke's Cognitive Examination - Revised Version (ACE-R), the Arizona Battery for Communication Disorders of Dementia (ABCD), and the Boston Naming Test (BNT). The results showed a statistically significant decline only in memory performance and in the total ACE-R score. In the language tests, a statistically significant decline was observed only in sentence reading comprehension. Final Comments: The speech-language intervention based on linguistic-cognitive strategies had a beneficial effect in maintaining expressive language skills despite the expected progression of a neurodegenerative disease.

TECHNICAL AREA: SPEECH THERAPY

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Intensive treatment of naming in semantic Primary Progressive Aphasia using semantic feature analysis: case report

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Case Presentation: We reported the case of a 61-year-old male patient with a university degree, diagnosed with the semantic variant of primary progressive aphasia (svPPA). For five years, he had experienced progressive anomia, initially affecting proper names and low-frequency nouns, gradually extending to more frequent items. The patient underwent an assessment involving 50 visual stimuli across five semantic categories. He was asked to name and describe each item. The stimuli were randomly assigned to trained (n=25) and untrained (n=25) items. The intervention lasted two months, with five weekly sessions (one

in-person and four remote via Google Forms), totaling 40 hours. A formal-semantic treatment based on Semantic Feature Analysis was implemented, incorporating spaced retrieval and errorless learning strategies. At baseline, naming accuracy was similar across groups. However, trained items showed a significant improvement (from 40 to 88%), whereas untrained items improved modestly (from 44 to 52%). In the definition task, both groups showed gains in semantic feature retrieval (30% for trained, 40% for untrained items). Discussion: Although naming gains did not generalize to untrained items, the overall improvement in semantic feature access in definition task suggests an improvement in the semantic network as a whole. This may support communication in daily life by enabling the patient to describe target concepts more effectively when word retrieval fails. Final Considerations: Intensive therapy based on Semantic Feature Analysis proved effective in improving naming performance and strengthening semantic network in a patient with svPPA.

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Longitudinal performance of a case of semantic variant primary progressive aphasia: a case report

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Case Presentation: The semantic variant of Primary Progressive Aphasia (svPPA) is marked by a progressive impairment in the comprehension of words and concepts. Longitudinal follow-up cases are relevant to understanding the clinical and functional progression of language over time. Patient SFL, male, 43 years old, right-handed, with 11 years of education, began multidisciplinary follow-up in 2022 with complaints of increasing word-finding difficulties, initially noticed during his sermons as a pastor. He was assessed until 2024 at three time points, approximately six months apart, using the Montreal-Toulouse Battery (oral and written sentence comprehension), the Cambridge Test (oral word comprehension), and the Boston Naming Test. A progressive decline was observed across all tasks, with notable decreases in the Cambridge Test (from 56/64 to 10/64) and written sentence comprehension (from 7/8 to 1/8). Discussion: The findings are consistent with the expected profile of svPPA, characterized by deterioration of conceptual and lexical knowledge. Notably, the abrupt decline observed in the final seven months in the Cambridge Test suggests a possible acceleration of semantic deterioration in that period, which warrants further investigation. The assessment data are better visualized in the accompanying graph. Final Comments: The findings reinforce the importance of language assessment and longitudinal monitoring to define the cognitive profile and identify changes in the clinical course, enabling the adaptation of interventions to communicative needs with the involvement of communication partners, aiming to maintain functionality throughout the progression of the disease.

TECHNICAL AREA: SPEECH THERAPY
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Frequency and association between hearing loss and cognitive decline in patients seen at audiology and cognitive neurology clinics

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Background and Objective: There may be limited identification and support for people with hearing loss possibly losing cognitive abilities or people with cognitive decline possibly suffering from limitations due to hearing loss. Hence, study aimed to investigate and discuss the prevalence and association between hearing loss and cognitive decline in patients from different outpatient clinics. **Methods:** Patients treated at a university hospital in Brazil formed three groups, totaling 183 subjects for this study: patients with cognitive impairment, including dementia due to Alzheimer's disease, treated at the Cognitive Neurology clinic (n=125); patients at the Audiology clinic (n=58) subdivided into: experienced hearing aid users (n=17) and new hearing aid users (n=41). All had their hearing thresholds tested and underwent cognitive screening. **Results/ Discussion:** In the overall, the MMSE performance per education level was significantly abnormal in 79.1%. In the group of patients treated at the Neurology clinic, the prevalence of MMSE alterations was 85.5%, and in the groups treated at the Audiology clinic, the alteration was 52.9% among experienced hearing aid users and 70% among new hearing aid users. Considering the responses of the best ear of the group of Cognitive Neurology patients, 75.2% presented altered average hearing thresholds. **Conclusion:** We found a high prevalence of patients with cognitive decline in the Audiology clinic and of patients with hearing loss in the Cognitive Neurology clinic. In the total sample, worse hearing thresholds were associated with abnormal MMSE results.

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Discourse impairment and inflammatory markers in long covid

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Background: While meta-analyses report moderate language deficits in individuals with long COVID-19, current evidence is limited to simple language tasks (e.g., verbal fluency, naming). Discourse analysis offers a more ecologically valid approach, capturing complex and functionally relevant aspects of language. **Objectives:** This study aims to (1) characterize discourse impairments in long COVID and (2) examine their associations with peripheral

inflammatory markers measured during both the acute phase and in long-COVID-19. Methods: We assessed 97 adults (mean age=56.6±13.7 years) previously hospitalized with RT-PCR-confirmed COVID-19 in Rio de Janeiro, Brazil, between May 2020 and March 2021. Neuropsychological and speech-language evaluations were conducted 3 to 8 months post-discharge (mean=168.5±90.3 days). Language was assessed using the Brief Montreal-Toulouse Language Assessment Battery and a narrative discourse task (“car accident” story). Blood samples collected during hospitalization and at follow-up were analyzed for inflammatory markers. Results: Discourse impairments were identified in 10.3% (conversational) and 15.5% (narrative) of participants. Notably, poorer discourse performance was associated with lower MCP-1 levels during the acute phase ($p=0.042$) and higher VEGF levels at follow-up ($p=0.015$). Discussion: Discourse impairments are present in a subset of individuals with long COVID and may be linked to dysregulated immune responses at acute phase and persistent endothelial dysfunction at follow-up. Conclusion: These findings underscore the relevance of ecologically valid assessments in post-COVID care and highlight neuroinflammation as a potential mechanism and therapeutic target.

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Performance of healthy adults and elderly individuals on the cognitive linguistic quick test: effects of age and education

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Background: In Brazil, there is a shortage of multidimensional cognitive-linguistic assessment tools. The Cognitive Linguistic Quick Test (CLQT) is internationally recognized for minimizing language influence on non-linguistic tasks, making it valuable for assessing cognition in vascular and primary progressive aphasia. However, understanding how age and education affect CLQT performance in healthy individuals is essential before applying it in clinical populations. Objective: To examine the influence of age and education on CLQT performance in cognitively healthy individuals across the lifespan. Methods: A cross-sectional study was conducted with 164 healthy participants, divided by age (young adults, middle-aged adults, young elderly, and older elderly, and education level (low, medium, and high). Participants were assessed using the CLQT, which assesses attention, memory, executive functions, language, and visuospatial skills. Results: Hierarchical regression showed that age significantly impacted performance on tasks involving attention, executive functions, and visuospatial skills, with marginal effects on memory tasks, while language remained more stable. Education influenced nearly all tasks, except time-spatial orientation. Interactions between age and education were observed in attention, executive functions, and visuospatial skills, with marginal effects on memory tasks. Discussion: Age had a greater impact on fluid cognitive functions, as supported by the literature, while language remained relatively stable throughout aging. Education showed a stronger overall influence on cognitive performance. Lower educational levels were associated with greater cognitive decline with age, suggesting a protective effect of education. Conclusion: Both age and education significantly influenced CLQT performance in healthy individuals.

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Measuring verb argument structure in nonfluent primary progressive aphasia: a one-year case study

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Primary progressive aphasia (PPA) is a neurodegenerative disorder characterized by progressive language decline, with nonfluent variant PPA (nfvPPA) typically affecting speech production and syntactic processing. Verb usage is particularly vulnerable, given its role in sentence construction and grammatical encoding. To examine longitudinal changes in verb production and argument structure in a patient with nfvPPA, using narrative discourse analysis and performance in a specific task, over a one-year interval. This case study forms part of a broader investigation involving neurological and language assessments in PPA. Narrative samples were elicited using the PICNIC scene from the Montreal-Toulouse Language Assessment Battery (MLT), and syntactic abilities were evaluated through the Argument Structure subtest of the Northwestern Assessment of Verbs and Sentences (NAVS). Data from 2023 and 2024 were compared. There was a 71.4% reduction in total verb usage in the narrative task, with intransitive verbs decreasing by 100% and transitive verbs by 60%. Greater hesitation and simplification of syntactic structures were observed, including substitution of complex verbs (e.g., “tentar”) with simpler gerund forms. NAVS results showed reduced accuracy in producing sentences with two or three obligatory arguments, declining by 40% and 100%, respectively. Findings reflect progressive syntactic degradation in nfvPPA. Verb argument structure analysis proved sensitive to subtle grammatical deficits, revealing impairments not captured by traditional tasks. This aligns with literature linking dorsal language tract degeneration to syntactic breakdown. Argument structure analysis is a valuable tool for assessing and monitoring syntactic decline in nfvPPA, with clinical implications for diagnosis and intervention.

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Adapting the Western Aphasia Battery revised for Brazil: concurrent validity with ACE-R in primary progressive aphasia assessment

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Background: The Western Aphasia Battery-revised (WAB-R) is a comprehensive tool for assessing aphasia, widely used and recommended. The WAB-R is currently being validated for Brazilian culture (WAB-R-Br). The battery provides an index of aphasia severity, the aphasia quotient (AQ), a weighted score derived from all oral subtests. Objectives: (1) To present preliminary data on

the application of the WAB-R-Br in Brazilian patients with primary progressive aphasia (PPA); (2) To analyse evidence of concurrent criterion validity of the WAB-R-Br. Methods: The local ethics committee approved the study, and the author formally authorized the adaptation. Eleven PPA patients (3 semantic; 4 nonfluent; 3 logopenic; and 1 mixed variant) were assessed with the WAB-R-Br and the ACE-R, a brief cognitive tool. Pearson's correlation analysis examined relationships between AQ-Br and ACE-R subtests. Results: A moderate positive correlation ($r=0.695$, $p<0.018$) was found between AQ-Br and ACE-R total score. AQ-Br strongly correlated with ACE-R Fluency subscore ($r=0.745$, $p<0.01$). Discussion: PPA is a progressive cognitive syndrome that affects primarily language skills. With its the progression, other cognitive domains are affected. ACE-R assesses distinct cognitive domains, including language. The moderate positive correlation between AQ-Br and ACE-R total score indicates that WAB-R-Br has good evidence of concurrent validity. The strong correlation between AQ-Br and Fluency suggest that verbal fluency may be a significant predictor of aphasia severity in PPA. The low AQ-Br and ACE-R Language correlation reflects the small sample size, especially of semantic PPA patients, as ACE-R emphasizes semantic memory. Conclusion: WAB-R-Br showed good evidence of concurrent validity.

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Prevalence of dysphagia and associated factors in institutionalized older adults with Alzheimer's disease or mild cognitive impairment

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Population aging leads to molecular and cellular decline, increasing susceptibility to chronic degenerative diseases such as mild cognitive impairment (MCI) and Alzheimer's disease (AD). Dysphagia, common in dementia, raises the risk of aspiration pneumonia, a leading cause of death in this population. This study aimed to evaluate dysphagia and its associated factors in institutionalized older adults with AD, MCI, and controls, as well as in community-dwelling controls. The sample included individuals aged 60 years or older, with or without a diagnosis of AD or MCI, institutionalized or not. The study was approved by the Research Ethics Committee (approval no. 5.923.008/UNESC). Participants completed a sociodemographic questionnaire and instruments assessing cognition, functionality, and symptoms of anxiety and depression, in addition to undergoing clinical swallowing evaluation. Cognitive test results showed significantly lower scores in the AD and MCI groups. Community-dwelling controls had better scores than institutionalized controls. Individuals with AD and MCI were significantly more dependent. Depressive symptoms were higher among institutionalized controls, while anxiety symptoms were lower in the community control group. Dysphagia prevalence was 71.4% in the AD group, 70.6% in MCI, 48.5% among institutionalized controls, and 4.0% in community controls. Dysphagia was associated with physical activity, medication use, smoking, cognition, functionality, depressive and anxiety symptoms, dentition, age, family history of cancer, and marital status. It is concluded that dysphagia is more prevalent in institutionalized older adults, especially those with dementia, and is associated with advanced age and poorer cognitive performance.

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Use of switching and clustering in patients with primary progressive aphasia.

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Background: Verbal fluency tasks are commonly used in the language assessment battery for patients with primary progressive aphasia (PPA). Deepening the analysis of free verbal fluency (FVL) quantitatively and qualitatively allows a better understanding of the strategies used to perform the task. Objective: To compare the use of clustering (semantic and phonological subcategories) and switching (alternation between subcategories) between PPA groups and analyze correlation between the parameters analyzed in patients with PPA. Methods: A cross-sectional study was conducted with 15 individuals with PPA [semantic PPA (n=5), non-fluent PPA (n=4), logopenic PPA (n=2), unclassifiable PPA (n=4)] using the FVL test (duration of 150 seconds). The number and size of semantic and phonological clusters and the number of switches were quantified. Results: The groups were matched by age and education. No significant differences were found between the groups regarding the parameters analyzed. In the total sample, it was observed that the number of switches correlated positively with the number of words produced ($p=0.888$, $p\leq 0.001$), number of semantic clusters ($p=0.840$, $p\leq 0.001$), and phonological clusters ($p=0.852$, $p<0.001$) and with the size of the phonological cluster ($p=706$, $p=0.003$). Discussion: The data presented contribute to patient monitoring and therapeutic plans that encompass strategies used by patients. Conclusion: Future studies with a larger sample of patients would allow the analysis of correlations between the groups for an even more detailed analysis.

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Onset of the pharyngeal phase in elderly individuals with Alzheimer's dementia: impact of different food consistencies and dementia stages

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Introduction: Alzheimer's disease (AD) is a progressive neurodegenerative condition that affects memory and higher cognitive functions, with increasing impact on communication and feeding. Studies have shown that the onset of this phase can occur at different sites within the oropharynx, even in individuals without swallowing complaints. The pharyngeal phase involves rapid events triggered by sensory receptors in the oropharynx. The Videofluoroscopic Swallowing Study (VFSS) is considered the gold standard for assessing swallowing, allowing dynamic analysis of all phases and the patient's response to different food consistencies. Objective: To analyze the onset location of the pharyngeal phase in elderly individuals with AD, con-

sidering different food consistencies and stages of dementia. **Methods:** This retrospective observational study analyzed 45 VFSS exams of patients with AD Clinical Dementia Rating [CDR] 1, 2, and 3, using food consistencies based on IDDSI levels 1, 4, and 7. The onset of the pharyngeal phase was assessed using the MBSImp protocol, based on the position of the bolus head at the first sign of hyoid excursion (scored from 0 to 4). **Results:** Across all groups, the vallecula was the most frequent onset site, especially with solid consistency (IDDSI 7). In more advanced stages of dementia (CDR 3), greater variability and delayed initiation of the pharyngeal phase were observed, including onset in the piriform recesses. **Conclusion:** The onset location of the pharyngeal phase varies according to food consistency and the stage of Alzheimer's dementia, with the vallecula being the most frequent site.

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Writing proficiency in different subtypes of primary progressive aphasia and its correlation with neuroimaging and demographic data

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Background: Primary Progressive Aphasia (PPA) is a neurodegenerative syndrome that primarily affects language function. Three recognized variants include the semantic (PPA-s), non-fluent (PPA-nf), and logopenic (PPA-l) types. Mixed cases are classified as PPA-nc. **Objective:** To assess the performance of PPA subtypes in the HFSP dictation test and correlate results with neuroimaging and demographic data. **Methods:** This observational, descriptive cross-sectional study was conducted at a university cognitive and behavioral neurology clinic. Thirty-three patients (aged 52–77, education 4–21 years) participated after signing Ethics Committee–approved consent forms. The HFSP test was used to assess and classify writing errors, followed by analysis of correlations with gender, age, education, and neuroimaging data. **Results:** No correlation was found between number of errors, age, or education. Most frequent errors were graphemic substitutions (M=4.8) in PPA-s, lexicalization (M=9.0) in PPA-nf, and omissions (M=4.5) in PPA-l. In the PPA-nc group, graphemic substitutions (M=3.5) were most frequent, along with 42 cases of task incompleteness (words not written). All patients showed impairment in at least one hemisphere: 16 had left hemisphere (LH) lesions only, 13 had bilateral lesions, 0 had right-only lesions, and 4 had unspecified encephalic damage. In LH lesion cases, lesion sites did not correlate with error type. **Discussion:** Findings support the literature associating PPA, regardless of subtype, with LH impairment mostly. **Conclusion:** Preliminary results suggest no significant correlations among the variables analyzed.

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Communicative engagement in speech therapy program for primary progressive aphasia: observational case report

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Background: Primary Progressive Aphasia (PPA) is a neurodegenerative syndrome characterized by progressive language decline, affecting functionality and quality of life. Structured speech and language therapy interventions can promote functional communication and coping with the disease. **Objective:** To analyze the engagement of PPA patients and their communication partners during a group speech therapy intervention. **Methods:** Dyads composed of PPA patients and their caregivers participated in eight weekly sessions conducted by speech therapists. The program addressed three main areas: education about PPA, language remediation, and compensatory strategies, including Augmentative and Alternative Communication (AAC). Sessions were interactive, encouraging experience sharing and strategy practice. Engagement was evaluated by two observers using the Communication Partner Engagement Assessment (EAV-PC) and Patient Engagement Assessment (AV-PA) scales, alongside qualitative clinical records. **Results:** Engagement levels fluctuated across sessions, with increases during functional training and AAC implementation phases. Caregivers evolved from moderate to higher engagement, while patients actively participated in practical activities, such as script training and AAC use. Clinical observations and spontaneous reports indicated improvements in communicative initiative and the use of strategies in daily routines. Variability between sessions reflected the nature and complexity of each activity. **Conclusion:** The program proved feasible and positively impacted functional communication and engagement among PPA patients and caregivers. The results highlight the importance of multifocal interventions integrating education, remediation, and compensation, with emotional and communicative benefits. This reinforces the value of group-based, interactive speech and language therapy in managing PPA.

GERIATRICS

TECHNICAL AREA: GERIATRICS
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Influence of cognition on the assessment of sarcopenia in Alzheimer's dementia

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Background: Alzheimer's disease is associated with functional impairment, part of which may be due to sarcopenia. **Objective:** Correlate the impact of cognitive decline on the assessment of sarcopenia. **Methods:** Cross-sectional study with individuals ≥55 years old, classified as having no cognitive decline, mild cognitive impairment (MCI), or Alzheimer's dementia (CDR 1 or 2). Patients with advanced dementia, stroke, or active psychiatric disorders were excluded. All participants underwent sarcopenia (TSL5x and handgrip strength) and functional (TUG and walking speed) assessments. **Results:** A total of 102 patients were evaluated: 33% without cognitive impairment, 23.5% with MCI, 31.4% with mild dementia, and 11.8% with moderate dementia. The mean age was 76.7±7.2 years. A significant correlation was observed between longer time on the TSL5x and increased

CDR ($p=6.41 \times 10^{-6}$; $r=0.45$), reduced handgrip strength ($p=0.01$; $r=-0.25$), longer time on the TUG ($p=8.2 \times 10^{-4}$; $p=0.34$), and slower walking speed ($p=1.68 \times 10^{-3}$; $r=-0.40$). **Discussion:** The moderate correlation between TSL5x, TUG, and walking speed shows the early functional impact of dementia progression. **Conclusion:** The findings indicate an association between cognitive decline and poorer functional performance, highlighting the value of simple physical tests as early indicators of dementia progression.

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Gait apraxia in an elderly patient with dementia: a case study

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Case Presentation: A 63-year-old male patient with higher education presented two years ago with gait disturbances, speech difficulties, and impaired writing. Magnetic resonance imaging showed asymmetric frontotemporal atrophy, resulting in a diagnosis of Corticobasal Degeneration. Physiotherapeutic evaluation identified visuospatial and visuomotor impairments, difficulty with postural transitions, and deficits in static and dynamic balance, with a tendency to fall backward. Specific testing revealed dynamic apraxia, limb-kinetic apraxia, gait apraxia, and incoordination in the right hemibody, along with spasticity. The patient scored 16 on the Berg Balance Scale, indicating high fall risk, and took over 30 seconds to complete the Timed Up and Go test, suggesting limited mobility and an inability to climb stairs without assistance. The Gait Apraxia Assessment Protocol by Della Sala, Spinnler, and Venneri (2004) was applied, in which the patient scored 26/42. Despite motor limitations, attention and cognitive flexibility were preserved. **Discussion:** Gait disturbances, particularly gait apraxia, are common in dementia. Gait apraxia is characterized by difficulty in ambulation despite intact motor, sensory, and comprehension systems. Physiotherapeutic evaluation plays a key role in early identification of such impairments, enabling timely interventions that may reduce fall risk and promote autonomy. **Final Comments:** This case exemplifies a typical manifestation of gait apraxia in an elderly patient with dementia, emphasizing the relevance of detailed functional assessment to guide individualized rehabilitation strategies that aim to improve independence in daily activities.

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Ventriculoperitoneal shunt in a patient with advanced atypical Alzheimer's Disease: a case report

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Case Presentation: T.A.S., a 75-year-old woman, began follow-up at a specialized outpatient clinic for advanced cognitive impairment in February

2020, with a prior diagnosis of atypical Alzheimer's disease (behavioral variant), evolving for about eight years. At admission, she had a Clinical Dementia Rating—Sum of Boxes (CDR-SB) score of 18, indicating severe functional impairment and significant caregiver burden. In the second year of follow-up, she experienced progressive gait decline—initially performed with assistance—leading to complete loss of ambulation, as well as cervical and trunk instability. Etiologic investigation identified newly developed normal pressure hydrocephalus (NPH). After multidisciplinary discussion, a ventriculoperitoneal shunt was placed, despite the advanced dementia stage. Postoperatively, the patient had modest motor improvement, with partial recovery of axial support, though she remained non-ambulatory. However, caregivers reported significant improvement in quality of life, including greater ease in daily care and reduced emotional strain. **Discussion:** NPH is often associated with neurodegenerative conditions. Its classic triad—gait disturbance, cognitive impairment, and urinary incontinence—can respond to shunting in selected cases. Patients with advanced dementia are usually excluded from this intervention due to unclear functional benefits. This case suggests that even modest motor gains may translate into meaningful quality-of-life improvements for patients and caregivers. It highlights the importance of individualized decision-making and considering broader outcomes, particularly in advanced dementia stages.

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Hepatic encephalopathy presenting as rapidly progressive dementia

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Case Presentation: Female patient, 63 years old, previously independent for basic and instrumental activities of daily living, with hypertension, diabetes, dyslipidemia, and osteoporosis. In January 2025, she began to experience memory impairment and temporal-spatial disorientation. Over the next three months, she developed gait disturbances, bradykinesia, resting tremor in the upper limbs, and involuntary eye movements characterized by a “scanning gaze” pattern. The patient became totally dependent for activities of daily living and was admitted in April 2025 for investigation of rapidly progressive dementia. Admission tests revealed normocytic and normochromic anemia and thrombocytopenia. Brain MRI showed mild global atrophy and mild microangiopathy. Lumbar puncture was unremarkable. The electroencephalogram showed mild and nonspecific disorganization of brain electrical activity. Abdominal ultrasound revealed signs of chronic liver disease, portal hypertension, and the presence of a splenorenal shunt, with normal liver function tests. Nasoenteral tube feeding was necessary due to dysphagia, and bronchoaspirative pneumonia was diagnosed and treatment with piperacillin-tazobactam was initiated. After completion of treatment and optimization of constipation management, significant clinical improvement and functional recovery were observed. **Discussion:** The presented case is in the context of hepatic encephalopathy secondary to chronic liver disease with splenorenal shunt. The absence of liver laboratory abnormalities does not exclude the possibility of encephalopathy, given the portal flow diversion and toxin accumulation. **Final considerations:** It is important to investigate potentially reversible etiologies in rapidly progressive dementias, even in the face of atypical clinical presentations.

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Basal ganglia calcifications and parkinsonism in an elderly woman with persistent hypocalcemia

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Case Presentation: A 68-year-old woman was evaluated at a referral center for elderly care due to progressive postural instability, bradykinesia, rigidity, and mild cognitive decline. Labs showed hypocalcemia (6.9 mg/dL), intact PTH at the lower limit (17.7 pg/mL), hypomagnesemia (1.4 mg/dL), normal vitamin D (59.5 ng/mL), and phosphorus of 4.5 mg/dL. Thyroid and renal function, blood count, and albumin were normal. Cranial CT revealed symmetric calcifications in the basal ganglia and cerebellar nuclei, along with cortical and subcortical deposits and mild atrophy. Parathyroid ultrasound showed normal volume and echogenicity. Bone densitometry revealed osteopenia; ECG showed normal sinus rhythm. **Discussion:** The combination of extensive brain calcifications and persistent calcium metabolism disorder, with low-normal PTH, suggests Fahr's syndrome secondary to parathyroid dysfunction. Possible causes include functional hypoparathyroidism or peripheral PTH resistance. Parkinsonism and cognitive decline required a broad differential diagnosis, including normal pressure hydrocephalus, Parkinson's disease, Lewy body dementia, vascular parkinsonism, and metabolic encephalopathies such as chronic hypomagnesemia or heavy metal toxicity. Clinical, biochemical, and imaging data were essential for diagnosis. **Final Remarks:** Neurological and metabolic conditions were managed, and the patient was referred to neurology and a multidisciplinary team. This case reinforces the importance of investigating metabolic causes in elderly patients with new-onset parkinsonism. Fahr's syndrome should be considered when basal ganglia calcifications coexist with persistent electrolyte disturbances, even when PTH is within normal limits.

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Management of neuropsychiatric symptoms: a palliative perspective

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In an attempt to understand the problems in the care of neuropsychiatric symptoms in elderly individuals with Neurocognitive Disorder (NCD), this study aimed to observe and recognize the actions of psychologists who work with this population in the context of palliative care. The objective was to define the forms of treatment of symptoms, seeking a common way of acting among professionals. Semi-structured interviews were conducted with psychologists who work in palliative care. The data showed that there are no clear rules for management, indicating a failure in the organization of professional practices. The treatment of symptoms focused on support, assessing the patient's discomfort and looking for their part of reality in the hallucination, as a means of articulating interventions. Contextual factors, such as conversations between professionals and family, were shown to interfere with the frequency and seve-

riety of symptoms. Communication between professionals and family members was considered important, providing information and skills for decision-making regarding symptoms. The study showed the need for multidisciplinary and psychoeducation of families, aiming at a more complete and effective care. It is concluded that, to deal with the symptoms of NCD, it is necessary to personalize and contextualize the intervention, emphasizing reception. In addition, the importance of environmental and functional factors in the care of symptoms was highlighted, as well as the need for more research to develop clear and adequate management rules.

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Psychomotor syndromes in aging: a systematic review

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Background: In aging, psychomotor syndromes are associated with neurodegeneration, cognitive decline and psychiatric disorders. These conditions involve motor manifestations such as bradykinesia, rigidity, tremors, catatonia, dyskinesias and psychomotor agitation. Understanding their mechanisms is essential for early diagnosis and appropriate management to reduce functional impact and improve the quality of life in elderly. **Objective:** To systematically review the available evidence on psychomotor syndromes in aging, their neurological and psychiatric causes and clinical implications. **Methods:** A systematic review was conducted following PRISMA guidelines. Prospective studies, reviews and meta-analyses published between 2000 and 2024 were included, using the Cochrane, LILACS, MEDLINE, PubMed, SpringerLink and PsycINFO databases. The search terms used were "psychomotor syndromes", "aging" and "psychiatric disorders". Only studies with representative samples of elderly were included. **Results:** Fifty-two studies met the inclusion criteria. Bradykinesia and rigidity were the most prevalent findings, mainly associated with Parkinson's disease and atypical parkinsonian syndromes. Tremors and tardive dyskinesias were related to prolonged antipsychotic use. Abnormal gait patterns occurred in both neurological and psychosocial contexts. **Discussion:** Psychomotor syndromes reflect dysfunctions in circuits connecting the prefrontal cortex, basal ganglia, cerebellum and limbic system. Dopaminergic and cholinergic pathways play a key role in the pathophysiology of Parkinson's and Alzheimer's diseases. Catatonia and psychomotor agitation highlight the interplay between motor, cognitive and emotional factors. Antipsychotic use requires careful monitoring due to extrapyramidal side effects. **Conclusion:** Early recognition and proper management of psychomotor syndromes are essential to reduce complications and improve care for older adults.

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A bedside cognitive tool to screen for dementia and detect risk of delirium and mortality in hospitalized older adults: a 5-country cohort study

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Background: Identifying cognitive impairment at the bedside is critical but challenging in older hospitalized adults. **Objective:** We assessed the diagnostic and prognostic performance of two easy-to-use tools—the 10-point Cognitive Screener (10-CS) and the Short Portable Mental Status Questionnaire (SPMSQ)—to detect preexisting dementia and predict incident delirium and 90-day mortality. **Methods:** We analyzed data from the CHANGE Study, a prospective cohort of adults aged ≥ 65 years admitted to 43 hospitals across Brazil, Angola, Chile, Colombia, and Portugal. Patients with delirium at admission, according to the Confusion Assessment Method (CAM), were excluded. Preexisting dementia was defined by a score ≥ 1 on the informant-based Clinical Dementia Rating (IB-CDR). The 10-CS and SPMSQ were administered within 48 hours of admission. We examined their diagnostic accuracy for preexisting dementia and prognostic associations with incident delirium—recorded at discharge by CAM-trained clinicians based on continuous monitoring during hospitalization—and 90-day mortality. **Results:** Of 1,978 participants (mean age=78.2 $\pm$ 8.5; women=56%; White=52%; mean education=6.3 $\pm$ 4.9 years), 409 (21%) had preexisting dementia by IB-CDR, 37% of which were undocumented in hospital records. The 10-CS showed greater accuracy than the SPMSQ in detecting dementia (AUC=0.89 vs. 0.87; $p=0.02$). A 10-CS score ≤ 5 , indicating probable cognitive impairment, yielded 90% sensitivity and 70% specificity. Patients with 10-CS ≤ 5 had increased odds of incident delirium (OR=3.3; 95%CI 2.5–4.2) and increased risk of 90-day mortality (HR=1.6; 95%CI 1.3–2.0), adjusted for sociodemographic and clinical factors. **Conclusions:** The 10-CS is a practical tool to identify preexisting dementia and stratify risk of adverse outcomes, facilitating bedside cognitive assessment and care planning in hospitalized older adults.

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Dementia and mental health during the covid-19 pandemic: a systematic review

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The COVID-19 pandemic raised significant concerns related to the management of care for people with dementia, but few studies have examined the mental health of older adults with dementia and their caregivers during the pandemic, when compared to other populations. This systematic review thus aims to compare and discuss the impact of the COVID-19 pandemic on people with dementia and on their caregivers' mental health. A search was performed in the PubMed/Medline and ISI databases according to the PRISMA methodology. We included studies published in 2020 and 2021 with the following combinations of Keywords: "COVID-19 and mental health and elderly," "COVID-19 and mental health and dementia," "COVID-19 and dementia and caregivers," "pandemic and mental health and elderly," and "pandemic and anxiety." The studies generally showed that social isolation can increase motor deficits and neuropsychiatric symptoms and caregivers' burden and anxiety. New avenues for care and intervention are thus needed for older adults with cognitive deficits and their caregivers to avoid the intensification of physical and psychological suffering. Technological initiatives and support should consider people with cognitive impairment and different levels of technology literacy.

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Probiotic supplementation in elderly with mild cognitive impairment or Alzheimer's disease: triple-blind randomized, placebo-controlled clinical trial

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The inflammatory processes related to aging are documented and recent research has focused on understanding the relationship between dysbiosis and gut-brain-axis. Based on this context, the study aimed to evaluate the effects of probiotic on cognitive, inflammatory, and neurotrophic parameters in older adults as cognitively healthy, mild cognitive impairment (MCI), or Alzheimer's disease (AD). This study was approved n° 7,041,292 and registered n° RBR-9gccx5c. This is a placebo-controlled, triple-blind, randomized clinical trial conducted with individuals aged 60 years and older. Initially, participants underwent a sociodemographic assessment, cognitive testing, and biological sample collection. They were randomly assigned to receive placebo or probiotic. After 12 weeks of treatment, all assessments were repeated to compare pre-(P1) and post-treatment (P2). The final sample included 53 individuals (16 controls, 18 MCI, and 19 AD). Participants in the AD group were older and had lower educational levels. When comparing cognitive test results, the control group performed significantly better than participants with MCI or AD. In terms of cytokine levels, an increase in IL-1 β , IL-6, and TGF- β , and a decrease in IL-4, IL-10, and BDNF were observed at P2 compared to P1. Additionally, interactions between time and treatment were found for TNF- α , GDNF, and NGF levels: TNF- α and GDNF increased in the control group receiving probiotics, while NGF increased in the MCI group. These findings suggest that probiotic supplementation influenced levels of TNF- α , GDNF, and NGF, underscoring the importance of clinical research into the role of gut microbiota in maintaining brain health.

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Adaptation of the Personalized Activities Program (TAP-BR) to virtual environment

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Introduction: Caring for people with dementia requires support for care partners, especially in the face of psychological and behavioral symptoms of dementia, which impact quality of life and increase the burden of caregiving. The Personalized Activities Program (TAP-BR) is an Occupational Therapy intervention with meaningful activities and management strategies. Adaptation to a virtual environment can expand access to care. **Objective:** To present the initial stages of adapting the TAP-BR to a virtual environment. **Methods:** Meetings between Occupational Therapy specialists and

the use of information technology to review the TAP-BR manual and procedures, adjust terms, and make necessary adaptations to the virtual environment. Results: The program structure remained in three phases. Two sessions were added to prepare for the use of information and communication technology at the beginning of the program, and one session to prepare the environment before the activity was implemented. The number of evaluation sessions increased from 2 to 3, including observation of activities from the routine tasks inventory. The program expanded from 9 to 13 sessions. Evaluations, mediation strategies, and video call organization were adapted.

Discussion: The adaptation of TAP-BR to a virtual environment was carried out while maintaining the original structure. The inclusion of preparatory sessions is crucial for supporting care partners in using technologies. **Conclusion:** The initial adaptation of TAP-BR was completed. The following steps include expert committee evaluation and a pilot study with dyads to make necessary adjustments.

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Multisensory stimulation intervention based on Snoezelen principles for older adults with dementia: a pilot study

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Objective: To evaluate the benefits of MSI based on the Snoezelen principles, in reducing behavioral and psychological symptoms in dementia (BPSD). Materials and **Methods:** MSI was performed in 12 individual sessions, lasting 45 minutes and occurring weekly. The sample consisted of 10 older adults with dementia allocated into two groups: Experimental and Control. The benefits were assessed by objective measures using the Neuropsychiatric Inventory (NPI). Descriptive and comparative analysis were carried out considering the pre- and post-intervention and the effect size. **Results:** Comparing the pre- and post-intervention using the NPI, there was an improvement in the EG and a worsening in the CG in all variables: Total score on the NPI (Cohen's $d=-1.55$), Frequency (Cohen's $d=-1.12$), Intensity (Cohen's $d=-1.42$), Caregiver exhaustion (Cohen's $d=-1.19$) and number of behaviors (Cohen's $d=-0.55$), adoption $p \leq 0.05$. **Conclusion:** The MSI Intervention based on Snoezelen's principles has a positive effect on reducing the frequency and intensity of BPSD in elderly participants, as well as on the level of wear and tear on caregivers related to these behaviors.

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"Olhares sobre as demências": an experience report on operative groups with elderly

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Background: The interdisciplinary project "Educação em Saúde: Olhares Sobre as Demências" conducts group interventions that address topics such as the general concept of dementia, major risk factors, treatment, prevention, and "caring for the caregiver" using active learning methodologies. The project began in May, and so far, four sessions have been held at a fitness center in Recife, with a group of Pilates participants. **Objective:** This study reports on interventions carried out by this project, which was led by a group of students from various health-related programs. **Methods:** The workshops were composed of 17 elderly and were based on the concept of operative groups from the perspective of Pichon-Rivière. **Results:** The elderly shared meaningful insights about the importance of lifestyle changes and raised questions related to dementia prevention. **Discussion:** The results validated the project's role as a facilitator of knowledge, particularly in controlling risk factors and promoting health. Among the challenges, a key issue was working with a pre-existing group that had its own established dynamics, which made it difficult to design effective activities without a prior understanding of those dynamics. **Conclusion:** The intervention demonstrated the potential of interdisciplinary educational initiatives with elderly in promoting knowledge about dementia and fostering changes in lifestyle habits. The planned expansion of the project in the coming semesters is expected to yield even more significant results. These actions aim to encourage both physical and intellectual engagement among elderly—key factors in preventing dementia and enhancing quality of life.

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Modifiable predictors of successful cognitive aging in Brazil: a sex-stratified analysis from ELSI-Brasil

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Successful cognitive aging varies across individuals and may be influenced by modifiable and non-modifiable factors. Understanding these associations in older adults is essential to develop prevention strategies, especially in low- and middle-income settings. To investigate 13 risk factors for dementia from the Lancet Commission in individuals with successful cognitive aging—defined as High-Performing Older Adults (HPOA)—in a nationally representative sample from the Brazilian Longitudinal Study of Aging (ELSI-Brasil). Older adults ≥ 75 years were classified as HPOA if they scored 1SD above the sample average in a global cognitive composite. Individuals scoring within 1SD for the sample average were defined as Typical Older Adults (TOA). Logistic regression models were used to estimate odds ratios (OR) for HPOA status, stratified by sex. In the included sample ($n=1,343$), higher education (OR=2.32; 95%CI 1.88–2.85) and physical activity (OR=0.53; 95%CI 0.36–0.77 for inactivity) were significantly associated with HPOA. In women, both education (OR=1.78; 95%CI 1.31–2.42) and physical activity (OR=0.38; 95%CI 0.21–0.69) remained significant. Among men, education

showed a stronger association (OR=3.41; 95%CI 2.14–5.43), while physical inactivity was not significant. Age was inversely associated with HPOA only in men (OR=0.86; 95%CI 0.76–0.97). These findings highlight education and physical activity as key protective factors, with distinct patterns by sex. Interventions tailored to sex-specific risk profiles may enhance cognitive resilience in aging populations.

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Pain prevalence and associated factors among older adults in Santa Catarina, Southern Brazil

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Introduction: Aging is one of the main transformations of the century. People are living longer due to improved living standards and advances. **Objective:** Determine the relationship between pain, mental health and functional capacity in the oldest old, characterize the pattern of pain in this population in the extreme south of Santa Catarina. **Methodology:** A questionnaire was applied to patients with sociodemographic and clinical data, also the visual analog pain scale (VAS), the Center for Epidemiologic Studies Depression Scale (CES-D), Generalized Anxiety Disorder (GAD-7) and Functional Capacity. Results and **Discussion:** The present study included 100 elderly; mean age was 84.9 years and 81.5% of the individuals classified their pain as mild to moderate. By the CES-D scale, 43% had depressive symptoms. There was a significant relationship between female gender and pain intensity. And a significant difference between the degrees of pain and marital status. Intense pain and depressive symptoms were strongly correlated, as was poor self-rated health. **Conclusion:** These findings suggest that severe pain in older adults is correlated with sex, marital status, symptoms of depression, and poor self-rated health. This highlights the multifaceted nature of pain and the importance of considering psychosocial factors in its assessment and treatment.

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A systematic review of the economic costs of dementia treatment across countries with varying income levels

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Background: Dementia is considered a term that encompasses a set of symptoms affecting the brain, leading to cognitive decline, memory impairment, and reduced ability to perform Activities of Daily Living. The total healthcare costs associated with individuals with Alzheimer's disease and other forms of dementia in the United States are estimated at \$384 billion. This highlights the need to understand the nature and extent of these

costs across different socioeconomic contexts. **Objective:** This study aims to identify, through scientific evidence, the costs associated with caring for individuals with dementia in High-Income Countries (HICs) and Low and Middle-Income Countries (LMICs). **Methods:** The search was conducted between February and June 2025 in the MEDLINE, Web of Science, BIREME, and SciELO databases. Studies reporting direct and indirect costs related to the treatment of dementia, diagnosed according to international guidelines, were selected. Evidence quality assessment tools were used to minimize the risk of bias in the studies included in this ongoing review. **Results:** A total of 2,629 were retrieved in the search strategy (747 duplicates), while 1952 were screened for title and abstract. A total of 260 articles were evaluated for inclusion. Only 8 studies evaluating patients in Latin America were included, while a minority of 3 studies evaluated cost analysis in Brazil. **Conclusion:** The systematized synthesis of studies on the cost profile of dementia treatment across different economic contexts provides a deeper understanding of its impacts on different perspectives of the health system, and they are essential for public health planning.

NEUROGENETICS

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Report of the TARDBP p.I383V pathogenic variant: a case series of frontotemporal dementia from Minas Gerais state, Brazil

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Background: Frontotemporal dementia (FTD) is a neurodegenerative disorder characterized by changes in behaviour and cognition. Future treatments depend on identifying specific genetic and biological pathways targets. Among FTD-related genes, TARDBP stands out. Its rare pathogenic variant p.I383V has not yet been described in Brazil. **Objectives:** Investigate and describe the profile of patients with FTD and the presence of the genetic variant p.I383V of the TARDBP gene. **Methods:** This is an observational, retrospective, and descriptive study, based on a sample of patients assessed between 2013 and 2024 in a neurological service. We included patients diagnosed with FTD some of whom carried the p.I383V pathogenic variant in the TARDBP gene, confirmed through genetic testing. **Results:** 33 patients with FTD were included (age: 59.6±11.3 years), and the average age of onset of symptoms was 56.2±11.5 years. The p.I383V.5 genetic variant was identified in five individuals (four women). Clinical presentations included two cases of behavioral variant FTD, two of semantic dementia, and one mixed variant with characteristics of the previous two. **Discussion:** The p.I383V variant in the TARDBP gene, linked to FTD, is rare but pathogenic. Identified in up to 3.5% of genetic FTD cases in Europe, to the best of our knowledge, this is the first report of this variant in Brazil, aiding genetic understanding. **Conclusion:** The identification of a potential cluster of patients with the pathogenic TARDBP p.I383V variant in Minas Gerais highlights an opportunity to explore region-specific genetic patterns.

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Nasu-Hakola disease: case reports

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Case Presentation: A 30-year-old college-educated man presented with psychomotor agitation, impulsivity and disinhibited behavior. He rapidly developed dementia with amnesic deficits and executive dysfunction. His medical history included chronic multifocal osteomyelitis, with surgical treatment of the left foot. Brain MRI revealed diffuse cerebral atrophy, predominant in frontotemporal regions with extensive white matter hyperintensities. SPECT showed bilateral posterior parietal hypoperfusion. Investigations for infectious, metabolic and autoimmune causes were negative, as was an initial genetic dementia panel. Alzheimer's disease biomarkers were inconclusive. During follow-up, his 40-year-old brother developed progressive cognitive impairment and also had a history of bone disease, with a femoral fracture at the age of 20. Long bone X-rays revealed multiple metaphyseal cysts consistent with membranous lipodystrophy. Given the clinical and radiological findings, Nasu-Hakola Disease (NHD) was suspected. Whole exome sequencing identified a homozygous pathogenic mutation in the TREM2 gene, confirming the diagnosis. **Discussion:** NHD, also known as polycystic lipomembranous osteodysplasia with sclerosing leukoencephalopathy, is a rare autosomal recessive disorder caused by mutations in TYROBP or TREM2 genes, affecting microglial function and bone remodeling. It is clinically characterized by early-onset behavioral changes, progressive cognitive decline, associated with typical cystic bone lesions. **Final Comments:** These cases highlight the importance of genetic investigation in young patients with rapidly progressive dementia, especially when atypical signs like skeletal manifestations are present. Bone abnormalities, often preceding neurological symptoms, are key to raising diagnostic suspicion. Early diagnosis allows for better symptom management and genetic counseling.

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Phenotypic variability of early-onset Alzheimer's disease associated with PSEN1: a case report

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Case Report: A 48-year-old female patient with a ten-year history of progressive attention difficulties and recurrent depression, rapidly progressing to parkinsonian symptoms and global poverty of speech. Six years after symptom onset, she became fully dependent for basic daily activities and developed unintelligible speech. Besides an aunt diagnosed with Alzheimer's at 45, family history was unremarkable. Neuroimaging and

general laboratory testing were unremarkable. A dementia genetic panel showed a Pathogenic Variant (PV) in PSEN1, and a Variant of Uncertain Significance (VUS) in GRN c.456_457delinsAT (p.Met152_Pro153delinsIleSer). The second patient, a 63-year-old female, had a seven-year history of amnesic symptoms. Her family history was strongly positive, with her mother, one sister, and a niece previously diagnosed with early dementia associated with PSEN1. Unlike patient 1, who mainly presented with executive dysfunction, patient 2 had severely affected amnesic domains, progressing to a complex profile of behavioral and neuropsychiatric symptoms, leading to complete dependence for daily activities. **Discussion:** Early-onset Alzheimer's disease (EOAD) may present with a heterogeneous spectrum phenotype, often difficult to recognize due to interindividual variability and incomplete penetrance. Notably, one patient exhibited primary progressive aphasia as the predominant symptom. Although a VUS in the GRN was identified, it presents low evidence supporting pathogenicity, and its role in modifying the phenotype is currently unknown, highlighting interpretation challenges for genetic findings in EOAD. **Conclusion:** Genotype-phenotype correlations may help elucidate this complex disorder. Further studies with Latin American patients and population-specific variants could improve the understanding of EOAD heterogeneity in our region.

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CSF1R-related leukoencephalopathy presenting as frontotemporal dementia plus gait apraxia

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Case report: A 54-year old woman presented with a two-year history of progressive behaviour, speech and gait change. She lost her abilities to do household chores, but was unconcerned, behaving childishly. On examination, she was apathetic and disinhibited, with a docile demeanor, perseveration and utilization behaviour. She scored 2/18 in the Frontal Assessment Battery and showed moderate anomia. Her speech was fast and dysarthric, tone mildly increased and gait apraxic. Brain magnetic resonance imaging (MRI) demonstrated important atrophy with an anterior-posterior gradient and extensive, frontotemporal predominant white matter hyperintensities. Subsequently, a fall caused her a traumatic right temporal injury, after which her condition rapidly deteriorated. Six months later, she was anarthric, severely demented and akinetic. MRI showed marked progression of the white matter lesions. The patient's biological family was unknown. CSF1R-related disorder was suspected. A panel identified a heterozygous mutation in CSF1R (NM_001288705.3) c.1765G>A (p.Gly589Arg). **Discussion:** Given the

behavioural and motor presentation, frontotemporal dementia, particularly corticobasal degeneration was considered, however imaging suggested a white matter disease. CSF1R-related disorder is an autosomal dominant leukoencephalopathy caused by mutations in the tyrosine kinase domain of the colony stimulating factor 1 receptor gene (CSF1R), whose phenotype has been increasingly recognised as frontotemporal dementia-like plus motor syndrome. It is often underdiagnosed or misdiagnosed. **Final Comments:** This case highlights the need to consider CSF1R mutations within the differential diagnosis of frontotemporal dementia. Early diagnosis allows genetic counselling, cascade testing and identification of asymptomatic carriers who may benefit from screening and potential disease modifying treatment.

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Impact of genetic variants in neurodevelopment-associated genes in sporadic Alzheimer's disease

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Neurodevelopmental factors have increasingly been recognized as potential contributors to Alzheimer's disease (AD), suggesting that early-life processes may influence the brain's vulnerability to later neurodegeneration. However, the mechanisms by which neurodevelopmental genes modulate AD pathophysiology remain poorly understood. In this study, we investigated the impact of single-nucleotide polymorphisms (SNPs) in neurodevelopment-related genes on amyloid- β (A β)-driven tau pathology. We first identified candidate genes involved in developmental processes through a comprehensive literature review and Gene Ontology annotations. Using the ADNI genetic database, we screened for SNPs within these genes. Our analysis included 17 genes (1854 SNPs) from 197 individuals with available genotyping, A β and tau PET imaging, and clinical data. To assess the relationship between neurodevelopment-related SNPs and tau pathology, we performed linear regression analyses adjusted for age, sex, and APOE ϵ 4 carrier status. Our results revealed that carriers of NLGN1 SNPs rs7638716 and rs116334566 exhibited significantly greater A β -associated tau pathology (β =+0.55 and +0.66, respectively; FDR-adjusted p <0.05). Of the cohort, 23 individuals carried rs7638716 and 27 carried rs116334566, with approximately half of the carriers presenting cognitive impairment. Participants (mean age: 78 years) included 87 women and 95 A β -positive individuals. NLGN1 encodes neuroligin, a synaptic adhesion molecule critical for synapse formation which has previously been implicated in neurodevelopmental disorders. These findings suggest that genetic variations in NLGN1 may enhance susceptibility to AD-related tau pathology, reinforcing the hypothesis that neurodevelopmental processes influence AD trajectory. Our study highlights NLGN1 as a promising genetic target for future research into early-life contributors to AD risk.

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A psychiatric-risk variant in TRIM26 exacerbates tau accumulation in the presence of amyloid- β in Alzheimer's disease

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Schizophrenia (SCZ) and bipolar disorder (BD) are among the most prevalent psychiatric conditions worldwide. Emerging evidence suggests that individuals with SCZ or BD are at an increased risk of developing neurodegenerative disorders, including Alzheimer's disease (AD). However, the biological mechanisms underlying this vulnerability to dementia remain unclear. We investigated whether single-nucleotide polymorphisms (SNPs), which are associated with psychiatric risk, could influence amyloid- β (A β)-driven tau accumulation. Using the GWAS Catalog, we identified SNPs with increased risk for SCZ or BD (odds ratios ranging from 1.1 to 2.1; p <0.05). We then examined the impact of the interaction between these risk variants and A β burden on tau pathology. We included 197 participants from the ADNI cohort with available genotyping, A β and tau PET imaging, and clinical data. Linear regression models adjusting for age, sex, APOE ϵ 4 status, and clinical diagnosis revealed that carriers of the SCZ-linked TRIM26 rs2021722 exhibited significantly greater A β -associated tau accumulation (β =+0.34; p =0.008). This variant was present in approximately 40% of the cohort (52% of women). Among TRIM26 rs2021722 carriers, 27% had mild cognitive impairment and 9% had dementia. Contrastingly, none of the BD-associated risk SNPs showed a significant effect on tau pathology. These findings suggest that the SCZ-associated TRIM26 rs2021722 may enhance susceptibility to tau-related neurodegeneration, suggesting that psychiatric disorders are risk factors for dementia at the genetic level. Our results highlight a potential genetic intersection between psychiatric and neurodegenerative disorders, underscoring the importance of further investigating psychiatric-risk genes in AD.

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APOE ϵ 4 is independently associated with plasma A β 42/40 levels in SCD and MCI

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Background: The APOE ϵ 4 allele is the major genetic risk factor for sporadic Alzheimer's disease (AD) and is strongly linked to brain amyloidosis. However, its influence on plasma biomarkers during the early clinical stages, subjective cognitive decline (SCD), and mild cognitive impairment (MCI) remains uncertain in the Brazilian population. **Objective:** To determine whether APOE ϵ 4 predicts plasmatic biomarkers in individuals with SCD, mild

MCI, and cognitively normal controls. **Methods:** Sixty-four older adults (13 controls, 12 with SCD, and 39 with MCI; mean age 65.8 ± 6.6 years) underwent plasma biomarker quantification (A β 42/40, A β 42/p-tau181, p-tau217, p-tau181, and neurofilament light [NfL]) using the SIMOA HD-X platform. The APOE genotype determination was based on analyzing single nucleotide polymorphisms rs429358 and rs7412. Binomial logistic regression with backwards stepwise selection was performed using APOE ϵ 4 status as the dependent variable, controlling for age and sex. **Results:** The A β 42/40 ratio was the only biomarker retained in the final model, showing a significant association with APOE ϵ 4 status ($B = -130.36$; 95%CI 0.30–0.97; $p = 0.044$). No associations were found for the other biomarkers. **Discussion and Conclusion:** Lower plasma A β 42/40 levels were independently associated with APOE ϵ 4 carriage, supporting its utility as a peripheral marker of amyloid-related genetic vulnerability in early-stage AD. These results reinforce the value of A β 42/40 as a candidate biomarker for preclinical and prodromal phases of the disease.

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Brazilian cohort brain age prediction model evaluation: potential biomarker for neurodegeneration

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Brain age models have been suggested as a promising neurodegeneration biomarker that could fill the gap of automatic metrics for accurate diagnosis. It highlights the dementia process that can be seen as a similar process to accelerated aging. Here, we verify its discriminative performance in a Brazilian population. 138 cross-sectional T1-weighted MRI scans were selected from a Brazilian dementia cohort: 33 with Alzheimer's Disease (AD, 75.9 ± 7.9 yo), 27 with Mild Cognitive Impairment (MCI, 72.1 ± 5.2 yo), and 78 with normal cognition (NC, 69.6 ± 7.2 yo), all with 8 or more years of education. A Bayesian regression model was built on cortical thickness features from FreeSurfer Destrieux parcellation (v.7.2.0) with education as a covariate. The brain age gap (BAG) was the difference between predicted and chronological ages. Mean BAG across groups was compared using Tukey post hoc tests. The model's performance for the cognitively normal group was MAE=5.6. Significant differences were found for AD (mean BAG 10.7 ± 12.1 years) and MCI group (diff=9.88 years; $p < 0.001$), and for AD and NC group (diff=8.64 years; $p = 0.0175$). Education had a negative effect on the predicted age ($\beta = -1.54$, 95%CI -2.6 to -0.8). This brain age model was effective in distinguishing AD from the other groups. Education's negative effect on predicted age legitimates its protection for global neurodegeneration, as shown in recent literature focusing on Latin American aging populations. We highlight the need to explore this neurodegenerative biomarker in more diverse age ranges and sociodemographic scenarios.

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Hypoparathyroidism with brain calcifications and microangiopathy: copathology or causality?

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Male patient, 71 years old, former accountant, with 14 years of education, diagnosed with autoimmune hypoparathyroidism since the age of 34, and a former smoker (98 pack-years). He experienced four symptomatic seizures due to hypocalcemia. In 2022, he was found unconscious, with suspected new seizure. Since then, he progressively developed repetitive speech, forgetfulness, spatial disorientation, jealousy-related delusions, disinhibition, hypersexuality, affective lability, and anxiety symptoms. He lost independence in instrumental activities of daily living, while basic activities remained preserved. Neurological examination revealed a broad-based gait. Neuropsychological assessment showed attentional deficits, executive dysfunction, and verbal memory impairment, with preserved visuospatial and language functions. Brain MRI revealed cortical, basal ganglia, and cerebellar calcifications, along with microangiopathy (Fazekas score 3). Given the executive-predominant cognitive syndrome, neuropsychiatric symptoms, and imaging findings, vascular and hypoparathyroidism were considered as mixed etiologies for the dementia. Correction of calcium and depression improved behavior. Hypoparathyroidism may cause cognitive dysfunction similar to "brain fog" through multiple mechanisms, with chronic hypocalcemia and disease duration being the most important. No clear correlation has been identified between intracerebral calcifications and cognitive impairment. However, cerebellar calcifications may lead to ataxia, as observed in this case. The relationship between intracranial calcifications and cerebrovascular disease remains uncertain. Vascular calcium deposition may contribute to cardiovascular pathology by reducing vessel elasticity and altering cerebral hemodynamics. We present the case of a patient with hypoparathyroidism and cognitive dysfunction beyond what is expected from "brain fog," with probable vascular contribution secondary to chronic hypocalcemia and a heavy smoking history.

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Manic episode unmasking posterior cortical atrophy

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A 63-year-old right-handed man with a college degree in electronics presented with progressive cognitive and behavioral complaints over the past three years. He reported difficulties with multitasking, inattention, repetitive speech, forgetfulness, and behavioral changes such as disinhibition, compulsive eating of sweets, and excessive spending. Previously unrecognized manic episodes were identified during follow-up, leading to a diagnosis of bipolar affective disorder (BD) and treatment initiation, with improvement in behavior. However, cognitive decline progressed, particularly in visuospatial abilities, as he began bumping into furniture and objects. There was functional loss in instrumental activities of daily living. Physical examination showed no signs of parkinsonism. Neuropsychological assessment revealed impairments in visuoconstructive praxis, episodic memory, and visual stimulus manipulation. Brain MRI showed prominent bilateral

parieto-occipital atrophy (Koedam score 3, MTA 0). Given the clinical and imaging findings, the main diagnostic hypothesis was Posterior Cortical Atrophy (PCA), despite the presence of amnesic symptoms. Although BD can be associated with diffuse cortical thinning, including posterior regions, there is no direct evidence linking it to PCA, which is characterized by selective and progressive degeneration. However, BD is associated with an increased risk of dementia, especially Alzheimer's disease (AD), through shared mechanisms such as neuroinflammation, oxidative stress, and metabolic dysfunction. Lithium use has been associated with a neuroprotective effect. This case highlights the importance of maintaining high suspicion for neurodegenerative syndromes in patients with BD, given the higher risk and that the disorder itself may cause cognitive symptoms, potentially mask coexisting degenerative conditions.

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Mapping the temporal and anatomical gradients in AD: association of blood and MRI biomarkers

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We aim to explore the recent increase in research on plasma biomarkers for dementia, along with the known power of sMRI in characterizing neurodegeneration. We verified whether an optimal combination of BBMs and sMRI biomarkers could benefit clinical and research investigations while controlling for age and education, known risk factors for dementia in the region. T1w MRI were processed with FreeSurfer v7.2.0 for 67 control subjects (NC, Age=71±5), 23 with amnesic Mild Cognitive Impairment (MCI, Age=72±13), 30 with Alzheimer's Disease (AD, Age=76±13), and 14 with other dementia (non-AD, Age=74±13) to extract common hippocampus, ventricles, cortical, and white matter volumes, along with average cortical thickness. Blood samples from the same session were processed using SIMOA to obtain concentrations of pTau217, pTau181, pTau217/αβ42, pTau181/αβ42, NFL, and GFAP. A stepwise model was obtained with the most relevant terms to discriminate diagnostic groups, covarying for age, sex, education, and intracranial volume. The resultant multinomial logistic linear model is composed of pTau 217/αβ42 and volumes of the Lateral Ventricle, Inferior Lateral Ventricle, Hippocampus, Cortex, and Cerebral White Matter. It has 76% accuracy, discriminating aMCI from the NC group with a significantly different coefficient pTau217/αβ42 ($p=0.015$), while AD is different from NC for pTau217/αβ42 and all imaging neurodegeneration terms except Cortical volume (all $p<0.05$). The model includes variables from the ATN framework types, being a summary of the AD trajectory. It comprises different time points in the AD evolution spectrum, including early disease stages, local neurodegeneration, and global neurodegeneration.

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Multimodal biomarker profiling for MCI staging and Alzheimer's conversion prediction using sustain modeling

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Background: Alzheimer's Disease (AD) involves neurodegeneration in specific brain regions. Mild cognitive impairment (MCI) patients show higher AD conversion risk than cognitively normal individuals (CN), where early diagnosis improves prognosis. In this perspective, machine learning using MRI and neuropsychological data enhances AD prediction. **Objective:** To analyze the best biomarkers for identifying MCI-to-AD conversion using machine learning models through stage and subtype inference by pySuStaIn. **Methods:** We analyzed 558 MCI and 21 CN subjects from ADNI with longitudinal neuropsychological (NPS), CSF (β -amyloid/pTAU), and MRI-T1w volume data (FreeSurfer-processed). For PySuStaIn modeling, four biomarker combinations were used: nSPC (NPS+MRI+CSF), NPS-only, MRI+CSF-only, and optimized biomarkers (bBMK). Finally, Conversion rates were tracked at one-two years. **Results:** The model identified two principal MCI subtypes: nSPC-SB1 dominated by WM_Hypointensities/Ventricles/GDTOTAL and nSPC-SB2 driven by TRABSCOR/pTAU in early stages. In NPS data, SB1 showed RAVLT_Forgetting/GDTOTAL alterations, while SB2 emphasized TRABSCOR. MRI+CSF analysis revealed SB1 with pTAU changes and SB2 with ventricular/WM_Hypointensities alterations. Using bBMK, SB1 and SB2 showed ventricular changes and pTAU dominance, respectively; while neuropsychological tests had minimal impact. Longitudinal analysis showed highest AD-progression in nSPC_SB2 and NPS_SB2, while bBMK_SB0 had the fewest converters. **Discussion:** The identification of distinct profiles suggests different pathophysiological pathways in disease progression, aligning with growing evidence of AD heterogeneity. Furthermore, we demonstrated the superiority of multimodal biomarkers over single-domain approaches for AD MCI-conversion prediction. **Conclusion:** Machine learning-driven stratification using combined biomarkers significantly improves MCI-to-AD conversion prediction and could improve AD diagnosis.

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Neuroimaging biomarkers associate with cortical atrophy progression in preclinical Alzheimer's disease

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Imaging biomarkers, especially PET for amyloid- β (A β -PET) and tau tangles (tau-PET), are important tools for the early biological diagnosis and staging of Alzheimer's disease (AD). However, the relationship between these diagnostic tools and other AD biomarkers, such as longitudinal cortical atrophy, remains unclear. We aimed to investigate the association between PET biological staging and longitudinal cortical atrophy in asymptomatic A β -positive (A+) individuals. We analyzed 162 A+ participants from the A4 Study placebo group with baseline MRI, and A β -PET ([18F]Florbetapir) and tau-PET ([18F]Flortaucipir), along with a follow-up MRI at least two years later. Tau positivity in the medial temporal lobe (TMTL+) and neocortex (TNEO+) was defined as a standardized uptake value ratio (SUVR) 2.5 standard deviations (SD) above the mean of a reference A β -negative population. Participants (mean age: 71.5 $\pm$ SD 4.8 years; 48.9% women) were classified as A+T- (n=113), A+TMTL+ (n=21), or A+TNEO+ (n=28). Analysis of covariance (ANCOVA) with Tukey's post hoc test revealed that, longitudinally, A+TNEO+ showed significantly greater cortical atrophy than A+T- ($P<0.001$) and A+TMTL+ ($P<0.01$). Regional analysis further revealed that a higher atrophy rate in the fusiform, temporal, and inferior parietal cortices is more closely associated with biological staging ($P<0.01$), particularly among A+TNEO+ individuals. These findings suggest that cortical atrophy, especially in temporoparietal regions, may serve as a practical alternative for capturing tau-related neurodegeneration. In conclusion, imaging-based biological staging schemes correlate with progressive cortical atrophy, highlighting their potential as sensitive markers for early AD detection.

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Evolution of magnetic susceptibility of Locus Coeruleus with tau pathology progression: a postmortem study

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Background: The Locus Coeruleus (LC) is one of the first structures to be affected in Alzheimer's Disease. However, neuroimaging studies are still limited due to difficulties in assessing quantitative maps. The Quantitative Susceptibility Mapping (QSM) is a MRI technique able to map magnetic composition of tissue. Separating paramagnetic (PCS) from diamagnetic (DCS) effects leads to improved evaluation of iron (PCS), myelin/proteins (DCS). Objectives: In this study, QSM, PCS and DCS maps were evaluated for postmortem (in situ and ex situ) subjects at different tau pathology stages. **Methods:** In situ MRI from 17 postmortem subjects were acquired at 7T scanner, with 12 also being imaged at ex situ condition. Images were processed to obtain QSM, PCS and DCS maps. LC was manually segmented and tau pathology staging was assessed by the Braak staging. Results and **Discussion:** At in situ condition no contrast was observed for the LC. A decline of QSM and DCS was observed at earlier stages, suggesting progressive demyelination. At the last stages, both maps showed an increase, suggesting significant protein aggregation. At ex situ condition LC was visible from QSM and PCS maps due

to iron oxidation. Significant increase of PCS was observed, suggesting continuous iron accumulation with progression of tau pathology. **Conclusion:** These results indicate that in situ and ex situ maps are able to show different magnetic susceptibility patterns and are complementary to each other in evaluating changes due to tau pathology progression.

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Brain morphometry and normative modeling with ScanOmetrics

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Morphometric analysis of brain MRI has significantly advanced our understanding of both healthy and pathological aging. However, translating research-grade morphometry tools into clinical practice demands time-efficient robust structural MRI reconstruction pipelines. In this context, deep learning (DL)-based methods have emerged as the leading approach for whole-brain segmentation and parcellation. In this work, we introduce ScanOmetrics, an open-source, research-grade tool designed to detect statistical anomalies in individual MRI scans using both surface- and voxel-based metrics. The tool leverages DeepSCAN, a deep learning-based brain segmentation and cortical parcellation method, as input to an 11-minute surface reconstruction pipeline adapted from FreeSurfer. We present the first validation results of ScanOmetrics, benchmarking its performance against standard FreeSurfer outputs. The analysis was focused on OASIS3, a large and freely available dataset containing clinical grade high-resolution isotropic T1-weighted MRI scans of patients with Alzheimer's disease (AD) and healthy controls (HC). We demonstrate that cortical thickness anomalies in patient scans were mainly detected in regions that are known as predilection areas of cortical atrophy in AD. In contrast, anomaly detections in HCs were up to twenty-fold reduced and spatially unspecific. Progression of the atrophy pattern with clinical dementia rating (CDR) was clearly observable. Notably, ScanOmetrics delivered results in under 25 minutes, over 15 times faster than FreeSurfer.

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How glucose metabolism mirrors neuropsychiatric symptoms in dementia: a PET/CT-FDG cross-sectional study

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Background: Behavioral symptoms, stress, and depression in dementia are commonly assessed using validated instruments such as the Neurop-

psychiatric Inventory Questionnaire (NPI-Q), the Hospital Anxiety and Depression Scale (HADS), the Geriatric Depression Scale (GDS), and the 10-item Perceived Stress Scale (PSS-10). However, few studies have explored associations between these clinical scales and cerebral metabolic patterns measured by 18F-fluorodeoxyglucose positron emission tomography (FDG-PET). **Objective:** To investigate the association between NPI-Q, PSS-10, GDS, and HADS scores and regional cerebral glucose metabolism on FDG-PET in patients with dementia. **Methodology:** This cross-sectional study evaluated 29 patients (12 men and 17 women; mean age: 69 years) followed at the Cognitive Neurology Clinic of HC-UFPE, who underwent brain FDG-PET within the last 18 months or during the study period. **Results:** On the NPI-Q, the mean severity score was 9.71 (± 6.3) (range: 0–36), and the mean caregiver distress score was 13.5 (± 8.9) (range: 0–60). Mean HADS scores were 7.08 (± 5.26) (range: 0–21) for anxiety and 6.56 (± 4.71) (range: 0–21) for depression. The average GDS score was 4.34 (± 3.67) (range: 0–15). Among the 27 brain regions analyzed, significant hypometabolism was observed in the right parietal cortex, bilateral medial parietal cortices, and the left caudate nucleus in 53.8% of cases. **Conclusion:** The findings underscore the potential association between neuropsychiatric symptoms and specific metabolic alterations in dementia. Expanding the sample size will be critical to strengthen these associations and enhance the understanding of the neurobiological substrates of behavioral symptoms in dementia.

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Cerebral atrophy in mild cognitive impairment and dementia: visual scales versus automated volumetry data from a reference center in Recife, Brazil

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Background: Magnetic resonance imaging (MRI) biomarkers have proven effective in evaluating patients with cognitive complaints. **Objective:** To describe the radiological profile of patients seen at a neurocognition outpatient clinic in a reference center in Northeast Brazil and to investigate how visual scales performed in contrast to automated volumetric analysis. **Methods:** A neuroradiologist was responsible for classifying MRI scans using visual atrophy rating scales: medial temporal lobe atrophy (MTA) and entorhinal cortex atrophy (ERICA). Automated volumetric analysis was performed using VolBrain. Spearman correlation coefficients were produced to assess the strength of association between visual and volumetric measures. **Results:** A total of 81 brain MRIs were reviewed, of which 45 included a volumetric T1 sequence suitable for MTA, ERICA, and automated volumetric assessment. Of the 45 patients, 55.6% were women, and the mean age was 69.9 years. According to neuroimaging, the mean MTA was 0.89 and the mean ERICA was 0.73. There was a moderately negative correlation between MTA/hippocampal volumetry ($p = -0.436$) and MTA/entorhinal cortex volumetry ($p = -0.480$). The ERICA score had a moderately negative correlation

with hippocampal volumetry ($p = -0.409$) but a strong negative correlation with entorhinal cortex volumetry ($p = -0.504$). **Discussion:** Visual atrophy scales are widely used in clinical practice and, as supported by previous studies, show a good correlation with automated brain volumetry. **Conclusion:** MTA and ERICA demonstrated moderate to strong correlation with automated volumetric measures, supporting their value in the radiological assessment of cognitive complaints.

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Clinical profile of patients undergoing florbetaben PET imaging: descriptive analysis of a clinical cohort

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Background: The biological diagnosis of Alzheimer's disease (AD) is increasingly relevant, enabling detection of amyloid pathology. Florbetaben (FBB) PET imaging provides visualization of amyloid plaques and enhance diagnostic confidence. Modifiable risk factors remain key targets for dementia prevention, highlighted by the 2024 Lancet Commission, while age remains the strongest non-modifiable risk factor. **Objective:** To describe the clinical profile of patients undergoing FBB PET imaging, focusing on age, sex, and body mass index (BMI), and to characterize subgroups with positive amyloid PET. **Methods:** We retrospectively analyzed 57 FBB PET scans performed between 2023 and 2024. Age, sex, and BMI (calculated and classified according to WHO guidelines) were extracted from electronic medical records. Age was categorized as ≤ 60 years, 61–85 years, and > 85 years. Descriptive statistics were used to characterize the sample and FBB-positive subgroup. Results Among 57 patients, 37 had positive and 20 had negative FBB PET results. The overall sample had a mean age of 73.3 years and mean BMI of 25.5. There were 30 males and 27 females. In the FBB-positive group ($n = 37$), mean age was 74.7 years and mean BMI was 25.2. Age distribution: ≤ 60 years ($n = 2$), 61–85 years ($n = 34$), > 85 years ($n = 1$). BMI classification: underweight ($n = 2$), normal weight ($n = 13$), overweight ($n = 19$), obesity grade I ($n = 3$), grades II–III ($n = 0$). **Discussion:** This analysis provides insights into the demographic and anthropometric profile of patients undergoing FBB PET imaging. **Conclusion:** The cohort was predominantly composed of older adults with normal to overweight BMI. Further studies are needed to explore clinical associations.

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How the use of FDG PET is improving the diagnosis of dementia in a reference center in Recife, Brazil

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Background: The use of FDG PET to differentiate atypical Alzheimer's disease (AD) from other conditions such as Lewy body dementia (LBD), frontotemporal dementia (FTD), and vascular cognitive impairment (VCI). **Objective:** To analyze the main indications and findings from the first five years of routine FDG PET use at a specialized center in Recife, Northeastern Brazil. **Methods:** This is a descriptive, retrospective study followed at Hospital das Clínicas, UFPE, from 2018 to 2023, who had dementia or mild cognitive impairment and underwent FDG PET scans. **Results:** 62 patients underwent FDG PET in the investigations of cognitive syndromes. While 15 scans were non-diagnostic and showed non-specific changes, 19 suggested AD, 8 LBD, 11 FTD, and 9 showed no changes. In 21/29 (72.4%) patients, the scan helped investigate types of atypical neurodegenerative dementias; in 14/24 (58.3%), FDG PET helped answer the clinical question. FDG PET did not help much in diagnosing AD versus FTD only answered in 50% of the cases. When considering the differential diagnosis of AD and LBD, the results were helpful in 75% of the cases. **Discussion:** FDG PET was effective in diagnosing atypical neurodegenerative dementias, answering clinical questions of atypical cases and early-onset cases. However, it showed limited accuracy in differentiating AD from FTD. FDG PET performed better in distinguishing AD from LBD. **Conclusion:** FDG PET proved valuable in diagnosing atypical and early-onset dementias, with high diagnostic relevance in these groups. However, its use remains limited to specialized centers, especially in Brazil.

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FDG PET/CT-Based comparative analysis of functional network involvement in dementias

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Background: ¹⁸F-FDG PET/CT is a validated neuroimaging tool for evaluating cerebral metabolic changes in neurodegenerative dementias. Recent approaches have proposed analyzing brain glucose metabolism based on functional network organization rather than isolated anatomical regions. **Objective:** To identify and compare metabolic alterations in intrinsic functional networks across patients with Alzheimer's disease (AD), frontotemporal dementia (FTD), and dementia with Lewy bodies (DLB), based on a cluster-wise analysis of FDG-PET/CT images. **Methods:** We retrospectively analyzed 21 patients (7 AD, 9 FTD, 5 DLB) who underwent ¹⁸F-FDG PET/CT at a Brazilian university hospital. Individual images were compared voxel-by-voxel with a control group from the ADNI database. Significant

clusters of hypo- and hypermetabolism were anatomically labeled and functionally categorized into resting-state networks (e.g., DMN, SN, ECN). Binary coding was applied to allow intergroup comparisons. **Results:** AD patients showed predominant hypometabolism in the default mode network (85.7%) and hypermetabolism in subcortical/cerebellar regions (71.4%). FTD cases exhibited more frequent involvement of the salience and executive networks, with hypermetabolism in the sensorimotor network. DLB presented a heterogeneous profile, with sensorimotor hypermetabolism (80%) and relative preservation of limbic metabolism. **Discussion:** These findings suggest disease-specific network-based metabolic signatures. Cluster-wise metabolic analysis may reflect compensatory activity or network disinhibition, offering added value to differential diagnosis even without direct connectivity mapping. **Conclusion:** This study introduces an innovative framework for interpreting FDG-PET/CT through functional network mapping, supporting its potential to improve dementia classification and enhance pathophysiological understanding.

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Cerebral perfusion SPECT in dementias: a naturalistic study of clinical and imaging associations

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Background: The clinical distinction between different types of dementia is a diagnostic challenge. **Objective:** To investigate the value of brain perfusion SPECT in the diagnostic framework of dementias. **Methods:** We included patients with cognitive disorders who underwent SPECT between 2013 and 2019. Medical data were collected and the clinical diagnosis were established independently of the SPECT. All SPECT images were blindly analyzed, by describing the topographical patterns of hypoperfusion. **Results:** We included 73 patients: typical Alzheimer's disease (tAD; n=25), behavioral variant frontotemporal dementia (bvFTD, n=16), primary progressive aphasia (n=8), mild cognitive impairment (n=8), atypical parkinsonisms (n=5), dementia due to Parkinson's disease (n=4), functional (psychiatric) disorders (n=4) and atypical AD (n=3). Inferior parietal hypoperfusion was detected in 44% of tAD patients and in no bvFTD patient, while superior parietal hypoperfusion was observed in 40 and 6% of tAD and bvFTD, respectively. Hypoperfusion in the lateral temporal cortex was observed in 52% and 19% of tAD and bvFTD, respectively. Medial temporal hypoperfusion was identified in 36% and 31% of tAD and bvFTD patients, respectively. Posterior cingulate hypoperfusion was found in 32% of tAD and in 12% of bvFTD. Hypoperfusion in the ventromedial prefrontal cortex was identified in 56% and 32% of tAD and bvFTD, respectively. **Discussion e Conclusion:** There was a significant overlap of hypoperfusion patterns between tAD and bvFTD. The diagnostic value of a single hypoperfusion pattern is limited, but a comprehensive analysis of all cerebral regions may be useful for the clinical distinction. Importantly, clinical information should always be considered.

TECHNICAL AREA: NEUROIMAGING
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White matter damage patterns in corticobasal syndrome due to different pathologies

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Background: Corticobasal syndrome (CBS) is an atypical and rare parkinsonian disorder, often associated with corticobasal degeneration, but can also be related to others neuropathologies, such as Alzheimer's disease (AD). Degeneration patterns of white matter tracts may aid in distinguishing between different etiologies, as seen in previous studies using diffusion tensor imaging (DTI). Clinical symptoms alone are unable to predict underlying pathology. **Novel Methods:** This study enrolled 31 patients with clinical diagnosis of CBS. All participants underwent MRI-based DTI and also performed [¹¹C]Pittsburgh Compound-B (PIB)-PET imaging on a 3T PET-MRI to assess their brain amyloid deposition. Patients were categorized in CBS associated with AD (CBS-AD, n=13) or CBS not associated with AD (CBS non-AD, n=18), based on their amyloid status. With matter damage evaluated through DTI on fractional anisotropy (FA) and mean diffusivity (MD). Images were quantified using the MRICloud platform, and statistical analyses were corrected for multiple comparisons using the Bonferroni method. **Results:** Significant white matter damage was observed in both groups in the corpus callosum, fornix, tapetum, reticular formation, sagittal stratum, and the left posterior thalamic radiation. Additionally, the CBS non-AD group presented further microstructural white matter alterations in the fornix stria terminalis, superior cerebellar peduncle, inferior fronto-occipital fasciculus, and cerebral peduncle. **Conclusion:** The observation of additional regional involvement in CBS non-AD highlights the potential of DTI as a diagnostic biomarker for identifying underlying neuropathological substrates in vivo.

CLINICAL NEUROLOGY

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Anticholinergic burden is associated with cognitive impairment in older adults at risk for dementia in latin america: the LatAm-FINGERS study

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Background: Comorbidities and polypharmacy are highly prevalent in older adults. Many medications have anticholinergic properties, whose cumulative effect, known as anticholinergic burden (AB), may impair cognitive health, especially among older adults. Most evidence comes from single-center studies in high-income countries, with no prior multicenter research in Latin America. **Objective:** To investigate whether AB is independently associated with questionable cognitive impairment and with cognitive and functional measures in older adults from the LatAm-FINGERS study. **Methods:** This study included 1,243 participants aged 60–77 from 12 Latin American countries, without dementia (CDR=0–0.5), and complete baseline clinical/neuropsychological data. Measures included the Anticholinergic Cognitive Burden (ACB) scale, Clinical Dementia Rating (CDR), Mini-Mental State Examination (MMSE), Instrumental Activities of Daily Living (IADL), and Cognitive Function Index (CFI). **Results:** The mean education level of the population studied was 13.28 years. ACB was negatively associated with MMSE and positively with CDR-SB and CFI ($p=0.030$, 0.007 , and 0.001 , respectively). No association was found with IADL ($p=0.403$). Each ACB point increased the odds of scoring 0.5 on CDR Global by 28% ($OR=1.28$; 95%CI 1.05–1.56; $p=0.015$). **Discussion:** Our findings support that AB plays a significant role in cognitive

performance, and subjective cognitive decline, and increases the likelihood of scoring in CDR, even when controlled for major confounders. **Conclusion:** AB might be a relevant predictor of cognitive impairment and probably mediates the negative effects of polypharmacy in the LatAm-FINGERS cohort.

TECHNICAL AREA: CLINICAL NEUROLOGY

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Cognitive performance in carriers of the transthyretin Val122Ile mutation: a cross-sectional study

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Background: While the cognitive effects of hereditary transthyretin amyloidosis (ATTRv) have been primarily described in carriers of the Val30Met (p.Val50Met) mutation, evidence suggests that other variants may also impair the central nervous system. **Objective:** To characterize the cognitive performance of individuals with the Val122Ile (p.Val142Ile) mutation. **Methods:** Observational, cross-sectional study involving individuals aged over 18 years, carriers of the Val122Ile mutation, who underwent comprehensive cognitive assessment. **Results:** A total of 22 individuals were included, 14 (63.6%) asymptomatic and eight (36.4%) symptomatic. The mean age was 53 years, with 12 (54.5%) male and a mean educational level of 8.45 years. Among symptomatic individuals, four had a cardiac phenotype and four a neuropathic phenotype due to carpal tunnel syndrome, with a mean disease duration of 3.44±2.9 years. MMSE performance was below average in 81.8% of participants. Verbal episodic memory was the most affected domain, with delayed recall deficits on the HVLIT in 93% of asymptomatic and 88% of symptomatic individuals. Executive dysfunction was frequent, especially among symptomatic individuals, with poor performance on TMT-B, Victoria Stroop Test and semantic fluency. Attention deficits emerged in tasks requiring processing speed and interference control, while working memory remained largely preserved. Visuospatial and constructive impairments were observed in 63.6% on the Clock Drawing Test. No significant differences were found between symptomatic and asymptomatic groups. **Conclusion:** Individuals with the Val122Ile mutation may exhibit early and significant cognitive impairment, similar to studies involving the Val30Met mutation, highlighting the need for further studies dedicated to this population.

TECHNICAL AREA: CLINICAL NEUROLOGY

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Progressive dementia in Parkinson's disease with associated psychosis

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Summary A 73-year-old female with Parkinson's Disease (PD) for nine years presented with progressive cognitive decline over the past 18 months, characterized by language impairments (anomia), visual agnosia, ideomotor apraxia, and prosopagnosia. She also developed affective lability, erratic behavior, and structured visual hallucinations, while preserving reading ability and basic arithmetic skills. The neurological exam showed disorientation in time and partial disorientation in space, hypophonic speech with monotone prosody, hyposmia, and oculomotor apraxia. Motor function assessments revealed bradykinesia, rigidity, resting tremor, mild postural instability, and a forward-leaning small-step gait, but no freezing episodes or falls. Cerebellar dysfunction was noted, with impaired diadochokinesia and mild dysmetria, along with visuoconstructive deficits (clock-drawing test, Rey complex figure test). Complementary exams indicated global cognitive impairment (MMSE: 10/30), bilateral striatal hypodensity (SPECT-TRODAT-1), and the absence of parieto-temporal atrophy (MRI). The final diagnosis was Dementia associated with Parkinson's Disease (PDD), with psychotic symptoms and an atypical cognitive pattern. Differential diagnoses included Dementia with Lewy Bodies (DLB) and Alzheimer's Disease (posterior variant).

Background: While the cognitive effects of hereditary transthyretin amyloidosis (ATTRv) have been primarily described in carriers of the Val30Met (p.Val50Met) mutation, evidence suggests that other variants may also impair the central nervous system. **Objective:** To characterize the cognitive performance of individuals with the Val122Ile (p.Val142Ile) mutation. **Methods:** Observational, cross-sectional study involving individuals aged over 18 years, carriers of the Val122Ile mutation, who underwent comprehensive cognitive assessment. **Results:** A total of 22 individuals were included, 14 (63.6%) asymptomatic and eight (36.4%) symptomatic. The mean age was 53 years, with 12 (54.5%) male and a mean educational level of 8.45 years. Among symptomatic individuals, four had a cardiac phenotype and four a neuropathic phenotype due to carpal tunnel syndrome, with a mean disease duration of 3.44±2.9 years. MMSE performance was below average in 81.8% of participants. Verbal episodic memory was the most affected domain, with delayed recall deficits on the HVLIT in 93% of asymptomatic and 88% of symptomatic individuals. Executive dysfunction was frequent, especially among symptomatic individuals, with poor performance on TMT-B, Victoria Stroop Test and semantic fluency. Attention deficits emerged in tasks requiring processing speed and interference control, while working memory remained largely preserved. Visuospatial and constructive impairments were observed in 63.6% on the Clock Drawing Test. No significant differences were found between symptomatic and asymptomatic groups. **Conclusion:** Individuals with the Val122Ile mutation may exhibit early and significant cognitive impairment, similar to studies involving the Val30Met mutation, highlighting the need for further studies dedicated to this population.

TECHNICAL AREA: CLINICAL NEUROLOGY

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Tinnitus-induced severe auditory psychosis: a clinical case managed with environmental measures

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Presentation: N.M., male, 82 years old, married, 4 years of education, works as street market vendor. Former smoker (100 pack-years). Active alcohol use: one dose of distilled alcohol and beer (300 mL) daily. The patient and his daughter reported that, over the past year, he has been hearing noises coming from the neighbor's house. He describes the sounds as metal clanging, as if the neighbor had "opened a factory." Four months ago, he began throwing stones at the neighbor's house during nighttime, causing material damage. A diagnosis of dementia was made at another facility, and risperidone 1 mg/day was prescribed. Mini-Mental State Examination: 23/30. Functional independence preserved for basic and instrumental activities. Brain MRI showed no selective atrophy. FDG-PET revealed mild hypometabolism in the left anterior temporal, mesial, and insular regions. Audiometry was requested, and an increase in risperidone to 1.5 mg along with cessation of alcohol use was recommended, without clinical improvement. Audiometry revealed severe hearing loss. A hypothesis of complex tinnitus secondary to hearing loss leading to hallucinations was considered, as they occurred exclusively during complete silence. After initiating white noise during nighttime, the patient improved significantly, ceased throwing objects at the neighbor's house, and reduced the antipsychotic dose. **Discussion:** Auditory Charles Bonnet Syndrome is not formally described in the literature, although a few case reports exist. **Conclusion:** This case highlights how sensory impairments can cause severe behavioral disturbances, and how individualized environmental assessment can aid in optimizing pharmacological and non-pharmacological management.

TECHNICAL AREA: CLINICAL NEUROLOGY

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Mild cognitive impairment and seizures evolve into Alzheimer's disease: a post-mortem clinical case

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Epilepsy and Alzheimer's disease (AD) are complex neurological disorders that affect the elderly with relative frequency. Seizures or epileptic episodes can occur at any stage of AD, including in cases of the mild cognitive impairment (MCI) phase. **Objective:** To describe a case of seizures that began in a healthy 88-year-old man and later progressed to Alzheimer's disease. **Methods:** A clinical and family history was taken, the Measurement of Functional Activities in Older Adults in the Community was administered, and a Mini-Mental State Examination (MMSE), neuroimaging studies, and an autopsy were conducted. **Results:** Neuroimaging showed signs of age-related atrophy. The Measurement of Functional Activities in Older Adults in the Community scored 20/30, the MMSE score was 6/30, and the autopsy confirmed a diagnosis of Alzheimer's disease. **Conclusion:** Seizures may play a role in the pathophysiology of AD and, consequently, cannot be ignored in its progression. In this context, we describe a clinical case of epilepsy in a healthy 88-year-old man with no prior history of seizures, who was in the MCI phase and later progressed to AD.

TECHNICAL AREA: CLINICAL NEUROLOGY

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Primary progressive aphasia with mixed features: diagnostic challenges in overlapping syndromes

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Presentation: I.A.M., 62-year-old female, university degree in psychology, hypertension, dyslipidemia and essential tremor. Symptoms began in 2021 with depression, anhedonia and emotional lability. In March 2022, she developed word-finding difficulties and circumlocutions. Progressive speech fluency reduction and slowed speech. By 2023, she presented with phonemic paraphasias and anomia, though semantic knowledge was preserved. She could semantically associate objects and gesture their use. Repetition of simple words was intact, but sentence repetition was impaired. Apraxia of speech and orobuccolingual apraxia became evident. Her husband reported new behavioral changes, disinhibition and increased tendency to approach and touch strangers, especially men. Brain MRI showed asymmetric temporal lobe atrophy, more pronounced on the left. FDG-PET revealed hypometabolism in the left precuneus and posterior cingulate, bilateral posterior

temporoparietal cortices (worse on the left), basal ganglia and left frontal cortex. Amyloid-PET (PIB-PET) was negative for cortical β -amyloid deposition. **Discussion:** The case is consistent with Primary Progressive Aphasia (PPA) associated with Frontotemporal Lobar Degeneration (FTLD), displaying mixed features of logopenic PPA (anomia, word-finding difficulties, circumlocutions, repetition deficits) and non-fluent/agrammatic PPA (reduced fluency, apraxia of speech and orobuccolingual apraxia), alongside behavioral changes. Structural and metabolic imaging did not help differentiating the pathologies, making it necessary to use a specific biomarker for Alzheimer's disease to exclude the diagnosis. **Conclusion:** Differentiating canonical PPA variants can be challenging, often requiring biomarker studies for accurate diagnosis. Etiological clarification is essential for patient management, treatment decisions, and prognostic counseling.

TECHNICAL AREA: CLINICAL NEUROLOGY

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Clinical and imaging findings in language-predominant dementia syndromes: a case series

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Case Presentation: Five female patients, with ages from 56 to 65 years (average 60.5 years), were followed at a cognitive neurology center. In 80% of the cases, isolated language impairment was the first symptom. Associated features included lexical access difficulty (n=3), paraphasia (n=3), echolalia (n=2), writing deficits (n=2), apraxia of speech (n=1), dysexecutive symptoms (n=1), and stuttering (n=1). MRI showed volumetric reduction predominantly in the parietal (60%) and frontotemporal (40%) lobes, mostly on the left. FDG-PET revealed hypometabolism patterns suggestive of Frontotemporal Dementia (n=2), Corticobasal Degeneration (n=2), and Alzheimer's Disease (n=1). The clinical variants identified were semantic (n=1), nonfluent (n=3), and logopenic (n=1). Speech therapy had high adherence (80%), and SSRIs were prescribed in most cases (80%). **Discussion:** Primary Progressive Aphasia (PPA) is a rare neurodegenerative condition, marked by progressive language impairment that impacts daily activities but initially spares behavior and episodic memory. While PPA is a prototype of language-dominant dementia, aphasia can also present in other conditions like Corticobasal Syndrome, which involves cortical and subcortical signs, often with asymmetric parietal lobe involvement. PPA progresses with language-dominant cognitive decline, 40% of cases develop movement disorders, and another 40% exhibit behavioral symptoms. **Final Comments:** PPA should be considered in middle-aged and elderly patients presenting with language deficits. Early diagnosis enables timely multidisciplinary care aimed at slowing cognitive decline and enhancing the quality of life for both patients and caregivers.

TECHNICAL AREA: CLINICAL NEUROLOGY

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Alzheimer's disease (frontal variant) or behavioral variant frontotemporal dementia? Insights from a challenging clinical case

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Presentation: A.M.F., a 75-year-old married male civil engineer (17 years of education). Depression. Oral cancer (2020). Non-Hodgkin lymphoma (2012). He and his wife reported cognitive changes starting 2 years ago, including forgetting he had sold his car, failing to recognize his home (especially in the late afternoon), and difficulty dressing. A neuropsychological assessment indicated executive deficits (inhibitory control and mental flexibility), memory impairment, and attentional deficits, with functional impact (FAQ - Pfeffer of 10/30). Behavioral symptoms included hyperorality, apathy, lack of empathy, and ritualistic behaviors (repeatedly organizing drawers and washing hands). He was diagnosed with Alzheimer's disease. The progression was atypically rapid; within 18 months, he became dependent for instrumental activities of daily living (IADLs, Lawton: 7/27) and partially dependent for basic activities of daily living (ADLs: 3/6). Given the rapid progression and pronounced initial behavioral symptoms, the etiological diagnosis was reconsidered, leading to two main hypotheses: frontal variant Alzheimer's disease (fvAD) and behavioral variant frontotemporal dementia (bvFTD). Key behavioral changes such as ritualistic acts and hyperorality favored bvFTD, while worsening episodic memory and hallucinations suggested fvAD. Brain MRI showed medial temporal atrophy (MTA III, right and MTA II, left) and PET-FDG indicated bilateral predominant temporoparietal involvement, sparing the caudate and putamen, supporting beta-amyloid pathology. **Discussion:** This case underscores the challenges in differentiating bvFTD from fvAD. **Conclusion:** PET-FDG can be useful guiding diagnosis when clinical presentation is ambiguous, impacting therapeutic approaches and potential genetic and prognostic evaluations.

TECHNICAL AREA: CLINICAL NEUROLOGY

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Cannabinoids: anti-tau therapy in neurodegenerative diseases?

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Background: Cannabinoid therapy is emerging as a critical approach for treating neurodegenerative diseases, especially dementia and cognitive impairments. With Alzheimer's disease in need of effective treatment, cannabinoid-based therapy appears to be viable and potentially beneficial option. **Case Presentation:** This case report aimed to examine the positive effects of cannabinoids treatment on laboratory and clinical tests in a 74-year-old white woman patient diagnosed with Alzheimer's disease. The patient presents memory lapses, difficulty retaining new information, and exhibits anxiety, agitation and mood swings that affect daily activities. The experimental therapeutic inter-

vention was carried out for 12 weeks with cannabidiol oil 60mg/day associated with curcumin and the terpene beta-caryophyllene. Clinical evaluations using Mini-Mental Stage Examination, Sunderland test and laboratory test for biomarkers (β-amyloid, Tau and Phospho-Tau) through cerebrospinal fluid samples were performed. After 12 weeks, a new measurement of biomarkers in the cerebrospinal fluid showed a 52% reduction in TAU protein, as well as improvements in the Mini-Mental State Examination and Sunderland test scores. **Conclusions:** This report presents original evidence suggesting that cannabinoids, curcumin and the terpene beta-caryophyllene may be an effective treatment for Alzheimer's disease potentially reducing the Tau protein biomarker. This finding marks a significant advancement in harnessing the therapeutic benefits of cannabinoids for neurodegeneration.

TECHNICAL AREA: CLINICAL NEUROLOGY

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A case of primary progressive apraxia of speech, prosodic subtype, and the case for non-Alzheimer's biomarkers

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Case report: A 75-year old retired teacher complained of progressive speech disturbance, which started six months earlier. Her voice had lost "rhythm", preventing her from singing or indeed reading in church, as it got "trembling and tired". On examination, fluency was reduced, speech slowed, monotone, and effortful. Neurological examination was otherwise notable only for hyperkinetic reflexes on the left arm. Cognition was normal. Detailed phonological assessment excluded aphasia and confirmed apraxia of speech, characterized by slow and constant-velocity speech, with segmentation within and between words, implying the prosodic subtype. Magnetic resonance imaging depicted disproportionate bilateral frontal and perisylvian atrophy, with mildly right predominance. Over one-year follow-up apraxia of speech worsened, remaining relatively isolated. FDG-PET demonstrated moderate bilateral frontal hypometabolism, more marked in prefrontal dorsomedial, paracentral and supplementary motor regions. Cerebrospinal fluid amyloid-beta-42 levels were reduced with normal tau levels. **Discussion:** The recognition of primary progressive apraxia of speech (PPAOS), i.e., occurring without aphasia or other neurodegenerative disease, is relatively recent. Two subtypes - phonetic and prosodic - have been described, which may only be distinguishable at early stages. They are relevant for prognosis. PPAOS often represents a tauopathy, while the prosodic subtype may evolve into the progressive supranuclear palsy phenotype. This patient has no parkinsonism or eye movement disorder thus far. **Final Comments:** This very early case of PPAOS - the first in Portuguese - may add on to the characterization of its phenotype, neural correlates and progression. In addition it illustrates the complexity of Alzheimer's biomarkers interpretation in the elderly.

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Bridging the gaps: addressing disparities in access to anti-amyloid therapies

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Background: Anti-amyloid therapies have emerged as a promising treatment for Alzheimer's disease (AD), targeting beta-amyloid plaques, a pathological hallmark of the disease. However, global access to these therapies remains unequal, raising concerns about equity and representation in real-world clinical applications. **Methods:** We analyzed global patient data from the TriNetX platform, focusing on individuals who received anti-amyloid therapies (lecanemab, donanemab, or aducanumab) between March and April 2025. Demographic and geographic characteristics were assessed by age, sex, race, ethnicity, and region. Additionally, the clinical stage at therapy initiation was evaluated to distinguish early versus late intervention. Descriptive statistics are presented as mean±standard deviation. **Results:** Among 900 patients (mean age 73±8 years, range 39-90 years), 55.11% were women. Racial analysis revealed that 90.66% were White, while Black or African American (1.55%), Asian (2.00%), and other races (5.55%) were significantly underrepresented. Regarding ethnicity, 84.11% were non-Hispanic, 3.56% were Hispanic or Latino, and 12.33% were categorized as unknown. Geographically, 98.3% (885 patients) were from the United States. Notably, 86.3% (777 patients) initiated therapy at the stage of dementia syndrome. **Conclusions:** The data highlight significant disparities in access to anti-amyloid therapies, with a pronounced bias toward White and non-Hispanic populations. The predominance of therapy initiation at the dementia stage suggests missed opportunities for early diagnosis, potentially limiting treatment benefits. These findings underscore the need for inclusive clinical trial designs, equitable public health policies, and cost-reduction strategies to ensure diverse populations benefit from disease-modifying therapies for AD.

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Inclusion matters: gender and racial disparities in anti-amyloid therapies for Alzheimer's disease

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Background: Anti-amyloid therapy represents a significant advancement in Alzheimer's disease (AD) treatment, targeting beta-amyloid plaques. However, inequities in access and representation persist, particularly

among women, who exhibit a higher prevalence of AD. This study examines gender disparities and underrepresentation in anti-amyloid therapy access and outcomes. **Methods:** Data from 933 patients treated with lecanemab, donanemab, or aducanumab were analyzed using the TriNetX platform between March and April 2025. Demographic variables, including age, sex, race, and ethnicity, were assessed to identify inequities. Patients ranged from 39 to 90 years (mean 73±8 years), with women comprising 55.11% of the cohort. **Results:** The racial distribution revealed marked underrepresentation of Black (1.55%) and Hispanic (3.56%) populations. Women were older at therapy initiation and exhibited a higher prevalence of comorbidities, such as hypertension and diabetes, which may influence treatment response. The cohort was predominantly from the United States (98.3%), further limiting generalizability. **Discussion:** The findings underscore significant disparities in access to anti-amyloid therapies and clinical trial representation. Biological factors, including estrogen's influence on AD progression, highlight the need for gender-specific research. The exclusion of underrepresented populations undermines the efficacy and equity of these therapies. Economic and infrastructural barriers disproportionately affect low-income women, exacerbating disparities, particularly in low- and middle-income countries. Inclusive clinical trial designs and equitable public health policies are essential to address these gaps and ensure that anti-amyloid therapies benefit diverse populations, especially those most affected by AD.

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Early diagnosis and risk factor management in mild cognitive impairment: impact on cognitive outcomes in a tertiary memory clinic

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Introduction: Mild Cognitive Impairment (MCI) is an intermediate stage between normal aging and dementia. Early identification and proactive management can stabilize or improve cognition, particularly in specialized care settings. **Objective:** To evaluate cognitive trajectories in patients with MCI compared to those with Alzheimer's disease or mixed dementia, emphasizing the benefits of early diagnosis and risk factor control. **Methods:** This retrospective study included 50 patients from a tertiary memory clinic in São Paulo, Brazil: MCI (n=20), Alzheimer's disease (n=20), and mixed dementia (n=10). Cognitive performance was assessed using the Mini-Mental State Examination (MMSE) at two time points, with a mean follow-up of 17 months. Statistical analyses included linear regression and the Wilcoxon rank-sum test. Results MCI patients showed a mean MMSE improvement of 2 points, while significant cognitive decline was observed in the dementia groups (p<0.001). Regression analysis identified MCI as an independent predictor of cognitive improvement (β=+2.01; p<0.001). Depressive symptoms and education level did not significantly affect outcomes. The model explained 42% of the variance in cognitive change. **Conclusion:** Early MCI diagnosis, structured follow-up, and targeted risk factor management were associated with significant cognitive improvements. These findings highlight the importance of early detection and intervention strategies in delaying or preventing dementia progression.

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Post-COVID-19 cognitive changes in low-income and low-educated adults: a longitudinal study

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Background: Cognitive impairment is frequently reported after moderate to severe COVID-19, but the long-term trajectory remains unclear in socioeconomically vulnerable populations. **Objective:** To investigate the longitudinal course of cognitive/neuropsychiatric performance two years after hospitalization due to moderate to severe COVID-19 in low-income/low-educated adults. **Methods:** Longitudinal cohort study in three public hospitals in Belo Horizonte, involving adults hospitalized for moderate to severe COVID-19. Neuropsychological and behavioral assessments occurred at 12–18 months (T1) and 24–30 months (T2) post-discharge. Global cognition was evaluated with the Mini-Mental State Examination (MMSE) and the Addenbrooke's Cognitive Examination – Revised (ACE-R). Cognitive domains were assessed using ACE-R subtests, Digit Symbol Test, and Picture-Free and Cued Selective Reminding Test. Behavioral symptoms were measured with Hospital Anxiety and Depression Scale and Impact of Event Scale – Revised. Analyses were conducted in Python using appropriate tests. Results Forty-two participants were evaluated. Median age was 63 years; 71.4% women, 85.7% non-White, and 59.5% with household income of 1–3 minimum wages. Significant improvement was observed in MMSE ($p=0.032$), ACE-R scores ($p<0.0001$), memory ($p<0.0001$), attention ($p=0.043$), and processing speed ($p=0.021$). Anxiety and depressive symptoms increased (28.6 to 40.5% and 9.5 to 16.7%, respectively), and post-traumatic stress symptoms declined (33.3 to 19.0%), all without statistical significance. **Discussion:** Cognitive performance improved after two years, suggesting a tendency of partial recovery. Behavioral symptoms, however, persisted. **Conclusion:** Cognitive improvement was observed, but neuropsychiatric symptoms underscore the need for long-term support in this population.

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Visuospatial impairment as an early marker of cognitive decline in Parkinson's disease: interim analysis of the VINDICATOR-PD study

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Background: Visuospatial dysfunction has been frequently observed in Parkinson's disease (PD), even in the absence of global cognitive impairment. Its clinical relevance as an early predictor of cognitive decline remains underexplored. **Objective:** To evaluate whether patients with PD

and visuospatial impairment perform worse on global and domain-specific cognitive tests compared to those with preserved visuospatial skills. **Methods:** This is an interim analysis of the VINDICATOR-PD study, an observational cohort of idiopathic PD patients without dementia. Participants ($n=48$) were categorized into normal or impaired visuospatial groups, based on performance in the Visual Object and Space Perception Battery (VOSP). Primary outcome was the Addenbrooke's Cognitive Examination Revised (ACE-R) total score. Secondary analyses included ACE-R visuospatial subdomain and stratification by VOSP subtypes (ventral, dorsal, both). Robust linear regression adjusted for age, sex, years of schooling, symptoms' duration, UPDRS-motor score, REM sleep behavior disorder symptoms, and apathy. **Results:** Participants with impaired VOSP ($n=18$) had significantly lower ACE-R total scores ($\beta=-4.61$; $p=0.013$). This group also showed worse visuospatial domain performance (median 14.0 vs. 15.0; $p=0.0012$). Patients with combined ventral and dorsal visual stream impairment had lower ACE-R scores ($p=0.015$) and longer symptoms' duration (median 8.0 vs. 4.0 years; $p=0.019$) than those with isolated impairments. **Conclusion:** Visuospatial dysfunction is associated with worse cognitive performance in this sample and may represent an early marker of cognitive decline. Longitudinal follow-up is ongoing to assess its prognostic.

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Correlation between functionality, sarcopenia, and cognitive function in patients with Alzheimer's disease: preliminary findings

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Background: Alzheimer's disease (AD) and sarcopenia are age-related conditions that impair cognition and physical function, respectively, affecting health and quality of life. Understanding their link is the key to enhancing patient care and treatment outcomes. **Objective:** To investigate functionality, sarcopenia, and their correlation with cognitive function in patients diagnosed with AD. **Methods:** This cross-sectional study involved patients aged 60 or older with mild cognitive impairment or mild/moderate dementia due to AD. Sarcopenia was assessed by the EWGSOP protocol, cognition function using the ACE-R and Mini-Mental State Examination (MMSE) tests, and sarcopenia was evaluated by handgrip strength, appendicular muscle mass and time up and go (TUG) test. Descriptive statistics and correlation analyses were conducted with significance at $p<0.05$. **Results:** Fourteen AD patients (71.4% women; mean age 76.7 years) were evaluated. Mean MMSE score was 19.8 ± 5.7 and the mean ACE-R score was 56.6 ± 19.5 . We observed that TUG showed a significant and negative correlation with ACE-R scores ($r=-0.543$;

$p=0.045$). The functionality parameters did not correlate with cognitive or sarcopenia variables. **Discussion:** The TUG assesses physical function and is used to screen for sarcopenia, with higher values indicating reduced functionality. The findings showed that lower TUG scores correlate with higher ACE-R scores, reflecting better cognitive function in AD patients with lower probability of sarcopenia diagnosis. **Conclusion:** These preliminary results indicate a negative correlation between TUG scores and ACE-R scores. Linking functionality and cognition in individuals with AD can enhance care and guide more effective monitoring and treatment strategies.

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Sensitivity and specificity between methods for assessing the risk of sarcopenia in patients with Alzheimer's disease: preliminary results

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Background: Studies suggest a link between Alzheimer's disease (AD) and sarcopenia, indicating that early sarcopenia treatment may help prevent age-related conditions, including AD. **Objective:** To compare the sensitivity and specificity between methods for assessing the risk of sarcopenia in patients with AD. **Methods:** This study included 14 patients aged 60 or older (71.4% women; mean age 76.7 years) with mild cognitive impairment or mild/moderate dementia due to AD. Patients underwent sarcopenia assessment considering: muscle strength assessed by handgrip strength, with a hand dynamometer; skeletal muscle mass estimated by bioimpedance; and physical performance by the Timed-Up and Go test, as defined by the European Working Group on Sarcopenia in Older People (EWGSOP). The risk of sarcopenia was assessed by instruments: SARC-F, SARC-Calf and SARC-Global. Statistical analysis included the determination of sensitivity and specificity between screening tests, considering the EWGSOP protocol as the gold standard instrument. **Results:** The mean MMSE score was 19.8 ± 5.7 . SARC-F showed a sensitivity of 22.2% and specificity of 100%. SARC-Calf sensitivity was 42.8% and specificity of 100%. The sensitivity of SARC-Global was 85.7% and specificity of 100%. **Discussion:** These data suggest that SARC-F and SARC-Calf presented lower sensitivity to detect sarcopenic AD patients, although they were able to detect 100% of the non-sarcopenic patients. In contrast, SARC-Global showed high specificity and proved to be more indicated in assessing the risk of sarcopenia in this group. **Conclusion:** SARC-Global is a better instrument to detect sarcopenic AD patients compared to SARC-F and SARC-Calf.

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Association sarcopenia and nutritional status in Alzheimer's patients attended at a specialized outpatient clinic: preliminary findings

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Background: Alzheimer's Disease (AD) is a progressive age-related condition, where poor nutrition and sarcopenia can worsen outcomes. Understanding this link is crucial for better patient care. **Objective:** To investigate the association between nutritional status, sarcopenia, and functionality in older individuals with AD. **Methods:** This cross-sectional study evaluated 14 older adults (mean age 76.7; 71.4% women) with mild cognitive impairment or mild/moderate dementia due to AD. Assessments included the Mini Nutritional Assessment (MNA), Mini-Mental State Examination (MMSE), and the EWGSOP criteria for sarcopenia, using handgrip strength (HGS), skeletal muscle mass via bioimpedance, and the Timed-Up and Go (TUG) test for physical performance. Statistical analyses involved descriptive measures and correlation tests, with a significance level set at $p < 0.050$. **Results:** The mean nutritional status was considered normal ($MNA = 24.5 \pm 2.7$ of 30 points), although close to the risk of malnutrition. MMSE mean score was 19.8 ± 5.7 . Six patients had sarcopenia, one had probable sarcopenia and seven did not present sarcopenia. A positive correlation was observed between HGS and MNA ($r = 0.599$; $p = 0.023$), and a negative correlation between the TUG test and MNA ($r = -0.588$; $p = 0.027$). Correlations between the MMSE and other variables were not significant. **Discussion:** These preliminary findings indicate that the worst nutritional status is correlated to low muscle strength and more time to complete the TUG test, which suggests compromised functional mobility. **Conclusion:** The combined use of MNA, HGS, and TUG effectively identifies nutritional and functional risks, highlighting the need for early intervention in AD patients.

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Assessment of dietary intake and body composition of Universidade Federal de Minas Gerais participants in the LatAm-FINGERS study

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Introduction: The Latin American Initiative for Lifestyle Intervention to Prevent Cognitive Decline (LatAm-FINGERS) is a randomized, multicenter, non-pharmacological clinical trial for dementia prevention in Latin America. This study investigates the influence of a multidomain intervention on cognitive impairment in older adults with associated cardiovascular risk factors. **Objective:** To assess body composition and food consumption of participants followed at the UFMG center, comparing them to reference values for macro and micronutrients. **Methods:** Participants in the systematic intervention group underwent initial assessments during individualized nutritional consultations. Measurements included anthropometric parameters. Dietary intake was evaluated through a 24-hour recall and quantified using the Brazilian food composition table (TACO). **Results:** Twenty-two participants were evaluated; mean age: 69.4±4.9 years. Average weight 71.4±17.4 kg; BMI 27.3±5.4 kg/m² (ref. 22–27); neck circumference 36.9±4.0 cm (ref. 34–37); calf circumference 37.3±4.2 cm (ref. >34); waist-to-hip ratio 0.9±0.1 (ref. 0.85–0.90); and waist-to-height ratio 0.6±0.1 (ref. <0.5). The average carbohydrates intake was 176.2±64.8 g/d (ref. 168–243); proteins 1.08±0.21 g/kg/w (ref. 1.2–1.5); lipids 60.9±24.6 g/d (ref. 33–58); fiber 18.0±9.0 g/d (ref. 25–30); calcium 575.0±267.7 mg/d (ref. 1,200); vitamin B12 1.9±3.0 µg/d (ref. >2.4); and vitamin D 1.6±2.1 µg/d (ref. >20). **Discussion:** The anthropometric measurements indicate higher adiposity and cardiovascular risk in this population. Besides, it was observed an inadequate consumption of macro and micronutrients. **Conclusion:** The findings reinforce the need for personalized nutritional interventions to improve the intake of essential nutrients and reduce metabolic outcomes in this population with cardiovascular risk.

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Use of the figure test of the brief battery of cognitive screening in hospitalized chronic alcoholics

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Background: Chronic alcohol abuse is strongly associated with cognitive impairment, affecting memory, attention, language, and executive functions due to neurotoxic effects on the central nervous system. Cognitive assessment is essential in this population to support diagnosis and therapeutic planning. **Objective:** This study evaluated cognitive performance in hospitalized chronic alcoholic patients using the Brief Cognitive Screening Battery (BCSB). **Methods:** This is a descriptive, cross-sectional study involving chronic alcoholic patients admitted to a referral hospital in Pernambuco, Brazil. The BCSB was applied, assessing the following domains: identification, incidental memory, immediate memory, learning, delayed recall, recognition, semantic verbal fluency, and visuospatial skills (clock

drawing test), in addition to the total score. **Results:** So far, nine patients have been examined. Cognitive impairment varied widely. Three patients showed severe deficits, three preserved performance, and others moderate. Memory, verbal fluency, and visuospatial abilities were most affected. **Discussion:** The heterogeneous cognitive impact of chronic alcoholism, from mild to severe, highlights alcohol's neurotoxic effects on executive functions and memory. Consistent deficits in memory, verbal fluency, and visuospatial abilities align with known brain damage. The BCSB effectively identified these impairments, validating its utility for early detection and guiding individualized cognitive rehabilitation. Comprehensive neuropsychological evaluations are crucial for personalized treatment. **Conclusion:** The BCSB is a sensitive tool for detecting cognitive deficits in hospitalized chronic alcoholic patients, supporting rehabilitation. Continued data collection is underway to expand the sample size, which is fundamental for more robust comparisons and to confirm these findings, allowing for better exploration of correlations with alcohol abuse severity.

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Classification of patients with dementia and healthy older adults using electroencephalogram data: a comparison between decision tree models

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Background: Population aging has led to an increase in chronic degenerative diseases. Therefore, it is necessary to identify brain activity patterns associated with dementia, contributing to early diagnosis. **Objective:** To compare the performance of machine learning (ML) models using a database containing information from patients with dementia and healthy older adults through electroencephalogram (EEG) data. **Methods:** Various ML models were tested using the GeronGnosis EEG database (including mild cognitive impairment - MCI, Alzheimer's disease - AD, frontotemporal dementia - FTD, and healthy controls). After extracting different signal features, the Random Tree and Random Forest models were tested, with various hyperparameter adjustments. **Results:** The Random Forest model with 500 trees demonstrated the best classification performance, achieving an accuracy of 98%±0.51, sensitivity of 96%±0.01, and specificity of 98%±0.01. These are relevant and promising results for the identification and differentiation of the analyzed clinical conditions. **Discussion:** The proposed method was able to correctly classify patients with dementia and healthy controls, achieving excellent overall performance across metrics. Decision tree-based models were effective in classification, indicating potential applicability in cognitive impairment screening. **Conclusion:** Decision tree-based ML models are effective in distinguishing between patients with dementia and healthy older adults, representing a promising tool for screening cognitive deficits in at-risk populations.

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Cognitive domain performance on the CCAS Scale: subitem-level analysis in patients with acquired cerebellar lesions

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Background: The Cerebellar Cognitive Affective Syndrome (CCAS) includes impairments in executive function, language, visuospatial abilities, and affect. The CCAS Scale was developed for structured screening of these domains in patients with cerebellar lesions, but domain-specific patterns remain underexplored. **Objective:** To identify which cognitive domains are most frequently impaired in patients with acquired cerebellar lesions, based on CCAS Scale subtest performance. **Methods:** Eighteen adults with non-genetic cerebellar lesions (vascular, tumoral, or traumatic) from a tertiary neurology service in Brazil were assessed using the Brazilian Portuguese version of the CCAS Scale. Subtests were grouped by domain. Descriptive analyses and associations with lesion characteristics were explored. **Results:** Subtests most frequently failed included verbal similarities (83.3%), backward digit span (61.1%), forward digit span (50%), and word recall (50%), highlighting executive dysfunction, especially in working memory. Fluency and category-switching were also commonly impaired. Visuospatial performance (cube copy) was least affected (11.1%). Right-sided lesions were more associated with language-related deficits. The total CCAS score moderately correlated with years of education ($p=0.496$, $p\leq 0.05$). **Discussion:** Executive function was the most affected domain, especially in patients with posterior and right-sided lesions. Visuospatial deficits were less prominent. **Conclusion:** Subitem-level analysis of the CCAS Scale suggests that executive functions, with emphasis on working memory, are particularly vulnerable in cerebellar damage. These findings support the use of the scale in Clinical Neurology to enhance cognitive screening and to guide individualized rehabilitation strategies in routine care.

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Cognitive screening through mini-examination of mental state and semantic verbal fluency in hospitalized elderly without previous complaints

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Background: Hospitalization in elderly patients can lead not only to cognitive decline but also to depression. **Objective:** To evaluate the cognition of hospitalized geriatric patients and to correlate the Mini-Mental State Examination (MMSE) and the semantic Verbal Fluency Test (VFT), taking into account the effects of education and age. **Methods:** Cross-sectional study conducted with patients aged 60 years or older and hospital stay longer than 24 hours. The MMSE, the semantic VFT and the 15-item Geriatric Depression Scale (GDS-15)

were applied. Spearman's test was used to assess the degree of correlation between the variables. **Results:** The sample consisted of 100 elderly individuals, 50% of whom were women, the mean age was 69.6 years, 47% had more than 4 years of schooling and 31% were in the clinical-surgical ward. The mean scores of the MMSE, VFT and GDS-15 were 21.2, 12.3 and 4.3, respectively. In the distinction between the wards analyzed, only the VFT screening instrument yielded a statistically significant result ($p=0.014$). A moderate correlation was observed between MMSE and education ($r=0.590$) and a low correlation between VFT and age ($r=-0.314$). The MMSE and VFT presented a low correlation with each other ($r=0.429$). **Discussion:** Education and age were factors that significantly influenced the cognitive performance of the elderly, which is in line with data in the literature. **Conclusion:** Cognitive screening in the hospital setting facilitates the early detection of potential dementia.

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Use of functional capacity data for classifying patients with dementia and healthy elderly through machine learning models

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Background: The rising incidence of dementia among the elderly compels the development of accessible and reliable screening methods for this condition. **Objective:** To compare the performance of machine learning (ML) models in classifying functional capacity data from both dementia patients and healthy elderly individuals evaluated through the Pfeffer Questionnaire and the Katz Index. **Methods:** A variety of ML models were assessed using functional capacity data from patients with MCI, AD, FTD, and healthy control subjects. After feature extraction from the data and hyperparameter tuning, the BayesNet, Random Forest, and SVM models were tested. **Results:** The analysis included data from 75 individuals (47 patients and 28 control subjects) aged 60 to 95 years, comprising 19% men and 81% women. The Random Forest (with 200 trees) and BayesNet models performed best, achieving an accuracy of 76.89 ± 11.98 and 74.81 ± 12.87 , specificity of 96.93 ± 0.07 and 90.17 ± 0.11 , and an area under the ROC curve of 84.75 ± 0.11 and 88.43 ± 0.14 , respectively, in the training data. These performance metrics were sustained in the test group. **Discussion:** The proposed method effectively classified patients with dementia and healthy controls through conditional probability and decision tree models. **Conclusion:** The machine learning models presented here offer a promising approach to screening cognitive deficits in the elderly using functional capacity assessments. Larger sample sizes are needed in future studies for more robust comparisons.

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Prevalence of chronic diseases and polypharmacy among older adults in latin america: the LatAm-FINGERS study

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Older adults often experience polypharmacy and comorbidities, which may lead to adverse clinical outcomes, such as cognitive impairment, falls, and increased mortality. Part of these outcomes is associated with the anticholinergic properties or adverse events of medications. To investigate the frequency of self-reported diseases, use of medications with anticholinergic properties, and the overall medication use pattern in a multicenter cohort of older adults from LatAm-FINGERS. Descriptive study of the complete baseline clinical data from participants aged 60–77 years without dementia from 12 Latin American countries. The anticholinergic burden was assessed using the Anticholinergic Cognitive Burden (ACB) scale. We included 1,243 participants (73.9% women, mean age=67.5 years). Hypertension (44.6%) and dyslipidemia (42.8%) were the most frequently reported diseases. The average number of medications was 2.58, from 0.31 (Bolivia) to 4.2 (Puerto Rico). Polypharmacy was observed in 18.4% of participants. Commonly reported medications included statins (including atorvastatin, rosuvastatin, and simvastatin; 31.2%), losartan (19.0%), and levothyroxine (17.4%). More frequently prescribed medications with anticholinergic properties were metformin (14.6%) and omeprazole (4.3%). Most frequent higher-ACB drugs were paroxetine (0.7%) and amitriptyline (0.4%). Our findings align with prior literature on the high prevalence of cardiovascular diseases and dyslipidemia in older adults, both important midlife risk factors for dementia. Even low ACB scores may contribute to cognitive decline, warranting clinical attention. There is a significant frequency of polypharmacy, including anticholinergic medications, as well as important clinical comorbidities in the studied population.

TECHNICAL AREA: CLINICAL NEUROLOGY

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Mild behavioral impairment checklist: assessment in individuals with subjective and mild cognitive decline from the BRASCODE cohort

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Background: Mild Behavioral Impairment (MBI) is defined as the later-life emergence of sustained neuropsychiatric symptoms not better explained by other psychiatric disorders. In Brazil, research on MBI in cognitively unimpaired older adults remains limited. This study aims to evaluate the Mild Behavioral Impairment Checklist (MBI-C) in individuals with Subjective Cognitive Decline (SCD) and Mild Cognitive Impairment (MCI) from the BRASCODE cohort in southern Brazil. **Methods:** This analysis included cognitively unimpaired individuals (≥ 65 years) with cognitive complaints from the BRASCODE cohort. Exclusion criteria included uncontrolled medical or psychiatric conditions, cerebrovascular disease, use of psychotropic medications. The applied scales were: MBI-C and Memory Complaint Scale (MCS) (participant and informant); Neuropsychiatric Inventory–Questionnaire (NPI-Q) (informant); Geriatric Depression Scale (GDS), Geriatric Anxiety Inventory (GAI), Mini-Mental State Examination (MMSE); and Clinical Dementia Rating (CDR) scale. In this analysis, we included the 12-month follow-up data. **Results:** A total of 113 participants completed the 12-month follow-up, of whom 13 were diagnosed with MCI. Median age was 71 years (IQR: 68–74), 74.3% were female, and median education was 15 years (IQR: 11–18). MBI-C participant scores correlated positively with anxiety (GAI, $\rho=0.551$, $p<0.0001$) and depressive symptoms (GDS, $\rho=0.635$, $p<0.0001$). MBI-C informant scores were strongly correlated with SCD-informant ($\rho=0.749$, $p<0.0001$) and NPI-Q ($\rho=0.749$, $p<0.0001$). **Discussion:** The correlations suggest that MBI-C identifies behavioral changes, especially anxiety and depression, and is associated with neuropsychiatric symptoms and cognitive decline. **Conclusion:** The MBI-C was useful in detecting neuropsychiatric symptoms in individuals with SCD and MCI.

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The role of anticoagulants and antiplatelet agents in reducing the risk of vascular dementia

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Background: Vascular dementia (VD), the second most common dementia in Brazil, is linked to cerebrovascular events. Oral anticoagulants prevent strokes in atrial fibrillation, potentially reducing VD risk. Antiplatelets' direct VD impact is unclear. Understanding these agents' roles is crucial due to population aging. **Objective:** To analyze evidence on anticoagulants and antiplatelet agents in reducing vascular dementia risk. **Methods:** An integrative review used Whittemore & Knaf's steps. The question, "Do anticoagulants/antiplatelets reduce vascular dementia

risk?" guided searches in PubMed, Cochrane, and VHL (2020-2025; English/Portuguese/Spanish). Six articles (clinical trials, systematic reviews) on anticoagulant/antiplatelet use and VD outcomes were analyzed. **Results:** Clopidogrel reduced dementia risk in small vessel disease (aOR 0.30); aspirin/cilostazol showed no significant impact. No robust evidence supported antiplatelet cognitive benefits in SVD without dementia. NOACs were linked to lower new dementia risk in atrial fibrillation. Critical gaps exist in RCTs for both agent types in VD prevention. Bleeding risks were noted. **Discussion:** Observational data suggest anticoagulants/antiplatelets may reduce VD risk in high-risk individuals. Clopidogrel's effect may extend beyond platelet inhibition. Findings are inconsistent; reviews note study limitations. Some agents show neuroprotective potential, but trial results are mixed. Bleeding risk, especially with CAA, requires caution. **Conclusion:** NOACs show promise in reducing dementia risk in specific groups. Clopidogrel shows some benefit. Significant gaps in robust RCTs remain. Future research needs well-designed studies on efficacy, safety, and optimal use for VD prevention.

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Impact of Anti-Beta-Amyloid monoclonal antibodies on the progression of Alzheimer's disease: An integrative literature review

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Background: Alzheimer's disease (AD) is a progressive neurodegenerative disorder driven by A β plaque deposition. Anti-A β monoclonal antibodies (aducanumab, lecanemab, donanemab) aim to reduce amyloid load and slow progression. **Objective:** To analyze anti-A β monoclonal antibody impact on AD progression, synthesizing evidence on cognitive outcomes, quality of life, and safety. **Methodology:** This is an integrative literature review, conducted between April and May 2025, using the PubMed database. Six studies published between 2020 and 2025 in English and Portuguese were selected. The analysis process followed the classic methodological steps of an integrative review and the articles were assessed by the Critical Appraisal Skills Program (CASP) and classified as high quality. **Results:** Monoclonal antibodies showed a modest but significant reduction in cognitive decline in patients with early AD, as well as a reduction in brain amyloid load. Lecanemab stood out for its more robust effect. However, adverse events, especially amyloid-related imaging abnormalities (ARIA), were reported, affecting approximately 14% of patients. **Discussion:** Anti-A β antibodies validate the amyloid hypothesis, offering modest clinical benefits and amyloid reduction. Lecanemab shows promise. ARIA, especially in ApoE4 carriers, demands careful monitoring. Long-term benefit sustainability and efficacy in diverse populations are uncertain. A β clearance alone may be insufficient. Future research needs long-term trials, combination therapies, and predictive biomarkers. Lecanemab's approval is significant, but cost and safety concerns persist. **Conclusion:** Anti-A β therapies advance AD treatment, providing modest benefits with notable risks. Combined strategies and long-term studies are essential to confirm efficacy and safety.

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Genetic Contributions to Alzheimer's disease and frontotemporal dementia in admixed Latin American populations: the Brazilian cohort

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Background: Brazil's diverse genetic makeup, shaped by centuries of admixture, presents a unique opportunity to study Alzheimer's disease dementia (AD) and frontotemporal dementia (FTD). **Methods:** The Multi-Partner Consortium to Expand Dementia Research in Latin America (ReDLat) recruited 2,162 participants with AD, FTD, and healthy controls from six Latin American countries (Argentina, Brazil, Chile, Colombia, Mexico, and Peru). To identify genes associated with autosomal dominant, autosomal recessive, or X-linked forms of adult-onset dementia, we searched the Online Mendelian Inheritance in Man database and analyzed pedigree information. **Results:** Global ancestry analysis of the entire Latin American ReDLat cohort revealed a predominant mix of American, African, and European ancestries. Uniquely, Brazil displayed an additional East Asian component accurately reflecting the historical admixture patterns from this region. The Brazilian cohort included forty-two patients with AD, 88 normal controls and 99 patients with FTD were included. Pathogenic variants were found in C9ORF72, GRN (p. Q257pfs*27), MAPT (p. L266V), and TARDBP (p.I383V). A kindred with an MAPT pathogenic variant will be presented. **Conclusions:** This study provides the first snapshot of genetic contributors to AD and FTD in Brazil. With the whole Latin American cohort, it will be critical to evaluate the generalizability of genetic risk factors for AD and FTD across diverse ancestral backgrounds, considering distinct social determinants of health and accounting for modifiable risk factors that may influence disease risk and resilience across different cultures.

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Frequency of possible limbic-predominant age-related TDP-43 encephalopathy (LATE) among Alzheimer's disease patients in a tertiary cognitive center

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Background: The increasing recognition of the high prevalence of Limbic-predominant Age-related TDP-43 Encephalopathy (LATE) as a differential diagnosis of Alzheimer's disease (AD) calls for a re-evaluation of clinically compatible cases in routine practice. **Objective:** To assess the frequency of possible LATE in a Brazilian outpatient tertiary public cognitive neurology center. **Methods:** We performed a retrospective analysis of medical records of clinically diagnosed AD patients using the

criteria for possible LATE as proposed by Wolk et al., 2024. **Results:** We identified 4 cases that fulfilled the possible LATE criteria in 95 AD-diagnosed patients (4.2% of the sample size). The median (interquartile range, IQR) age of possible LATE cases (80, 78.25–82.25) was not different from AD cases (77, 71–80). Similarly, the frequency of women, LATE: 2/4 (50%) vs AD: 53/91 (58.2%), and years of formal education (LATE: 7.5, 5.5–8; AD: 5, 3–8), were not different between the groups. **Discussion:** Our results suggest that a subset of patients with AD might instead present with underlying LATE pathology or AD/LATE co-pathology. Reasons for a low frequency of possible LATE cases include the referral of many patients with advanced dementia, which limits accurate etiological differentiation; and incomplete assessments or follow-up loss in some entries, which hinders the application of the minimal diagnostic criteria for LATE. **Conclusion:** The frequency of possible LATE is low in our tertiary outpatient setting, compared to what is expected from the literature.

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Clinical characterization of a Brazilian cohort of patients with Lewy body dementia: a foundational step towards deeper understanding

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Lewy Body Dementia (LBD) is often misdiagnosed due to symptom overlap, fragmented assessments, and limited biomarker access. This misdiagnosis severely impacts treatment and prognosis, as LBD requires distinct management. This study aimed to characterize LBD in a Brazilian cohort using a systematized clinical evaluation, crucial for improving diagnostic accuracy. This cross-sectional study, conducted at Hospital das Clínicas FMRP, assessed 120 patients with cognitive complaints or dementia syndromes. The comprehensive evaluation included detailed anamnesis, motor assessment (MDS-UPDRS), evaluation of visual hallucinations, cognitive fluctuations (Mayo Fluctuations Scale), RBD symptoms, neuroleptic sensitivity, and dysautonomia. Neuropsychological testing, MRI, PET-FDG, and EEG were also utilized. LBD diagnosis followed established clinical criteria. Of 120 patients, 27 (22.5%) were diagnosed with LBD. Importantly, 9 (33.3%) of these were initially misclassified as AD or other dementias, highlighting the protocol's value. The LBD cohort had a mean age of 76.16 years, 5.24 years of education, and 60% male predominance. Memory complaints were the most frequent initial symptom (42.3%), followed by executive dysfunction (30.8%). Significant parkinsonism was observed (UPDRS mean 44.92). Anosmia was reported in 50% of cases. Delayed neuroimaging is the main challenge. Our findings indicate that LBD is significantly underdiagnosed in routine Brazilian clinical practice. Parkinsonism and executive dysfunction are key markers, even with predominant memory complaints. This initial characterization will inform future research and improve patient care in Brazil.

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Pharmacological interventions in patients with posterior cortical atrophy: a systematic review

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Background: Posterior Cortical Atrophy (PCA) is a rare dementia syndrome, typically caused by an atypical variant of Alzheimer's disease, and is characterized by progressive visuospatial decline. Cholinesterase inhibitors (AChEIs), used in typical Alzheimer's disease, are sometimes prescribed for PCA, but evidence supporting their efficacy in this population remains limited. **Objective:** To summarize current clinical evidence on pharmacological treatments for PCA. **Methods:** A systematic review was conducted according to PRISMA guidelines. Searches were performed in PubMed, EMBASE, Scopus, and Web of Science for studies published between January 2010 and May 2025. Keywords included "posterior cortical atrophy," "pharmacological treatment," and related terms. Studies were excluded if they addressed other causes of cognitive decline, focused solely on non-pharmacological approaches, were case reports, preclinical studies, or lacked full-text access. **Results:** Only one study met inclusion criteria: a randomized, double-blind, placebo-controlled trial (Ridha et al., 2018) involving 18 PCA patients treated with donepezil. MMSE scores improved significantly at six weeks (difference of 2.5 points; $p < 0.05$), but not at 12 weeks. No improvements were observed in visuospatial tests. Adverse effects were frequent, especially gastrointestinal symptoms (72%) and vivid dreams (44%). **Discussion:** The study highlighted recruitment challenges and a high rate of side effects, which may have affected blinding. The small sample limits the generalizability of findings. No other neuropsychological outcomes improved. **Conclusion:** Evidence on pharmacological treatment for PCA is scarce. High-quality clinical trials are needed to guide therapeutic strategies.

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Accuracy of the Brazilian Portuguese version of the Mini-Linguistic State Examination

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Background: The Mini-Linguistic State Examination (MLSE) is a screening tool for Primary Progressive Aphasia (PPA) and its variants. Initially developed in English and Italian, it has since been adapted to Spanish. The MLSE includes tasks like naming, comprehension, repetition, reading, writing, and description, with errors scored across five domains (total: 100 points): motor speech (30), phonology (30), semantics (20), syntax (10), and working memory (10). Semantic variant PPA (svPPA) has impaired semantics. Non-fluent variant PPA (nfvPPA) has impaired motor speech and/or syntax. Logopenic variant PPA (lvPPA) has impaired

phonology and working memory. **Objective:** Translate, culturally adapt, standardize, and validate the English MLSE for Brazilian Portuguese. **Methods:** The MLSE was translated and culturally adapted: changes of pictures followed criteria of familiarity, category, and visual complexity. Controls and PPA patients were assessed with the Brazilian MLSE, ACE-R, and MMSE. A machine learning algorithm (MATLAB®) was used to differentiate groups based on MLSE performance. **Results:** Controls (n=60): age 61.35 (± 11.14); schooling 15.01 (± 4.35); MLSE 97.6; MMSE 28.59 (± 1.35); ACE-R 89.25 (± 6.95). PPA (n=22): age 66.04 (± 7.62); schooling 14.54 (± 3.92); MLSE score of 74.23 (± 11.77); MMSE 20.14 (± 7.09); ACE-R 58.89 (± 19.17). MLSE had ROC accuracy of 93.5%. Decision-making tree profiles: controls (semantics >16.5 and motor speech >28.5); nfvPPA (semantics >16.5 and motor speech <28.5); svPPA and lvPPA had semantics <16.5, with lvPPA presenting phonology <28.5 while svPPA had phonology >28.5. **Conclusion:** the Brazilian Portuguese MLSE is an accurate tool for PPA screening.

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A data-driven ensemble method for feature selection in the differentiation of AD and bvFTD

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Background: High-dimensional clinical data challenges machine learning models, increasing overfitting risk and hindering clinical translation, particularly when differentiating early Alzheimer's disease (AD) from behavioral variant frontotemporal dementia (bvFTD). **Objective:** To develop a robust feature selection pipeline to identify a parsimonious subset of clinical variables for distinguishing AD from bvFTD in mild dementia (CDR=1). **Methods:** From 112 clinical psychiatric tests variables in the NACC UDS for 10,452 participants (9,691 AD, 761 bvFTD), we applied an ensemble pipeline combining four ranking methods (SHAP, RFE, Lasso, Mutual Information) within a cross-validated framework. A sequential feature-addition experiment then identified the smallest feature set achieving 95% of the maximum observed average precision. **Results:** The method successfully identified an optimal and parsimonious subset of 29 features. This selection was justified by a clear performance plateau in the sequential addition experiment, balancing high discriminative power with model efficiency. **Conclusion:** Our data-driven strategy effectively reduced the initial variables to a concise 29-item profile without significant loss of performance. This approach provides a clinically translatable feature set that reduces assessment burden, forming a robust foundation for accurate diagnostic models in this complex differential diagnosis.

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An interpretable XGBoost model for differentiating early Alzheimer's disease and bvFTD

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Background: Differentiating early Alzheimer's disease (AD) from beha-

vioral variant frontotemporal dementia (bvFTD) is clinically challenging due to overlapping symptoms. While machine learning (ML) can improve diagnostic accuracy, model performance and interpretability are crucial for clinical adoption. **Objective:** To develop, optimize, and interpret a high-performing ML model to differentiate AD from bvFTD in mild dementia (CDR=1) using data-driven clinical variables. **Methods:** Using a 29-variable set derived from data of 10,452 NACC participants, we trained eight ML classifiers. Models were evaluated with and without hyperparameter optimization (HPO) via nested cross-validation. The optimal model, selected based on the area under the precision-recall curve (PR AUC), was interpreted using SHapley Additive exPlanations (SHAP). **Results:** The optimized XGBoost model demonstrated superior discrimination (PR AUC 0.787; ROC AUC 0.957). SHAP analysis revealed lower age and higher behavioral impairment scores as the top predictors for bvFTD, while memory and orientation deficits were key predictors for AD. **Conclusion:** An optimized and interpretable XGBoost model accurately differentiates these syndromes using only clinical data. The model's decision-making process, clarified by SHAP, aligns with established clinical knowledge, highlighting its potential as a transparent data-driven tool to support diagnostic decisions.

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Clinical and epidemiological baseline characterization of dementia cases in a specialized center in Recife

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Background: Dementia is rising globally, particularly in low- and middle-income countries. In Brazil, data remain limited and concentrated in more developed Southeastern regions. Recife, a large Northeastern city with 1.5 million inhabitants, faces deep social disparities and widespread low literacy. **Objectives:** To describe the demographic, clinical, and social characteristics of patients followed at a dementia outpatient clinic in a tertiary center in Recife, Brazil. **Method:** A descriptive, single-center study based on medical records from patients evaluated between 2023 and 2024 at the dementia outpatient clinic of the Neurology and Neurosurgery Unit. **Results:** A total of 256 consultations were conducted for 107 patients. The mean age at follow-up onset was 63.9 years; 53.3% were women, 49.5% mixed race. Most (73%) were from the coastal region, 2.8% from the hinterland (sertão). Average education was 8.28 years (SD=5.16). Alzheimer's disease was the most common diagnosis (28%), followed by frontotemporal dementia (5.6%), Lewy body disease (2.8%), and pure vascular dementia (2.8%). Advanced dementia was observed in 28.6%. Hypertension was present in 61.2%, diabetes in 31.5%, smoking in 14%, and alcoholism in 15.1%. Depressive disorder affected 41.2%, and anxiety disorder 36%. **Discussion:** The predominance of advanced stages and non-Alzheimer's dementias suggests potential delays in diagnosis and barriers to early specialized care, possibly reflecting social vulnerabilities and referral patterns. **Conclusion:** Despite sample limitations, this study provides valuable insight into dementia profiles in Northeast Brazil and highlights the need for earlier access to specialized services.

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AI-based emotion recognition from EEG and voice for personalized therapy

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The world is facing significant changes in the demographic profile. A related phenomenon is the rise in the elderly population, which is associated with the growing incidence of age-related diseases such as dementia. Studies indicate that music therapy has the potential to slow the progression of dementia through sound stimuli, promoting the activation of memory-related brain areas mediated by emotions. However, the effectiveness of music therapy is closely related to identifying and evoking correct emotions. Recent advances in artificial intelligence and signal processing have eased the application of deep artificial neural networks to solve complex problems. This study focused on building a model to recognize emotions in elderly. Our approach employs the Wavelet Transform to convert signals into images, used as input for a hybrid architecture based on pretrained convolutional network for feature extraction and a shallow classifier. To assess the method's ability to interpret emotional data from elder individuals, especially those with dementia, we built a multimodal database, including electroencephalography, voice, facial expression, demographic data, and diagnosis from 39 participants, being 21 healthy controls and 18 with mild to moderate dementia. Computational experiments were conducted for the automatic recognition of positive, negative, and neutral emotional states. The proposed method achieved accuracy up to 73% when using a LeNet-Random Forest architecture. This model will serve as the core for a human-machine interface for biofeedback to assess emotions evoked by specific therapy stimuli. This technology can be employed to support the personalization of music therapy and other therapeutic approaches. Dementia, AI

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Distinguishing Alzheimer disease and frontotemporal dementia: insights from a large Brazilian study

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Background: Differentiating Alzheimer's disease (AD) and frontotemporal dementia (FTD) is challenging due to overlapping clinical features. **Objective:** To investigate key clinical, neuropsychological, and behavioural discriminators in a large Brazilian cohort. **Methods:** This descriptive study included

245 cognitively healthy controls (69% women; age=58.5[48.0;68.8]years; education=15.0[11.0;17.0]years; MMSE=29[28.0;30.0]), 58 AD patients (56.9% women; age=70.0[62.8;75.5]years; education=11.0[4.0;16.0]years; age of onset=65.0[55.0;70.3]years; MMSE=20.0[17.0;24.5]), and 45 FTD patients (61.4% men; age=66.0[60.3;70.8]years; education=12.5[5.0;16.0]years; age of onset=62.0[53.0;66.0]years; MMSE=23[18.0;27.0]). Participants were recruited from two Brazilian centers (USP-SP/UFGM). Assessments included demographics, clinical evaluation, neuropsychiatric symptoms, neuropsychological testing, and genetic analysis. Statistical analyses used non-parametric tests. **Results:** FTD cases included: bvFTD (68.9%), svPPA (17.8%), nfvPPA (6.7%), PPA-non-specified (4.4%), and PSP(2.2%). At evaluation, AD exhibited greater memory impairment ($p<0.001$), while FTD had higher language ($p=0.004$) and behavioural ($p<0.001$) deficits. Initial symptoms differed significantly between groups ($p<0.001$), with memory complaints predominating in AD and behavioural/language symptoms in FTD. Total NPI scores were higher in FTD than AD ($p<0.001$), driven by disinhibition ($p<0.001$), motor disturbance ($p<0.001$), apathy ($p=0.013$), irritability ($p=0.004$), and anxiety ($p=0.007$). Both groups scored worse than controls on all NPI subscales and most neuropsychological tests. AD performed worse than FTD on delayed visuospatial recall (Benson Figure Delayed, $p<0.001$). Two familial FTD cases carried the TARDBP I383V pathogenic mutation. **Discussion and Conclusion:** In a large Brazilian AD/FTD cohort, we identified that initial symptoms, neuropsychiatric profiles, and visuospatial memory patterns are key to distinguishing AD from FTD. Two unrelated carriers of the same TARDBP mutation (p.I383V) suggest a rare "de novo" mutation or a founder effect.

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Dementia diagnosis inequality between high-income countries and Brazil: how challenging the diagnosis can be

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Background: Early diagnosis is crucial for effective dementia management, but low- or middle-income countries, including Brazil, face disparities in diagnostic resources. **Objectives:** To assess diagnostic tools in specialized dementia care centers across Brazil. **Methods:** An online survey was conducted in July 2024, targeting medical specialists working in Brazilian dementia care centers. The survey collected data on infrastructure, cognitive tests, biomarker availability, and patients' sociodemographic profiles. **Results:** Dementia specialists from 24 outpatient clinics across 10 states participated in this study. Most settings (87%) are part of the Public Health System. While patients' age, sex, and education were consistently recorded, ethnicity (67%) and socioeconomic status (27%) were often omitted. Cognitive assessments rely heavily on the Mini-Mental State Examination (100%), Montreal Cognitive Assessment (66.7%), and Brief Cognitive Screening Battery (54.2%). Functional assessments rely on the Pfeffer Questionnaire (33.3%) and KATZ Scale (20.8%). Neuropsychiatric evaluations occur in 58.3%, mainly via the Geriatric Depression Scale. Only 29.2% assess dementia staging, and Clinical Dementia Rating is the primary tool. Neuropsychological and functional assessments were unavailable in 33.3 and 66.7% of clinics, respectively. CSF biomarkers were available in 29.2%; brain MRI was universal, while PET-FDG and PET-amyloid were rare. **Discussion:** Significant gaps and heterogeneity exist in dementia diagnostic practices in public facilities in Brazil. The predominance of cognitive screening over comprehensive evaluations underscores the need for standardized protocols. **Conclusion:** Expanding access to advanced tools and standardizing assessments are essential for improving dementia care in Brazil.

TECHNICAL AREA: CLINICAL NEUROLOGY

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Risk factors for behavioral variant frontotemporal dementia: insights from the NACC database

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Background: Modifiable risk factors for dementia are associated with half of dementia cases. However, risk factors for behavioral variant frontotemporal dementia are widely unexplored. **Objective:** To investigate whether risk factors associated with AD are also linked to bvFTD diagnosis in a large multicenter real-world cohort. **Methods:** We analyzed data from the National Alzheimer's

Coordinating Center (NACC) from 2005 to 2023. Participants clinically diagnosed with bvFTD were included and compared with other diagnoses (Alzheimer's disease, Mild Cognitive Impairment and Lewy Bodies Dementia). Logistic regression was performed to examine associations between bvFTD and 18 risk factors, including cardiovascular, psychiatric, metabolic, and demographic variables. Analyses were adjusted for age at baseline, sex, and education. **Results:** A total of 21197 individuals were included in this analysis, comprehending 1,563 bvFTD (mean age 63.7±9.8 years, 944 females) and 19,634 with other diagnoses (mean age 74.6±10.3 years, 9272 females). The following factors were significantly associated with bvFTD diagnosis: obsessive-compulsive disorder ($\beta=0.17$), bipolar disorder ($\beta=0.14$), diabetes ($\beta=0.03$), insomnia ($\beta=0.03$), recent depression ($\beta=-0.03$), and alcohol abuse ($\beta=0.02$), $p<0.05$ for all. **Discussion:** While AD and bvFTD may share some common risk factors, our findings suggest that bvFTD has a distinct risk profile, particularly emphasizing psychiatric and behavioral components. This supports the need for differential preventive strategies and reinforces the relevance of addressing modifiable psychiatric risk factors. **Conclusion:** Previous psychiatric and behavioral disturbances emerged as significant risk factors for bvFTD.

TECHNICAL AREA: CLINICAL NEUROLOGY

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Association of cardiovascular risk factors, aging and incidence of dementia score with cognitive tests in older adults without dementia

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Background: The CAIDE (Cardiovascular Risk Factors, Aging and Incidence of Dementia) score was created to relate neurovascular risk factors, important due to their modifiability, to subsequent development of dementia. Evidence associating CAIDE score and specific cognitive domains remains limited, particularly in the early stages of cognitive decline, such as mild cognitive impairment (MCI). **Objective:** to evaluate the association between CAIDE score and performance on neuropsychological tests in a cohort of participants with MCI. **Methods:** A Cross-sectional analysis was performed using data from 98 participants in the Brazilian Aging and Memory Study (BRAMS) cohort. Participants were divided between MCI (56.1%) and cognitively unimpaired (CU). Data included CAIDE score, Mini-Mental State Examination (MMSE), brief cognitive screening battery (BCSB) and a comprehensive neuropsychological tests battery. **Results:** The MCI group had higher CAIDE scores (6.76 ± 2.12 vs. 5.88 ± 1.76 ; $p=0.048$). CAIDE score displayed negative correlation with the MMSE in the total sample ($r=-0.238$; $p=0.020$), negative correlation with phonemic verbal fluency in the MCI group ($r=-0.327$; $p=0.015$) and negative correlation with delayed recall of the Rey Auditory Verbal Learning Test ($r=-0.338$; $p=0.031$) and trail making B ($r=-0.348$; $p=0.030$) in the CU group. **Conclusion:** Although the CAIDE score was higher in participants with MCI, the data suggests CAIDE score may be associated with subtle cognitive changes even in individuals without diagnosis of clinical impairment, which reinforces its potential as an early risk marker for cognitive decline.

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Translation and cross-cultural adaptation of the DAPHNE scale for the assessment of behavioral variant of frontotemporal dementia

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Background: Frontotemporal Dementia (FTD) is a clinically diverse neurodegenerative disease often underdiagnosed and misdiagnosed. The behavioral variant of FTD (bvFTD) is the most common presentation, and its diagnosis is challenging due to overlap with psychiatric and neurological conditions. Current diagnostic criteria have limitations in specificity, prompting the development of assessment scales for the diagnostic evaluation of bvFTD. The DAPHNE behavioral inventory is one such instrument and comprises a 6-item screening version (DAPHNE-6) and a 40-item diagnostic version (DAPHNE-40), both of which achieved 92% sensitivity and specificity for bvFTD diagnosis. **Objectives:** This study aimed to translate and cross-culturally adapt the DAPHNE scale for the Brazilian context, thus increasing the availability of culturally appropriate instruments for clinicians in Brazil. **Methods:** The study followed established guidelines described by Beaton et al. and Herdman et al. for instrument translation and adaptation, comprising five stages: (1) translation; (2) back-translation; (3) formal assessment of equivalence and development of a synthesized version by an expert committee; (4) pilot application of the synthesized version in 22 patients (10 with bvFTD and 12 with Alzheimer's Disease) and (5) review of the pre-testing results and finalization. **Results:** Cultural and linguistic adaptations were rigorously implemented, ensuring the adapted version's conceptual, semantic, and operational consistency and high comprehensibility across diverse Brazilian social classes. Pre-testing showed easy administration and a seven-minute completion time. **Conclusion:** This study successfully adapted the DAPHNE scale for Brazil, offering a promising tool for structured bvFTD diagnosis and differentiation. Future studies are important to validate its psychometric properties.

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Clinical, cognitive, and neuroimaging profiles of patients with transient global amnesia

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Background: Transient Global Amnesia (TGA) is characterized by sudden anterograde amnesia, without other neurological deficits. TGA typically lasts up to 24h and resolves spontaneously. There is a paucity of studies investigating TGA in Brazil. **Objective:** To characterize the clinical, cognitive, and neuroimaging features of patients with TGA. **Methods:** We selected consecutive TGA patients admitted to the Emergency Care service from

2023–2025. All participants underwent the Brief Cognitive Battery and the Facial Emotion Recognition Test (FERT). Brain MRI was performed 24–96h after the ictus. Results were compared with cognitively healthy controls (n=40), matched for age, sex, and education. **Results:** We included 41 TGA patients (28 women/13 men; mean age=60.8 years). Among them, 45.8% had hypertension, 25% had obstructive sleep apnea syndrome, 16.7%, migraine. None had history of stroke, or epilepsy. Compared to controls, TGA patients had lower scores on the MMSE (p<0.0001), on the immediate memory (p<0.0001), delayed recall (p<0.0001) and in the recognition scores (p<0.0001). Patients did not differ from controls in the recognition of facial emotions (total score) or in the subscores of each emotion. Brain MRI found bilateral hippocampal signal alterations 16%, unilateral changes (left or right hippocampus) in 18%, and no abnormalities in 48% of the cases.

Discussion and Conclusions: Patients with TGA have spared recognition of facial emotions, thus highlighting the strict memory involvement in TGA. While cognitive findings showed a relatively homogeneous profile, neuroimaging was heterogeneous. Longitudinal studies are essential for elucidating the long-term cognitive outcome of TGA.

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Sociodemographic profile of patients at the cognitive neurology outpatient clinic of a tertiary hospital

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Background: Sociodemographic analysis is fundamental to understanding health needs and optimizing services in cognitive neurology outpatient clinics, where neurodegenerative conditions are prevalent in the elderly. **Objective:** To outline the sociodemographic profile of patients seen at the cognitive neurology outpatient clinic at Fortaleza General Hospital, using data from patient records. **Methods:** Retrospective descriptive study of 60 patients seen between July 2024 and June 2025. Variables included age, education, clinical diagnosis, MMSE subscores, and BCB performance. Descriptive statistics and group comparisons were conducted. **Results:** The predominant profile was elderly patients (majority over 60), mostly female (67.3%). There was a low level of formal schooling (19.2% without schooling/illiterate) and a high proportion of retired people (69.2% with benefits, 7.7% without benefits). A significant proportion were widows (28.8%). Geographically, patients come from Fortaleza (55.8%), other cities in Ceará (30.8%) and other states. **Discussion:** This profile points to socioeconomic and educational vulnerabilities, with direct implications for clinical communication, the need for social support and barriers to accessing treatment. Low schooling suggests lower cognitive reserve and the need for simplified communication approaches. The predominance of widowers and retirees without benefits indicates support and financial challenges. **Conclusion:** The sociodemographic profile outlined reveals a population with specific needs that require adapting communication strategies, strengthening psychosocial support and optimizing access, especially for patients with low levels of education and economic and support vulnerabilities.

NEUROPATHOLOGY

TECHNICAL AREA: NEUROPATHOLOGY

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Association of LATE-NC with advanced neuropathological lesions: a population-based autopsy study

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Background: Limbic-predominant age-related TDP-43 encephalopathy neuropathological change (LATE-NC) is characterized by amnesic dementia and often coexists with other neuropathological lesions. We aimed to investigate the frequency of other neuropathological lesions and their associations with LATE-NC. **Method:** This cross-sectional study analyzed a population-based sample of 1,434 Brazilians from the Biobank for Aging Studies at the University of São Paulo Medical School. Neuropathological data were collected for Braak for Parkinson's Disease (PD), Braak for neurofibrillary tangles (NFT), amyloid- β burden (CERAD), LATE-NC, lacunar infarcts, cerebral amyloid angiopathy, and hyaline arteriosclerosis. Ordinal logistic regression evaluated the relationship between LATE-NC and the staging of other neuropathological lesions. For descriptive analyses, LATE-NC was dichotomized into binary categories: absent (stages 0–1) and present (stages 2–3). Models were adjusted for sociodemographic factors, clinical variables, and neuropathological lesions. In each analysis, the outcome lesion was excluded from the adjustments. **Result:** Sociodemographic and clinical variables varied significantly across participants at different LATE-NC stages. Participants in LATE-NC stages 1 and 2 exhibited a higher frequency of advanced neuropathological changes compared to those without LATE-NC, while participants in stage 3 showed lower frequencies. Dichotomized LATE-NC demonstrated an overall higher frequency of all neuropathological lesions associated with LATE-NC. Adjusted analyses for sociodemographic and clinical factors along with other neuropathological lesions, demonstrated a significant association only between LATE-NC presence and advanced Braak PD stages (OR=1.46, 95%CI 1.09–1.94, $p=0.008$). **Conclusion:** Higher LATE-NC stages were correlated with more neuropathological lesions, but only Braak PD remained independently associated after adjustments.

TECHNICAL AREA: NEUROPATHOLOGY

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Hippocampal subfield volume and tau pathology in PART and AD

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Background: The term Primary age-related tauopathy was proposed for cases with low to moderate neurofibrillary pathology associated to sparse or absent A β plaques. Despite the increased number of studies, questions about PART as a distinct pathology or a subtype of AD remain. **Method:** to investigate the impact of tau neurofibrillary pathology in the hippocampal volume, we analyzed 7T postmortem MRI and neuropathological data from 16 individuals. Demographic and clinical data were collected through semistructured interview with a next of kin. The FreeSurfer automated segmentation tool was used to provide data for the entire hippocampus and subregions. In addition, immunohistochemistry using antibodies against amyloid- β and hyperphosphorylated tau was performed to evaluate AD-type pathology. **Result:** The mean age was 74 years (± 13.8). The presence of some degree of neurofibrillary pathology was found in 93%, while neuritic plaques were found in 52%. The presence of cognitive impairment (CDR >0) was associated with reduced hippocampal total and subregion volumes. Higher burden of neurofibrillary pathology was associated with lower hippocampal total and subregion volumes. In contrast, no association of total and subregion hippocampal volumes and neuritic plaques was observed. AD was associated with reduced volume in CA1 compared to PART. **Conclusion:** Greater burden of tau pathology was associated with smaller hippocampal volume, whereas no detectable effect of neuritic plaque was observed. Compared with PART, AD had significantly reduced CA1 volume.

TECHNICAL AREA: NEUROPATHOLOGY

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Sex differences in co-pathologies in Alzheimer's disease

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Background: Sex differences in Alzheimer's disease neuropathological changes (ADNC) have been explored in several studies, but few have examined the occurrence of copathologies. This study investigates the associations of sex with AD pathology, the number of co-pathologies, and cognitive abilities in a Brazilian sample. **Method:** Participants were from the Biobank for Aging Studies, a large population-based autopsy study in Brazil. The neuropathological assessment includes AD pathology (neuritic plaques and neurofibrillary tangles), TDP-43, Lewy body pathology, lacunar infarct, hyaline arteriosclerosis, and cerebral amyloid angiopathy. Cognitive abilities were evaluated using the Clinical Dementia Rating Sum of Boxes (CDR-SOB). Ordinal logistic and linear regressions were used to examine the associations between neuropathology and cognitive abilities, considering interaction by sex and adjusting for sociodemographic and clinical variables. **Result:** Among 894 participants, 52.7% were women, with a mean age of 73 $\pm$ 14 years, and 65.1% were white. Women were older, less educated, and had poorer cognitive abilities than men. They also had a higher frequency of Alzheimer's disease (AD) pathology, co-pathologies, and worse cognitive abilities ($\beta=10.5$, 95%CI 9.0–12.0, $p<0.001$). **Conclusion:** Women had poor cognitive abilities and higher association of ADNC and co-pathologies than men.

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Neuroprotective effects of extracellular vesicles from human adipose-derived mesenchymal stem cells in Alzheimer's disease models

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Alzheimer's disease (AD) is a progressive neurodegenerative disorder and the most common cause of dementia, significantly impacting patients, families, and healthcare systems. Despite extensive research, effective therapeutic options remain limited, highlighting the urgent need for novel approaches. In this study, we evaluated the neuroprotective potential of extracellular vesicles (EVs) derived from human adipose tissue mesenchymal stem cells (adMSCs) in preclinical models of AD. EVs offer a promising alternative to stem cell therapy by delivering therapeutic cargo while minimizing risks associated with cell transplantation, such as immune rejection and tumorigenicity. Two distinct AD mouse models—one induced by amyloid- β oligomer administration and another using APP/PS1 transgenic mice—received intravenous injections of adMSC-derived EVs (EVsMSCs) or saline as a control. Following treatment, animals were subjected to behavioral assays to assess memory performance, using the novel object recognition (NOR) test and the Morris water maze (MWM). Results demonstrated that EVsMSC-treated animals in both models showed significant improvements in memory-related tasks compared to controls. These findings suggest that EVs derived from adMSCs can mitigate cognitive deficits in AD models, supporting their therapeutic potential. Our study contributes to the growing evidence that cell-free strategies, particularly those utilizing stem cell-derived EVs, may represent a safer and effective approach for AD treatment.

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Neuroprotective effects of a glycosylated peptide: a study with an acute model of seizures in mice

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Alzheimer's disease (AD) and epilepsy show a bidirectional relationship, sharing common pathophysiological mechanisms. Research indicates that epilepsy in early adulthood might be correlated to the development of Alzheimer's disease later in life. However, current available antiepileptic drugs often cause adverse effects or fail to control seizures, highlighting the need for new therapeutic options, such as neurovespina, a peptide bioinspired in wasp venom. It showed considerable neuroprotective effects against chemically induced seizures in mice, but exhibited hydrophobicity and a short half-life. Peptide glycosylation emerges as an interesting

strategy in this context, enhancing molecular stability, aqueous solubility, and blood brain barrier permeability. This study focused on Tyr-O-glycosylation of neurovespina, aiming to explore its neuroprotective effects. The glycosylated neurovespina was synthesized, purified, and characterized using reverse-phase liquid chromatography (HPLC) and mass spectrometry (MS). In vivo assays evaluated the neuroprotective effects of the peptides in an acute seizure model induced by pentylenetetrazol (PTZ) in Swiss mice. Three doses of the peptides were administered intraperitoneally, and effects were recorded via video-electroencephalogram (EEG). Behavioral data were analyzed using the Racine scale adapted for PTZ, while electrophysiological data were processed using Sirenia Seizure Pro. The 4 mg/kg dose was the most effective in controlling seizures for all tested peptides. At this dose, the glycosylated peptide exhibited the longest latency to both maximum seizures and mortality. The results indicate that glycosylated peptide exhibits greater solubility, bioavailability and neuroprotection representing a promising therapeutic candidate. Further studies should explore its neuroprotection applied to AD models.

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In vitro assays of neurotoxicity, neuroprotection, and hemolysis of fraternine-10 and octovespin, peptides with prior β -amyloid interaction evidence

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Alzheimer's disease (AD) and Parkinson's disease (PD) are the two most prevalent neurodegenerative disorders, with no current cure or effective treatment capable of slowing their progression. In the search for new therapeutic alternatives, peptides have emerged as promising candidates due to their unique properties and efficacy. Fraternine-10 and octovespin are bioinspired peptides that showed encouraging results in AD and PD animal models, though their mechanisms remain under investigation. In vitro research plays a crucial role in early drug development, providing a controlled environment to evaluate therapeutic potential. This study aims to assess the pathophysiological relevance of fraternine-10 and octovespin using cellular models that replicate neurodegenerative scenarios. To evaluate their neuroprotective properties, human neuroblastoma cells (SH-SY5Y) were exposed to 6-OHDA to induce neurotoxicity. DCFDA and resazurin fluorescence were measured to assess ROS production and cell viability, respectively. For neurotoxicity assessment, SH-SY5Y cells were incubated with the peptides at similar doses. Hemolytic activity was tested using a 3% suspension of murine erythrocytes and analyzed via spectrophotometry. No change in resazurin fluorescence was observed in 6-OHDA-treated cells co-incubated with peptides, suggesting a lack of neuroprotection. However, when applied alone, both peptides reduced DCFDA fluorescence, indicating antioxidant activity without affecting cell viability. Hemolysis assays showed no hemolytic effect at tested doses. In conclusion, while fraternine-10 and octovespin were not neuroprotective at tested doses, they exhibited antioxidant activity and were non-hemolytic. Further studies and optimization are needed to better understand their therapeutic potential.

TECHNICAL AREA: NEUROPATHOLOGY
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Evaluation of the effect of nanoencapsulated neuroccidentalis peptide in a model of Alzheimer's disease in mice

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Neuroprotection, Drug Delivery System, nanotechnology Alzheimer's disease is a progressive neurodegenerative condition and the leading cause of dementia in the elderly. Despite advances in understanding the disease, treatments still have limitations. In this context, compounds derived from animal venoms have emerged as promising sources of bioactive molecules with high specificity for CNS targets. However, crossing the blood-brain barrier remains a challenge, for which nanocarriers appear to be a potential solution, as they improve drug release profiles and enable targeted delivery, ensuring therapeutic activity with increased safety. This study aims to evaluate the therapeutic effect of neuroccidentalis combined with a nanoemulsion in a mouse model of Alzheimer's disease. To assess neuroprotective action, male mice underwent surgery to induce the Alzheimer's model. After surgery, the animals were treated with the nanoencapsulated peptide, subjected to behavioral tests, and euthanized. Brains were collected for immunohistochemical analysis. In both Novel Object Recognition Test and the Morris Water Maze, the nano dose group showed cognitive improvement compared to the β A group, similarly to the SHAM group. This suggests that the major nanoencapsulated dose may effectively disaggregate β A and reverse its induced deficits, improving cognition. Other doses did not show cognitive improvement in any of the tests compared to the β A group. However, to better clarify the results, it is important to elucidate the mechanism of action of this compound, as well as pharmacokinetic studies, permeabilization of the blood-brain barrier, duration of action. In addition to studies on the stability of nanoparticles and how to improve them.

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Development of a non-motor model of Parkinson's disease and evaluation of antiparkinsonian of the nanoencapsulated peptide Fraternine-10

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Parkinson's disease (PD) is a neurodegenerative disease, mainly caused by the loss of dopaminergic neurons in the Substantia Nigra. Around 10 million people are affected by the disease worldwide. PD is composed by motor and non-motor symptoms. Non-motor symptoms impacts more than half of patients, nevertheless, current treatment does not take them into consideration most of the time. Hence, the objective of this work is to generate a 6-OHDA non-motor symptom model of PD, with the purpose of testing

the nanoparticle involved peptide. For the non-motor model two doses of 6 hydroxydopamine (6-OHDA) were tested 40 ug and 30 ug, the chosen dose was 30 ug, since it showed anxiety-like symptoms in the elevated plus maze test. After reaching a dose the following test were made elevated plus maze (EPM) to assess anxiety, open field to assess motor and behavioral chances, and new object recognition test (NORT) to assess long memory consolidation. In this protocol were utilized the following groups, vehicle, 6-OHDA (30 ug) and NF-10 in its therapeutic dose (10 ug). The NF-10 group showed difference to 6-OHDA in the EPM, in spite of that, it was not similar to the vehicle. No motor difference was observed in the open field between the 6-OHDA and vehicle. For the TRON the 6-OHDA showed memory deficit in comparison to the vehicle group, however, the NF-10 group showed no improvement in comparison to the 6-OHDA group. Results suggest that it was possible to mimic non-motor symptoms in a 6-OHDA model.

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Spontaneous cerebral amyloid angiopathy associated intracerebral hemorrhages in aged capuchin monkey

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Background: Cerebral amyloid angiopathy (CAA) is characterized by accumulation of β -amyloid (β A) protein in cerebral vessels. Animals models are important for studies in CAA. However, among these models none developed CAA-related intracerebral hemorrhage. We analyze the brain of a 29-year-old female capuchin monkey who developed CAA-associated vascular changes. **Methods:** The brain was fixed in 4% buffered paraformaldehyde and 7T MRI were acquired before and after brain slicing. Consecutive coronal sections from the fixed brain were embedded in paraffin, and 5 μ m sections were used for staining and immunohistochemistry evaluation. All brain sections were immunostained with HE, and antibodies against the following proteins: β A, phospho tau and ubiquitin-proteasome. Also, additional immunostaining analysis using TDP-43, α -synuclein, GFAP, Iba-1, β A40 and β A42 antibodies, and Masson Trichrome and Perls stains were performed in selected areas. **Results:** β A deposits were noted in meningeal and parenchymal arterioles as well in numerous parenchymal capillaries of several areas of the brain, predominantly in the neocortex. Parenchymal and meningeal vessels were immunoreactive for β A40 and β A42. Whereas both isoforms were found in arterioles and capillaries, there was a predominance of β A40 compared to β A42. 7T MRI demonstrated an old hematoma, characterized as low foci in SWI and T2 images with central cavitation, alongside multiple smaller areas of old hemorrhage. HE staining revealed cystic microcavitation with hemosiderin deposit. **Conclusions:** The presence CAA, neuroinflammation markers and β A40 and 42 isoforms suggest that this New World monkey could be a model not only for AD but also for CAA studies.

NEUROPSYCHOLOGY

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A interdisciplinary rehabilitation program for people with dementia and their caregivers

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Caregivers of people with dementia face significant emotional, physical, and financial burdens, especially in low- and middle-income countries where underfunded health systems, limited training, and fragmented care intensify these challenges. Integrated, person-centered interventions may help mitigate such effects. This descriptive study involved 30 dyads (person with Alzheimer's disease and primary caregiver) who participated in an eight-week group intervention at a neurorehabilitation center in Brazil. Weekly 120-minute sessions were conducted by an interdisciplinary team (psychologist, nurse, pharmacist, dance teacher, physical educator, physiotherapist, and geriatrician). Activities included psychoeducation, physical and cognitive stimulation, music, reminiscence, relaxation, and caregiver skills training. After each session, the team met to discuss cases and adjust interventions. Caregivers' testimonials were analyzed through IRaMuTeQ software (Descending Hierarchical Classification method). The Chi-square test assessed associations between words and lexical classes. Six categories emerged. The study was approved by the Ethics Committee of the Institute of Psychiatry at the Federal University of Rio de Janeiro (IPUB/UFRJ), Brazil, CAAE: 92966318.3.0000.5263. The mean attendance rate was 89.02% ($\pm 10.65\%$). Caregivers reported improved understanding of dementia, positive restructuring of disease perception, and practical skills. Highlights included shared experiences, peer support, and involvement in person-centered care. The program demonstrated strong engagement and cultural appropriateness, with promising initial outcomes. Further studies are needed to evaluate its long-term effectiveness and generalizability to other settings.

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Anticholinergic burden and cognitive aging: investigating the protective role of digital cognitive training

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Association between anticholinergic burden and cognitive decline in older adults remains unclear, particularly regarding the modulatory effect of digital cognitive training. This study investigated the relationship between anticholinergic burden, cognitive impairment, and cognitive remediation in healthy older adults (HOA) and individuals with mild cognitive impairment (MCI). A total of 165 participants aged 60 and older (37 men, 128 women; mean age=69.1 years) from two randomized controlled trials were assessed. Baseline data included sociodemographic, clinical status, medication, and cognitive measures. Anticholinergic burden was quantified using the An-

ticholinergic Cognitive Burden (ACB) scale. The mean Montreal Cognitive Assessment (MoCA) score was 24.1 ± 3.5 , and mean ACB score was 0.77 ± 1.1 . Participants with MCI were older and used more medications, with higher ACB scores, compared to HOA. Each additional medication increased the odds of MCI by 1.01 times ($p < 0.001$). However, these associations lost significance after adjusting for age and education ($OR = 0.8$, $p = 0.60$; $OR = 0.9$, $p = 0.69$, respectively). Among MCI participants who completed 20 hours of digital cognitive training, higher ACB scores correlated with lower cognitive gains (Spearman's $r = -0.55$, $p < 0.01$). No significant correlations were found in the HOA group. Findings suggest that anticholinergic burden may hinder cognitive training benefits in MCI. Although not a significant predictor of MCI after adjustments, clinicians should consider minimizing anticholinergic load to optimize interventions.

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Behavioral outcomes of cognitive training in patients with amnesic MCI ($A\beta+$ and $A\beta-$) and healthy controls

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Objective: To evaluate the effectiveness of a unimodal cognitive training (CT), based on news reports with and without images, on memory functions in older adults with amnesic MCI (single or multiple domain), with and without β -amyloid positivity ($A\beta+$ and $A\beta-$), and healthy controls (HC). **Method:** Fifty-five older adults participated: 19 $A\beta+$, 16 $A\beta-$, and 16 HC. All underwent six individual online CT sessions (1.5 h each). Neuropsychological assessments were conducted pre- and post-training and included Logical Memory (LM1 – immediate recall, LM2 – delayed recall) and the Rey Auditory Verbal Learning Test (RAVLT A1–A5 – immediate recall, A7 – delayed recall). A Memory Complaints and Strategy Use Questionnaire was administered before and after CT. Statistical analyses included ANOVA or Kruskal-Wallis, based on normality (Shapiro-Wilk) and homogeneity (Levene), with Dunn's or Tukey's post hoc tests. Intra-group: Wilcoxon. **Results:** Memory complaints were reported by 94.74% of $A\beta+$ and 85% of $A\beta-$ (vs. 6.5% in HC), decreasing to 15%, 20%, and 0% after CT. Strategies were initially used by 5.26% ($A\beta+$) and 10% ($A\beta-$), increasing post-training. In immediate recall, HC outperformed $A\beta+$ ($p < 0.05$). After training, all groups improved. In delayed recall, HC maintained superiority ($p < 0.01$). **Conclusion:** CT improved memory and strategy use in all groups. Gains were observed in both immediate and delayed recall, with the greatest benefit in $A\beta-$. $A\beta+$ improved especially in immediate recall, while HC showed strong verbal learning. $A\beta+$ benefited most from visually supported, structured training.

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Digital cognitive training for functionality in mild cognitive impairment: a randomized controlled clinical trial

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Digital cognitive training (DCT) may improve cognition in people with mild cognitive impairment (MCI); however, its effect on functionality remains poorly defined. Despite preserved independence in Instrumental Activities of Daily Living (IADL), individuals with MCI perform them with increased effort. Conventional IADL measures may miss specific difficulties and crucial data. In contrast, the Canadian Occupational Performance Measure (COPM) is an individualized tool designed to identify challenges in IADLs. This study used the COPM to investigate the effects of DCT on functionality in individuals with MCI. Participants aged 60 or older with MCI were recruited for a double-blinded, randomized clinical trial comparing DCT to an active control group using commercial computer games. **Results:** Fifty-five participants were evaluated before and after 10 hours of intervention. The DCT group showed significant improvement in functionality, measured by COPM performance (pre: 5.66 [SD=2.29]; post: 6.87 [SD=1.72]), compared to the control group (pre: 5.64 [SD=1.64]; post: 5.72 [SD=2.23]). A main effect of time ($F(1, 53)=7.07$, $p=0.01$) and a group-time interaction ($F(1, 53)=4.35$, $p=0.04$) were observed. Additionally, learning over trials improved in the DCT group and decreased in controls. Significant effects of time ($F(1, 53)=12.41$, $p<0.001$) and group-time interaction ($F(1, 53)=7.05$, $p=0.01$) were observed. DCT improved functionality and learning in MCI, emphasizing the need for sensitive tools like the COPM to assess changes.

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Early cognitive dispersion and risk of cognitive impairment: insights from the ELSA-Brazil study

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Background: Neuropsychological assessments rely on mean-level performance. However, middle-aged and older adults may exhibit increased intraindividual cognitive variability (IICV), also seen as dispersion of scores, even when classified as normal. IICV reflects inconsistencies in cognitive function and has been linked to higher dementia risk. **Objective:** We aimed to investigate whether baseline IICV predicts cognitive impairment over eight years. **Methods:** We analyzed data from 9,914 ELSA-Brazil participants. Baseline assessments included measures of memory and executive function. Participants with two or more baseline regression corrected-based norms scores below -1.5SD were excluded for possible deficits. IICV was measured using the coefficient of variation (CoV), which adjusts for global cognitive ability differences. CoV scores were scaled to enhance clinical interpretation (higher CoV indicates lower IICV). Cognitive impairment was determined using education-adjusted MMSE cutoffs. Logistic regression models, adjusted for sociodemographic factors and clinical confounders, estimated the odds ratios for cognitive impairment. **Results:** Participants had a mean age of 50.7 ± 8.7 years; 55% were women and 7% had less than a high school education. At baseline, 2.4% had possible atypical IICV. After eight years, 13% was classified as having cognitive impair-

ment. Logistic regression indicated that for every one-unit increase in CoV, the odds of developing cognitive impairment decreased by 7% (OR 0.93, 95%CI 0.91–0.95). **Conclusion:** Higher IICV at baseline was associated with increased risk of cognitive impairment eight years later, underscoring the importance of considering cognitive dispersion in neuropsychological assessments. IICV may be a valuable target for identifying individuals at risk for impairment.

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Effectiveness of a cognitive stimulation program for cognitively unimpaired older adults: a randomized controlled clinical trial – 24 month follow-up

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Background: The SUPERA® Cognitive Stimulation Study was a randomized, single-blinded clinical trial that evaluated a multicomponent cognitive stimulation program in cognitively unimpaired older adults. **Objective:** To investigate the effectiveness and maintenance of cognitive, emotional, and functional benefits from an 18-month cognitive stimulation intervention. **Methods:** A total of 207 participants were randomized into three groups: Training Group (TG), Active Control Group (ACG), and Passive Control Group (PCG). The TG received 72 weekly sessions (120 minutes each) over 18 months. The ACG received health education with equivalent frequency, and the PCG received no intervention. Assessments occurred at baseline, 6, 12, 18, and 24 months. Outcomes included cognitive, emotional, and functional measures. Analyses followed the intention-to-treat principle. **Results:** The TG showed significant improvements from baseline to 6 months (T0–T1), with continued memory gains up to 12 months and executive function gains up to 18 months. Improvements were maintained at 24 months. TG participants also reported fewer cognitive complaints post-intervention compared to controls. **Discussion:** The results suggest early and sustained cognitive benefits from the intervention, with memory and executive functions improving over different timeframes. Maintenance effects observed at 24 months reinforce the program's long-term potential. Self-reported cognitive improvement highlights perceived benefits. Further investigation is needed to explore mediators of efficacy and applicability in broader populations. **Conclusion:** An 18-month multicomponent cognitive stimulation program appears effective in enhancing and maintaining cognitive function in healthy older adults, with promising implications for preventive gerontological care.

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Functional independence and cognition associated with loneliness in a nationwide Brazilian cohort

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Background: With rising dementia rates, identifying modifiable risk factors is essential. Loneliness has emerged as a key contributor to cognitive decline. This study investigated the relationship between loneliness, social isolation, and cognitive performance in Brazilians, considering demographic differences. **Methods:** Data from 5,284 individuals aged 50+ (mean age: 62.8; 50.1% male) from the ELSI-Brasil population-based study were analyzed. Only non-depressed participants were included. A global cognitive composite score was derived from tests on memory, orientation, and verbal fluency. Functional independence was measured using a modified Lawton scale. Loneliness was assessed with the question: "How often do you feel lonely?" Responses of "always" or "sometimes" indicated loneliness. Analyses were conducted in R (v4.1.0). **Results:** Loneliness predicted lower cognitive scores ($\beta=-0.07$, $p=0.006$) and decreased functional independence ($\beta=0.18$, $p=0.0004$). When stratified by age, the link between loneliness and cognition was significant only in older adults ($\beta=-0.09$, $p<0.001$). Among men, loneliness moderated the association between age and cognition ($\beta=-0.09$, $p=0.006$), and between age and functional independence ($\beta=-0.23$, $p=0.006$), even after adjusting for social isolation. **Conclusion:** Lonely men appear particularly vulnerable to cognitive decline and functional impairment with age. These findings highlight the importance of addressing loneliness in public health strategies and for prevention efforts to risk groups, especially older males.

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Differences of pattern of impairment between decision-making capacity in people with mild cognitive impairment and Alzheimer's disease

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Background: Mild Cognitive Impairment (MCI) is an intermediate state between normal aging and dementia. People with MCI may be aware that their memory or mental ability has worsened. Alzheimer's Disease (AD) is the most common form of dementia, characterized by gradually progressive cognitive and functional deficits and behavioral changes. The decision-making capacity is the process of making choices by identifying a decision, gathering information and assessing alternative resolutions, and selecting an appropriate action from several possible actions. There is a lack of research on differences between decision-making capacity in People with MCI and People with AD. **Method:** A cross-sectional study included 137 participants, being 37 people with MCI, 50 people with mild DA and 50 people with moderate DA. **Result:** People with MCI were more cognitively aware and more

aware of cognitive deficits and health conditions, with moderate effect sized. All people with AD presented deficits in the domains of decision-making capacity, with more impairment in understanding. There was a relationship between understanding and awareness domains, such that awareness was particularly important for decision-making capacity in the AD group. **Conclusion:** Better awareness involved better understanding in the MCI group. Clinically, our findings shed light on the need to consider the differences in the domains of decision making capacity and their relationship with other clinical aspects. Furthermore, our data can suggest hypotheses for larger and longitudinal studies.

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Decision-making capacity and awareness in people with young-onset Alzheimer's disease: the differences between

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Background: Young-onset AD (YOAD) typically occurs before the age of 65 and affects less than 6% of all people diagnosed with AD. There is a lack of research on differences between decision-making capacity and awareness according to age at onset of dementia. We investigated the relationship between decision-making capacity and awareness domains in people with young- (YOAD) and late-onset Alzheimer's Disease (LOAD). **Method:** A cross-sectional study included 169 consecutively selected people with AD and their caregivers (124 people with LOAD and 45 people with YOAD). **Result:** People with YOAD were more cognitively impaired, but more aware of their cognitive deficits and health condition, with moderate effect sizes. All people with AD presented deficits in the domains of decision-making capacity, with more impairment in understanding. There was a relationship between understanding and awareness domains, such that awareness was particularly important for decision-making capacity in the YOAD group. **Conclusion:** Better awareness involved better understanding in the YOAD group. Clinically, our findings shed light on the need to consider the differences in the domains of awareness and their relationship with other clinical aspects such as decision-making capacity according to age at onset of AD. Furthermore, our data can suggest hypotheses for larger and more robust prospective studies.

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Longitudinal perspectives on awareness domains in Alzheimer's disease

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Background: Awareness refers to the ability to recognize deficits associated with disease progression. In Alzheimer's Disease (AD), it is a multifaceted and relational concept encompassing different domains of the

condition. **Objectives:** This study aimed to longitudinally examine the pattern of impairment across domains of awareness at three time points and explore their associations with clinical aspects of AD. **Methods:** A total of 158 individuals with mild-to-moderate AD and their caregivers were recruited from an outpatient clinic. After 24 months, 41 participants completed the follow-up assessments. Awareness was evaluated across five domains using the Assessment Scale of Psychological Impact of the Diagnosis of Dementia (ASPIDD): cognitive functioning and health condition, functional impairments, emotional state, social functioning and relationships, and awareness of the disease itself. **Results:** Post-hoc analysis revealed significant decline in most domains of awareness over time, except for the emotional state domain, which remained stable. Principal Component Analysis showed overlap between emotional state and social functioning, both associated with neuropsychiatric symptoms and caregiver burden at the first two time points. In contrast, cognitive and functional factors were more strongly associated with all awareness domains by the third assessment. **Conclusion:** Awareness impairment in AD follows a non-linear trajectory, with some domains remaining stable over time. Distinct clinical factors influence each domain, underscoring the complexity of awareness in the progression of AD.

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Analysis of errors in the verb fluency task in individuals with Alzheimer's disease

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Background: The verb fluency task has emerged as a tool in the assessment of cognition. However, most studies in Alzheimer's Disease (AD) focused on the total verbs produced, without considering qualitative aspects. Analyzing error patterns can provide information about the mechanisms underlying language difficulties in AD and improve the sensitivity of the task as a cognitive marker. **Objective:** To analyze the types of errors produced by individuals with AD during the verb fluency task. **Methods:** Case-control study with individuals aged 50 or over with AD and healthy controls. AD diagnosis was confirmed by clinical and neuropsychological assessment. Verb fluency was assessed using a task adapted to Brazilian Portuguese. Responses were recorded, transcribed, and error types were categorized. Statistical analysis was performed using R software. **Results:** Participants with AD produced fewer correct verbs and more errors in the task, such as repetitions and circumlocutions. Performance decreased with increasing dementia severity. Significant correlations were found between responses and MMSE scores and education. Total verb fluency showed 86.2% sensitivity and 68.1% specificity for AD diagnosis. **Discussion:** Errors among individuals with AD highlight the influence of executive and semantic impairments on language production. The progression of error frequency suggests that qualitative aspects deteriorate with disease severity. **Conclusion:** Error analysis in verb fluency offers important insights into the linguistic and cognitive deficits in AD. Qualitative features may serve as sensitive markers of disease severity and should be considered in clinical assessments and future research.

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Neuropsychological profiles in cognitive disorders: A diagnostic battery analysis

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Neuropsychological assessment (NPA) in a behavioral neurology outpatient clinic can help characterize cognitive profiles and differentiate clinical groups. The study aimed to compare patient performance across six diagnostic groups on an NPA battery to identify tests that best differentiate between conditions. Eighty-nine patients from this outpatient clinic were evaluated and categorized into Parkinson's Disease (PD, n=13), Subjective Cognitive Decline (SCD, n=3), Mild Neurocognitive Disorder (MdND, n=32), Major Neurocognitive Disorder (MND, n=29), Intellectual Disability (n=1), and Others (n=11). A standardized NPA was conducted, including the CERAD word list (CERAD), Figure Memory Test (FMT), digit span, Five-Digit Test (FDT), Frontal Assessment Battery (FAB), Trail-Making Test A (TMT-A), CLOX clock drawing test, semantic verbal fluency, and the abbreviated Boston Naming Test (BNT). Data analysis involved the Kruskal-Wallis test and Dwass-Steel-Critchlow-Fligner post-hoc comparisons, with significance at $p < 0.05$. Groups were matched for age and education. Episodic memory tests (CERAD, FMT) and executive function tests (FAB, FDT) showed greater discriminative power among the groups. MND showed lower performance in memory tasks when compared to the other groups. FAB effectively differentiated the groups, especially motor series and verbal fluency tasks. MdND and SCD showed better functionality. Findings align with existing literature. Patients with MND showed severe global cognitive deficits, while those with MdND had more localized impairments. Neuropsychological assessment has proven to help distinguish cognitive profiles and is an important tool for supporting diagnostic clarification in a specialized neurology outpatient clinic.

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Neuropsychological assessment in the diagnosis of mild cognitive impairment

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Background: New studies aim to standardize the neuropsychological assessment of Mild Cognitive Impairment (MCI). **Objective:** to analyze a neuropsychological assessment battery (NPA) in MCI. **Methods:** 89 individuals were referred for NPA at a behavioral neurology outpatient clinic. The NPA included tests: Digit Span, Five Digit, Frontal Assessment Battery, Trail Making A, Boston Naming-15, Verbal Fluency (semantic), Word List Memory (CERAD), Figure Memory (BBRC), Clock Drawing (CLOX), and Pfeffer (FAQ), administered in a 90-minute session. MCI was diagnosed according to Bondi et al. (2014) criteria, which required an impaired score on both measures within at least one cognitive domain (memory, language, or speed/executive function); or one impaired

score in each of the three cognitive domains sampled; or a score on the FAQ of 9. Impairment was defined $>1SD$ below the age-corrected normative mean. After the initial diagnosis, MCI was classified as either amnesic or non-amnesic and single or multiple cognitive domains. **Results:** 32 participants (50% women, mean age 70 ± 10 years, average education 6 ± 4 years) were diagnosed with Mild Cognitive Impairment (MCI). MCI subtypes included 9 cases of non-amnesic single-domain impairment, 11 cases of non-amnesic multiple-domain impairment, 9 cases of amnesic multiple-domain impairment, 2 cognitively healthy individuals, and 1 case with incomplete data. **Conclusion:** 34% of patients in this outpatient clinic were diagnosed with MCI. Most of these cases exhibited non-amnesic subtypes, with impairments across multiple cognitive domains. Some observed differences may be attributed to low educational levels and the common comorbidities associated with such socio-educational backgrounds.

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Verbal memory test performance in patients with cognitive impairment associated with long COVID: a descriptive analysis based on the HVLt-R

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This descriptive and exploratory study aimed to characterize the performance on verbal memory tests in patients with Long COVID Syndrome, based on the Hopkins Verbal Learning Test-Revised (HVLt-R). The study included 15 individuals seen at a university outpatient clinic, aged between 38 and 81 years, with varying levels of education. The results indicated below-average performance compared to normative data in the delayed recall and recognition stages, suggesting possible cognitive impairment associated with Long COVID. Participants with lower educational levels generally showed low performance. The group aged 60 or older with ≥ 12 years of education showed the best scores, even outperforming younger participants with the same level of education. In contrast, the group aged 60 or older with < 12 years of education showed the lowest performance in the delayed recall stage. Younger participants with low educational levels also presented difficulties, especially in the recognition stage. These findings highlight the role of age and education as influential factors in verbal memory task performance, reinforcing the hypothesis that higher levels of education are associated with greater cognitive reserve. The small sample size and low heterogeneity limit the generalizability of the data, suggesting that future studies with larger and more diverse samples should further explore the cognitive impacts of Long COVID on verbal memory.

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Cognitive performance in long COVID and its association with sociodemographic factors, self-perceived health, and vaccination

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Cognitive complaints are common among patients affected by COVID-19, and it is known that some type of persistent cognitive impairment is highly prevalent in the presence of prior vulnerability. Therefore, this study investigated the association between cognitive performance and sociodemographic factors, self-perceived health, vaccine doses, and number of infections in patients with long COVID. A cross-sectional study was conducted in a specialized outpatient clinic. A questionnaire was applied to collect age, gender, education level, infections, COVID-19 vaccine doses, and self-perceived health. The Montreal Cognitive Assessment (MoCA) was used for cognitive screening. Spearman's correlation was tested between variables. Thirty-three individuals participated: 25 women (75.7%) and 8 men (24.4%). The mean age was 51.2 years ($SD=10.84$). Most participants had completed secondary (18.2%) or higher education (48.5%). A statistically significant correlation was found between MoCA scores and education ($r=0.640$, $p<0.01$), and self-perceived health ($r=0.433$, $p<0.007$). Higher education and greater subjective well-being are associated with better cognitive performance, possibly due to cognitive reserve, which delays onset of neurodegenerative disease symptoms. It has been suggested that individuals vaccinated against COVID-19 have better cognitive performance than unvaccinated ones. On the other hand, negative self-perception of health and repeated infections have been associated with persistent cognitive impairments. In the present study, no associations were observed between cognition, vaccination, and reinfections, which may be due to the small sample size. Further studies are needed to clarify these relationships.

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Immersed memories: the potential of virtual reality in Alzheimer's disease

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The use of virtual reality (VR) in healthcare has garnered growing interest as a complementary strategy in the care of individuals with dementia. By enabling more realistic and engaging experiences, VR emerges as an innovative alternative to traditional cognitive stimulation approaches. Despite its potential, significant gaps remain regarding its clinical applicability, particularly among older adults diagnosed with Alzheimer's disease (AD). This study aimed to synthesize, through a systematic review and meta-analysis, the available evidence on the impact of VR-based interventions on cognition in individuals with AD. A comprehensive search was conducted across MEDLINE, EMBASE, Web of Science, BVS, and Scopus, including randomized controlled trials (RCTs) that compared VR strategies to conventional methods. Studies employing any level of immersion and reporting cognitive outcomes were included. Methodological quality was assessed using the Cochrane RoB 2.0 tool. Three RCTs were included, comprising a total of 75 participants. Results showed a significant improvement in memory performance among participants who underwent VR-based interventions (Mean Difference=0.99; 95%CI 0.33–1.66). However, the effects on executive functions did not reach statistical significance (Mean Difference=1.36; 95%CI -1.12 to 3.85). Variations in intervention protocols and duration likely contributed to data heterogeneity. Adverse effects such as nausea or discomfort were rarely reported. In conclusion, VR represents a promising avenue, particularly for memory rehabilitation in patients with AD. Nevertheless, further research is needed with larger samples, standardized intervention protocols, and systematic monitoring of tolerability.

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Assessing social cognition in Brazilian people with Alzheimer disease: findings and clinical implications

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The aim of our study was to investigate the pattern of social cognition impairment using the Social and Emotional Questionnaire (SEQ) in people with Alzheimer Disease (AD), differentiating between early-onset (YOAD) and late-onset (LOAD) cases, as well as considering different stages of the disease (mild and moderate). **Method:** We conducted two cross-sectional studies involving individuals with AD and their informants. The first study focused on age at onset, including 48 individuals with YOAD and 118 with LOAD. The second study was organized by disease stage, with 87 participants in the mild stage and 50 in the moderate stage. We assessed several aspects, including social cognition, global cognition, quality of life, functioning, neuropsychiatric symptoms, and caregiver burden. **Results:** The YOAD group exhibited greater impairment in general cognition, reported poorer quality of life, and experienced more neuropsychiatric symptoms compared to the LOAD group. However, social cognition did not show the same degree of decline when compared with individuals with LOAD. A similar trend was observed when analyzing disease stages; while the moderate AD group demonstrated an overall decline compared to the mild group, there were no significant differences in social cognition between these two groups. **Conclusion:** The results indicate a different progression of social cognition impairment, which leads us to infer that more direct and spontaneous tasks may be associated with automatic processes and appear more stable, whether in a moderately advanced stage of the disease or in a more aggressive progression, such as YOAD.

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Self-perception of emotional recognition in individuals with mild cognitive impairment and Alzheimer's disease: a comparative analysis

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Background: There is limited evidence on the ability of people with Alzheimer disease (AD) to perceive their own emotions. **Objective:** compare emotional recognition in participants with mild cognitive impairment (MCI) and mild to moderate Alzheimer's disease (AD) against caregivers' perceptions of these participants' emotional states, while exploring the influence of clinical variables. **Methods:** We included 141 participants (32 with MCI, 50 with mild AD, and 59 with moderate AD) and their primary caregivers. We employed tasks assessing emotional decoding, identification, and correspondence, along with objective evaluations. **Results:** Participants across all groups showed significant differences in cognition and functionality. However, emotional recognition abilities did not significantly differ between MCI and mild or moderate AD groups. Most cogni-

tive and neuropsychiatric variables had no significant impact on emotion recognition or social functioning. No differences emerged in patients' self-evaluations of social and emotional functioning. Caregiver assessments revealed significant differences only between the MCI and moderate AD groups. **Conclusion:** Participants with MCI and AD displayed expected clinical progression while retaining some emotional recognition and social functioning capabilities.

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Metamemory functioning in people with dementia

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Metamemory is the ability to monitor and control one's own memory. The objective of our study was to investigate the estimated and actual performance of metamemory in older adults with dementia compared to healthy older controls. Additionally, we examined whether cognition, awareness, neuropsychiatric symptoms, functionality, and executive functions influence metamemory functioning. We conducted a cross-sectional evaluation of 126 participants. Memory and metamemory were assessed in each group and compared using a memory test combined with a prediction task administered before and after exposure to the scale. Individuals with dementia presented impaired metamemory functioning in both short- and long-term memory evaluations. There was a general disconnect between perceived and actual performance across both groups, albeit to a lesser extent in the control group. In the control group, we found impaired metamemory when evaluating short-term memory. However, no alterations in metamemory were observed regarding long-term memory after this group had been exposed to the task. Regression analysis revealed that only awareness influenced metamemory when associated with short-term memory. Individuals with dementia tend to present impaired metamemory functioning, and awareness of disease appears to be a factor that can influence this impairment.

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Variations in awareness and neuropsychiatric symptoms in dementia: what changes with age of disease onset?

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Background: Impairments in awareness of disease and neuropsychiatric symptoms presents a major challenge in managing dementia. Young-onset dementia (YOD) is a significant clinical and psychosocial issue, with characteristics and evolution different from late-onset dementia (LOD) and specific adverse impacts. **Objectives:** This study aimed to explore neuropsychiatric symptoms and their relationship with awareness and its domains: awareness cognitive functioning and health condition, functional activity

impairments, social functioning and relationships and emotional state in YOD and LOD. **Method:** 185 dyads of people with dementia (51 YOD and 134 LOD) and their caregivers were assessed for global cognition, dementia severity, functionality, awareness, and neuropsychiatric symptoms. **Results:** The YOD group was more cognitively compromised ($p=0.012$), more anxious ($p=0.031$), and presented more aberrant motor behaviors than the LOD group ($p=0.032$). However, people with YOD were more aware of their cognitive functioning and health conditions than the LOD group ($p=0.001$). Multivariate linear regressions indicated that anxiety and aberrant motor behavior had a significant relationship with awareness in people with YOD. Hallucinations and apathy impacted awareness in the LOD group. **Conclusions:** The age at onset interfere in the grading of how awareness and its various domains are manifested. Our findings clarify how neuropsychiatric symptoms' influence on awareness and its domains in dementia in both groups studied.

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APOE and memory in latin america: Baseline Findings from the LatAm-FINGERS trial

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Background: Detecting early memory decline is key for dementia prevention, especially in Latin America, where risk exposure is high. FCSRT-IR offers a sensitive approach, yet how sociodemographic and genetic factors affect its performance in this context remains poorly understood. **Objective:** To investigate sociodemographic factors and APOE $\epsilon 4$ genotype influence on mnemonic performance in high-risk LA individuals. **Methods:** Cross-sectional analysis of LatAm-FINGERS baseline data, a randomized multicenter trial evaluating non-pharmacological interventions for cognitive decline. Inclusion: age 60–77; CAIDE ≥ 6 ; $-1.5 \leq z \leq 0$ on MMSE or CERAD list. Exclusion: MMSE < 20 ; dementia; illiteracy. Participants from Argentina, Brazil, Chile, and Uruguay. Memory was assessed via FCSRT-IR and SOMI—a staging model using FCSRT-IR scores for preclinical memory impairment. APOE genotyping used PCR-RFLP. Statistical analyses performed using Jamovi(v2.3); significance set at $p < 0.05$. **Results:** Sample ($n=358$): age 67.75 ± 4.86 ; education 13.41 ± 3.14 years; 72.3% female; 61.2% white; 22.3% APOE $\epsilon 4$ carriers. FCSRT-IR scores correlated with sociodemographics ($p < .001$) and differed across countries. SOMI classification: 58% SOMI-0 (preserved memory), 27.6% SOMI-1/2 (mild impairment). No sociodemographic differences between $\epsilon 4$ carriers/non-carriers. Non-carriers had better FCSRT-IR scores, significant only for identification ($p=0.017$). No $\epsilon 4$ -SOMI association. **Discussion:** SOMI score may capture early memory changes not explained by APOE $\epsilon 4$, highlighting the value of staging tools in diverse sociocultural contexts. **Conclusion:** Sociodemographic profile significantly influences memory. APOE $\epsilon 4$ shows limited association—suggesting social/environmental factors may outweigh genetic risk in LA.

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Subjective cognitive decline and cognitive performance among older adults

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Background: Subjective Cognitive Decline (SCD) may represent a preclinical stage of mild cognitive impairment, but its prevalence is underexplored

in developing countries, and standardized assessment tools are lacking. **Objective:** To examine the relationship among SCD, cognitive performance, and mood variables in older adults. **Method:** A total of 207 older adults from community centers and retiree associations were assessed using the Cognitive Function Instrument (CFI) for SCD, the Addenbrooke's Cognitive Examination-Revised (ACE-R) for cognitive performance, and the Geriatric Depression Scale-15 (GDS15) and Depression, Anxiety and Stress Scale (DASS-21) for mood symptoms. **Results:** Participants were categorized as No-SCD (CFI: $M=1.34$) or SCD (CFI: $M=5.15$). The groups were similar in age, education, and global cognitive performance. However, significant differences were found in retirement status ($p=0.047$), ACE-R visuospatial scores ($p=0.009$), DASS-21 ($p<0.001$), and GDS15 scores ($p<0.001$). Multiple logistic regression showed that higher GDS15 scores were significantly associated with increased odds of belonging to the SCD group ($p=0.008$; coefficient=0.290). **Discussion:** These findings suggest that subjective cognitive complaints in older adults may be more closely related to emotional and psychosocial factors than to objective cognitive decline. The elevated mood symptoms observed in the SCD group highlight the potential of SCD as a marker of emotional distress rather than neurodegeneration alone. **Conclusion:** Depressive and anxious symptoms were associated with SCD, underscoring the importance of mental health interventions for older adults with subjective cognitive complaints.

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Cultural readaptation and content validity the activities of direct assessment of functional status – revised (DAFS-BR) for Brazilian culture

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Introduction: The Direct Assessment of Functional Status - Revised (DAFS-BR) is the only ecological assessment measure for the performance of older adults in basic (ADL) and instrumental activities (IADL) daily living available in Portuguese. The DAFS-BR presents performance in activities that no longer reflect the functional activities performed in current Brazilian culture. **Objective:** To carry out the cultural readaptation and evaluate the content validity of the readapted activities in DAFS-BR. **Methods:** The cultural readaptation and content validity analysis of the readapted activities of the DAFS-BR were accomplished out in 4 stages: readaptation of the DAFS-BR activities; evaluation of equivalences of readapted activities by a committee of experts ($n=5$); readaptation of the application manual and scoring sheet and pre-test assessments of the new version with occupational therapists ($n=5$). **Results:** Three rounds of equivalence assessment were accomplished, and in the 3rd round, all proposed activities and adjustments obtained 100% agreement. In the pre-test phase, 139 suggestions for updating terms, formatting and spelling were incorporated into the manual and the scoring sheet. **Conclusion:** The "Direct Assessment of Functional Status - revised - Portuguese version (DAFS-BR (BR)) which has been readjusted in 3 domains (Communication Abilities, Financial Abilities and Shopping) and new activities have been proposed that can be used as an alternative to those already existing, considering the functional performance profile of the older adult to be assessed.

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Social support, burden, and symptoms of anxiety, depression, and stress in caregivers of older adults with dementia: preliminary findings

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According to PAHO and WHO, 40% of older adults in Latin America require long-term care, mostly provided by family members. Additionally, dementias account for over 12% of years lived with disability among them.

Objectives: To identify the profile of caregivers of older adults with dementia and the associations between burden, social support, anxiety, depression, and stress. **Methods:** Cross-sectional study with 47 caregivers of older adults with dementia, through an online questionnaire including sociodemographic data; caregiving time; burden (ZARIT); social support (MOS-SSS); depression, anxiety, and stress (DASS-21). **Results:** Among the 47 caregivers: 21 (45.7%) provide care for more than 10 hours a day; 37 (80.4%) are family members; 34 (73.91%) reported moderate to severe burden; 7 (15.91%), 7 (15.91%), and 9 (20.45%) reported severe symptoms of anxiety, depression, and stress, respectively; 40 (93%) share caregiving responsibilities. Nineteen answered the MOS-SSS, with positive association found between tangible support and division of care ($\rho=0.456$, $p<0.05$), and negative association between caregiving hours and emotional/informational support ($\rho=-0.644$, $p<0.01$). **Discussion:** Despite high burden and long caregiving hours, these conditions were not associated with severe anxiety, depression, or stress symptoms. Only stress was significantly associated with caregiver burden ($p=0.017$). Greater tangible support was associated with greater division of care, and greater emotional/informational support with fewer caregiving hours, suggesting that shared care and perceived social support may serve as protective emotional factors. Further research is needed to deepen these findings and guide interventions aimed at caregivers' well-being.

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Understanding quality of life in Alzheimer's: the role of awareness and emotional well-being

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Introduction: Quality of life (QoL) is a complex concept that can be influenced by subjective well-being. This study aimed to examine which clinical characteristics are associated with QoL evaluations across different levels of disease awareness, and to determine whether self-reported QoL in people with Alzheimer's disease is more closely related to emotional or cognitive/functional domains. **Methods:** We assessed the QoL of 179 individuals with Alzheimer's disease, categorized according to different levels of disease awareness: preserved awareness ($n=32$), mildly impaired awareness ($n=88$), and severely impaired/absent awareness ($n=59$). Participants

and their caregivers completed questionnaires assessing QoL, awareness of disease, cognition, dementia severity, functionality, neuropsychiatric symptoms, and depression. **Results:** Individuals with preserved awareness reported lower QoL than those with mildly impaired awareness. The group with severely impaired/absent awareness exhibited greater cognitive impairment ($p=0.010$), more frequent neuropsychiatric symptoms ($p<0.001$), greater depression severity ($p=0.001$), lower functional ability in activities of daily living ($p<0.001$), and higher QoL ($p<0.001$), compared to the other groups. In participants with preserved or mildly impaired awareness, QoL was negatively associated with the presence of neuropsychiatric symptoms ($p=0.029$ and $p=0.004$, respectively). In the group with severely impaired/absent awareness, QoL was associated with marital status ($p=0.002$). **Conclusions:** Awareness of disease may inversely influence self-perceived QoL. As awareness decreases, QoL becomes more closely associated with psychosocial factors and emotional well-being, whereas in individuals with greater awareness, functional aspects play a more prominent role.

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Is there a relationship of beliefs and attitudes towards older age with performance on global cognition in healthy older adults?

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Background: Cognition and beliefs about aging are important elements in the process of healthy aging. This study investigates the relationship between these variables in a sample of older adults. **Objective:** To examine the correlation between beliefs about old age and cognitive functions in older adults. **Method:** A cross-sectional study was conducted with 189 older adults. Data collection included sociodemographic variables, beliefs and attitudes towards aging, and cognitive performance (ACE-R, MMSE, SKT). Spearman correlation coefficients were calculated. **Results:** Participants were mostly female (76.72%) and white (73.02%), with a mean age of 67.33 years (± 4.95) and 17.12 years of education (± 4.77). No significant correlation was found between general beliefs about aging and global cognitive performance (ACE-R: $\rho=0.067$, $p=0.359$) or specific domains. However, age showed a negative correlation with cognition (ACE-R: $\rho=-0.274$, $p<0.001$), and education was positively associated with fluency ($\rho=0.166$, $p=0.023$) and language ($\rho=0.147$, $p=0.044$). **Discussion:** While beliefs about aging are conceptually important for understanding older adults' well-being, no direct relationship with cognitive function was found. The influence of age and education aligns with existing literature. Indirect effects of beliefs through psychosocial or behavioral factors should be explored in future studies. **Conclusion:** Beliefs about old age were not directly associated with cognitive performance. Age and education are relevant variables. Longitudinal designs may clarify causal pathways.

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Is there a relationship between cognitive performance and sense of general life satisfaction?

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Introduction: General life satisfaction (GLS) is a subjective evaluation of one's personal life trajectory and is associated with better health outcomes, including cognitive performance. Evidence suggests that higher levels of GLS contribute to healthier aging. **Objective:** To investigate the influence of GLS on cognitive performance in older adults over a 24-month period. **Method:** This was a longitudinal study with 60 participants enrolled in a multicomponent cognitive stimulation program. Sociodemographic data, GLS (measured using a ladder scale), and cognitive performance—assessed through the Addenbrooke's Cognitive Examination – Revised (ACE-R) and the Mini-Mental State Examination (MMSE)—were collected at baseline and after 24 months (T4). Analyses included descriptive statistics, the Wilcoxon test to compare cognitive performance between baseline and T4, and Spearman correlations to examine the relationship between GLS and cognitive outcomes. **Results:** There was a significant improvement in global cognitive performance (ACE-R, $p<0.001$), memory ($p<0.001$), verbal fluency ($p<0.001$), and attention/orientation ($p=0.004$). GLS was positively correlated with MMSE total score ($\rho=0.449$; $p<0.001$) and attention/orientation ($\rho=0.341$; $p=0.008$). No significant correlations were found between GLS and other cognitive domains such as memory, language, fluency, or visuospatial abilities. **Conclusion:** General life satisfaction was positively associated with improvements in overall cognitive performance, particularly in attention and orientation, among older adults undergoing cognitive stimulation. These findings highlight the potential impact of subjective aspects of quality of life on healthy cognitive aging.

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Home environments of multisensory stimulation based on Snoezelen principles: pilot study.

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Background: Multisensory Stimulation Interventions (MSI) aims to stimulate the senses such as sight, hearing, touch, taste, smell, vestibular and proprioceptive systems, which can be areas affected by dementia.

Studies with MSI in the home environment, both for older adults with dementia and for their family caregivers, are scarce. **Objective:** To evaluate the effects of MSI based on the principles of Snoezelen in home environments for older adults with dementia living in the community. **Materials and Methods:** MSI in home environments was carried out in 8 sessions, with a duration of 1.5 hours, weekly. The sample was allocated into two groups: experimental (EG n=10) and control (CG n=7). The impacts of the intervention were assessed by Neuropsychiatric Inventory (NPI) and Anxiety, Depression and Stress Scale (DASS-21) through analyses considering pre- and post-intervention and effect size. **Results:** Comparing the pre- and post-intervention through the NPI, there were an improvement of great magnitude in the EG in all the variables: total score in the NPI (d Cohen=0.90), Frequency (d Cohen=0.91), Intensity (d Cohen=-1.54), Caregiver distress (d Cohen=-1.68) and quantity of behaviors (d Cohen=-0.82). In the DASS-21 there was an improvement in Anxiety (Cohen's d=-0.96). **Conclusion:** The MSI intervention based on the principles of Snoezelen in home environments has a positive effect on reducing the frequency and intensity of BPSD in the older adults, as well as on the caregiver distress and anxiety of the caregiver family member.

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Evaluating episodic memory screening tests for differentiating mild and major neurocognitive disorders

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There is currently no consensus on the best tool to assess memory in outpatient behavioral neurology clinics, highlighting the need for easily administered and quickly scored assessment tools. This study utilized receiver operating characteristic (ROC) curve analysis to evaluate two episodic memory screening tests: the Word List Test (WL, from CERAD) and the Figure Memory Test (FMT), aiming to differentiate between Mild Neurocognitive Disorder (MnND) and Major Neurocognitive Disorder (MND). We compared the test performances of 26 patients diagnosed with MnND and 38 with different subtypes of MND, with no significant differences in age or education. All patients were diagnosed through clinical assessments and neuropsychological evaluations at a university hospital's neurology outpatient service. Mann-Whitney U tests were used for performance comparison, while ROC analyses determined the area under the curve (AUC), specificity and sensitivity values, with optimal cut-off scores identified using Youden's index. Results showed significant differences between the two groups for most subtests, except for the naming and recognition subtests of the FMT. The WL memory subtest had the highest accuracy, with an AUC of 0.80 (95%CI 0.69–0.91) and an optimal cut-off point below 10.5 to differentiate MnND from MND patients. Overall, the WL memory subtest showed the least likelihood of incorrect classifications, indicating its potential as a useful tool for identifying levels of neurocognitive impairment while underscoring the need for more accurate cognitive screening methods for differential diagnosis.

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Norms and diagnostic validity of the letter p verbal fluency in Brazilian older adults with heterogeneous educational levels

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Background: Verbal fluency tasks are widely used in neuropsychological assessment due to their sensitivity to a range of cognitive functions and are useful for detecting cognitive impairment. **Objective:** The aim was to provide normative data for the Letter P verbal fluency test in older adults with heterogeneous levels of education, using stratification by age and education and to assess the diagnostic accuracy of the task in differentiating dementia and cognitive impairment no dementia (CIND) from cognitively normal (CN) in this population. **Methods:** Participants were part of an epidemiological study in Tremembé, Brazil. A total of 246 individuals were assessed: 141 CN (mean age=70.1 years; education=6.09 years; MMSE=25.7), 68 CIND (mean age=72.2; education=5.76; MMSE=23.5), and 37 with dementia (mean age=77.5; education=4.00; MMSE=16.1). **Results:** ROC analysis revealed modest discriminatory power for identifying CIND (AUC=0.658; cut-off ≤8; sensitivity=53.62%; specificity=76.47%) and moderate accuracy for detecting dementia (AUC=0.726; cut-off ≤6; sensitivity=68.84%; specificity=72.41%). **Discussion:** The Letter P verbal fluency test showed moderate diagnostic accuracy in distinguishing dementia from CN but demonstrated limited utility in identifying CIND. **Conclusion:** Stratified normative data by age and education are essential for accurate interpretation in clinical and research settings that serve low-education populations and should prioritize culturally and educationally appropriate norms to improve diagnostic accuracy and promote equitable cognitive assessment.

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Coping styles among caregivers of young-onset Alzheimer's patients: associations with psychological and behavioral symptoms

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Background: Young-onset Alzheimer's disease (YOAD) affects individuals aged 65 or younger, leading to early cognitive decline in attention, visuospatial skills, executive functions, and language. Caregivers often experience emotional distress, with dysfunctional coping strategies exacerbating anxiety, depression, and stress. The relationship between coping styles, dementia severity, and caregiver well-being remains underexplored. **Objective:** To examine associations between coping strategies and psychological burden in caregivers with neuropsychiatric symptoms of YOAD patients. **Methods:** Thirty caregiver-patient dyads were assessed. Patients underwent evaluations for cognition, depression, and disease awareness. Caregivers completed standardized measures for depression,

anxiety, burden, coping, and resilience. They also reported on patients' neuropsychiatric symptoms, dementia severity (CDR), and functional abilities. Correlation analyses were conducted ($p < 0.05$). **Results:** Caregivers showed moderate burden ($M = 38.94$), low anxiety ($M = 9.11$), moderate coping ($M = 65.71$), and mild depression ($M = 10.72$). Patients had moderate dementia in 50% of cases, mild depression ($M = 12.35$), moderate neuropsychiatric symptoms ($M = 36.2$), and partial disease awareness impairment ($M = 9.06$). Problem-focused coping correlated positively with caregivers' education and cognition and negatively with anxiety. No significant associations were found for emotion-focused or dysfunctional coping. **Discussion:** Higher education and cognitive resources may enable caregivers to use more adaptive coping strategies. Dysfunctional strategies like self-distraction may signal emotional strain, despite lacking statistical significance. **Conclusion:** Dementia severity influences caregiver coping. Promoting problem-focused strategies may protect caregiver mental health. Further research should guide targeted interventions to enhance resilience and reduce burden.

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Supporting Alzheimer's caregivers: the critical role of coping strategies

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Background: Caring for individuals with Alzheimer's disease (AD) involves complex demands that can lead to high psychological distress in caregivers. Coping strategies—cognitive and behavioral efforts to manage stress—may influence caregivers' mental health and their response to patients' dysfunctional behaviors. **Objective:** To evaluate coping strategies used by caregivers of individuals with AD and examine their relationship with patients' behavioral and neuropsychiatric symptoms. **Methods:** Thirty-three caregiver-patient dyads diagnosed with AD were enrolled in the study. Patients were evaluated for cognition, functionality, depression, disease awareness, and quality of life. Caregivers completed measures of coping (Brief COPE), resilience, depression (BDI), anxiety (BAI), burden (Zarit), and locus of control. Caregivers also reported on patients' dementia severity (CDR), mood, neuropsychiatric symptoms (NPI), and daily functioning (Pfeffer). **Results:** Caregivers were mostly women (82%) with a mean age of 56.36. They presented moderate burden ($M = 37.61$), high resilience ($M = 141.9$), and moderate coping use ($M = 66.55$). Patients showed mild dementia in 60% of cases and moderate neuropsychiatric symptoms. Coping strategies positively correlated with resilience and internal locus of control and were associated with lower anxiety, depression, and burden in caregivers. **Discussion:** Emotion-focused coping appeared more effective for caregivers of patients with mild dementia, while problem-focused strategies were more beneficial in moderate stages. **Conclusion:** Tailored interventions that enhance adaptive coping and resilience may improve caregiver well-being and care quality.

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Transient global amnesia and emotional recognition memory

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Background: Transient Global Amnesia (TGA) is characterized by a sudden temporary inability to form new memories (anterograde amnesia), while other cognitive abilities remain preserved. The impact of emotional valence on recognition processes during the acute phase remains poorly understood. **Objective:** To examine whether TGA affects the recognition of emotional stimuli and whether such effects are associated with changes in response bias and confidence. **Methods:** An experimental recognition memory task with emotionally negative, neutral, and positive pictures as stimuli was administered during the ictal phase of 20 TGA patients and 20 healthy controls matched by sex, age, and education. After an initial presentation of all pictures for encoding, they were presented again among new pictures, and participants indicated whether each image was "old" or "new", and rated their confidence on their responses. Participants also completed cognitive assessments, including the Mini-Mental State Examination (MMSE), episodic memory tasks, semantic fluency, and digit span. **Results:** Compared to controls, TGA patients showed poorer performance on episodic and working memory tasks, as well as on the MMSE. TGA patients showed significantly lower recognition accuracy relative to controls. No effects of emotion on memory performance were found. TGA patients adopted a response bias more conservative than controls, indicating they might metacognitively observe their memory difficulties, and thus judged pictures as "old" only in rare occasions, resulting in fewer "false alarms." **Conclusion:** Although TGA affects recognition, this effect is not modulated by emotion. The response bias suggests preserved metamemory.

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Impact of social determinants of health on executive functions and language: an analysis from the promote study

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Background: Social determinants of health (SDH) are environmental factors linked to increased risk for several conditions, including dementia. However, their impact across different cognitive domains remains unclear. This study aimed to examine how SDH influence distinct cognitive domains in a South American population, addressing a gap in research predominantly based on European and North American cohorts. **Methods:** Baseline data

from the PROMOTE trial, conducted in Brazil between 2022 and 2023, were analyzed. Participants underwent clinical evaluations and the Montreal Cognitive Assessment (MoCA), with scores divided into six cognitive domains: Memory (MIS), Executive (EIS), Attention (AIS), Language (LIS), Visuospatial (VIS), and Orientation (OIS). SDH variables included years of education, ethnicity, family income, and neighborhood income. Regression models, adjusted for age and sex, assessed SDH impacts on total MoCA scores and subscores. **Results:** Among 147 participants (mean age: 59; mean education: 13.2 years), most were White ($n=139$). No SDH variable was associated with total MoCA scores. However, distinct associations were found for subscores. Years of education were significantly associated with EIS ($\beta=0.11$, p -adjusted $=0.002$), while lower family income was also linked to lower EIS ($\beta=-0.32$, $p=0.04$). Neighborhood income was significantly associated with LIS ($\beta=0.03$, $p=0.04$). No significant associations were observed between ethnicity and cognitive performance. **Conclusion:** SDH selectively influence cognitive domains. Education and family income affect executive function, while neighborhood income influences language. These findings highlight the importance of addressing socioeconomic disparities in cognitive health.

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Relationship between the use of memory strategies and cognitive performance in older adults without dementia or depression

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Introduction: Internal and external memory strategies can help compensate for age-related cognitive decline, potentially benefiting the cognitive performance of older adults. However, the relationship between the number of strategies used and cognitive performance still requires further investigation. **Objective:** To examine the association between the number of memory strategies used and cognitive performance in older adults. **Method:** This cross-sectional study included 189 participants. Sociodemographic variables were assessed, along with cognitive performance using the Addenbrooke's Cognitive Examination – Revised (ACE-R) and the Mini-Mental State Examination (MMSE), and self-reported use of memory strategies. Analyses included descriptive statistics and Spearman correlations between the number of strategies used and cognitive domains. **Results:** Participants used an average of 2.06 (± 0.97) memory strategies. The most frequently reported strategies were associating information (77.25%) and creating mental images (59.26%). No significant associations were found between the number of strategies and total cognitive scores (ACE-R: $p=0.211$; MMSE: $p=0.433$), nor with attention/orientation, memory, or verbal fluency domains. However, weak negative correlations were observed with language ($\rho=-0.183$; $p=0.012$) and visuospatial skills ($\rho=-0.150$; $p=0.040$). **Con-**

clusion: A higher number of memory strategies was not associated with better overall cognitive performance and showed negative correlations with language and visuospatial abilities. These findings suggest that the impact of memory strategies may vary depending on the type of cognitive task and individual differences in strategy use.

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SOMI score: contributions of the FCSRT-IR to the detection of cognitive decline

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Background: The Stages of Objective Memory Impairment (SOMI) system classifies performance on the Free and Cued Selective Reminding Test with Immediate Recall (FCSRT-IR) into stages which have been associated with Alzheimer's disease (AD) biomarkers. It relies on Immediate Free Recall (IFR, sum of three free recall trials) and Immediate Total Recall (ITR free and cued recall). **Objective:** To investigate the correlation between SOMI scores and delayed recall on the Rey Auditory Verbal Learning Test (RAVLT) in cognitively unimpaired (CU) participants, with Subjective Cognitive Decline (SCD), and amnesic Mild Cognitive Impairment (aMCI). **Method:** 59 participants (43 women, 16 men), 15 CU, 24 DCS and 20 aMCI were assessed with the FCSRT, RAVLT and additional tests. Clinical diagnosis was reached by experienced neurologists according to international criteria. **Results:** Mean age was 74.0 (± 7.6) and mean education was 12.7 (± 4.5). Groups differed significantly in education but not age. SOMI scores differed significantly across groups ($p=0.003$), with higher frequencies of SOMI 1 and 2 in SCD and aMCI. SOMI scores were significantly associated with RAVLT delayed recall ($p=0.006$), with SOMI 0 participants showing mostly average or above-average scores in the RAVLT, while SOMI 1 and 2 more frequently had below-average performance, indicating greater memory impairment. **Conclusion:** Results support the utility of the SOMI classification in distinguishing stages of memory impairment across clinical groups.

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Psychoeducation focusing on cognitive stimulation for caregivers of elderly people diagnosed with Alzheimer's disease

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Introduction: The literature reports that caregivers of older people with dementia face major challenges concerning mental health and wellbeing. **Objectives:** To implement a psychoeducational intervention in combination with

cognitive stimulation for caregivers of people with dementia. **Methods:** A total of 45 caregivers took part, randomly stratified by age and education into 2 groups: a Training Group (TG) with 22 participants; and a Control Group (CG) with 23 participants. Protocol: the sociodemographic and health questionnaire, the Alzheimer's Disease Knowledge Scale (ADKS), the Mini-Mental State Exam for telephone application (MEEM-Braztel), the Subjective Cognitive Decline Questionnaire (SCD-Q), the Geriatric Depression Scale (GDS-15), the Geriatric Anxiety Inventory (GAI), the Depression, Anxiety and Stress Scale (DASS-21), the Caregiver Burden Scale (ZARIT), the Control, Autonomy, Self-Realization and Pleasure (CASP-19) scale, questionnaire on positive and negative mood states, and general Satisfaction with life domains scale. **Results:** The caregivers in the TG had positive impacts for some aspects after participating in the intervention, such as for the perceived quality of life variable, as measured by the CASP-19, the CG showed a decrease of -3.52 and the TG an increase of 1.86 ($p=0.014$). On the ADKS for the domains "Treatment and Management" ($p=0.005$) and "Course" ($p=0.056$). **Conclusion:** The study found that caregivers who participated in the 12 intervention sessions held greater knowledge on Alzheimer's Disease, showing better management for dealing with dementia-related situations.

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Relationship of levels of stress, depression and anxiety with purpose in life among active socially-engaged older adults

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Introduction: Purpose in life (PiL) has been associated with positive mental health outcomes and well-being in older adults. High levels of stress, anxiety and depression can adversely impact this sense of purpose, a relationship which warrants further investigation. **Objective:** To examine the association of levels of stress, anxiety and depression (DASS-21) with PiL in a sample of healthy older adults. **Method:** A cross-sectional study of 189 participants (76.72% women; mean age=67.33±4.95 years) was conducted. Sociodemographic data were collected and the Purpose in Life Scale (Total PiL) and DASS-21 scale (total and subdomains) were applied. Spearman correlations were used to analyze the relationships between variables. **Results:** Purpose in life showed a significant negative correlation with total levels of stress, anxiety and depression. The strongest correlation was between PiL and depression ($\rho=-0.499$; $p<0.001$), followed by stress ($\rho=-0.264$; $p<0.001$) and anxiety ($\rho=-0.193$; $p=0.008$). The relationship with total score on the DASS-21 was also significant ($\rho=-0.381$; $p<0.001$), indicating that higher levels of psychological distress are associated with lower sense of PiL. **Conclusion:** The findings suggest that healthy older adults with higher levels of stress, anxiety and depression tend to report lower purpose in life. This result underscores the importance of interventions aimed at improving mental health and strengthening sense of purpose as a means of promoting well-being in this population.

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Social cognition indicators in behavioral variant frontotemporal dementia, Alzheimer's disease and bipolar disorders: preliminary results

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Background: Social cognition (SC) and social decision-making (SDM) are relevant for distinguishing cognitive profiles in older adults, considering the overlapping cognitive deficits in conditions such as behavioral variant frontotemporal dementia (bvFTD), Alzheimer's disease (AD), and bipolar disorder (BD). **Objectives:** To investigate the diagnostic accuracy of the Ultimatum Game (UG), Reading the Mind in the Eyes Test (RMET), and Mini-SEA in differentiating patients with bvFTD, AD, BD, and healthy controls (HC). **Methods:** A total of 48 HC, 24 bvFTD, 26 AD, and 10 BD patients were assessed using the UG, RMET, and Mini-SEA. **Results:** In the RMET, participants' performance was HC>BD>AD>bvFTD, $p>0.001$. In the facial emotion recognition test, HC performed best, followed by BD, with bvFTD showing the lowest scores (HC>BD>AD>bvFTD). In the Faux Pas test, HC and BD showed similar and better performance than AD and bvFTD (HC=BD>AD>bvFTD). In the Mini-SEA, HC outperformed all groups, followed by BD, AD, and bvFTD (HC>BD>AD>bvFTD). In the UG (baseline), HC and BD accepted more fair offers than AD and bvFTD (HC=BD>AD=bvFTD); bvFTD accepted more unfair offers (bvFTD>BD>AD=HC). In the prosocial condition, HC and BD accepted more unfair prosocial offers (HC=BD>AD>bvFTD); no group differences emerged for fair prosocial offers. For fair punishment offers, HC had the highest acceptance rate (HC>AD=BD>bvFTD); for unfair punishment offers, HC and BD accepted more than AD and bvFTD (HC=BD>AD>bvFTD). **Conclusion:** SC may contribute to the differential diagnosis between bvFTD and BD, remains limited due to small BD sample size.

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The frontal assessment battery as a clinical tool in the assessment of patients with primary progressive aphasia

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Background: Primary Progressive Aphasia (PPA) is a neurodegenerative syndrome characterized by alterations in linguistic components. However, cognitive deficits are also found in different domains, such as in the executive functions, from the early stages of the condition, challenging the previous understanding that such impairments could be secondary to language. Therefore, exploring executive function can assist in the differential diagnosis and shed light on the cognitive network affected by PPA. **Objective:** To assess executive function by using the Frontal Assessment

Battery (FAB) in patients with PPA and comparing their results with other neurodegenerative diseases. **Methods:** Different clinical groups were compared: APP (n=19); Frontotemporal Dementia (FTD n=23); Progressive Supranuclear Palsy (PSP n=23); Amyotrophic Lateral Sclerosis (ALS n=15); Alzheimer's Disease (AD n=20) and a healthy control group (HC n=23). Group comparisons and post-hoc analysis were performed using Jasp software.

Results: The performance of the PPA (M=8.90, SD=3.81) group was significantly lower than the HC (M=15.74, SD=1.66, $p<0.001$), ALS (M=13.2, SD=3.01, $p=0.021$), and AD (M=13.1, SD=2.69, $p=0.039$) groups, but similar to the FTD (M=11.23, SD=3.99, $p=0.637$) and PSP (M=11.30, SD=3.25, $p=0.637$) groups. **Discussion:** The similar executive performance observed in patients with PPA, FTD and PSP reinforces how such conditions affect similar cognitive networks, and demonstrates that dysexecutive syndromes may not be secondary to language dysfunction. **Conclusion:** FAB shows to be a useful tool for differentiating PPA from HC and other neurological conditions, such as DA and ALS.

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Executive functions and social support network of older adults without dementia or depression

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Introduction: Executive functions (EF) are key for autonomy in aging. Social support may help preserve EF, offering cognitive and emotional benefits. This study examined links between EF and social relationships in older adults. **Objective:** To analyze the association between executive functions and social support, measured by MMRI entries, in a sample of older adults.

Method: Cross-sectional study with 189 older adults (mean age=67.3). EF was assessed with standardized tests, and social support was measured via MMRI. Spearman's correlation was used. **Results:** Higher MMRI entries correlated with better EF performance (FAS and composite score) and shorter Trail B time, suggesting that greater social support is linked to better executive functioning. **Conclusion:** Better EF, especially verbal fluency and cognitive control, was linked to greater social support. Findings highlight the role of cognition in sustaining social ties in older age.

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The use of artificial intelligence in investigating social cognition in patients with dementia

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Population aging has led to a rising prevalence of dementias, particularly dementia related to Alzheimer's Disease. Social cognition, a key domain in neuropsychological assessment of AD, is linked to early stages of cognitive decline. When evaluated early, it becomes a valuable tool for screening cognitive complaints, but clinical application remains unclear. To apply artificial intelligence methods to the integrated analysis of neuropsychological, electroencephalographic, and clinical data to identify patterns of social cognition impairment throughout progression from Mild Cognitive Impairment to AD. Older adults diagnosed with MCI and AD were assessed using the Mini-SEA, MMSE, and EEG. A total of 18 participants aged between 60 and 80 years were included, with 61.1% female and 38.9% male. Data were analyzed using AI algorithms—Support Vector Machines (SVM) with polynomial and RBF kernels, and Random Forests with 400 and 500 trees. SVM classifiers (polynomial and RBF) and Random Forest models (400 and 500 trees) achieved outstanding performance (accuracy ~100%, kappa=1, AUC=1), demonstrating precision in distinguishing between groups. These findings highlight the potential of AI in early detection of social cognition deficits across the cognitive decline continuum. The classifiers showed robust performance in differentiating MCI from AD, indicating high sensitivity of this AI-based approach. Integrating clinical, neuropsychological, and electrophysiological data through AI proved effective in identifying social cognition impairment. AI-driven models integrating multidimensional data effectively distinguished between MCI and AD, suggesting a promising tool for early screening of social cognition in older adults.

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Cross-cultural validation of the memory alteration test for detecting amnesic mild cognitive impairment in low-educated older Brazilians

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Reliable instruments for early cognitive decline screening in low-literacy populations remain scarce. To adapt the Memory Alteration Test (M@T), a brief episodic memory-focused screening tool, to Brazilian Portuguese and evaluate its psychometric performance in older adults with ≤ 12 years of schooling. Following translation, back-translation, expert review, and cognitive debriefing, we enrolled community-dwelling adults ≥ 55 years, classifying them as cognitively unimpaired controls or amnesic mild cognitive impairment (aMCI) through multidisciplinary consensus. The M@T (maximum=50) and complementary battery including California Verbal Learning Test-II, Symbol Digit Modalities Test, Trail-Making Tests A/B were administered. Thirty-nine participants (32 aMCI, 7 controls) completed the protocol; mean age 66 ± 6 years, median schooling 6 years. Internal consistency showed Cronbach's $\alpha=0.68$. M@T total scores were 34.7 ± 4.7 for aMCI versus 42.9 ± 2.5 for controls (Welch's $t=6.5$, $p<0.001$), with large effect size (Hedges' $g=2.1$). Controls significantly outperformed aMCI participants on CVLT-2 total learning (43.7 ± 8.6 vs. 35.0 ± 9.5), SDMT correct pairs (33.3 ± 3.1 vs. 22.5 ± 10.2), and Trail-Making A completion time (58.4 ± 13.1 s vs. 85.2 ± 35.5 s). Convergent validity demonstrated significant correlations with CVLT-2 total learning

($r=0.35$, $p=0.03$) and SDMT ($r=0.53$, $p<0.001$). The Brazilian M@T effectively discriminated aMCI from cognitively intact peers despite limited formal education, demonstrating clinical utility for screening in low-education populations with adequate psychometric properties.

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Profile of older adults diagnosed with dementia: functionality, multimorbidity, and mental disorders

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Background: According to the WHO, it is estimated that more than 55 million people live with dementia, making it one of the leading causes of disability and dependency among older adults. Furthermore, up to 90% of people with dementia are estimated to have multimorbidity, which increases the risk of functional decline. **Objectives:** To analyze functional levels in older adults diagnosed with dementia, multimorbidity, and mental disorders. **Methods:** This was a cross-sectional study involving 41 older adults with a caregiver-reported diagnosis of dementia, including Alzheimer's disease, Parkinson's disease, Lewy body dementia, frontotemporal dementia, or vascular dementia. Data were collected using a standardized online questionnaire completed by caregivers. The variables analyzed included the total number of chronic diseases, diagnosis of dementia and mental disorders, as well as basic functionality (Katz Index) and instrumental functionality (P-FAQ). **Results:** Among the 41 older adults with dementia, 12 (30%) had a diagnosis of depression or anxiety, and 39 (95%) had at least one diagnosed chronic disease. Regarding functionality, 33 (79%) had scores consistent with functional dependence in instrumental activities of daily living, while 24 (57%) showed some level of impairment in basic activities. **Discussion:** The results indicate that most older adults with dementia present with multiple comorbidities, with a high prevalence of chronic diseases and depressive or anxious symptoms. Additionally, there was significant functional impairment, especially in instrumental activities of daily living, highlighting the need for ongoing care and multidimensional support.

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Free and cued selective reminding test with immediate recall: a resilient test for neuropsychological assessment in a neurological service

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Background: Neuropsychological assessment in outpatient settings poses significant challenges due to case heterogeneity, low patient education, limited evaluation time, and scarce resources. Identifying the most suitable tools is crucial for accurate screening and diagnosis. **Objectives:** This study assessed a neuropsychological protocol to identify the most accurate instruments. **Methods:** Data from 34 patients: 19 men; age 66.2 ± 9.9 years; education 9.8 ± 4.9 years with cognitive complaints. Participants were categorized as: subjective cognitive decline (SCD, $n=8$), mild cognitive impairment (MCI, $n=14$), and dementia ($n=12$). No statistically significant differences in age, sex, or education. Cognitive assessment: MMSE, ACE-R, Digit Span, FCSRT-IR, Verbal Fluency (VF; Animals, Fruits, FAS), Five Digit Test, Block Design Test, the Semantic Memory Assessment Battery, and Ekman's Facial Emotion Recognition Test. Statistical analyses included ANOVA, Kruskal-Wallis, chi-square, Spearman correlation, and regression. **Results:** Only VF Fruits (dementia<SCD; $p<0.05$) and FCSRT-IR scores differed significantly between groups. FCSRT-IR free and total recall scores, as well as free/guided/total delayed recall, effectively distinguished SCD from dementia. FCSRT-IR scores showed mild to moderate correlations with memory and executive measures. Guided recall was more associated with executive performance ($p<0.01$). Education explained less than 11% of the variance in Free Delayed Recall ($p=0.034$). **Discussion:** The FCSRT-IR is a robust cognitive measure in heterogeneous populations, engaging both memory and executive domains. **Conclusion:** The FCSRT-IR's low educational bias reinforces its suitability for neurological outpatient settings.

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Life satisfaction among active socially engaged community-dwelling older adults from São Paulo city, Brazil

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Abstract: National and international literature reports a high prevalence of depressive and anxious symptoms among older adults, negatively affecting their mental health and well-being, particularly in contexts of social vulnerability. These factors also impact cognition and social interactions. **Objective:** To investigate associations between mental health and cognitive performance in active, socially engaged older adults residing in São Paulo. **Method:** A cross-sectional study was conducted through telephone interviews with 576 older adults. Instruments used were the Brazilian telephone version of the Mini-Mental State Examination (Braztel-MMSE), the 15-item Geriatric Depression Scale (GDS-15), the Geriatric Anxiety Inventory (GAI), and a single question on mood and/or anxiety changes. **Results:** Participants had a mean age of 67.7 ± 5.6 years, were predominantly female, married, and had 15.2 ± 3.8 years of education. Mean scores were 20.3 ± 1.5 on the Braztel-MMSE, 2.6 ± 2.3 on the GDS-15, and 3.6 ± 4.1 on the GAI. Approximately 70.5% reported mood and anxiety changes related to factors such as community insecurity and social inequalities. A significant correlation between depressive symptoms and cognitive performance was observed only among those reporting such changes ($\rho=-0.14$; $p=0.004$). **Conclusion:** Older adults living in social vulnerability, with chronic health conditions, and lacking protective factors—higher income, greater educational attainment, lower medication use, and fewer self-reported diseases—are at increased risk of psychological disorders. These findings underscore the need for targeted public policies that provide comprehensive care for the older population, especially in challenging contexts such as the pandemic.

NEUROPSYCHIATRY

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Cogniform disorder: expanding the spectrum of cognitive presentations in generalized anxiety disorder

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Patient 1, a 52-year-old female, reported recurrent episodes of topographical disorientation since adolescence and difficulties with working memory under stress, while maintaining functional capacity at work and in daily activities. Clinical evaluation revealed anxiety and fluctuating attention, accompanied by impairments in working memory and sustained attention. Patient 2, a 45-year-old female with a history of depression and Generalized Anxiety Disorder (GAD), reported memory lapses and difficulties with verbal recall, yet preserved functional autonomy. Neuropsychological assessments indicated deficits in attention, visual memory, and planning, whereas language and praxis remained intact. The patient also exhibited behavioral impulsivity. Patient 3, a 43-year-old male programmer with recent polytrauma, complained of memory and attention alterations. Neuroimaging findings were unremarkable; however, cognitive testing demonstrated deficits in attention, processing speed, and executive functions, alongside impulsive behaviors. Generalized Anxiety Disorder may adversely impact cognitive functioning, particularly attention, working memory, and executive functions. This case series describes patients diagnosed with GAD who presented cognitive complaints without corroborating neuropsychological test impairments. Cognitive dimensions of anxiety disorders have been associated with diminished performance in higher-order neural functional domains. Literature suggests that anxiety correlates with deficits in inhibitory control and reduced performance on tasks requiring cognitive control, thereby reflecting executive function alterations. In the aforementioned cases, a diagnosis of Cognitive Disorder Not Otherwise Specified (Cogniform Disorder) was considered, with GAD exacerbating the reported cognitive symptoms. These findings contribute to a deeper understanding of the cognitive implications of GAD and underscore the importance of integrating emotional factors within the differential diagnostic process.

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Characterization of cognitive patterns and Alzheimer's disease in attention deficit hyperactivity disorder

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Background: The way ADHD relates to Alzheimer's has been increasingly

studied, with some studies indicating that there may be a direct relationship between these two conditions **Objective:** To characterize how ADHD is associated with cognitive decline and AD and how individuals with ADHD and AD are described along the clinical continuum between subjective cognitive decline (SCD), mild cognitive impairment (MCI), and AD. To evaluate genetic and environmental factors involved in the relationship between ADHD and AD **Methods:** A longitudinal, prospective, and observational cohort study will be carried out on 500 patients from the city of São Paulo with memory complaints from the age of 55. Patients will be followed for 5 years. Structured interviews will be conducted for cognitive complaints and ADHD (DIVA-5, CDR, ADASCog). Blood samples will also be collected, and the polygenic risk score for ADHD and Alzheimer's will be evaluated. **Results:** Ongoing study **Discussion:** The scientific literature on this topic has shown that close relatives of people with ADHD may have a higher chance of Alzheimer's disease, and that patients with biological markers of Alzheimer's and polygenic risk score have a worse decline in executive function. However, there is still much to be understood about this topic. **Conclusion:** In this ongoing cohort study, we hope to contribute to a better understanding of the pattern of patients on the cognitive continuum with ADHD and the relationship between these two comorbidities.

TECHNICAL AREA: NEUROPSYCHIATRY
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Feasibility, tolerability, and preliminary effects of combined tACS and digital cognitive training in older adults with mild cognitive impairment

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Mild Cognitive Impairment (MCI) represents a critical window for intervention in the trajectory of cognitive decline leading to Alzheimer's disease. Non-pharmacological strategies such as digital cognitive training (DCT) and transcranial alternating current stimulation (tACS) have shown promise for enhancing cognitive function in older adults. This pilot study aimed to assess the feasibility and tolerability of combining tACS with DCT in individuals with MCI, while also exploring preliminary cognitive and neurophysiological outcomes. Ten participants underwent five consecutive daily sessions of 6 Hz tACS applied over the left dorsolateral prefrontal cortex (DLPFC) while engaging in DCT. Feasibility was evaluated through adherence and attrition rates, and tolerability was assessed via self-reported side effects. Cognitive performance and oculomotor behavior were measured before and after the intervention. All ten participants completed the protocol, with mild side effects reported in 67% and moderate in 23% of sessions. No severe adverse events were observed. No correlation was found between side effect intensity and DCT performance, suggesting that tACS did not impair engagement in cognitive training. Improvements were observed in episodic memory (RAVLT, $p=0.03$), visuospatial construction and recall (ROCF copy and recall, $p=0.02$ and $p=0.04$), and in inhibitory control as indicated by reduced antisaccade errors ($p=0.002$). These preliminary findings suggest that the combined tACS+DCT intervention is feasible, well tolerated, and may support cognitive enhancement in older adults with MCI. Further research with larger samples and randomized controlled designs is warranted to evaluate its efficacy and long-term impact.

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Behavioral symptoms of dementia and their association with caregiver burden

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Background: In Brazil, caregiving by family members is predominant and caregivers often experience high levels of psychological, emotional, and financial distress. Neuropsychiatric symptoms contribute to caregiver burnout. **Objective:** Identify the most burdensome neuropsychiatric symptoms reported by caregivers of dementia patients in a tertiary clinic and examine the relationship between neuropsychiatric symptoms and caregiver burden. **Methods:** Participants answered the RMBPC and BEHAVE-AD scales and the Zarit Caregiver Burden Scale, as well as demographic characteristics. Data was analyzed using descriptive statistics at this point. **Results:** The project is ongoing, and to date, 15 interviews have been conducted, of which 11 were with female caregivers, aged 30 to 72 years. Most participants were the patients' children. The most frequently reported neuropsychiatric symptom on the BEHAVE-AD scale was visual hallucinations, and the scores on the Zarit Caregiver Burden Scale ranged from 12 to 62 (maximum score of 88). The behaviors most frequently reported on the RMBPC scale as having a moderate to severe impact on the caregiver were repeated questions and behavior dangerous to self or others. **Discussion:** So far, the interviews confirmed the impact of neuropsychiatric symptoms on caregiver's well-being. It is important for clinicians and researchers to understand behavior issues faced by patients and the ones that are most troublesome for caregivers as well. **Conclusion:** This study aims to shed light on the reality in a Brazilian reference center and has contributed to a better understanding of patient's reality at home and caregiver's burden.

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Behavioral changes in patients living with dementia and their relationship with caregiver burden

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Background: Dementia is a progressive neurodegenerative disease that often causes significant behavioral and psychological symptoms, contributing to increased suffering for caregivers. **Objective:** This qualitative study aims to investigate the relationship between behavioral changes in individuals with dementia and the perceived burden of their primary caregivers in a public hospital in Niterói, Brazil. **Methods:** To date, ten focus groups have been conducted with family caregivers, with an average of 2 to 3 participants per group, totaling approximately 25 participants. **Results:** Preliminary results indicate that behavioral symptoms such as aggression, agitation, apathy and hallucinations are strongly associated with caregivers' emotional overload and physical fatigue. Participants report feelings of helplessness, social isolation and psychological distress, especially in the face of unpredictable or socially inappropriate behavior. **Discussion:** These findings highlight the importance of support strategies that address both the physical demands and the emotional distress resulting from the behavioral symptoms of dementia.

Psychoeducational interventions and emotional support mechanisms can help reduce the burden on caregivers and improve the quality of care provided. **Conclusion:** This research contributes to understanding the impact of behavioral symptoms of dementia on the well-being of caregivers and the reality faced by them in a large city of Rio de Janeiro.

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Cognitive and neuropsychiatric changes related to covid-19 in patients evaluated at a referral center in São Paulo, Brazil.

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Background: Cognitive dysfunction and brain fog, characterized by fatigue, impaired memory and lack of concentration, are one of the most common symptoms of long covid. **Objectives:** To evaluate cognitive and neuropsychiatric changes in patients with neurological disease after COVID-19. **Methods:** A longitudinal prospective study designed to compare the cognitive performance in patients after COVID-19, based on cognitive complaints. Patients aged 18 to 65 years old (y.o.) without a history of neurological or psychiatric disorder were included, illiterate individuals were excluded. The Beck Anxiety and Depression Inventories and the Addenbrooke's Cognitive Examination-Revised (ACE-R) were applied at 2 moments—0 and 6 months. **Results:** Fifty-eight patients were included, 69% women, 70.7% with cognitive complaints. The mean age was 46.1 y.o. (SD 9.8), median of 45.50 y.o., mean educational level was 14.2 years (SD 3.9), median of 15 years. There was no difference in cognitive complaints whether the patient was hospitalized or not because of COVID-19. In the first assessment, patients with cognitive complaints had higher scores on the depression ($p=0.045$) scale and in the third assessment, patients with cognitive complaints had higher scores on anxiety ($p=0.026$) and depression ($p=0.018$) scales. In the ACE-R score, regardless of cognitive complaints and education, patients had a lower performance than the standard for the Brazilian population in the 2 assessments, but with improvement after 6 months—education ≥ 12 years ($p=0.019$) and education 4–11 years ($p=0.020$). **Conclusions:** Cognitive complaints after COVID-19 infection were not a predictor of cognitive decline, but were related to neuropsychiatric changes.

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Exploring the interface of psychiatric symptoms and cognitive decline: a retrospective study from a reference center in Recife, Brazil

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Psychiatric symptoms, such as depression, anxiety, and behavioral changes, are common and may precede or accompany cognitive decline, representing a significant clinical challenge. This retrospective study aimed to describe the sociodemographic profile of patients with psychiatric manifestations and cognitive decline who were treated at a specialized neurocognition outpatient clinic. A total of 254 medical records were reviewed for patients followed between January 2022 and April 2025 at a university hospital in the Northeast region of Brazil. Data collection included socio-demographic information (age, sex, educational level, marital status, and place of origin) and the etiological diagnosis recorded by the attending physician. Clinical manifestations were documented during specialized evaluations, with the Mini-Mental State Examination (MMSE) applied as a cognitive screening tool. Depressed mood was identified in 50.95% of patients, anxiety in 27%, and sleep disturbances in 46.26% of cases. The overall average MMSE score was 17.30. The majority of patients were female (63.8%), of mixed race (72%), and had low educational levels. Most were residents of Recife and the surrounding metropolitan area. In 204/254 cases, the cognitive impairment was classified as secondary to a major psychiatric condition. The findings underscore the importance of recognizing and characterizing psychiatric manifestations associated with cognitive complaints, especially in specialized outpatient settings where the overlap between neurological conditions and mental disorders may complicate diagnosis. The study highlights the need for integrated clinical approaches to better understand the interface between psychiatric symptoms and cognitive decline in neurology clinics, thereby promoting more appropriate diagnostic and therapeutic strategies.

TECHNICAL AREA: NEUROPSYCHIATRY

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Behavioral and psychological symptoms of dementia associated with vascular impairment: a systematic review and meta-analysis

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Background: Behavioral and psychological symptoms of dementia (BPSD) are common in vascular cognitive impairment (VCI). **Objective:** Analyze the prevalence of BPSD in different VCI subtypes. **Methods:** We conducted a systematic review and meta-analysis per PRISMA guidelines. Eligible studies were retrieved from PubMed, Embase, and Web of Science and they used the Neuropsychiatric Inventory (NPI) to assess symptoms. A random-effects model calculated pooled prevalence rates for each NPI domain. **Results:** Thirty-five studies encompassing 5,805 patients were included. In unspecified VCI, apathy (54.29%), depression (43.48%), and irritability

(38.76%) were the most common BPSD. Subcortical VCI showed a similar profile, with more frequent apathy (62.01%), depression (52.11%), and irritability (44.73%). Mixed dementia presented distinctively high apathy (61.65%), depression (45.68%), and sleep disturbances (44.63%), with elevated hallucinations (26.64%) compared to other VCI types. VCI non-dementia (VCI-ND) presented depression (44.97%), irritability (32.75%), and anxiety (30.07%) as the most common symptoms. **Discussion:** Hallucinations and sleep disturbances were more common in mixed cases, likely reflecting overlapping vascular and neurodegenerative pathology. High prevalences of apathy and agitation in subcortical VCI may reflect vascular burden. Depression was common across all groups. In VCI-ND, apathy, disinhibition, hallucinations, and agitation/aggression rates were significantly lower compared to other groups, suggesting an association of these symptoms with dementia progression and possibly increased vascular burden. **Conclusion:** BPSD are prevalent across VCI subtypes, with distinct profiles suggesting varying underlying mechanisms.

TECHNICAL AREA: NEUROPSYCHIATRY

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Cognitive, functional and socioemotional impairment in patients with brain microangiopathy: a case series

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Background: Cerebral microangiopathy is a major etiology of cognitive decline in older adults, impacting not only cognitive functions but also daily functionality and social cognition abilities. **Objective:** To describe the cognitive, functional, and social cognition profile of patients with microangiopathic lesions identified on neuroimaging. **Methods:** This is an observational, cross-sectional case series. Ten patients with neuroimaging findings of cerebral microangiopathy (Fazekas ≥ 1) were evaluated. Cognitive and functional assessments included MMSE, CDR (global, sum of boxes, and functional), Pfeffer Functional Activities Questionnaire, and Mini-SEA (Faux-Pas and Ekman Facial Emotion Recognition). **Results:** The mean age was 71.6 years, with an average education of 9.8 years. Seven patients had Fazekas 3, two Fazekas 2, and one Fazekas 1 (perivascular enlargement). Mean MMSE was 24. Average scores were: CDR global 1.1, CDR-SB 5, CDR functional 3.31, Pfeffer 9, Faux-Pas 20.4, and Ekman 23.5/35, with the greatest impairment in negative emotions recognition. Patients with higher microangiopathic burden exhibited lower Ekman scores, suggesting an association between lesion severity and social cognition impairment. **Discussion:** Findings indicate that cerebral microangiopathy impacts not only cognitive and functional domains but also social cognition, particularly in recognizing negative emotions as fear, anger, and sadness. This aligns with evidence of cortico-subcortical and frontostriatal network dysfunction frequently associated with vascular burden. **Conclusion:** Cerebral microangiopathic lesions may significantly affect cognition, functionality, and social cognition, reinforcing the importance of comprehensive assessment in this population.

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Efficacy of rTMS in primary dementias: a meta-umbrella review of cognitive and behavioral outcomes

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Background: Neurodegenerative dementias are a growing global burden, with Alzheimer's disease (AD) as the leading cause. Repetitive transcranial magnetic stimulation (rTMS) has emerged as a promising non-pharmacological intervention for cognitive and behavioral symptoms. **Objective:**

To evaluate the efficacy of rTMS in primary neurodegenerative dementias through an umbrella review of meta-analyses. **Methods:** This PRISMA-compliant review was registered in PROSPERO (CRD42023489100). A systematic search was conducted in PubMed/MEDLINE, Embase, SciELO, Web of Science, and Cochrane CENTRAL. Two reviewers independently selected studies, extracted data, and assessed quality using AMSTAR-2, ROBIS, and GRADE. A meta-umbrella synthesis was performed using R software. **Results:** From 2,288 records, 16 meta-analyses on AD were included. For global cognitive and behavioral outcomes (16 studies; n=350), rTMS showed a small but significant effect (eG=0.40; 95%CI 0.19–0.60; I²=42.9%) with moderate certainty. Behavioral outcomes (14 studies; n=461) showed a moderate effect (eG=0.37). Cognitive outcomes (39 studies; n=2,896) showed a large effect (eG=0.60), but high heterogeneity (I²=88.1%) and very low certainty. **Conclusion:** High-frequency rTMS over the left dorsolateral prefrontal cortex appears effective for cognitive and behavioral symptoms in AD. Despite encouraging results, especially for cognition, further standardized trials and robust reviews are needed to strengthen the evidence and guide recommendations for broader dementia subtypes.

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